MODULI OF GIESEKER (SEMI)STABLE SHEAVES

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ABSTRACT. The MVC notes are about moduli of vector bundles on curves. They follow a topics course taught by Riccardo Carini at Bonn, Summer semester 2025, titled "Moduli of vector bundles on curves and Hitchin systems." The GMS section is more general and is about moduli spaces of semistable sheaves, following [HL]. At some points, proofs in MVC are subsumed as special cases of proofs given in the GMS section.

CONTENTS

1. MVC 4/9/25: Defining a moduli functor

For the first three to four classes we will focus on more general moduli theory. Suppose we have a collection of geometric objects that we'd like to classify. For example, it could be lines in a plane or vector bundles on a curve. While this collection starts off as a set, we will also have some notion of families of such objects in the set. We'd have a notion of when two objects can be deformed to one another, or are "closed to each other." We would like for the geometry of our set to reflect such families of objects, and we encode this geometry through a moduli functor.

Here is some philosophical motivation. Suppose we'd like to classify lines in $\mathbb{C}^2$ going through the origin. Immediately, one should think of $\mathbb{P}^1$. But in what sense is $\mathbb{P}^1$ a moduli space? How does the geometry of $\mathbb{P}^1$ reflect families of lines in $\mathbb{C}^2$?

Let's consider these questions in terms of deciding the topological structure of $\mathbb{P}^1$. For example, there is a sequence of lines $\ell_n = \{Y = nX\}$ in $\mathbb{C}^2$ such that in our minds geometrically we can see they should 'converge' to the line $\ell_\infty = \{X = 0\}$. This means that whatever topological structure we endow $\mathbb{P}^1$ with, for it to be consistent with our geometric intuition, we should have the convergence

$$\lim_{n\to\infty} [1 : n] = [0 : 1] \in \mathbb{P}^1.$$

In other words, there should be a continuous map from the one point compactification of the natural numbers to $\mathbb{P}^1$:

$$\mathbb{N} \cup \infty \to \mathbb{P}^1,\ n \mapsto \ell_n.$$

Similarly, the map $\mathbb{C} \to \mathbb{P}^1$ where $z \mapsto [z : 1]$ should also be continuous. Then endow $\mathbb{P}^1$ with its usual topology. The claim is that all the continuous maps $X \to \mathbb{P}^1$... if we suddenly had amnesia and forgot the topology of $\mathbb{P}^1$ but remembered these maps as maps of sets, would uniquely determine the topology on $\mathbb{P}^1$. Thus, the geometry of $\mathbb{P}^1$ as a topological moduli space for the lines in $\mathbb{C}^2$ through the origin, is determined by all the maps $T \to \mathbb{P}^1$ which give rise to a family of lines parameterized over T. In line with this philosophy, we claim that the scheme structure of $\mathbb{P}^1$ is determined by all the maps $T \to \mathbb{P}^1$ for every $\mathbb{C}$-scheme T.

Definition 1.1. *A moduli functor $\mathcal{M}$ is a presheaf of sets on the category of $\mathbb{C}$-schemes; in other words, a contravariant functor*

$$\mathcal{M} : (Sch_{\mathbb{C}})^{op} \to Sets$$

Here we are saying: okay, suppose we have a moduli problem in algebraic geometry over $\mathbb{C}$. We want to classify some geometric objects in a set A. In order to get a moduli space, at minimum we should have a notion of families of such objects parameterized by a base $\mathbb{C}$-scheme T. The collection of the families over T is denoted $\mathcal{M}(T)$. So the objects we are classifying are $\mathcal{M}(\operatorname{Spec}\mathbb{C})$. The fact that $\mathcal{M}$ is a contravariant functor means we should also have a notion of pulling back families along some morphism $S \to T$ which provides a map $\mathcal{M}(T) \to \mathcal{M}(S)$. Given a moduli functor $\mathcal{M}$, we now have a well-posed moduli problem: does there exist a $\mathbb{C}$-scheme whose $\mathbb{C}$-points (closed points) are $\mathcal{M}(\operatorname{Spec}\mathbb{C})$? To seriously think about such a question, a fundamental fact we should familiarize ourselves with is the Yoneda lemma.

Definition 1.2. *Let X be a $\mathbb{C}$-scheme. Then the functor of points of X is defined as*

$$h^X : (Sch_{\mathbb{C}})^{op} \to Sets$$

where $T \mapsto \operatorname{Hom}_{Sch_{\mathbb{C}}}(T, X)$. We call $h^X(T)$ the T-points of X.

The functor of points of X is a moduli functor. What moduli problem does it pose? It poses the moduli problem of classifying the $\mathbb{C}$-points of X. All the morphisms in $\operatorname{Hom}_{\mathrm{Sch}_{\mathbb{C}}}(T, X)$ should be thought of as all the ways to parameterize some $\mathbb{C}$-points of X over T. Recall that the presheaves of sets on $\mathrm{Sch}_{\mathbb{C}}$ form a functor category $\mathrm{Psh}(\mathrm{Sch}_{\mathbb{C}}) := \mathrm{Fun}(\mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}}, \mathrm{Sets})$, where morphisms are natural transformations. This is the category of moduli functors, and the functor of points of X is an element in this category. Recall our philosophical claim from earlier: if X is a $\mathbb{C}$-scheme and we suddenly forgot its $\mathbb{C}$-scheme structure but remembered all the morphisms $T \to X$, which may be thought of as all the families of $\mathbb{C}$-points of X parameterized over T, then these would uniquely determine the $\mathbb{C}$-scheme structure of X. The Yoneda lemma formalizes this philosophical claim.

Proposition 1.3 (Yoneda)**.** *The Yoneda embedding is the covariant functor*

$$Sch_{\mathbb{C}} \to Psh(Sch_{\mathbb{C}})$$

sending X to its functor of points h^X, which satisfies canonical isomorphisms

$$\mathrm{Hom}_{Psh(Sch_{\mathbb{C}})}(h^X, \mathcal{M}) \cong \mathcal{M}(X)$$

so that the Yoneda embedding is a fully faithful embedding. Furthermore, it preserves isomorphisms so that $h^X \cong h^Y \iff X \cong Y$.

Proof. Suppose $\theta : h^X \to \mathcal{M}$. Then we send θ to the element $\theta_X(Id_X) \in \mathcal{M}(X)$. Now we claim any $\alpha \in \mathcal{M}(X)$ uniquely determines a morphism $\psi : h^X \to \mathcal{M}$. Indeed, for any element $T \to X$ in $h^X(T)$, this element must map to the image of α in $\mathcal{M}(T)$ to make the diagram

$$\begin{tikzcd} h^X(X) \arrow[r] \arrow[d] & \mathcal{M}(X) \arrow[d] \\ h^X(T) \arrow[r] & \mathcal{M}(T) \end{tikzcd}$$

commute. Since α uniquely determines this morphism $h^X \to \mathcal{M}$, we see that

$$\mathrm{Hom}_{\mathrm{Psh}(\mathrm{Sch}_{\mathbb{C}})}(h^X, \mathcal{M}) \cong \mathcal{M}(X).$$

In particular, we see

$$\mathrm{Hom}_{\mathrm{Psh}(\mathrm{Sch}_{\mathbb{C}})}(h^X, h^Y) \cong h^Y(X) = \mathrm{Hom}_{\mathrm{Sch}_{\mathbb{C}}}(X, Y).$$

Thus, we see that the Yoneda functor is a fully faithful embedding (injective on objects, isomorphisms on Hom sets). The fact that $h^X \cong h^Y \iff X \cong Y$ follows easily. □

Now we are ready to define what a fine moduli space is.

Definition 1.4. *Let $\mathcal{M} : (Sch_{\mathbb{C}})^{op} \to Sets$ be a moduli functor. We say $X \in Sch_{\mathbb{C}}$ is a fine moduli space for $\mathcal{M}$ if there exists a morphism $\alpha : h^X \to \mathcal{M}$ such that for any other morphism $\beta : h^Y \to \mathcal{M}$, there exists a unique map $h^X \to h^Y$ so that the diagram*

$$\begin{tikzcd} h^X \arrow[r, "\alpha"] & \mathcal{M} \\ h^Y \arrow[u, dashed, "\exists!"] \arrow[ur, "\forall\beta"'] & \end{tikzcd}$$

commutes.

In fact, any such $\alpha : h^X \to \mathcal{M}$ with this universal property must be an isomorphism. If we apply Yoneda to the universal property, we find the following rephrasing: there exists $\alpha \in \mathcal{M}(X)$ such that for any $\beta \in \mathcal{M}(Y)$, there exists a unique $Y \to X$ such that the induced map $\mathcal{M}(X) \to \mathcal{M}(Y)$ sends α to β: using this diagram

$$\begin{tikzcd} h^X(X) \arrow[r] \arrow[d] & \mathcal{M}(X) \arrow[d] \\ h^X(Y) \arrow[r] & \mathcal{M}(Y) \end{tikzcd}$$

we see that $h^X(Y) \cong \mathcal{M}(Y)$ for every Y. So $\alpha : h^X \to \mathcal{M}$ is an isomorphism. So $h^X \cong \mathcal{M}$. Perhaps more suggestively, $\alpha : h^X \to \mathcal{M}$ corresponds to $\alpha_X(Id_X) \in \mathcal{M}(X)$, which should be thought of as some family of objects parameterized by X. The universal property says that for any other family of objects over some T, i.e an element of $\mathcal{M}(T)$, this family comes from pullback along a unique map $T \to X$. The fact that this universal property implies $h^X \cong \mathcal{M}$ says that this family over X given by $\alpha_X(Id_X)$ should be thought of as the universal family for $\mathcal{M}$.

Remark 1.5. *We have that X is a fine moduli space for the moduli functor h^X. This formalizes our philosophical claim in the beginning. One might be tempted to say X is a fine moduli space for the $\mathbb{C}$-points of X, but be careful: there are moduli functors with fine moduli spaces with the same $\mathbb{C}$-points but their fine moduli spaces are not isomorphic $\mathbb{C}$-schemes. For example, one can consider the projective line versus a cupsidal line. This emphasizes that not only are the objects themselves important, but also the notion of families of such objects.*

Let's dive into some examples. First, we claimed that $\mathbb{P}^1$ was a moduli space for lines in $\mathbb{C}^2$ through the origin. We'll do slightly better and see that $\mathbb{P}(V)$ is the fine moduli space for lines of V through the origin, for any finite dimensional $\mathbb{C}$-vector space V. Consider the moduli functor $\mathcal{M} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$ given by

$$T \mapsto \{(\mathcal{L}, \psi)\}/\sim$$

where $\mathcal{L} \in \mathrm{Pic}(T)$ and $\psi : V^* \to H^0(T, \mathcal{L})$ such that $V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{L}$ is surjective. We say $(L_1, \psi_1) \sim (L_2, \psi_2)$ if there exists $\theta : L_1 \cong L_2$ such that ψ_1 and ψ_2 commute with $\theta^* : H^0(T, L_2) \cong H^0(T, L_1)$. Using the theory of linear systems, we see $\mathbb{P}(V)$ is a fine moduli space for this moduli functor. The universal family of this moduli functor is given by the element $(\mathcal{O}_{\mathbb{P}(V)}(1), \mathrm{Id})$.

On the other hand, we can obtain an isomorphic moduli functor in the following way: given any such $(\mathcal{L}, \psi) \in \mathcal{M}(T)$, note dualizing $V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{L}$ yields an injection $\mathcal{L}^\vee \hookrightarrow V \otimes \mathcal{O}_T$, which will have locally free cokernel. If we consider the actual vector bundles involved here which these sheaves are sheaf of sections of, we get an injection $L^\vee \hookrightarrow V \times T$. Then we have an isomorphic moduli functor sending

$$T \mapsto \left\{ \begin{tikzcd} L \arrow[r, hook] \arrow[d] & V \times T \arrow[dl] \\ T & \end{tikzcd} \right\}/\sim$$

where two diagrams are equivalent if their images in $V \times T$ are the same. The universal family for this is $\mathcal{O}_{\mathbb{P}(V)}(-1)$ as a $\mathbb{C}$-scheme. Again, $\mathbb{P}(V)$ is a fine moduli space for this moduli functor, which illustrates more clearly that $\mathbb{P}(V)$ classifies lines in V through the origin.

2. MVC 4/16/25 Part 1: Zariski sheaves, subfunctors, representability of Grassmannian

In this section, we begin by discussing an important obstruction to representability.

Theorem 2.1. *Let $\mathcal{M} \in Psh(Sch_{\mathbb{C}})$ be a moduli functor. We say $\mathcal{M}$ is a Zariski sheaf if, for every $X \in Sch_{\mathbb{C}}$, its restriction to the category of open sets of X is a sheaf in the usual sense.*

Note that for each $X \in \mathrm{Sch}_{\mathbb{C}}$, its functor of points h^X is a Zariski since morphisms of schemes, when compatible on overlaps, glue uniquely. Thus, any moduli functor which is representable is a Zariski sheaf. This presents a basic obstruction for a moduli functor to be representable. This is, however, not a sufficient condition to representability. There are Zariski sheaves which are not representable. Thus, we have a chain of strict fully faithful embeddings

$$\mathrm{Sch}_{\mathbb{C}} \subset \mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Sch}_{\mathbb{C}}) \subset \mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}}).$$

To the forgetful inclusion, $\mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Sch}_{\mathbb{C}}) \subset \mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}})$, there is a left adjoint called the sheafification functor. It does what you expect it to. Given any moduli functor $\mathcal{F} \in \mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}})$, the sheafification functor sends it to $\mathcal{F}^{sh} \in \mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Sch}_{\mathbb{C}})$. For every $X \in \mathrm{Sch}_{\mathbb{C}}$, the Zariski sheaf $\mathcal{F}^{sh}$ restricted to the subcategory of open sets of X is the sheafification of the restriction of $\mathcal{F}$ to the same subcategory. For a morphism of schemes $X \to Y$, the map $\mathcal{F}^{sh}(Y) \to \mathcal{F}^{sh}(X)$ is determined uniquely by the local maps provided by $\mathcal{F}$ on local open sets of Y and X combined with sheafification. There is a canonical map $\mathcal{F} \to \mathcal{F}^{sh}$ which does what you expect it to, carrying the universal property that for any other $E \in \mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Sch}_{\mathbb{C}})$ with a map of presheaves $\mathcal{F} \to E$, there is a universal property:

$$\begin{tikzcd} \mathcal{F} \arrow[r] \arrow[dr] & \mathcal{F}^{sh} \arrow[d, dashed, "\exists!"] \\ & E \end{tikzcd}$$

which expresses the left adjointness of the sheafification functor.

Okay, so suppose we have a moduli functor. The first test of whether it is representable is to sheafify it. If it's a Zariski sheaf already, we get the same thing back, no problem. Otherwise, we're one step closer. Now a useful thing about Zariski sheaves is that checking whether they're representable is equivalent to checking this on affines.

Proposition 2.2. *Let $Sh^{Zar}(AffSch_{\mathbb{C}})$ denote Zariski sheaves as contravariant functors $AffSch_{\mathbb{C}}^{op} \to Sets$. There is an equivalence of categories between*

$$Sh^{Zar}(Sch_{\mathbb{C}}) \cong Sh^{Zar}(AffSch_{\mathbb{C}}).$$

Proposition 2.2 holds because usual Zariski sheaves on a scheme are determined by their local sections over affines. And given a Zariski sheaf $\mathcal{F}$, the map $\mathcal{F}(Y) \to \mathcal{F}(X)$

corresponding to some morphism of schemes $f : X \to Y$ is also uniquely determined by $\mathcal{F}(U) \to \mathcal{F}(V)$ for affine opens $U \subset Y$ and $V \subset f^{-1}(U)$. The same can be said about morphisms of Zariski sheaves i.e. natural transformations of these Zariski-sheafy moduli functors.

In other words, suppose $\mathcal{F} \in \mathrm{Sh}^{\mathrm{Zar}}(\mathrm{Sch}_{\mathbb{C}})$. Then to check whether $\mathcal{F}$ is represented by $X \in \mathrm{Sch}_{\mathbb{C}}$, it suffices to check in the category $\mathrm{Sh}^{\mathrm{Zar}}(\mathrm{AffSch}_{\mathbb{C}})$ whether there is some morphism $h^X \to \mathcal{F}$, i.e. a natural transformation giving diagrams

$$\begin{tikzcd} h^X(\operatorname{Spec} B) \arrow[r] \arrow[d] & \mathcal{F}(\operatorname{Spec} B) \arrow[d] \\ h^X(\operatorname{Spec} A) \arrow[r] & \mathcal{F}(\operatorname{Spec} A) \end{tikzcd}$$

such that $h^X(\operatorname{Spec} R) \cong \mathcal{F}(\operatorname{Spec} R)$ for every affine $\operatorname{Spec} R \in \mathrm{AffSch}_{\mathbb{C}}$. Equivalently, the following is another way to check representability of $\mathcal{F}$ on affines: if there is a morphism $h^X \to \mathcal{F}^{sh}$ on affines, i.e. a morphism in the category $\mathrm{Sh}^{\mathrm{Zar}}(\mathrm{AffSch}_{\mathbb{C}})$, such that for any other morphism $h^T \to \mathcal{F}$ on affines, for $T \in \mathrm{Sch}_{\mathbb{C}}$, we have

$$\begin{tikzcd} h^X \arrow[r] & \mathcal{F} \\ h^T \arrow[u, dashed, "\exists!"] \arrow[ur] & \end{tikzcd}$$

there exists a unique morphism $h^T \to h^X$ in the category $\mathrm{Sh}^{\mathrm{Zar}}(\mathrm{AffSch}_{\mathbb{C}})$ making the diagram above commute. This is simply combining Proposition 2.2 with Definition 1.4.

Let's introduce one more tool for proving representability of a Zariski sheaf.

Definition 2.3. *Let $\mathcal{M} \in PSh(Sch_{\mathbb{C}})$ be a moduli functor. We say $\mathcal{N} \subseteq \mathcal{M}$ is an open subfunctor if for every $T \in Sch_{\mathbb{C}}$, we have*

$$\begin{tikzcd} \mathcal{N} \times_{\mathcal{M}} h^T \arrow[r] \arrow[d] & h^T \arrow[d] \\ \mathcal{N} \arrow[r] & \mathcal{M} \end{tikzcd}$$

the fiber product $\mathcal{N} \times_{\mathcal{M}} h^T$ is representable by an open subscheme of T.

In the category $\mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}})$, fiber products exist. Given any morphisms $\mathcal{M}_1 \to \mathcal{M}$ and $\mathcal{M}_2 \to \mathcal{M}$ of moduli functors, the fiber product $\mathcal{M}_1 \times_{\mathcal{M}} \mathcal{M}_2$ is defined as, for every $T \in \mathrm{Sch}_{\mathbb{C}}$, the fiber product of sets

$$\mathcal{M}_1 \times_{\mathcal{M}} \mathcal{M}_2(T) := \mathcal{M}_1(T) \times_{\mathcal{M}(T)} \mathcal{M}_2(T).$$

The universal property of the fiber product of sets makes $\mathcal{M}_1 \times_{\mathcal{M}} \mathcal{M}_2$ into a moduli functor, and lifts to the universal property of $\mathcal{M}_1 \times_{\mathcal{M}} \mathcal{M}_2$ as a fiber product of moduli functors. If we think of moduli functors as generalizations of schemes, then an open subfunctor is a generalization of an open subscheme. We can also generalize covers by open subschemes to covers of moduli functors:

Definition 2.4. *Let $\mathcal{M} \in PSh(Sch_{\mathbb{C}})$ be a moduli functor. Let $\{\mathcal{N}_i\}$ be a collection of open subfunctors of $\mathcal{M}$, such that for every $T \in Sch_{\mathbb{C}}$, the open subschemes*

$$\{\mathcal{N}_i \times_{\mathcal{M}} T\}$$

provide a Zariski open cover of T. Then we say $\{\mathcal{N}_i\}$ is a Zariski cover of $\mathcal{M}$.

Proposition 2.5. *Let $\mathcal{M}$ be a moduli functor. Then $\mathcal{M}$ is representable if and only if $\mathcal{M}$ is a Zariski sheaf and there exists a Zariski open cover $\{\mathcal{N}_i\}$ of $\mathcal{M}$ by open subfunctors such that each $\mathcal{N}_i$ is representable.*

Proof. The forward direction is immediate, so consider the reverse direction. Note we are abusing notation here and sometimes refer to a scheme and its functor of points as the same thing.

What scheme should represent $\mathcal{M}$? Note that we have the following diagrams

$$\begin{tikzcd}
\mathcal{N}_i \times_{\mathcal{M}} \mathcal{N}_j \arrow[r, hook] \arrow[d, hook] & \mathcal{N}_j \arrow[d] \\
\mathcal{N}_i \arrow[r] & \mathcal{M}
\end{tikzcd}$$

for every i, j. Since $\mathcal{N}_i$ and $\mathcal{N}_j$ are both open subfunctors which are also representable, then the fiber product $\mathcal{N}_i \times_{\mathcal{M}} \mathcal{N}_j$ is identified with an open subscheme of both $\mathcal{N}_j$ and $\mathcal{N}_i$. This fiber product diagram then tells us how to glue together the schemes $\mathcal{N}_i$ and $\mathcal{N}_j$. That the gluings are compatible on overlaps and thus the $\mathcal{N}_i$ altogether actually glue to a scheme X, is seen by the "fiber product cube"

$$\begin{tikzcd}
 & \mathcal{N}_z \times_{\mathcal{M}} \mathcal{N}_y \arrow[dl] \arrow[dd] & & \mathcal{N}_z \times_{\mathcal{M}} \mathcal{N}_y \times_{\mathcal{M}} \mathcal{N}_x \arrow[ll] \arrow[dl] \arrow[dd] \\
\mathcal{N}_z \arrow[dd] & & \mathcal{N}_z \times_{\mathcal{M}} \mathcal{N}_x \arrow[ll] \arrow[dd] & \\
 & \mathcal{N}_y \arrow[dl] & & \mathcal{N}_y \times_{\mathcal{M}} \mathcal{N}_x \arrow[ll] \arrow[dl] \\
\mathcal{M} & & \mathcal{N}_x \arrow[ll] &
\end{tikzcd}$$

Here, $\mathcal{N}_z \times_{\mathcal{M}} \mathcal{N}_y \times_{\mathcal{M}} \mathcal{N}_x$ is the scheme identified with the triple overlap of gluings.

Since $\mathcal{M}$ is a Zariski sheaf, the morphisms $\mathcal{N}_i \to \mathcal{M}$ then glue to a morphism $X \to \mathcal{M}$. We'd like to show that this morphisms makes X a fine moduli space for $\mathcal{M}$. Thus our task is to demonstrate the universal property as given in Definiton 1.4.

Let $T \in \mathrm{Sch}_{\mathbb{C}}$ and let $T \to \mathcal{M}$ be some morphism. Note we have the following diagram

$$\begin{tikzcd}
\mathcal{N}_i \times_{\mathcal{M}} T \arrow[r] \arrow[d] & T \arrow[d] \\
\mathcal{N}_i \arrow[r] & \mathcal{M}
\end{tikzcd}$$

for each $\mathcal{N}_i$. The $\mathcal{N}_i \times_{\mathcal{M}} T$ cover T, and each maps into $\mathcal{N}_i$ respectively. Some more diagrams should be drawn to show those maps are compatible across overlaps, but indeed they are. This implies that the map $T \to \mathcal{M}$ factors through a map $T \to X$ glued together from the $\mathcal{N}_i \times_{\mathcal{M}} T \to \mathcal{N}_i$. Thus, $h^X \cong \mathcal{M}$. □

Just for fun, one application of Proposition 2.5 and Proposition 2.2, we'll show that fiber products exist in the category of schemes.

Corollary 2.6. *Fiber products exist in $Sch_{\mathbb{C}}$.*

Proof. Let $X \to Z$ and $Y \to Z$ be morphisms of schemes. In what follows we will abuse notation between scheme and its functor of points. We know that $X \times_Z Y \in \mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}})$. We'd like to show it is representable. First, let's show it is a Zariski sheaf.

$$\begin{tikzcd}
X(U) \times_{Z(U)} Y(U) \arrow[drrr] \arrow[ddrr] \arrow[dd] & & & & \\
& & & Y(U) \arrow[dd] \arrow[ddr] & \\
\prod X(U_i) \times_{Z(U_i)} Y(U_i) \arrow[drrr] \arrow[ddrr] \arrow[dd, shift left] \arrow[dd, shift right] & & X(U) \arrow[drr] \arrow[dd] & & \\
& & & \prod Y(U_i) \arrow[dd, shift left] \arrow[dd, shift right] \arrow[ddr] & Z(U) \arrow[dd] \\
\prod X(U_i \cap U_j) \times_{Z(U_i \cap U_j)} Y(U_i \cap U_j) \arrow[drrr] \arrow[ddrr] & & \prod X(U_i) \arrow[drr] \arrow[dd, shift left] \arrow[dd, shift right] & & \\
& & & \prod Y(U_i \cap U_j) \arrow[ddr] & \prod Z(U_i) \arrow[dd, shift left] \arrow[dd, shift right] \\
& & \prod X(U_i \cap U_j) \arrow[drr] & & \\
& & & & \prod Z(U_i \cap U_j)
\end{tikzcd}$$

For any $U \in \mathrm{Sch}_{\mathbb{C}}$ with open cover $U = \bigcup U_i$, using the universal property of each set-wise fiber product, we see that we get a diagram expressing $(X \times_Z Y)(U)$ as an equalizer, thereby showing that $X \times_Z Y$ is a sheaf on the subcategory of open sets of any $T \in \mathrm{Sch}_{\mathbb{C}}$, which means the fiber product $X \times_Z Y$ is a Zariski sheaf.

Identify affine open sets $U_i \subseteq X, V_j \subset Z$ and $W_k \subseteq Y$, so that the morphisms $X \to Z$ and $Y \to Z$ restrict to morphisms on these affine opens.

$$\begin{tikzcd}
U_i \times_{V_j} W_k \arrow[rrr] \arrow[dddd] \arrow[dr, hook] & & & W_k \arrow[dl, hook'] \arrow[dddd] \\
& X \times_Z Y \arrow[r] \arrow[dd] & Y \arrow[dd] & \\
& & & \\
& X \arrow[r] & Z & \\
U_i \arrow[rrr] \arrow[ur, hook] & & & V_j \arrow[ul, hook']
\end{tikzcd}$$

Then running over all such U_i, V_j, W_k, we see the collection of $U_i \times_{V_j} W_k$ are open subfunctors of $X \times_Z Y$ which provide a Zariski cover. Then to show $X \times_Z Y$ is representable, it suffices to show that each $U_i \times_{V_j} W_k$ is representable.

Let $U_i = \operatorname{Spec} A$, $V_j = \operatorname{Spec} B$ and $W_k = \operatorname{Spec} C$. To show $U_i \times_{V_j} W_k$ is representable, since it is a Zariski sheaf, by Proposition 2.2 it suffices to check representability on affines. We claim that $\operatorname{Spec} A \otimes_B C$ represents $U_i \times_{V_j} W_k$ on affines. Indeed, we see by universal property of the tensor product $A \otimes_B C$, there is an isomorphism $(U_i \times_{V_j} W_k)(\operatorname{Spec} D) \cong h^{\operatorname{Spec} A \otimes_B C}(\operatorname{Spec} D)$. $\square$

Finally, as an application of Proposition 2.5, we'll give a moduli functor description of the Grassmannian. We want to generalize the moduli problem of classifying lines in a vector space, and classify all the r-dimensional subspaces of a $\mathbb{C}$-vector space V, where $0 < r < \dim V$.

We enhance this classification problem to the following moduli functor:

$$T \in \mathrm{Sch}_{\mathbb{C}} \mapsto \{(E, \psi) \mid E \text{ rank r vector bundle over T}, \psi : V^* \to H^0(T, E) \text{ such that } V^* \otimes \mathcal{O}_T \twoheadrightarrow E\}/\sim$$

Here, $(E, \psi) \sim (E', \psi')$ if and only if there exists a vector bundle isomorphism $\pi : E \cong E'$ such that

$$\begin{tikzcd}
& H^0(T,E) \\
V^* \arrow[ur, "\psi"] \arrow[dr, "\psi'"'] & \\
& H^0(T,E') \arrow[uu, "\pi^*"']
\end{tikzcd}$$

commutes. To make this slightly less mysterious, note that first it generalizes the moduli functor of $\mathbb{P}(V)$ to a rank r setting. Perhaps more enlighteningly, note if we dualize $V^* \otimes \mathcal{O}_T \twoheadrightarrow E$, which is an honest morphism of vector bundles and not just of sheaves, we obtain $E^\vee \hookrightarrow V \otimes \mathcal{O}_T$. Taking total spaces, our moduli functor is then identical to the moduli functor sending $T \in \mathrm{Sch}_\mathbb{C}$ to the collection of

$$\begin{tikzcd} \mathcal{E} \arrow[r, hook] \arrow[d] & V \times T \arrow[dl] \\ T & \end{tikzcd}$$

where $\mathcal{E}$ is the total space of a rank r vector bundle over T which embeds into $V \times T$, modulo the right notion of equivalence. Let's call this moduli functor we've defined $\mathcal{M}$. One can check that $\mathcal{M}$ is indeed a Zariski sheaf.

We'd like to show that $\mathcal{M}$ is represented by some scheme $\mathrm{Gr}(r, V) \in \mathrm{Sch}_\mathbb{C}$, which we'll define at the end as the Grassmannian parameterizing r-dimensional subspaces of V. In order to do this, we'll show that $\mathcal{M}$ admits a Zariski cover of open subfunctors, each of which are themselves representable.

Let $V \twoheadrightarrow W$ be an r-dimensional quotient vector space of V. Let $\mathcal{N}_W$ denote the subfunctor of $\mathcal{M}$ where

$$T \mapsto \{(E, \psi) \in \mathcal{M}(T) \mid \text{ The composition } W^* \otimes \mathcal{O}_T \hookrightarrow V^* \otimes \mathcal{O}_T \twoheadrightarrow E \text{ is an isomorphism } \}/\sim$$

In other words, this is the subfunctor of $\mathcal{M}$ where $T \in \mathrm{Sch}_\mathbb{C}$ is sent to the subset of $\mathcal{M}(T)$ such that

$$\begin{tikzcd} \mathcal{E} \arrow[r, hook] \arrow[d] & V \times T \arrow[r] \arrow[dl] & W \times T \arrow[dll] \\ T & & \end{tikzcd}$$

the composition $\mathcal{E} \to W \times T$ is surjective, and thus an isomorphism, over T. One sees that $\mathcal{N}_W$ is a Zariski sheaf.

We claim that $\mathcal{N}_W$ is an open subfunctor of $\mathcal{M}$. Then we need to verify that for every $X \in \mathrm{Sch}_\mathbb{C}$, the fiber product

$$\begin{tikzcd} \mathcal{N}_W \times_\mathcal{M} X \arrow[r] \arrow[d] & X \arrow[d] \\ \mathcal{N}_W \arrow[r] & \mathcal{M} \end{tikzcd}$$

$\mathcal{N}_W \times_\mathcal{M} X$ is representable by some open subscheme of X. Note that $(\mathcal{N}_W \times_\mathcal{M} X)(T)$ is by definition the fiber product of sets $\mathcal{N}_W(T) \times_{\mathcal{M}(T)} X(T)$. Unpacking what this means, and in particular noting that $X \to \mathcal{M}$ must come from an element of $\mathcal{M}(X)$, implies that $(\mathcal{N}_W \times_\mathcal{M} X)(T)$ are the elements of $\mathcal{N}_W(T)$ which come from pullback of an element of $\mathcal{M}(X)$ along some morphism $T \to X$:

$$
\begin{tikzcd}
 & & & & W\times T \arrow[drrr] \arrow[dd] & & & \\
 & & V\times T \arrow[urr] \arrow[drrr] \arrow[dd, dashed] & & & & & W\times X \arrow[dd] \\
\mathcal{E} \arrow[urr, hook] \arrow[drrr] \arrow[dd] & & & & T \arrow[drrr, dashed] & V\times X \arrow[urr] \arrow[dd] & & \\
 & & T \arrow[urr, dashed, "Id"] \arrow[drrr, dashed] & \mathcal{E}^0 \arrow[urr, hook] \arrow[dd] & & & & X \\
T \arrow[urr, "Id"] \arrow[drrr, "\pi"] & & & & & X \arrow[urr, "Id"] & & \\
 & & & X \arrow[urr, "Id"] & & & &
\end{tikzcd}
$$

Any such $\pi : T \to X$ must map closed points $z \in T$ to $\pi(z) \in X$ such that the vector bundle fiber $\mathcal{E}^0|_{\pi(z)} \subset V$ projects surjectively onto W. The locus of closed points $w \in X$ such that $\mathcal{E}|_w \subset V$ projects surjectively onto W are the closed points of an open subscheme of X. And any such morphism $T \to X$ to this open subscheme of X produces an element of $(\mathcal{N}_W \times_{\mathcal{M}} X)(T)$. We see then that $\mathcal{N}_W \times_{\mathcal{M}} X$ is represented by an open subscheme of X.

As W varies among all r-dimensional quotients $V \twoheadrightarrow W$, the $\mathcal{N}_W$ form a Zariski cover of $\mathcal{M}$, since for any $h^X \to \mathcal{M}$ corresponding to some family $\mathcal{E}^0 \to X$, any point $z \in X$ is contained in the open subscheme of X representing $\mathcal{N}_{\mathcal{E}^0|_w}$.

Finally, we claim that $\mathcal{N}_W$ is representable by $\mathbb{A}^{r(n-r)}$. Note that for any $[\mathcal{E} \to T] \in \mathcal{N}_W(T)$, we have that

$$
\begin{tikzcd}
\mathcal{E} \arrow[r] \arrow[d] & W \times T \arrow[dl] \\
T &
\end{tikzcd}
$$

is an isomorphism over T. So each fiber of $\mathcal{E}$ is a r-dimensional subspace of V which projects isomorphically onto W. So we are asking for a fine moduli space parameterizing all the r-dimensional subspaces of V which project isomorphically onto W. We claim this fine moduli space is isomorphic to $\mathbb{A}^{r(n-r)}$. Here's one way to see why. Let's suppose for ease of example that we chose a basis $e_1, \cdots, e_n$ for V, and $W = \mathbb{C}\langle e_1, \cdots, e_r\rangle$. Now consider embeddings $\mathbb{C}^r \to V$ which project isomorphically onto W. Any such linear map is given by a matrix $r \times n$ matrix L with respect to the basis $\{e_1, \cdots, e_n\}$. But if it projects isomorphically onto W, there exist a matrix $M \in \mathrm{GL}_r(\mathbb{C})$ such that the $r \times n$ matrix LW looks like

$$
\begin{pmatrix} Id_r \\ B \end{pmatrix}
$$

where Id_r is the $r \times r$ identity matrix and B is any $(n-r) \times r$ matrix. Each distinct r-dimensional subspace of V which projects isomorphically onto W is given by a distinct choice of matrices B. So we see that indeed, $\mathbb{A}^{r(n-r)}$ parameterizes such subspaces of V. In particular, there is a universal family $\chi \to \mathbb{A}^{r(n-r)}$ which is a subvariety of $V \times \mathbb{A}^{r(n-r)}$. Then any element of $\mathcal{N}_W(T)$ comes pulling back this universal family along a unique morphism $T \to \mathbb{A}^{r(n-r)}$.

Since the $\mathcal{N}_W$ are representable, so is $\mathcal{M}$, and we done this representing scheme $\mathrm{Gr}(r, V)$. Note these moduli functors $\mathcal{N}_W$ are represented by the usual affines $\mathbb{A}^{r(n-r)}$ which are glued together to get $\mathrm{Gr}(r, V)$.

3. MVC 4/16/25 Part 2: Corepresentability, coarse moduli space, group schemes

We've spent some time discussing representability, ways of showing representability, and examples of representable moduli functors (most notably, the Grassmannian moduli functor). There are, however, many meaningful moduli functors which are not representable. Let's relax the notion of representability to include a larger class of moduli functors. If representability is approximation of a moduli functor from below, then co-representability is approximation of a moduli functor from above.

Definition 3.1 (Co-representable)**.** *Let $\mathcal{M} \in PSh(Sch_{\mathbb{C}})$ be a moduli functor. Then we say $\mathcal{M}$ is co-represented by $X \in Sch_{\mathbb{C}}$ if there is some morphism $\mathcal{M} \to h^X$ of presheaves such that for all $\mathcal{M} \to h^Y$, there exists a unique map $h^X \to h^Y$ satisfying:*

$$\begin{tikzcd} \mathcal{M} \arrow[r] \arrow[dr, "\forall"'] & h^X \arrow[d, dashed, "\exists!"] \\ & h^Y \end{tikzcd}$$

Note co-representability generalizes representability, since if $\mathcal{M}$ is representable by h^X, then $\mathcal{M} \cong h^X$.

Definition 3.2. *Let $\mathcal{M} \in PSh(Sch_{\mathbb{C}})$ be a moduli functor. We say $X \in Sch_{\mathbb{C}}$ is a coarse moduli space for $\mathcal{M}$ if X is a coarse moduli space for $\mathcal{M}$ and $\mathcal{M}(\operatorname{Spec}\mathbb{C}) \cong h^X(\operatorname{Spec}\mathbb{C})$.*

Thus, a coarse moduli space parameterizes the right objects. Throwing in the notion of families, it is the best approximation from above of the moduli functor, in terms of being initial among morphisms to other schemes. We'll encounter many fruitful examples of coarse moduli spaces by quotienting schemes by group actions.

Definition 3.3. *We say $G \in Sch_{\mathbb{C}}$ is a group scheme if its functor of points*

$$h^G : Sch_{\mathbb{C}}^{op} \to Sets$$

factors through the category of groups:

$$\begin{tikzcd} & Grps \arrow[d] \\ Sch_{\mathbb{C}}^{op} \arrow[ur, dashed] \arrow[r, "h^G"'] & Sets \end{tikzcd}$$

In particular, for every morphism of schemes $X \to Y$, we have $G(Y) \to G(X)$ is a group homomorphism. Given a group scheme G, note that on T-points we have the group operation $G(T) \times G(T) = (G \times G)(T) \to T$, which is functorial with respect to $T \in \mathrm{Sch}_{\mathbb{C}}$. Then by Yoneda lemma 1.3, this map on T-points corresponds to a map

$$m : G \times G \to G.$$

There is also an identity map $e : \operatorname{Spec} \mathbb{C} \to G$, which on T-points maps to the identity element of $G(T)$. There is also an inverse map $\iota : G \to G$, so that $x \in G(T) \mapsto x^{-1} \in G(T)$. We see by Yoneda's lemma there is an equivalent way of defining a group scheme:

Definition 3.4. *Let $G \in Sch_{\mathbb{C}}$ be a scheme. We say G is a group scheme if there exist maps*

$$e : \operatorname{Spec} \mathbb{C} \to G, m : G \times G \to G, \iota : G \to G$$

satisfying the expected commutative diagrams.

Definition 3.5. *An algebraic group G is a finite type group scheme.*

Remark 3.6. *Algebraic groups over characteristic 0 algebraically closed fields are automatically smooth and quasi-projective.*

Let's see some examples of group schemes (which are also all algebraic groups).

Example 3.7 (Multiplicative group scheme)**.** *Consider the moduli functor*

$$\mathbb{G}_m : Sch_{\mathbb{C}}^{op} \to Grps$$

which sends $T \mapsto (H^0(T, \mathcal{O}_T)^{\times}, \cdot)$. One sees that this is indeed a Zariski sheaf. In fact it is representable by $\operatorname{Spec} \mathbb{C}[t, t^{-1}] \cong \mathbb{A}^1 \setminus \{0\}$.

Since $\mathbb{G}_m$ is a Zariski sheaf, it suffices to check this representability on affines. Indeed, there is a functorial isomorphism $\mathbb{G}_m(\operatorname{Spec}(B)) = B^{\times} \cong Mor_{Sch_{\mathbb{C}}}(\operatorname{Spec} B, \operatorname{Spec} \mathbb{C}[t, t^{-1}])$.

Note the group operation $m : \mathbb{G}_m \times \mathbb{G}_m \to \mathbb{G}_m$ is given by the map of $\mathbb{C}$-algebras

$$\operatorname{Spec} \mathbb{C}[t, t^{-1}] \to \operatorname{Spec} \mathbb{C}[x, x^{-1}] \otimes_{\mathbb{C}} \operatorname{Spec} \mathbb{C}[y, y^{-1}]$$

where $t \mapsto x \otimes y$.

Example 3.8 (Additive group scheme)**.** *Consider the moduli functor*

$$\mathbb{G}_a : Sch_{\mathbb{C}}^{op} \to Grps$$

such that $T \mapsto (H^0(T, \mathcal{O}_T), +)$. *One can check that this is a Zariski sheaf. In fact, it is represented by the affine line* $\operatorname{Spec}\mathbb{C}[t] \cong \mathbb{A}^1$.

Since $\mathbb{G}_a$ *is a Zariski sheaf, it suffices to show representability on affines. Indeed there is a funtorial isomorphism*

$$\mathbb{G}_a(\operatorname{Spec} B) = B \cong Mor_{Sch_{\mathbb{C}}}(\operatorname{Spec} B, \operatorname{Spec}\mathbb{C}[t]).$$

Example 3.9 (GL_n). *Consider the moduli functor*

$$\mathrm{GL}_n : Sch_{\mathbb{C}}^{op} \to Grps$$

such that $T \mapsto \operatorname{Aut}_T(T \times \mathbb{A}^n)$, *i.e. the automorphisms of the trivial rank* n *vector bundle* $\mathcal{O}_T^{\oplus n}$ *over* T. *Note that* GL_n *is clearly a Zariski sheaf. We claim this functor is representable by*

$$\operatorname{Spec} \frac{\mathbb{C}[a_{11}, a_{12}, \cdots, a_{nn}, t]}{\langle \det(a_{ij}) \cdot t - 1\rangle}.$$

It suffices to check representability on affines. Let $T = \operatorname{Spec} B$. *Then* $\operatorname{Aut}_T(T \times \mathbb{A}^n)$ *is all automorphisms*

$$\begin{tikzcd}
B \otimes_{\mathbb{C}} \mathbb{C}[z_1, \cdots, z_n] & & B \otimes_{\mathbb{C}} \mathbb{C}[z_1, \cdots, z_n] \arrow[ll] \\
& B \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

over B. *We see the automorphisms are exactly given by morphisms* $\frac{\mathbb{C}[a_{11}, a_{12}, \cdots, a_{nn}, t]}{\langle \det(a_{ij}) \cdot t - 1\rangle} \to B$, *so* $\operatorname{Aut}_{\operatorname{Spec} B}(\operatorname{Spec} B \times \mathbb{A}^n) \cong Mor_{Sch_{\mathbb{C}}}(\operatorname{Spec} B, \operatorname{Spec} \frac{\mathbb{C}[a_{11}, a_{12}, \cdots, a_{nn}, t]}{\langle \det(a_{ij}) \cdot t - 1\rangle})$.

If we want to write the algebraic group GL_n *in a coordinate free way, then we define* GL_V *to be the algebraic group representing the moduli functor*

$$T \in Sch_{\mathbb{C}}^{op} \mapsto \operatorname{Aut}_T(T \times V)$$

Example 3.10 (PGL_n). *Consider the moduli functor*

$$\mathrm{PGL}_n : Sch_{\mathbb{C}}^{op} \to Grps$$

such that $T \mapsto \operatorname{Aut}_T(T \times \mathbb{P}^{n-1})$. *We see that this is a Zariski sheaf, and clearly* $\operatorname{Aut}_T(T \times \mathbb{P}^{n-1})$ *is functorially isomorphic to morphisms* $T \to \operatorname{Spec} \frac{\mathbb{C}[a_{11}, \cdots, a_{nn}]}{\langle \det(a_{ij}) - 1\rangle}$. *If we want to write the algebraic group* PGL_n *in a coordinate-free way, we define* $\mathrm{PGL}(V)$ *to be the algebraic group representing the moduli functor*

$$T \in Sch_{\mathbb{C}}^{op} \mapsto \operatorname{Aut}_T(T \times \mathbb{P}(V)).$$

Now let's discuss group actions on schemes.

Definition 3.11. *Let* G *be a group scheme and* X *a scheme. Then* G *acts on* X *if there is a map*

$$G \times X \to X$$

such that for every $T \in Sch_{\mathbb{C}}$,

$$G(T) \times X(T) \to X(T)$$

is a left group action on $X(T)$ by $X(T)$. In other words, for every $g, h \in G(T)$ and $x \in X(T)$, we have $g(h(x)) = (gh)(x)$.

Definition 3.12. *Let G be a group scheme, and let G act on schemes X and Y. We say $f : X \to Y$ is G-equivariant if the diagram*

$$\begin{tikzcd} G \times X \arrow[r, "(id{,}f)"] \arrow[d] & G \times Y \arrow[d] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

commutes. We say f is G-invariant if the action of G on Y is trivial.

Example 3.13. *The algebraic group GL_n acts on $\mathbb{A}^n$. The action*

$$\mathrm{GL}_n \times \mathbb{A}^n \to \mathbb{A}^n$$

is given on T-points by the expected action

$$\mathrm{GL}(H^0(T, \mathcal{O}_T)) \times H^0(T, \mathcal{O}_T)^{\oplus n} \to H^0(T, \mathcal{O}_T)^{\oplus n}.$$

Example 3.14. *The algebraic group PGL_n acts on $\mathbb{P}(V)$. The action*

$$\mathrm{PGL}_n \times \mathbb{P}^{n-1} \to \mathbb{P}^{n-1}$$

is given on T-points by the expected action

$$\mathrm{Aut}_T(T \times \mathbb{P}(V)) \times \mathbb{P}^{n-1}(T) \to \mathbb{P}^{n-1}(T)$$

where given $M \in \mathrm{Aut}_T(T \times \mathbb{P}(V))$ and a family of lines

$$\begin{tikzcd} \mathcal{X} \arrow[r, hook] \arrow[d] & V \times T \arrow[dl] \\ T & \end{tikzcd}$$

parameterized by T, the action of M sends the family of lines to the new family lines

$$\begin{tikzcd} \mathcal{X} \arrow[r, hook] \arrow[d] & V \times T \arrow[r, "M"] \arrow[dl] & V \times T \arrow[dll] \\ T & & \end{tikzcd}$$

where we have technically lifted $M \in \mathrm{Aut}_T(T \times \mathbb{P}(V))$ to $\mathrm{Aut}_T(T \times V)$.

One might wonder how the algebraic groups GL_n and PGL_n are related. On $\mathbb{C}$-points, there is an isomorphism

$$1 \to \mathbb{G}_m(\mathbb{C}) \to \mathrm{GL}_n(\mathbb{C}) \to \mathrm{PGL}_n(\mathbb{C}) \to 1.$$

In fact, there a short exact sequence of Zariski sheaves

$$1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1.$$

Nevertheless, the exactness only holds on stalks and not on T-points for every T. In particular, there exists $T \in \mathrm{Sch}_{\mathbb{C}}$ such that the injection $\mathrm{GL}_n(T)/\mathbb{G}_m(T) \hookrightarrow \mathrm{PGL}_n(T)$ fails to be surjective. How do we know this? Because the quotient moduli functor

$$[\mathrm{GL}_n/\mathbb{G}_m]^{\mathrm{pre}} : \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}} \to \mathrm{Sets}, \text{ where } T \mapsto \frac{\mathrm{GL}_n(T)}{\mathbb{G}_m(T)}$$

fails to be a Zariski sheaf. Certainly, then it could not be isomorphic to PGL_n[1].

In general, given some group scheme G with a group action on X, we can form the quotient moduli functor

$$[X/G]^{\mathrm{pre}} : \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}} \to \mathrm{Sets}.$$

But this fails to be a Zariski sheaf in even the simplest cases. We already saw one example; let's see another.

Example 3.15. *Note $\mathbb{G}_m$ acts on $\mathbb{A}^2 \setminus \{0\}$. The action on $\mathbb{A}^2 \setminus \{0\}$ is the restriction of the action*

$$\mathbb{G}_m \times \mathbb{A}^2 \to \mathbb{A}^2,$$

given on affines by the map

$$\mathbb{C}[x_0, x_1] \to \mathbb{C}[x_0, x_1] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}], \text{ where } x_0 \mapsto x_0 \otimes t, \text{ and } x_1 \mapsto x_1 \otimes t,$$

to $\mathbb{A}^2 \setminus \{0\}$. Alternatively, we can describe the action on T-points: the action

$$\mathbb{G}_m(T) \times (\mathbb{A}^2 \setminus \{0\})(T) \to (\mathbb{A}^2 \setminus \{0\})(T)$$

is the action

$$H^0(T, \mathcal{O}_T)^\times \times \{(f, g) \in H^0(T, \mathcal{O}_T)^{\oplus 2} \mid 1 \in (f, g)\} \to \{(f, g) \in H^0(T, \mathcal{O}_T)^{\oplus 2} \mid 1 \in (f, g)\},$$

where $(\lambda, (f, g)) \mapsto (\lambda f, \lambda g)$.

To see that

$$[\frac{\mathbb{A}^2 \setminus \{0\}}{\mathbb{G}_m}]^{pre} : Sch_{\mathbb{C}}^{op} \to Sets$$

fails to be a Zariski sheaf, it suffices to show that there are some elements which agree on an overlap which fail to glue uniquely. Take $T = \mathbb{P}^1_{[z_0:z_1]}$ and let $U_0 = \mathrm{Spec}\,\mathbb{C}[\frac{z_1}{z_0}]$ and $U_1 = \mathrm{Spec}\,\mathbb{C}[\frac{z_0}{z_1}]$ be the distinguished affines which glue together to give $\mathbb{P}^1$.

Consider the section $(\frac{z_1}{z_0}, 1) \in [\frac{\mathbb{A}^2\setminus\{0\}}{\mathbb{G}_m}]^{pre}(U_0)$ and $(1, \frac{z_0}{z_1}) \in [\frac{\mathbb{A}^2\setminus\{0\}}{\mathbb{G}_m}]^{pre}(U_1)$. On $U_0 \cap U_1$, $(\frac{z_1}{z_0}, 1)$ maps to $(\frac{z_1}{z_0}, 1) = \frac{z_1}{z_0} \cdot (1, \frac{z_0}{z_1})$, so we have agreement on overlaps.

So if $[\frac{\mathbb{A}^2\setminus\{0\}}{\mathbb{G}_m}]^{pre}$ were a Zariski sheaf, these elements would glue uniquely. But instead, note $[\frac{\mathbb{A}^2\setminus\{0\}}{\mathbb{G}_m}]^{pre}(\mathbb{P}^1)$ will be simply pairs of invertible scalars, up to scaling.

In this case, the correct thing to do will be to sheafify $[\frac{\mathbb{A}^2\setminus\{0\}}{\mathbb{G}_m}]^{pre}$.

[1]It turns out, however, that PGL_n is the sheafification of this moduli functor.

Note that any G-invariant morphism $X \to Y$, must factor through a unique map

$$\begin{tikzcd} h^X \arrow[r] \arrow[dr, "\forall"'] & {[X/G]^{\mathrm{pre}}} \arrow[d, dashed, "\exists!"] \\ & h^Y \end{tikzcd}$$

$[X/G]^{\mathrm{pre}} \to Y$. This means that we have an isomorphism of covariant functors $\mathrm{PSh}(\mathrm{Sch}_{\mathbb{C}}) \to$ Sets such that

$$\mathrm{Mor}_{\mathrm{PSh}}([X/G]^{\mathrm{pre}}, -) \cong \{\text{G -invariant morphisms from X}\}.$$

Definition 3.16 (Categorical quotient)**.** *Let G be a group scheme acting on a scheme X. A categorical quotient for the action is a pair (Q, q) where $q : X \to Q$ is a G-invariant morphism which is initial among all G-invariant morphisms $X \to Y$.*

In particular, (Q, q) is a categorical quotient for the action of G on X if and only if Q co-represents $[X/G]^{\mathrm{pre}}$. We can actually impose a condition on the categorical quotient so that it is actually a coarse moduli space for $[X/G]^{\mathrm{pre}}$:

Definition 3.17. *Let G be a group scheme acting on scheme X. Then for a closed point $z \in X(\mathbb{C})$, the orbit map is the composition*

$$G \times \{z\} \hookrightarrow G \times X \to X.$$

The orbit of $z \in X(\mathbb{C})$ is the image of this composition, regarded as a topological subspace of X. Since it is locally closed, we may endow it with the induced reduced scheme structure.

Definition 3.18. *Let (Q, q) be a categorical quotient for the action of G on X. We say Q is an orbit space if $q : X \to Q$ separates the $G(\mathbb{C})$-orbits, i.e. if $z_1, z_2 \in X(\mathbb{C})$ and $q(z_1) = q(z_2)$, then z_1, z_2 are in the same $G(\mathbb{C})$-orbit, and $X(\mathbb{C})$ surjects onto $Q(\mathbb{C})$.*

We see that Q is an orbit space for the action of G on X if and only if Q is a coarse moduli space for $[X/G]^{\mathrm{pre}}$.

Example 3.19. *Recall from before that we claimed there was a short exact sequence*

$$1 \to \mathbb{G}_m \to \mathrm{GL}_n \to \mathrm{PGL}_n \to 1$$

of group schemes. In particular, we claimed that although the quotient moduli functor $[\mathrm{GL}_n/\mathbb{G}_m]^{pre}$ failed to be a Zariski sheaf, actually its sheafification is exactly PGL_n.

Then the universal property of sheafification implies that PGL_n co-represents $[\mathrm{GL}_n/\mathbb{G}_m]^{pre}$. Moreover, we know that $\mathrm{GL}_n(\mathrm{Spec}\,\mathbb{C})/\mathbb{G}_m(\mathrm{Spec}\,\mathbb{C}) \cong \mathrm{PGL}_n(\mathbb{C})$. So actually PGL_n is a coarse moduli space for $[\mathrm{GL}_n/\mathbb{G}_m]^{pre}$. In other words, PGL_n is an orbit space for the action of $\mathbb{G}_m$ on GL_n.

4. MVC 4/23/25 Basic geometric invariant theory, Picard functor

Previously, we saw examples of group schemes and saw situations where G can act on a scheme X but $[X/G]^{\mathrm{pre}}$ is not a Zariski sheaf and thus not representable. What these examples were supposed to lead us to, was the notion of co-representability and coarse moduli spaces. In doing so, we also encountered the notions of categorical quotients and orbit spaces, which is the concept of co-representability and coarse moduli spaces in the context of studying $[X/G]^{\mathrm{pre}}$.

In fact, under some assumptions, there is a general technique for producing such categorical quotients, which is called Geometric Invariant Theory[2], developed by David Mumford. We deal primarily with the affine case. Let's first see an example.

Example 4.1. *Consider $\mathbb{G}_m$ acting on $\mathbb{A}^2$ via $\lambda\cdot(z_1, z_2) = (\lambda z_1, \lambda z_2)$. So scheme-theoretically, the map*

$$\mathbb{G}_m \times \mathbb{A}^2 \to \mathbb{A}^2$$

is given on schemes by the map

$$\mathbb{C}[x, y] \to \mathbb{C}[x, y] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$$

where $x \mapsto x \otimes t$ and $y \mapsto y \otimes t$.

Note that, at least on $\mathbb{C}$-points, the $\mathbb{C}$-orbits of $\mathbb{A}^2$ under the action of $\mathbb{G}_m(\mathbb{C})$ are the punctured lines through the origin, and the origin itself. But remember, if we have a categorical quotient for this action of $\mathbb{G}_m$ on $\mathbb{A}^2$, then any map which is $\mathbb{G}_m$-invariant must be constant on the $\mathbb{G}_m(\mathbb{C})$-orbits... but looking at the orbits we have here, it actually must be constant everywhere on $\mathbb{A}^2$. So the claim is that the categorical quotient is just $\mathbb{A}^2 \to \operatorname{Spec}\mathbb{C}$.

There is an algebraic way of obtaining this categorical quotient. Namely, one should think of the categorical quotient as being the spectrum of the ring of functions which are $\mathbb{G}_m$ invariant. More specifically, note any $f \in H^0(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2}) = \mathbb{C}[x, y]$ corresponds to a map

$$\mathbb{A}^2 \to \mathbb{A}^1$$

given by $\mathbb{C}[t] \to \mathbb{C}[x, y]$ where $t \mapsto f$. To be invariant with respect to the $\mathbb{G}_m$ action on $\mathbb{A}^2$ means that the diagram

$$\begin{tikzcd}
\mathbb{G}_m \times \mathbb{A}^2 \arrow[r, "{(id, f_*)}"] \arrow[d] & \mathbb{G}_m \times \mathbb{A}^1 \arrow[d] \\
\mathbb{A}^2 \arrow[r, "f_*"'] & \mathbb{A}^1
\end{tikzcd}$$

[2]A more thorough set of notes on GIT is [Hos].

which on the level of affines is

$$\begin{tikzcd} \mathbb{C}[\lambda^{\pm 1}] \otimes_{\mathbb{C}} \mathbb{C}[x,y] & \mathbb{C}[\lambda^{\pm 1}] \otimes_{\mathbb{C}} \mathbb{C}[t] \arrow[l] \\ \mathbb{C}[x,y] \arrow[u] & \mathbb{C}[t] \arrow[l] \arrow[u] \end{tikzcd}$$

In order for this diagram to commute, one must have $f(x,y) = f(\lambda x, \lambda y)$ in the ring $\mathbb{C}[\lambda^{\pm 1}, x, y]$. This can only happen if $f \in \mathbb{C}$. Thus, we have an inclusion of the $\mathbb{G}_m$ invariant functions $\mathbb{C} \hookrightarrow \mathbb{C}[x,y]$, which induces the categorical quotient.

Example 4.2. *Consider $\mathbb{G}_m$ acting on $\mathbb{A}^2$ via $\lambda\cdot(z_1,z_2) = (\lambda z_1, \lambda^{-1} z_2)$. Scheme-theoretically, the map $\mathbb{G}_m \times \mathbb{A}^2 \to \mathbb{A}^2$ is given on rings by the map*

$$\mathbb{C}[x,y] \to \mathbb{C}[x,y] \otimes_{\mathbb{C}} \mathbb{C}[t,t^{-1}]$$

where $x \mapsto x \otimes t$ and $y \mapsto y \otimes t^{-1}$.

Consider what the ring of invariant functions are. The appropriate diagram for $f(x,y) \in H^0(\mathbb{A}^2, \mathcal{O}_{\mathbb{A}^2}) = \mathbb{C}[x,y]$ commutes if and only if $f(x,y) = f(\lambda x, \lambda^{-1} y)$ in the ring $\mathbb{C}[\lambda^{\pm 1}, x, y,]$. This holds only if f is generated by the monomial xy. So the functions invariant under this action by $\mathbb{G}_m$ is

$$\mathbb{C}[xy] \subseteq \mathbb{C}[x,y],$$

so we can express this categorical quotient as

$$\mathbb{A}^2 \to \mathbb{A}^1$$

where on closed points $(z_1,z_2) \mapsto z_1 z_2$.

Let's see what's going on with the $\mathbb{G}_m(\mathbb{C})$-orbits. The origin itself is an orbit, the punctured horizontal axis is an orbit, the punctured vertical axis, and then for $a \neq 0$, the conics $\{(z_1,z_2) \mid z_1 z_2 = a\}$ are orbits. We see that each of the conics map to $a \in \mathbb{A}^1(\mathbb{C})$, and the origin, punctured horizontal axis and the punctured vertical axis all collapse to the origin in $\mathbb{A}^1(\mathbb{C})$.

These situations generalize to a technique called affine GIT.

Theorem 4.3 (Affine GIT)**.** *Let G be a reductive algebraic group*[3] *acting on* $\operatorname{Spec} A$. *Let $A^G \subseteq A$ denote the regular functions on* $\operatorname{Spec} A$ *invariant under the action of G. Then*

$$q : \operatorname{Spec} A \to \operatorname{Spec} A^G$$

is the categorical quotient for the action of G on $\operatorname{Spec} A$. *Moreover:*

[3]The reason for the reductive assumption is the following: we are looking for an affine quotient. By Hilbert's Nullstellensatz, a k-algebra is the coordinate ring of an affine variety if and only if it is finitely generated over k and has no nilpotent elements. We have the candidate for the coordinate ring of the categorical quotient, namely A^G. But it turns out that A^G is only guaranteed to be finitely generated over k when G is reductive. This is Nagata's work on Hilbert's 14th problem. Familiar examples of reductive groups include $\mathbb{G}_m, \mathrm{GL}_n, \mathrm{PGL}_n$ and finite groups.

– $(\operatorname{Spec} A^G, q)$ *is a universal categorical quotient*[4]*, i.e. for any* $T \to \operatorname{Spec} A^G$*, the map* $T \times_{\operatorname{Spec} A^G} \operatorname{Spec} A \to T$ *is a categorical quotient for the induced action of* G *on* $T \times_{\operatorname{Spec} A^G} \operatorname{Spec} A$.
– q *is surjective on sets, and the topology on* $\operatorname{Spec} A^G$ *is quotient topology. If* $Z \subseteq \operatorname{Spec} A$ *is a* G*-invariant closed subscheme, then* $q(Z) \subseteq \operatorname{Spec} A^G$ *is closed subscheme.*
– *For* $x, y \in (\operatorname{Spec} \mathbb{A})(\mathbb{C})$*, then* $q(x) = q(y) \iff \overline{G(\mathbb{C})x} \cap \overline{G(\mathbb{C})y} \neq \emptyset$*. If* $Z_1, Z_2 \subseteq \operatorname{Spec} A$ *are* G*-invariant closed subschemes, then* $q(Z_1 \cap Z_2) = q(Z_1) \cap q(Z_2)$.

In particular, if the $G(\mathbb{C})$-orbits of $\operatorname{Spec} A$ are all closed, then for any $x, y \in (\operatorname{Spec} A)(\mathbb{C})$ such that $q(x) = q(y)$, we must have $\overline{G(\mathbb{C})x} \cap \overline{G(\mathbb{C})y} \neq \emptyset \implies G(\mathbb{C})x \cap G(\mathbb{C})y \neq \emptyset \implies G(\mathbb{C})x = G(\mathbb{C})y$. Thus, if all $G(\mathbb{C})$-orbits are closed, then $q : \operatorname{Spec} A \to \operatorname{Spec} A^G$ is an orbit space. And if it is an orbit space, then clearly orbits are closed.

Corollary 4.4. *Let* G *be a reductive group acting on* $\operatorname{Spec} A$*, and apply affine GIT to obtain*

$$q : \operatorname{Spec} A \to \operatorname{Spec} A^G.$$

Then $(\operatorname{Spec} A^G, q)$ *is an orbit space if and only if all* $G(\mathbb{C})$*-orbits are closed.*

The following is an important example where affine GIT can be used to find a categorical quotient for the action of a reductive group acting on a non-affine scheme.

Example 4.5 (Projective space is categorical quotient)**.** *We claim that* $\mathbb{P}^n$ *is the categorical quotient of* $\mathbb{G}_m$ *acting on* $\mathbb{A}^{n+1} \setminus \{0\}$*. Recall the action of* $\mathbb{G}_m$ *on* $\mathbb{A}^{n+1} \setminus \{0\}$ *is simply by scaling:* $\lambda \cdot (z_0, \cdots, z_n) = (\lambda z_0, \cdots, \lambda z_n)$.

We can cover $\mathbb{A}^{n+1} \setminus \{0\}$ *by* $\mathbb{G}_m$*-invariant affine opens* $U_i = \operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n]_{z_i}$*, for* $0 \leq i \leq n$*, where* $\mathbb{C}[z_0, \cdots, z_n]_{z_i} = \mathbb{C}[z_0, \cdots, z_n, \frac{1}{z_i}]$*. To see that each* U_i *is* $\mathbb{G}_m$*-invariant, note the action of* $\mathbb{G}_m$ *on* $\mathbb{A}^{n+1} \setminus \{0\}$ *is the restriction of the action of* $\mathbb{G}_m$ *on* $\mathbb{A}^{n+1}$*, and so it suffices to prove that* U_i *is a* $\mathbb{G}_m$*-invariant affine open of* $\mathbb{A}^{n+1}$*. Indeed this is true because the morphism*

$$\mathbb{C}[z_0, \cdots, z_n] \to \mathbb{C}[z_0, \cdots, z_n] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$$

where $z_j \mapsto z_j \otimes t$ *extends naturally to a morphism*

$$\mathbb{C}[z_0, \cdots, z_n, \frac{1}{z_i}] \to \mathbb{C}[z_0, \cdots, z_n, \frac{1}{z_i}] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}]$$

where $z_i^{-1} \mapsto z_i^{-1} \otimes t^{-1}$*. Then we have a commutative diagram*

$$\begin{tikzcd}
\mathbb{C}[z_0, \cdots, z_n] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}] \arrow[d] & \mathbb{C}[z_0, \cdots, z_n] \arrow[l] \arrow[d] \\
\mathbb{C}[z_0, \cdots z_n, z_i^{-1}] \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}] & \mathbb{C}[z_0, \cdots z_n, z_i^{-1}] \arrow[l]
\end{tikzcd}$$

[4]This relies on the characteristic 0 hypothesis [MFK94, Theorem 1.1].

and taking Spec *of the diagram, we see that U_i is $\mathbb{G}_m$ invariant.*

Next we apply affine GIT to each U_i. The $\mathbb{G}_m$-invariant subring of $\mathbb{C}[z_0, \cdots, z_n, z_i^{-1}]$ will be $\mathbb{C}[\frac{z_0}{z_i}, \cdots, \frac{z_n}{z_i}]$ and its inclusion induces the universal categorical quotient

$$q_i : U_i \to U_i^{\mathbb{G}_m} := \operatorname{Spec} \mathbb{C}[\frac{z_0}{z_i}, \cdots, \frac{z_n}{z_i}].$$

For each $j \neq i$, consider the open $\operatorname{Spec} \mathbb{C}[\frac{z_0}{z_i}, \cdots, \frac{z_n}{z_i}, (\frac{z_j}{z_i})^{-1}]$. Base changing q_i over this open subscheme, yields the open subscheme

$$\begin{tikzcd}
\operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n, z_i^{-1}, z_j^{-1}] \arrow[r] \arrow[d] & \operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n, z_i^{-1}] \arrow[d, "q_i"'] \\
\operatorname{Spec} \mathbb{C}[\frac{z_0}{z_i}, \cdots, \frac{z_n}{z_i}, (\frac{z_j}{z_i})^{-1}] \arrow[r] & \operatorname{Spec} \mathbb{C}[\frac{z_0}{z_i}, \cdots, \frac{z_n}{z_i}]
\end{tikzcd}$$

Note that this base change is again a categorical quotient, and it is exactly the categorical quotient obtained by applying affine GIT again to the action of $\mathbb{G}_m$ on $\operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n, z_i^{-1}, z_j^{-1}] \cong \operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n, (z_i z_j)^{-1}]$.

Note $\operatorname{Spec} \mathbb{C}[z_0, \cdots, z_n, z_i^{-1}, z_j^{-1}] \subseteq U_i$ is exactly the open subscheme of U_i which glues to the corresponding overlap on U_j. In summary, this means we have each $U_i \cap U_j$ admits categorical quotient $U_i \cap U_j \to (U_i \cap U_j)^{\mathbb{G}_m}$ by affine GIT, and $(U_i \cap U_j)^{\mathbb{G}_m}$ is identified with the images of $U_i \cap U_j$ under both quotient maps $q_i : U_i \to U_i^{\mathbb{G}_m}$ and $q_j : U_j \to U_j^{\mathbb{G}_m}$. Then by universal property of these categorical quotients as coarse moduli spaces, all the identifies satisfy cocycle conditions and so we get can glue these categorical quotients to obtain the usual map

$$\mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n.$$

In particular, we see now that $\mathbb{P}^n$ is a categorical quotient for the action of $\mathbb{G}_m$ on $\mathbb{A}^{n+1} \setminus \{0\}$ because for any $\mathbb{G}_m$-invariant map $\mathbb{A}^{n+1} \setminus \{0\} \to T$, we get $\mathbb{G}_m$-invariant maps $\{U_i \to T\}_{i=0}^n$, which descend to maps $\{U_i^{\mathbb{G}_m} \to T\}_{i=0}^n$ (and maps $(U_i \cap U_j)^{\mathbb{G}_m} \to T$), which by universal property glue compatible, so that we obtain a unique map $\mathbb{P}^n \to T$.

Example 4.6. *If X is quasi-projective over $\mathbb{C}$ and G is a finite group acting on X, then there exists a categorical quotient of X by G, which is also an orbit space. See Proposition 6.10.*

This situation of using affine GIT to identify $\mathbb{P}^n$ as the categorical quotient for the scaling action of $\mathbb{G}_m$ on $\mathbb{A}^{n+1} \setminus \{0\}$ is quite nice. In general, it is not guaranteed that a scheme X will admit even a covering by G-invariant affine opens. And even if it did, whether one can glue together the categorical quotients is subtle.

But at least when X is a projective variety, we can perform the following procedure. Suppose G is a reductive group acting on X. We hope for an embedding

$$X \hookrightarrow \mathbb{P}^n$$

along with a morphism of group schemes $G \to \mathrm{GL}_{n+1}$ such that the embedding is G-equivariant:

$$\begin{tikzcd} G \times X \arrow[r, hook] \arrow[dd] & G \times \mathbb{P}^n \arrow[d] \\ & \mathrm{GL}_{n+1} \times \mathbb{P}^n \arrow[d] \\ X \arrow[r, hook] & \mathbb{P}^n \end{tikzcd}$$

If we find ourselves in such a situation[5], we can consider the affine cone $\widetilde{X} \subseteq \mathbb{A}^{n+1}$. Since G admits a homomorphism to GL_{n+1}, it acts linearly on $\mathbb{A}^{n+1}$ i.e. respects the grading of $\mathbb{C}[z_0, \cdots, z_n]$. Furthermore, the G-equivariant embedding implies that $\tilde{X} \subseteq \mathbb{A}^{n+1}$ is G-invariant. In particular,

$$\tilde{X} = \operatorname{Spec} S = \operatorname{Spec} \frac{\mathbb{C}[z_0, \cdots, z_n]}{I},$$

where $S \cong \bigoplus_{m=0}^{\infty} H^0(X, \mathcal{O}_X(m))$. Then we define Proj S^G to be the GIT quotient of X. In particular, there is an open G-invariant subset $X^{\mathrm{ss}} \subseteq X$ called the semistable locus such that there is a map

$$X^{ss} \to \operatorname{Proj} S^G =: X//G$$

which is a universal categorical quotient and has all the properties of Theorem 4.3. A closed point $z \in X(\mathbb{C})$ is called semistable and $z \in X^{ss}$ if there exists some $s \in S^G$ such that $s(z) \neq 0$. Moreover, there is a finer G-invariant open X^s, called the stable locus, such that the restriction of the above map to X^s is an orbit space.

Example 4.7. *Consider the action of $\mathbb{G}_m$ on $\mathbb{P}^2_{[z_0:z_1:z_2]}$ defined by*

$$t \cdot [z_0 : z_1 : z_2] \mapsto [tz_0 : tz_1 : z_2] = [z_0 : z_1 : t^{-1} z_2].$$

There are two ways to linearize this action:

(1) *First way: define $\mathbb{G}_m \to \mathrm{GL}_3$ where $t \mapsto \begin{pmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 1 \end{pmatrix}$. On affine rings, this is the map*

$$\mathbb{C}[a_{ij}, \lambda]/\langle \det(a_{ij}) \cdot \lambda - 1 \rangle \to \mathbb{C}[t, t^{-1}]$$

where $a_{11} \mapsto t, a_{22} \mapsto t, a_{33} \mapsto 1$, $a_{ij} = 0$ for $i \neq j$, and $\lambda = t^{-2}$. Then considering the action of $\mathbb{G}_m$ on $\mathbb{A}^3$, the invariant subring is $f(z_0, z_1, z_2) \in \mathbb{C}[z_0, z_1, z_2]$ such that $f(tz_0, tz_1, z_2) = f(z_0, z_1, z_2) \implies f \in \mathbb{C}[z_2]$. So the categorical quotient is

[5]We call such a situation a linearisation of the action of G on X. The procedure is highly sensitive to choices of linearisations.

$\mathbb{A}^3 \to \mathbb{A}^1$ induced by $\mathbb{C}[z_2] \hookrightarrow \mathbb{C}[z_0, z_1, z_2]$. Then the claim is that there is a universal categorical quotient

$$\mathbb{P}^{2,ss} = \{[z_0 : z_1 : 1]\} \to \operatorname{Spec} \mathbb{C},$$

where the locus $\mathbb{P}^{2,ss} \subseteq \mathbb{P}^2$ is the one defined by $z_2 \neq 0$. If we set all points in $\mathbb{P}^{2,ss}$ to have coordinate $z_2 = 1$, then $\mathbb{P}^{2,ss} \cong \mathbb{A}^2$ and we see that $\mathbb{P}^{2,ss} \to \operatorname{Spec} \mathbb{C}$ is indeed the universal categorical quotient for the scaling action of $\mathbb{G}_m$ on $\mathbb{A}^2$.

(2) *Second way: define $\mathbb{G}_m \to \mathrm{GL}_3$ where $t \mapsto \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & t^{-1} \end{pmatrix}$. On affine rings, this is the map*

$$\mathbb{C}[a_{ij}, \lambda]/\langle \det(a_{ij}) \cdot \lambda - 1 \rangle \to \mathbb{C}[t, t^{-1}]$$

where $a_{11}, a_{22} \mapsto 1$, and $a_{33} \mapsto t^{-1}$ and $a_{ij} = 0$ for $i \neq j$, and $\lambda = t$. Then considering this action of $\mathbb{G}_m$ on $\mathbb{A}^3$, the invariant subring is comprised of those $f(z_0, z_1, z_2) \in \mathbb{C}[z_0, z_1, z_2]$ such that $f(z_0, z_1, t^{-1}z_2) = f(z_0, z_1, z_2) \implies f \in \mathbb{C}[z_0, z_1]$. Then the categorical quotient obtained is $\mathbb{A}^3 \to \mathbb{A}^2$ induced by $\mathbb{C}[z_0, z_1] \hookrightarrow \mathbb{C}[z_0, z_1, z_2]$. Then by the GIT procedure, there should be a categorical quotient

$$\mathbb{P}^{2,ss} = \{[z_0 : z_1 : z_2] \mid z_0, z_1 \text{ not both } 0\ \} \cong \mathbb{A}^2 \setminus \{0\} \sqcup \mathbb{A}^2 \setminus \{0\} \to \mathbb{P}^1$$

given by $[z_0 : z_1 : z_2] \mapsto [z_0 : z_1]$. A closed point $[z_0 : z_1 : z_2]$ is in $\mathbb{P}^{2,ss}$ if at least one of $z_0, z_1 \neq 0$. Then at least set-wise, the closed points of $\mathbb{P}^{2,ss} = \{[z_0 : z_1 : 0] \mid (z_0, z_1) \in \mathbb{A}^2 \setminus \{0\}\} \sqcup \{[z_0 : z_1 : 1] \mid (z_0, z_1) \in \mathbb{A}^2 \setminus \{1\}\}$. But note that for the action nof $\mathbb{G}_m$ on $\mathbb{P}^2$, we have $[z_0 : z_1 : 1]$ and $[z_0 : z_1 : \frac{1}{t}]$ are in the same orbit. As $t \to \infty$, the limit is $[z_0 : z_1 : 0]$. This implies that $\mathbb{G}_m(\mathbb{C})$ orbits of $\mathbb{A}^2 \setminus \{0\} \sqcup \mathbb{A}^2 \setminus \{0\}$ are really identified with the $\mathbb{G}_m(\mathbb{C})$ orbits of $\mathbb{A}^2 \setminus \{0\}$. So that's why the universal categorical quotient is $\mathbb{P}^1$ instead of $\mathbb{P}^1 \sqcup \mathbb{P}^1$.

The previous example illustrates that the GIT procedure for actions of reductive G on projective X is very sensitive to the choice of linearisation for the G-action. For example, a $\mathbb{C}$-point which is semistable under one linearisation may fail to be semistable in a different linearisation. Different linearisations can yield different universal categorical quotients for the semistable locus.

Here's an example where we need to actually choose an embedding.

Example 4.8. *Consider the action of $\mathbb{Z}/2\mathbb{Z}$ on $\mathbb{P}^1_{[z_0:z_1]}$ where*

$$[z_0 : z_1] \mapsto [-z_0 : z_1].$$

If we chose the obvious choice of linearisation of the action of $\mathbb{Z}/2\mathbb{Z}$ with respect to the identity map $id_{\mathbb{P}^2}$, we run into some issues. The invariant subring will be $\mathbb{C}[z_0^2, z_1] \subseteq \mathbb{C}[z_0, z_1]$. We want to do something a little easier.

Consider the Veronese embedding $\mathbb{P}^1 \hookrightarrow \mathbb{P}^2$ *given on coordinates by*

$$[z_0 : z_1] \mapsto [z_0^2 : z_0 z_1 : z_1^2].$$

This is induced by a linear map $\mathbb{A}^2 \to \mathbb{A}^3$ *where on functions we have*

$$\mathbb{C}[x, y, z] \to \mathbb{C}[z_0, z_1], x \mapsto z_0^2, y \mapsto z_0 z_1, z \mapsto z_1^2.$$

Then consider the homomorphism $\mathbb{Z}/2\mathbb{Z} \to \mathrm{GL}_3$ *where the nontrivial element maps to* $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. *This data gives a linearisation of the* $\mathbb{Z}/2\mathbb{Z}$ *action on* $\mathbb{P}^1$.

In particular, the cone of the image of $\mathbb{P}^1$ *under this Veronese is the closed subscheme* $\operatorname{Spec} \mathbb{C}[x, y, z]/\langle xz - y^2 \rangle$. *The invariant subring will be those elements* $f(x, y, z)$ *such that* $f(x, y, z) = f(x, -y, z)$. *Since any such* f *can be simplified so that the highest degree of* y *appearing is 1, we see that the invariant subring is* $\mathbb{C}[x, z] \subseteq \mathbb{C}[x, y, z]/\langle xz - y^2 \rangle$. *Then in particular, there should be a universal categorical quotient*

$$\mathbb{P}^{1,ss} \to \mathbb{P}^1, [z_0 : z_1] \mapsto [z_0^2 : z_1^2]$$

for the action of $\mathbb{Z}/2\mathbb{Z}$ *on* $\mathbb{P}^1$ *such that. Note* $\mathbb{P}^{1,ss}$ *is comprised of those points* $[z_0^2 : z_0 z_1 : z_1^2]$ *such that* z_0 *and* z_1 *are not both equal to 0... so* $\mathbb{P}^{1,ss} = \mathbb{P}^1$.

We end this section by discussing Picard functors. A very thorough reference is [Kle]. Suppose we have a scheme X. We might wonder whether we can turn the group $(\mathrm{Pic}(X), \otimes)$ into a group scheme. A naive attempt would be to define the absolute Picard functor

$$\underline{\mathrm{Pic}}_X : \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}} \to \mathrm{Grps}$$

in the following way: for $T \in \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}}$, define $\underline{\mathrm{Pic}}_X(T) = \mathrm{Pic}(X \times T)$. While this defines a moduli functor, it is not even a Zariski sheaf! To show that $\underline{\mathrm{Pic}}_X$ fails to be a Zariski sheaf, it's enough to show that for some $T \in \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}}$ with Zariski cover $\bigcup U_i$, we have

$$\underline{\mathrm{Pic}}_X(T) \to \prod \underline{\mathrm{Pic}}_X(U_i)$$

fails to be injective. Indeed, we can pick any $T \in \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}}$ such that $\mathcal{L} \in \mathrm{Pic}(T)$ is nontrivial and $\{U_i\}$ is a cover of T such that $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$. Then $\pi_T^*(\mathcal{L}) \in \underline{\mathrm{Pic}}_X(T)$ is nontrivial but its image in $\underline{\mathrm{Pic}}_X(U_i)$ is $\mathcal{O}_{X \times U_i}$ for every i.

The good news, however, is that with some assumptions the image of $\mathrm{Pic}(T)$ in $\underline{\mathrm{Pic}}_X(T)$ is exactly what obstructs $\underline{\mathrm{Pic}}_X$ from being a Zariski sheaf.

Definition 4.9. *Let* $X \to S$ *be a morphism of schemes. The relative Picard functor is the moduli functor*

$$\underline{\mathrm{Pic}}_{X/S} : (Sch/S)^{op} \to Grps$$

defined as sending $[T \to S]$ *to the group* $\frac{\mathrm{Pic}(X \times_S T)}{\pi_T^*(\mathrm{Pic}(T))}$.

Proposition 4.10. *Let $X \xrightarrow{f} S$ be a morphism of $\mathbb{C}$-schemes of finite type which admits a section $S \xrightarrow{s} X$. Suppose that for all $[T \to S] \in Sch/S$, the morphism*

$$\mathcal{O}_T \to (\pi_T)_* \mathcal{O}_{X \times_S T}$$

is an isomorphism. Then the relative Picard functor $\underline{\mathrm{Pic}}_{X/S}$ is a Zariski sheaf.

Proof. We want to show that for every $[T \to S] \in \mathrm{Sch}/S$ and every Zariski open cover (over S) $\{[U_i \to S]\}$ of $T \to S$, which is just equivalent to Zariski open cover $\{U_i\}$ of T, the sequence

$$0 \to \underline{\mathrm{Pic}}_{X/S}([T \to S]) \xrightarrow{\phi} \prod \underline{\mathrm{Pic}}_{X/S}([U_i \to S]) \xrightarrow{\psi} \prod \underline{\mathrm{Pic}}_{X/S}([U_i \cap U_j \to S])$$

is exact.

First things first. Recall $\underline{\mathrm{Pic}}_{X/S}([T \to S]) = \frac{\mathrm{Pic}(X \times_S T)}{(\pi_T)^* \mathrm{Pic}(T)}$. The reason for assuming the existence of a section $S \xrightarrow{s} X$ and for $\mathcal{O}_T \to (\pi_T)_* \mathcal{O}_{X \times_S T}$ to being an isomorphism, is to ensure that the functor

$$(\pi_T)^* : \mathrm{Pic}(T) \to \mathrm{Pic}(X \times_S T)$$

is injective and an isomorphism on Hom sets. Injectivity follows because the composition $S \xrightarrow{s} X \xrightarrow{f} S$ is identity, which induces a composition $T \to X \times_S T \xrightarrow{\pi_T} T$ which is also identity, forcing $(\pi_T)^* : \mathrm{Pic}(T) \to \mathrm{Pic}(X \times_S T)$ to be injective.

Now we claim that for $\mathcal{L}, \mathcal{L}' \in \mathrm{Pic}(T)$, we have

$$\mathrm{Hom}_T(\mathcal{L}, \mathcal{L}') \cong \mathrm{Hom}_{X \times_S T}((\pi_T)^* \mathcal{L}, (\pi_T)^* \mathcal{L}').$$

We have isomorphisms

$$\mathcal{L}' \cong (\pi_T)_* \mathcal{O}_{X \times_S T} \otimes_{\mathcal{O}_T} \mathcal{L}' \cong (\pi_T)_* (\pi_T)^* \mathcal{L}',$$

where the first is by tensoring our assumption and the second is by projection formula. Then

$$\mathrm{Hom}_T(\mathcal{L}, \mathcal{L}') \cong \mathrm{Hom}_T(\mathcal{L}, (\pi_T)_* (\pi_T)^* \mathcal{L}) \cong \mathrm{Hom}_{X \times_S T}((\pi_T)^* \mathcal{L}, (\pi_T)^* \mathcal{L}')$$

where the second isomorphism is by adjunction between π_* and π^*.

Now let's show ϕ is injective. Suppose we have $\mathcal{L} \in \frac{\mathrm{Pic}(X \times_S T)}{(\pi_T)^* \mathrm{Pic}(T)}$ such that $\mathcal{L}|_{X \times_S U_i} \in \frac{\mathrm{Pic}(X \times_S U_i)}{(\pi_{U_i})^* \mathrm{Pic}(U_i)}$ is trivial. Then $\mathcal{L}|_{X \times_S U_i} \cong \pi_{U_i}^* \mathcal{R}_i$ for some $\mathcal{R}_i \in \mathrm{Pic}(U_i)$. On overlaps $X \times_S (U_i \cap U_j)$ we have transition functions

$$(\pi_{U_i}^* \mathcal{R}_i)|_{X \times_S (U_i \cap U_j)} \cong (\pi_{U_j}^* \mathcal{R}_j)|_{X \times_S (U_i \cap U_j)}$$

$$\implies \pi_{U_i \cap U_j}^* \mathcal{R}_i|_{U_i \cap U_j} \cong \pi_{U_i \cap U_j}^* \mathcal{R}_j|_{U_i \cap U_j}$$

which satisfies cocycle conditions. But by what we deduced from our assumptions, this implies we must have transition functions

$$\mathcal{R}_i|_{U_i \cap U_j} \cong R_j|_{U_i \cap U_j}$$

also satisfying cocycle conditions, so that we get a line bundle $\mathcal{R} \in \mathrm{Pic}(T)$ such that $\pi_T^*\mathcal{R} \cong \mathcal{L}$, because the transition functions of $\mathcal{R}$ get identified with the transition functions of $\mathcal{L}$, per the isomorphism on Hom sets. This shows that ϕ is injective.

Finally, we claim $\ker\psi = \mathrm{im}\phi$. It is clear that we have $\ker\psi \supseteq \mathrm{im}\phi$. So it remains to show $\ker\psi \subseteq \mathrm{im}\phi$.

Then suppose we have $\mathcal{L}_i \in \mathrm{Pic}(U_i \times_S T)$ such that over $(U_i \cap U_j) \times_S T$, we have identifications

$$\theta_{ij} : \mathcal{L}_i|_{(U_i\cap U_j)\times_S T} \cong \mathcal{L}_j|_{(U_i\cap U_j)\times_S T} \otimes \pi^*_{U_i\cap U_j}\mathcal{R}_{ij},$$

where $\mathcal{R}_{ij} \in \mathrm{Pic}(U_i \cap U_j)$, for every i, j distinct (or if $i = j$, set $\mathcal{R}_{ij}$ to be trivial and the $\theta_{ij} = \mathrm{id}$). We want to find a $\widetilde{\mathcal{L}} \in \mathrm{Pic}(X\times_S T)$ such that the equivalence class of its restriction is

$$[\widetilde{\mathcal{L}}|_{(U_i\cap U_j)\times_S T}] = [\mathcal{L}_i] \in \underline{\mathrm{Pic}}_{X/S}([U_i \cap U_j \to S]).$$

Conveniently, recall that we have a section $S \xrightarrow{s} X$. Consider the diagram

$$\begin{tikzcd}
(U_i \cap U_j) \times_S X \arrow[r] \arrow[d] & U_i \cap U_j \arrow[r] \arrow[d] & (U_i \cap U_j) \times_S X \arrow[d] \\
U_i \times_S X \arrow[r] \arrow[d] & U_i \arrow[r] \arrow[d] & U_i \times_S X \arrow[d] \\
X \arrow[r, "f"'] & S \arrow[r, "s"'] & X
\end{tikzcd}$$

Let $\mathcal{N}_i \in \mathrm{Pic}(U_i \times_S X)$ denote the pullback of $\mathcal{L}_i^{-1} \in \mathrm{Pic}(U_i \times_S X)$ induced by $s \circ f$ (and be careful to note that $f \circ s$ is the one that is identity). Then define

$$\widetilde{\mathcal{L}_i} := \mathcal{L}_i \otimes \mathcal{N}_i \in \mathrm{Pic}(U_i \times_S X).$$

Note that $[\widetilde{\mathcal{L}_i}] = [\mathcal{L}_i] \in \underline{\mathrm{Pic}}_{X/S}([U_i \to S])$, since $\mathcal{N}_i \in \pi^*_{U_i}\mathrm{Pic}(U_i) \subseteq \mathrm{Pic}(U_i \times_S X)$. Furthermore, over $(U_i \cap U_j) \times_S X$, we have

$$\widetilde{\mathcal{L}_i}|_{(U_i\cap U_j)\times_S X} \cong \mathcal{L}_i|_{(U_i\cap U_j)\times_S X} \otimes \mathcal{N}_i|_{(U_i\cap U_j)\times_S X}$$

$$\cong \mathcal{L}_j|_{(U_i\cap U_j)\times_S T} \otimes \pi^*_{U_i\cap U_j}\mathcal{R}_{ij} \otimes \mathcal{N}_i$$

$$\cong \mathcal{L}_j|_{(U_i\cap U_j)\times_S T} \otimes \pi^*_{U_i\cap U_j}\mathcal{R}_{ij} \otimes \mathcal{N}_j \otimes \pi^*_{U_i\cap U_j}(\mathcal{R}_{ij}^{-1})$$

$$\mathcal{L}_j|_{(U_i\cap U_j)\times_S T} \otimes \mathcal{N}_j \cong \widetilde{\mathcal{L}_j}|_{(U_i\cap U_j)\times_S X},$$

where here the identification $\mathcal{N}_i \cong \mathcal{N}_j \otimes \pi^*_{U_i\cap U_j}(\mathcal{R}_{ij}^{-1})$ crucially relies on the fully faithfulness and injectivity of pullback of lines bundles that we proved in the beginning. Altogether, this shows that the $\widetilde{\mathcal{L}_i}$ glue together to a $\widetilde{\mathcal{L}}$ which restricts to the data we started with, thereby showing what we want. $\square$

The following is beyond our reach, but is an important result.

Theorem 4.11. *Let $X \to S$ be a morphism of $\mathbb{C}$-schemes of finite type which admits a section $S \to X$, such that for all $T \to S$ we have*

$$\mathcal{O}_T \cong \pi_{T,*}\mathcal{O}_{T\times_S X}.$$

Moreover, suppose $X \to S$ is projective and has integral fibers. Then $\underline{\mathrm{Pic}}_{X/S}$ is representable by a locally of finite-type, separated scheme $\mathrm{Pic}_{X/S}$ with quasi-projective connected components.

5. Classifying conics in the plane

The conics in the plane $\mathbb{P}^2_{\mathbb{C}}$ are those subvarieties cut out by elements in $H^0(\mathcal{O}_{\mathbb{P}^2}(2))$. We'd like to classify these conics. Then we enhance our classification to the following moduli functor:

$$\mathcal{M} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $\mathcal{M}(T) = \{ \begin{tikzcd} \mathcal{X} \arrow[r, hook] \arrow[d] & \mathbb{P}^2 \times T \arrow[dl] \\ T \end{tikzcd} \mid \mathcal{X}$ effective Cartier[6] of $\mathbb{P}^2\times T, \mathcal{X}_t \subseteq \mathbb{P}^2$ conic for every t $\in T(\mathbb{C})\}$. It turns out that $\mathbb{P}(H^0(\mathcal{O}_{\mathbb{P}^2}(2))$ is the fine moduli space representing $\mathcal{M}$. The obvious incidence correspondence defines the universal family $\subseteq \mathbb{P}^2_{[x_0:x_1:x_2]} \times \mathbb{P}(H^0(\mathcal{O}_{\mathbb{P}^2}(2))$.

Now, we could try to spice things up a little. Suppose we were trying to classify plane conics up to projective equivalence. More precisely, we consider conics in $\mathbb{P}^2$ isomorphic if there is an action of $\mathrm{PGL}_2 = \mathrm{Aut}(\mathbb{P}^2)$ mapping one conic to the other. Then we'd enhance our classification problem to the moduli functor

$$\widetilde{\mathcal{M}} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $\widetilde{\mathcal{M}}(T) = \{ \begin{tikzcd} \mathcal{X} \arrow[r, hook] \arrow[d] & \mathbb{P}^2 \times T \arrow[dl] \\ T \end{tikzcd} \mid \mathcal{X}$ effective Cartier divisor of $\mathbb{P}^2\times T, \mathcal{X}_t \subseteq \mathbb{P}^2$ conic for every t $\in T(\mathbb{C})\}/\sim$ where two families $\mathcal{X}$ and $\mathcal{X}'$ are equivalent if there exists an automorphism in $\mathrm{Aut}_T(\mathbb{P}^2\times T)$ transforming one to the other. Every equation of a plane conic can be written as

$$\begin{pmatrix} x_0 & x_1 & x_2 \end{pmatrix} M \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix},$$

where M is a symmetric 3×3 matrix. It happens to be that there are exactly 3 isomorphism types of plane conics. An easy way to see this is to note that every symmetric matrix over $\mathbb{C}$ can be diagonalized, so one can change basis so that M is diagonal. Then one sees that the 3 isomorphism types are

[6]Asking for effective cartier is equivalent to asking for flatness over T, since the fibers are effective carter [Sta25, Tag 062Y]

(1) Nonsingular $(x_0^2 + x_1^2 + x_3^2)$
(2) Two lines intersecting nodally $(x_0^2 + x_1^2)$
(3) A double line (x_0^2)

Then if $\widetilde{M}$ had a fine moduli space, we would expect it to be $\operatorname{Spec}\mathbb{C} \sqcup \operatorname{Spec}\mathbb{C} \sqcup \operatorname{Spec}\mathbb{C}$; a point for each plane conic isomorphism type. The family over $\mathbb{A}^1$, given by

$$\{2x_0x_1 - tx_2^2 = 0\} \subseteq \mathbb{A}_t^1 \times \mathbb{P}^2_{[x_0:x_1:x_2]}$$

shows that $\widetilde{M}$ fails to admit a fine moduli space, since in this family all fibers over $t \neq 0$ are smooth but the fiber over $0 \in \mathbb{A}_t^1$ is singular.

Instead, we could try to restrict ourselves to smooth conics up to projective equivalence, and modify our moduli functor to

$$\widetilde{\mathcal{M}}^{\mathrm{sm}} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $\widetilde{\mathcal{M}}^{\mathrm{sm}}(T) = \left\{ \begin{tikzcd} \mathcal{X} \arrow[r, hook] \arrow[d] & \mathbb{P}^2 \times T \arrow[dl] \\ T & \end{tikzcd} \;\middle|\; \begin{array}{c} \mathcal{X} \text{ effective Cartier divisor of } \mathbb{P}^2 \times T \\ \\ \mathcal{X}_t \subseteq \mathbb{P}^2 \text{ smooth conic for every t} \in T(\mathbb{C}) \end{array} \right\}/\sim$ But if $\widetilde{\mathcal{M}}^{\mathrm{sm}}$ were to be finely represented, it would need to be represented by $\operatorname{Spec}\mathbb{C}$, as there is only one nonsingular plane conic, up to projective equivalence. Then every element $[\mathcal{X} \to T]$ of $\widetilde{\mathcal{M}}^{\mathrm{sm}}(T)$ would need to be isomorphic to a trivial family, i.e. $\mathcal{X} \to T$ is isomorphic to $C \times T$ for some nonsingular plane conic C. It turns out, however, that the family

$$\{2x_0x_1 - tx_2^2 = 0\} \subseteq \mathbb{A}_t^1 \setminus \{0\} \times \mathbb{P}^2_{[x_0:x_1:x_2]}$$

is not trivializable. Thus, $\widetilde{\mathcal{M}}^{\mathrm{sm}}$ also does not admit a fine moduli space.

While $\operatorname{Spec}\mathbb{C}$ is not a fine moduli space for $\widetilde{\mathcal{M}}^{\mathrm{sm}}$, we claim it is an orbit space for the natural action of PGL_3 on $\widetilde{\mathcal{M}}^{\mathrm{sm}}$. Note that $\widetilde{\mathcal{M}}^{\mathrm{sm}} = [\mathcal{M}^{\mathrm{sm}}/\mathrm{PGL}_3]^{\mathrm{pre}}$. Note that $\mathcal{M}^{\mathrm{sm}}$ is finely represented by an affine scheme. To see this, note $\mathcal{M}$ is finely represented by $\mathbb{P}(V)$, and $\mathcal{M}^{\mathrm{sm}} \subseteq \mathbb{P}(V)$ is the complement of a degree 3 hypersurface in $\mathbb{P}(V)$. Then we see it is affine because if we take a degree 3 Veronese embedding, the image of the zero locus of the degree 3 hypersurface is the intersection of a hyperplane in $\mathbb{P}(\mathrm{Sym}^3 V)$ with the image of $\mathbb{P}(V)$ under the Veronese embedding; then the complement is the affine complement of the hyperplane intersecting the image of $\mathbb{P}(V)$, so we get a closed subscheme of an affine space. Furthermore, $\mathcal{M}^{\mathrm{sm}}$ is PGL_3 invariant and there's only one $\mathrm{PGL}_3(\mathbb{C})$ orbit since the action is transitive on smooth plane conics, so by affine GIT the categorical quotient is $\operatorname{Spec}\mathbb{C}$. Since there's only one orbit, it is an orbit space. This shows that $\operatorname{Spec}\mathbb{C}$ is an orbit space for the action of PGL_3 on $\widetilde{\mathcal{M}}^{\mathrm{sm}}$.

Finally, note that the moduli functor of all plane conics up to projective equivalence may be identified as

$$\widetilde{\mathcal{M}} \cong [\mathcal{M}/\mathrm{PGL}_3]^{\mathrm{pre}} \cong [\mathcal{M}/\mathrm{SL}_3]^{\mathrm{pre}}.$$

The linearly reductive group SL_3 acts on $\mathbb{P}(V)$, where V is the $\mathbb{C}$-vector space of symmetric 3×3 matrices, by $g\cdot M := g^T Mg$ where $g \in \mathrm{SL}_3$ and $[M] \in \mathbb{P}(V)$. Then let's apply projective GIT to this situation and see what happens.

We can linearize the action of SL_3 on $\mathbb{P}(V)$ by choosing a lift $\mathrm{SL}_3 \to \mathrm{GL}(V)$ to act on $\mathrm{Sym}^* V$. To make this concrete, note that $\mathrm{Sym}^* V$ is the $\mathbb{C}$-algebra $\mathbb{C}[a, b, c, d, e, f]$, where the matrix $\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}$ represents the conic with equation $ax_0^2 + cx_1^2 + fx_2^2 + 2bx_0x_1 + 2dx_0x_2 + 2ex_1x_2$. The group scheme multiplication

$$\mathrm{SL}_3 \times \mathrm{Spec}\,\mathrm{Sym}^* V \longrightarrow \mathrm{Spec}\,\mathrm{Sym}^* V$$

is induced by the map on rings

$$\phi : \mathrm{Sym}^* V \longrightarrow \mathrm{Sym}^* V \otimes_{\mathbb{C}} \frac{\mathbb{C}[e_{ij}]}{\langle \det(e_{ij}) - 1\rangle},$$

and where a, b, c, d, e, f should be mapped to can be read off from multiplying

$$\begin{pmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{pmatrix}\begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}\begin{pmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{pmatrix}.$$

In particular, the SL_3 invariant functions of $\mathbb{C}[a, b, c, d, e, f]$ should be those polynomials $\mathcal{P}(a, b, c, d, e, f)$ such that

$$\mathcal{P}(a, b, c, d, e, f) = \mathcal{P}(\phi(a), \phi(b), \phi(c), \phi(d), \phi(e), \phi(f)).$$

If one thinks about how SL_3 acts on the conics, it has three orbits: those with nonzero determinant, the nodal intersecting lines, and the double lines. On each of these, it acts transitively. This makes it believable that the SL_3-invariant functions are

$$\mathbb{C}[a, b, c, d, e, f]^{\mathrm{SL}_3} = \mathbb{C}[\det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}] \subseteq \mathbb{C}[a, b, c, d, e, f].$$

Then by the projective GIT procedure, there should be a semistable locus

$$\mathbb{P}(V)^{ss} \longrightarrow \mathrm{Proj}\ \mathbb{C}[\det \begin{pmatrix} a & b & d \\ b & c & e \\ d & e & f \end{pmatrix}] = \mathrm{Spec}\,\mathbb{C}.$$

The closed points in the semistable locus are exactly those with determinant $\neq 0$. Thus, $\mathcal{M}^{\mathrm{sm}}$ is the semistable locus. The singular conics are unstable under this choice of linearization. This is another way of showing that $\mathrm{Spec}\,\mathbb{C}$ is the orbit space for the action of $\mathrm{SL}_3 = \mathrm{PGL}_3$ on the locus of nonsingular plane conics.

6. MVC 4/30/25: Picard scheme and symmetric products of curves

Recall that last time we ended with asserting the following representability result:

Theorem 6.1. *Let $X \xrightarrow{f} S$ be a projective morphism with integral fibers, admitting a section $S \xrightarrow{s} X$ and such that for every $T \to S$, we have $\mathcal{O}_T \to \pi_{T,*}\mathcal{O}_{T\times_S X}$ is an isomorphism. Then the relative Picard functor*

$$\underline{\mathrm{Pic}}_{X/S}$$

is representable by the Picard scheme $\mathrm{Pic}_{X/S}$, which has quasi-projective connected components.

In particular, note that any integral projective $\mathbb{C}$-scheme $X \subseteq \mathbb{P}^n$ admits the structure morphism $X \to \operatorname{Spec}\mathbb{C}$. This admits a section, and is trivially a projective morphism with integral fibers. Furthermore, for any $T \to \operatorname{Spec}\mathbb{C}$, note the regular functions of X are the constants so $\pi_{T,*}\mathcal{O}_{T\times X}$ is a trivial line bundle on T; thus, $\mathcal{O}_T \to \pi_{T,*}\mathcal{O}_{T\times X}$ is an isomorphism. Thus, the hypotheses of Theorem 6.1 are satisfied in this un-relative setting.

Theorem 6.2. *Let $X \subseteq \mathbb{P}^n$ be a projective integral $\mathbb{C}$-scheme. Then the Picard functor $\underline{\mathrm{Pic}}_{X/\mathbb{C}}$ is represented by a group scheme $\mathrm{Pic}_{X/\mathbb{C}}$ with quasi-projective connected components.*

If we take the analytification of X, i.e. take the $\mathbb{C}$-points and consider them as a complex analytic space, then we can take the long exact sequence of the exponential exact sequence

$$0 \to \underline{\mathbb{Z}}_X \to \mathcal{O}_X \to \mathcal{O}_X^\times \to 0$$

to derive that

$$\frac{H^1(X,\mathcal{O}_X)}{\mathrm{im}[H^1(X;\mathbb{Z}) \to H^1(X,\mathcal{O}_X)]} \cong \ker[c_1 : \mathrm{Pic}(X) \to H^2(X;\mathbb{Z})].$$

In particular, one sees that $\mathrm{Pic}(X)$ as an analytic space has tangent space at the identity

$$T_{[\mathcal{O}_X]} = H^1(X,\mathcal{O}_X).$$

The same result holds true in the algebraic category.

Proposition 6.3. *Let X be a projective $\mathbb{C}$-scheme.*

1. *There is a universal family $\mathcal{L} \in \mathrm{Pic}(\mathrm{Pic}_X \times X)$ such that $\mathcal{L}_{[L]\times X} \cong L$, well-defined up to tensor products by line bundles in $\pi^*_{\mathrm{Pic}_X}\mathrm{Pic}(\mathrm{Pic}_X)$.*
2. *$T_{[\mathcal{O}_X]}\mathrm{Pic}_{X/} = H^1(X,\mathcal{O}_X)$. The Picard scheme is smooth if its tangent space has dimension $h^1(X,\mathcal{O}_X)$ everywhere.*
3. *The connected component of the identity $\mathrm{Pic}^0_X \subseteq \mathrm{Pic}_X$ is quasi-projective. If X is normal, then it is projective and an abelian variety.*

(4) *The Neron-Severi group of* X *is the abelian group*

$$NS(X) := \mathrm{Pic}_X(\mathbb{C})/\mathrm{Pic}_X^0(\mathbb{C})$$

parameterizing the connected components of Pic_X*. It is finitely generated of rank* $\rho(X)$*, which we call the Picard number of* X*.*

Example 6.4. *Let* $X = \mathrm{Spec}\,\mathbb{C}$*. Then for every* $T \in Sch_{\mathbb{C}}$*, note* $\underline{\mathrm{Pic}}_X(T) = \{1\}$*, so* $\mathrm{Pic}_{\mathrm{Spec}\,\mathbb{C}} = \mathrm{Spec}\,\mathbb{C}$*.*

Example 6.5. *Let* $X = \mathbb{P}^n$*. Then* $H^1(X, \mathcal{O}_X) = 0$*, so* $\mathrm{Pic}_{\mathbb{P}^n}$ *is a 0-dimensional scheme. Then we see that*

$$\mathrm{Pic}_{\mathbb{P}^n} = \bigsqcup_{k\in\mathbb{Z}} [\mathcal{O}_{\mathbb{P}^n}(k)]$$

the Picard scheme of $\mathbb{P}^n$ *is the group scheme associated to* $\mathbb{Z}$*.*

Note that this implies that if T *is connected, then*

$$\underline{\mathrm{Pic}}_{\mathbb{P}^n}(T) = \frac{\mathrm{Pic}(T\times\mathbb{P}^n)}{\pi_T^*\,\mathrm{Pic}(T)} \cong \mathrm{Hom}(T, \mathbb{P}^n) \cong \mathrm{Pic}(\mathbb{P}^n) \implies \mathrm{Pic}(T\times\mathbb{P}^n) \cong \mathrm{Pic}(T)\times\mathrm{Pic}(\mathbb{P}^n).$$

Corollary 6.6. *Let* X *be projective* $\mathbb{C}$*-scheme such that* $H^1(X, \mathcal{O}_X) = 0$*. Then its Picard scheme* Pic_X *is 0-dimensional and discrete, namely it equals* $\bigsqcup_{\mathrm{Pic}(X)} \mathrm{Spec}\,\mathbb{C}$*, and for any connected* T *of finite type, we have*

$$\mathrm{Pic}(T\times X) \cong \mathrm{Pic}(T) \cong \mathrm{Pic}(X).$$

For example, X *could be a projective K3 surface, such as the Fermat quartic surface.*

In addition, for any such X *with* $H^1(X, \mathcal{O}_X) = 0$*, note* $\mathrm{Pic}(C) = \mathrm{Pic}_X(\mathbb{C})$ *is torsion free and in particular the only degree 0 line bundle is the trivial one* $\mathcal{O}_X$*.*

A non-example is the following: $\mathbb{A}^1$ is not projective as it has non constant regular functions. In particular, one can show that the Picard functor $\underline{\mathrm{Pic}}_{\mathbb{A}^1}$ is not representable.

Example 6.7. *Let* C *be a smooth curve of genus* g*. Smooth curves are always projective. So* Pic_C *will be a* g*-dimensional variety, and in particular the degree map*
$\deg : \mathrm{Pic}_C(\mathbb{C}) \to \mathbb{Z}$ *implies that the Picard scheme of the curve is*

$$\mathrm{Pic}_C = \bigsqcup_{k\in\mathbb{Z}} \mathrm{Pic}_C^d,$$

where Pic_C^d *is the quasi-projective component whose* $\mathbb{C}$*-points are the line bundles of degree* d *on* C*.*

For curves, a moduli functor related to $\underline{\mathrm{Pic}}_C$ is the moduli functor parameterizing effective divisors.

Definition 6.8. *Define*

$$\underline{Div}_C : (Sch_{\mathbb{C}})^{op} \to Sets$$

to be the moduli functor sending

$$T \mapsto \{ \begin{tikzcd} D \arrow[r, hook] \arrow[d] & T \times C \arrow[dl] \\ T & \end{tikzcd} \mid D \text{ is a relative effective Cartier divisor}\}.$$

Recall that D is a relative effective Cartier divisor if D is an effective Cartier divisor of $\mathcal{O}_{T\times C}$ and $D \to T$ is a flat morphism. This algebraic definition is capturing the notion of a family of divisors of C parameterized by T. Note for every $t \in T(\mathbb{C})$, $D_t \subseteq C$ is an effective Cartier divisor, and will have a degree $\deg D_t$. By upper-semicontinuity, $\chi(\mathcal{O}_X(D_t)) = \chi(\mathcal{O}_{T\times C}(D)|_{\{t\}\times C})$ is locally constant as a function of $t \in T(\mathbb{C})$, which by Riemann-Roch implies $\deg D_t$ is locally constant as a function of t. Then the following subfunctors are both open and closed subfunctors of $\underline{\mathrm{Div}}_C$:

Definition 6.9. *The moduli functor*

$$\underline{Div}^d_C : (Sch_{\mathbb{C}})^{op} \to Sets$$

where

$$T \mapsto \{ \begin{tikzcd} D \arrow[r, hook] \arrow[d] & T \times C \arrow[dl] \\ T & \end{tikzcd} \mid D \text{ is a relative effective Cartier divisor and} \deg D_t = d, \forall t \in T(\mathbb{C})\}$$

is both an open and closed subfunctor of $\underline{Div}_C$.

The moduli functor $\underline{\mathrm{Div}}^d_C$ classifies the effective Carter divisors of degree d on C, i.e. sums of points $\sum_{i=1}^d x_i$ where $x_i \in C(\mathbb{C})$. There are natural examples of relative effective Cartier divisors in $\underline{\mathrm{Div}}_C(T)$. For example, given any morphism $\phi : T \to C$, we can consider the fiber product

$$\begin{tikzcd} \Gamma_\phi \arrow[r] \arrow[d, hook] & \Delta \arrow[d, hook] \\ T \times C \arrow[r, "(\phi,\mathrm{id})"'] & C \times C \end{tikzcd}$$

Then $\Gamma_f \subseteq T \times C$ is a relative effective Cartier divisor. In particular, fixing $\phi_j : T \to C$ for $1 \leq j \leq d$, we have $[\sum_{j=1}^d \Gamma_{\phi_j} \subseteq T \times C] \in \underline{\mathrm{Div}}^d_C$.

A natural candidate for a scheme representing $\underline{\mathrm{Div}}^d_C$ is the d-th symmetric product of a curve $\mathrm{Sym}^d C$, which is defined to be the orbit space for the natural action of Σ_d on C^d. To show that $\mathrm{Sym}^d C$ exists, let us show the following more general statement.

Proposition 6.10. *Let G be a finite group acting on a quasiprojective $\mathbb{C}$-scheme X. Then the categorical qoutient exists $X//G$, and in particular is an orbit space.*

Proof. For any point $p \in X$, let U denote some affine open containing all the (finitely many) G-orbits of p. Then the action of G will transport U around; take the intersection of the orbit of U, which will be nonempty because of the Zariski topology/they all contain the G-orbits of p. This intersection will again be an affine open of our point since X is separated, but furthermore is G-invariant. Do this for all points in X. Then we have covered X by G-invariant affine opens.

Let U_i and U_j be two G-invariant affine opens. Let $q_i : U_i \to U_i^G$ and $q_j : U_j \to U_j^G$ be the affine GIT quotients 4.3. The goal is to glue U_i^G and U_j^G together compatibly. Consider the intersection $U_i \cap U_j$. This is again G-invariant, and since X is separated, is a G-invariant affine open. Consider the image $q_i(U_i \cap U_j) \subseteq U_i^G$ of $U_i \cap U_j$ with respect to q_i. Note that $q_i^{-1}q_i(U_i \cap U_j) \supseteq U_i \cap U_j$. But note that we must actually have equality; otherwise there is some $\mathbb{C}$-point in $q_i^{-1}q_i(U_i \cap U_j) \setminus U_i \cap U_j$ which maps to the same point in $q_i(U_i \cap U_j)$ as some point in $U_i \cap U_j$. But then the closure of their G-orbits must intersect. But since G is discrete, the G-orbits are already closed, which means that the points are in the same G-orbit, which would contradict $U_i \cap U_j$ being G-invariant. So $q_i^{-1}q_i(U_i \cap U_j) = U_i \cap U_j$. By the quotient topology, $q_i(U_i \cap U_j) \subseteq U_i^G$ is an open subset, and taking fiber products and using the universality of $q_i : U_i \to U_i^G$ as categorical quotient, we have

$$U_i \cap U_j \to q_i(U_i \cap U_j) \subseteq U_i^G$$

is the universal categorical quotient with respect to the action of G on $U_i \cap U_j$ obtained via affine GIT. The same holds for

$$U_i \cap U_j \to q_j(U_i \cap U_j) \subseteq U_j^G.$$

This implies there is an isomorphism $q_j(U_i \cap U_j)$ and $q_i(U_i \cap U_j)$ which is compatible with the G-action on $U_i \cap U_j$. This implies that we can glue together the categorical quotients

$$q_i : U_i \to U_i^G \text{ and } q_j : U_j \to U_j^G$$

along the categorical quotient

$$q_{ij} : U_i \cap U_j \to (U_i \cap U_j)^G.$$

We can do this for all U_i and their intersections with other U_j's, and there is compatibility on triple overlaps by universal properties. The result is a map

$$X \to X^G$$

which we claim is a categorical quotient for the action of G on X. Indeed, if $X \to T$ is a G-invariant morphism, then it restricts to G-invariant $U_i \to T$, which must factor uniquely thorugh $U_i \to U_i^G$. The uniqueness implies that the morphisms $U_i^G \to T$ must glue together to a unique $X^G \to T$ such that $X \to X^G \to T$ gives back $X \to T$. This shows that the categorical quotient for the action of G on X exists. Furthermore, since the G orbits of the $\mathbb{C}$-points of X are closed since G is finite, we have $X \to X^G$ is an orbit space. $\square$

By proposition 6.10, there is an orbit space for the action of Σ_d on C^d. We call this orbit space $\operatorname{Sym}^d C$.

Proposition 6.11. *Let C be a smooth curve. Consider the natural action of Σ_d on C^d. The orbit space $\operatorname{Sym}^d C$ is a smooth variety of dimension d.*

Proposition 6.12. *Let C be a smooth $\mathbb{C}$-curve. The symmetric product $\operatorname{Sym}^d C$ represents the moduli functor $\underline{\mathrm{Div}}^d_C$.*

Proof. First, we claim the following construction is the universal family on $\operatorname{Sym}^d C$: note there are projection morphisms $\pi_i : C^d \to C$, where the index i denotes projection onto the i-th factor. Then there is a relative effective Cartier divisor $\sum_{i=1}^d \Gamma_{\pi_i} \subseteq C^d \times C$. Under the map

$$C^d \times C \to \operatorname{Sym}^d C \times C,$$

$\sum_{i=1}^d \Gamma_{\pi_i}$ remains well-defined and we can call its image $D \subseteq \operatorname{Sym}^d C \times C$; it is again a relative effective Cartier divisor, so that the fiber of D over $[\sum_{i=1}^d x_i]$ is the closed subscheme

$$D_{[\sum_{i=1}^d x_i]} = \sum_{i=1}^d x_i \subseteq C.$$

Then we claim that this $D \subseteq \operatorname{Sym}^d C \times C$ is the universal family over $\operatorname{Sym}^d C$.

Consider some element in $\underline{\mathrm{Div}}^d_C(T)$. Let us first suppose that it is of the form $\sum_{i=1}^d \Gamma_{\phi_i}$ for some morphisms $\phi_i : T \to C$. Then we see that we have a pullback diagram

$$\begin{tikzcd}
\sum_{i=1}^d \Gamma_{\phi_i} \arrow[rr] \arrow[d, hook] & & D \arrow[d, hook] \\
T \times C \arrow[r, "{((\phi_1,\cdots,\phi_d),\mathrm{id})}"'] & C^d \times C \arrow[r] & \operatorname{Sym}^d C \times C
\end{tikzcd}$$

Next, we claim that for any relative effective Cartier divisor $[\mathcal{X} \to T] \in \underline{\mathrm{Div}}^d_C(T)$, there exists finite flat morphism $T' \to T$ such that the pullback of $\mathcal{X}$ along $T' \to T$ is a relative effective Cartier divisor $\sum_{i=1}^d \Gamma_{\theta_i}$ for some $\theta_i : T' \to C$. Once this is proven, one can use faithfully flat descent so that the morphism $T' \to \operatorname{Sym}^d C$ descends to a morphism $T \to \operatorname{Sym}^d C$. □

Having now discussed symmetric products of curves and the moduli functors they represent, we turn to the Abel-Jacobi maps.

Let $z_0 \in C$. Consider the morphism of schemes

$$\mu^1 : C \to \operatorname{Pic}^0_C$$

defined by the morphism on T-points where $[T \xrightarrow{f} C] \in C(T)$ is mapped to the element

$$\mathcal{O}_{T\times C}(\Gamma_f - T \times \{z_0\}) \in \operatorname{Pic}^0_C(T).$$

On $\mathbb{C}$-points, this morphism sends

$$z \in C(\mathbb{C}) \mapsto \mathcal{O}_C(z - z_0) \in \mathrm{Pic}^0(C).$$

Since Pic^0_C is a group scheme, there is an induced morphism

$$(\mu^1)^{\times d} : C^d \to \mathrm{Pic}^0_C$$

where on T-points, we have $(f_1, \cdots f_d) \in C^d(T)$ are mapped to

$$\bigotimes_{i=1}^{d} \mathcal{O}_{T\times C}(\Gamma_{f_i} - T \times \{z_0\}) \in \mathrm{Pic}^0_C(T).$$

Note that this morphism is Σ_d-invariant with respect to the action of Σ_d on C^d. Then by the universal property of $\mathrm{Sym}^d C$ as an orbit space for this action, we have that $(\mu^1)^{\times d}$ factors through a unique morphism called the Abel-Jacobi map:

$$\mu^d : \mathrm{Sym}^d C \to \mathrm{Pic}^0_C,$$

where on $\mathbb{C}$-points we have $\sum_{i=1}^d z_i \in \mathrm{Sym}^d C(\mathbb{C})$ is mapped to

$$\mathcal{O}_C(\sum_{i=1}^{d} z_i - dz_0) \in \mathrm{Pic}^0(C).$$

7. MVC 5/7/25: Abel-Jacobi maps and theta divisors

Let C be a smooth curve of genus g. Fixing a basepoint $z_0 \in C(\mathbb{C})$, recall in the previous section we defined the Abel-Jacobi maps

$$\mu^d : \mathrm{Sym}^d C \to \mathrm{Pic}^0_C,$$

which are obtained by using the universal property of $\mathrm{Sym}^d C$ as a coarse moduli space for the action of Σ_d on C^d, with respect to the map

$$C^d \to \mathrm{Pic}^0_C$$

given on T-points by $(\phi_1, \cdots, \phi_d) \mapsto \mathcal{O}_{T\times C}(\sum_{i=1}^d \Gamma_{\phi_i} - T \times \{z_0\})$. Then the Abel-Jacobi map on T-points

$$\mu^d(T) : \mathrm{Sym}^d C(T) \to \mathrm{Pic}^0_C(T)$$

sends relative effective Cartier divisor $[D \subseteq T\times C] \in \mathrm{Sym}^d C(T)$, where $\deg D_t = d$ for every $t \in T$, to the element $\mathcal{O}_{T\times C}(D) \otimes \pi_C^* \mathcal{O}_C(-dz_0) \in \mathrm{Pic}^0_C(T)$.

Remark 7.1. *Note that* $\bigsqcup_{d\geq 0} \mathrm{Sym}^d C$ *represents the moduli functor* $\underline{Div}_C$*, where* $\mathrm{Sym}^0 C :=$ $\mathrm{Spec}\,\mathbb{C}$*. The Abel-Jacobi maps then define*

$$\underline{Div}_C \to \mathrm{Pic}^0_C.$$

This is all with respect to a base point $z_0 \in C(\mathbb{C})$*. But if we remove the base point, then we obtain canonical maps*

$$\mathrm{Sym}^d C \to \mathrm{Pic}^d_C,$$

which on T-points sends $D \mapsto \mathcal{O}_{T\times C}(D)$, which yield a canonical map

$$\bigsqcup_{\geq 0} \operatorname{Sym}^d C = \underline{Div}_C \to \operatorname{Pic}_C,$$

which is the scheme-theoretic upgrade of the usual map $Div(C) \to \operatorname{Pic}(C)$

Let's see what we can say about these Abel-Jacobi maps.

Proposition 7.2. *Let C be a smooth $\mathbb{C}$-curve and fix a base point $z_0 \in C(\mathbb{C})$. The image subfunctor $W_d \subseteq \operatorname{Pic}^0_C$ of the Abel-Jacobi map*

$$\mu^d : \operatorname{Sym}^d C \to \operatorname{Pic}^0_C$$

is the subfunctor defined on T-points by

$$W_d(T) = \{L \in \operatorname{Pic}^0(T\times C) \mid H^0(\{t\}\times C, L_t(dz_0)) \neq 0, \forall t \in T(\mathbb{C})\}/\pi_T^* \operatorname{Pic}(T).$$

Proof. Note that $W_d(T)$ is well-defined with respect to $\pi_T^* \operatorname{Pic}(T)$, as for any $L \in \pi_T^* \operatorname{Pic}(T)$, note L_t is trivial on $\{t\}\times C$ for any $t \in T(\mathbb{C})$.

Suppose we have some element $D \in \operatorname{Sym}^d C(T)$, so that $D \subseteq T \times C$ is a relative effective Cartier divisor and $\deg D_t = d$ for all $t \in T(\mathbb{C})$. Then the Abel-Jacobi map sends it to the element $L = \mathcal{O}_{T\times C}(D) \otimes \pi_C^* \mathcal{O}_C(-dz_0)$. Indeed, we see that for every $t \in T(\mathbb{C})$, we have $L_t \otimes \mathcal{O}_C(dz_0) = D_t - dz_0 + dz_0 = D_t$, which is effective of degree d, so $H^0(\{t\}\times C, L_t(dz_0)) \neq 0$. Thus, the image is contained in $W_d(T)$.

Now suppose $L \in W_d(T)$ nontrivial. So $L \in \operatorname{Pic}^0(T\times C)$ such that $H^0(\{t\}\times C, L_t(dz_0)) \neq 0$ for all $t \in T(\mathbb{C})$. Then we see that it is in the image of $L \otimes \pi_C^* \mathcal{O}_C(dz_0)$, which defines an element of $\operatorname{Sym}^d C(T)$. □

Proposition 7.3. *Let C be a smooth $\mathbb{C}$-curve and fix a base point $z_0 \in C(\mathbb{C})$. Consider the Abel-Jacobi map*

$$\mu^d : \operatorname{Sym}^d C \to \operatorname{Pic}^0_C.$$

The image subfunctor $W_d \subseteq \operatorname{Pic}^0_C$ is a closed subfunctor, and in particular represented by a closed subscheme of Pic^0_C.

Proof. Consider the universal line bundle

$$\mathcal{L}_{\mathrm{univ},d} \in \operatorname{Pic}(C \times \operatorname{Pic}^d_C)$$

and consider the isomorphism

$$\theta : C \times \operatorname{Pic}^0_C \xrightarrow{(\mathrm{id},\times \mathcal{O}_C(dz_0))} C \times \operatorname{Pic}^d_C$$

which is identity on C, and which identifies $\operatorname{Pic}^0_C \cong \operatorname{Pic}^d_C$ via the isomorphism induced by multiplication by $\mathcal{O}_C(dz_0)$. On T-points, this isomorphism is the map

$$\operatorname{Pic}^0_C(T) \to \operatorname{Pic}^d_C(T)$$

where $L \in \mathrm{Pic}^0_C(T) \mapsto L \otimes \pi_C^* \mathcal{O}_C(dz_0) \in \mathrm{Pic}^d_C(T)$. Then we can consider the pullback

$$\theta^* \mathcal{L}_{\mathrm{univ}} \in \mathrm{Pic}(C \times \mathrm{Pic}^0_C),$$

which is the line bundle such that its fiber over $[\ell] \in \mathrm{Pic}^0_C(\mathbb{C})$ is the line bundle $\ell \otimes \mathcal{O}_C(dz_0)$. With respect to the projection $C \times \mathrm{Pic}^0_C \to \mathrm{Pic}^0_C$, note $\theta^*\mathcal{L}_{\mathrm{univ}}$ is Pic^0_C-flat and by upper-semicontinity, there is an open subscheme $U_0 \subseteq \mathrm{Pic}^0_C$ whose $\mathbb{C}$-points are those $[\ell]$ such that $H^0(C, \ell(dz_0)) = 0$. Then we claim that the complement $V := \mathrm{Pic}^0_C \setminus U_0$ is the closed subscheme which represents $W_d \subseteq \mathrm{Pic}^0_C$.

Indeed, for any element $L \in W_d(T)$, we have $L \in \mathrm{Pic}^0(T \times C)$ such that $H^0(C, L_t(dz_0)) \neq 0$ for every $t \in T(\mathbb{C})$. This $L \in W_d(T)$ must come from pullback of the universal line bundle $\mathcal{L}_{\mathrm{univ},0} \in \mathrm{Pic}(C \times \mathrm{Pic}^0_C)$, along some morphism $T \to \mathrm{Pic}^0_C$. But since $H^0(C, L_t(dz_0)) \neq 0$ for every $t \in T(\mathbb{C})$, the morphism $T \to \mathrm{Pic}^0_C$ must factor through the locus $V := \mathrm{Pic}^0_C \setminus U_0$. □

Remark 7.4. *Note we have $W_d \subseteq W_{d+1}$.*

Theorem 7.5. *Let C be a smooth $\mathbb{C}$-curve and fix a basepoint $z_0 \in C(\mathbb{C})$. Consider the Abel-Jacobi map*

$$\mu^d : \mathrm{Sym}^d C \to \mathrm{Pic}^0_C$$

and its image $W_d \subseteq \mathrm{Pic}^0_C$. The scheme-theoretic fiber of μ^d over $[L] \in W_d(\mathbb{C})$ is identified with

$$(\mu^d)^{-1}([L]) = \mathbb{P}H^0(C, L(dz_0)).$$

Proof. For $[L] \in W_d(\mathbb{C}) \subseteq \mathrm{Pic}^0_C(\mathbb{C})$, note that $[L](T)$ will be the image of $[L] \in W_d(\mathbb{C}) \subseteq \mathrm{Pic}^0_C(\mathbb{C})$ in $W_d(T) \subseteq \mathrm{Pic}^0_C(T)$ under the map induced by the morphism $T \to \mathrm{Spec}\,\mathbb{C} = [L]$. Thus, $[L](T)$ will be the line bundle $\pi_C^* L \in \mathrm{Pic}(T \times C)$. Then the T-points of the scheme-theoretic fibre

$$\begin{tikzcd} (\mu^d)^{-1}(L)(T) \arrow[r] \arrow[d] & \mathrm{Spec}\,\mathbb{C}(T) \arrow[d] \\ \mathrm{Sym}^d C(T) \arrow[r] & \mathrm{Pic}^0_C(T) \end{tikzcd}$$

is the set

$$(\mu^d)^{-1}(T) = \left\{ \begin{tikzcd} D \arrow[r, hook] \arrow[d] & T \times C \arrow[dl] \\ T & \end{tikzcd} \text{rel eff. div. deg d } \;\middle|\; \mathcal{O}_{T\times C}(D) \otimes \pi_C^*\mathcal{O}_C(-dz_0) \cong \pi_C^* L \otimes \pi_T^* N, N \in \mathrm{Pic}(T) \right\}$$

Note the $N \in \mathrm{Pic}(T)$ is uniquely determined because $T \times C \to T$ admits a section, so $\pi_T^* : \mathrm{Pic}(T) \to \mathrm{Pic}(T \times C)$ is injective. We would like to naturally identify $(\mu^d)^{-1}(T)$ with

$$\mathbb{P}H^0(C, L(dz_0))(T) = \{\mathcal{E} \in \mathrm{Pic}(T) \mid \mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \twoheadrightarrow \mathcal{E}\}/\sim.$$

Note that given $D \in (\mu^d)^{-1}(T)$ with the data

$$\mathcal{O}_{T\times C}(D) \otimes \pi_T^* N^{-1} \cong \pi_C^* L(dz_0),$$

we can pushforward the aforementioned equation along π_T. Then we obtain

$$\pi_{T,*}\mathcal{O}_{T\times C}(D) \otimes N^{-1} \cong \mathcal{O}_T \otimes H^0(C, L(dz_0))$$

where the left hand side is by projection formula and the right hand side is by using flat base change [Sta25, Tag 02KH] with respect to the diagram

$$\begin{tikzcd} T \times C \arrow[r, "\pi_C"] \arrow[d, "\pi_T"'] & C \arrow[d] \\ T \arrow[r] & \operatorname{Spec} \mathbb{C} \end{tikzcd}$$

So then we find

$$\pi_{T,*}\mathcal{O}_{T\times C}(D) \cong \mathcal{O}_T \otimes H^0(C, L(dz_0)) \otimes N.$$

Note that D is the zero locus of a global section $s \in H^0(T\times C, \mathcal{O}_{T\times C}(D))$. The aforementioned isomorphism implies that we have an isomorphism on global sections:

$$H^0(T \times C, \mathcal{O}_{T\times C}(D)) \cong H^0(C, L(dz_0)) \otimes H^0(T, N).$$

Furthermore, note that any $s' \in H^0(T\times C, \mathcal{O}_{t\times C}(D))$ defines a morphism $\operatorname{Hom}_{\mathbb{C}}(H^0(C, L(dz_0))^\vee, H^0(T, N))$. In particular, these maps to global sections of N yield maps $\mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \to N$. But note that specifically for $s \in H^0(T \times C, \mathcal{O}_{T\times C}(D))$, for every $t \in T$ we have that D_t is effective of degree d and thus nonzero. This means that over every $t \in T$, the map $\mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \to N$ on stalks is surjective, and thus $s \in H^0(T \times C, \mathcal{O}_{T\times C}(D))$ corresponds to a surjection

$$[\mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \twoheadrightarrow N] \in \mathbb{P}H^0(C, L(dz_0))(T).$$

On the other hand, given some surjection

$$\mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \twoheadrightarrow \mathcal{E},$$

the line bundle on $T \times C$

$$\pi_C^* L(dz_0) \otimes \pi_T^* \mathcal{E}.$$

Pushing forward, by flat base change and projection formula we have

$$\pi_{T,*}(\pi_C^* L(dz_0) \otimes \pi_T^* \mathcal{E}) \cong \mathcal{O}_T \otimes H^0(C, L(dz_0)) \otimes \mathcal{E}.$$

Then $H^0(T\times C, \pi_C^* L(dz_0) \otimes \pi_T^* \mathcal{E}) \cong H^0(C, L(dz_0)) \otimes H^0(T, \mathcal{E})$. Then $\mathcal{O}_T \otimes H^0(C, L(dz_0))^\vee \twoheadrightarrow \mathcal{E}$ defines an element $s \in H^0(T \times C, \pi_C^* L(dz_0) \otimes \pi_T^* \mathcal{E})$ which defines a relative effective divisor D such that $\deg D_t = d$ for every $t \in T(\mathbb{C})$. □

Using Theorem 7.5, we can say when $\operatorname{Sym}^d C \to \operatorname{Pic}^0_C$ is a closed embedding, surjective, or birational to W_d, depending on what d is.

Definition 7.6. *Let C be a smooth curve of genus g. We define the gonality $\gamma(C)$ to be the smallest degree possible of a non-constant map $C \to \mathbb{P}^1$.*

So $\mathbb{P}^1$ has gonality $\gamma(\mathbb{P}^1) > 1$. For curves of genus $g(C) \geq 1$, we have $\gamma(C) \geq 2$. A hyperelliptic curve has gonality $\gamma(C) = 2$, so all genus 1 curves have gonality $\gamma(C) = 2$.

Furthermore, we always have $\gamma(C) \leq g + 1$. This is because a map $C \to \mathbb{P}^1$ witnessing the gonality $\gamma(C)$ must be given by a degree $\gamma(C)$ line bundle L with $h^0(C, L) = 2$. Then by Riemann-Roch, $2 - h^1(C, L) = \gamma(C) - g + 1 \implies \gamma(C) = g + 1 - h^1(C, L) \leq g + 1$.

Theorem 7.7. *Let C be a smooth curve of genus $g \geq 1$. Fix a base point $z_0 \in C(\mathbb{C})$. Let*

$$\mu^d : \operatorname{Sym}^d C \to \operatorname{Pic}_C^0$$

denote the Abel-Jacobi map of degree d and let $W_d \subseteq \operatorname{Pic}_C^0$ denote the image.

- *When $1 \leq d \leq g$, μ^d is birational to W_d.*
- *When $1 \leq d < \gamma(C) \leq g + 1$, μ^d is a closed immersion into Pic_C^0.*
- *When $d \geq g$, μ^d is surjective.*
- *When $d > 2g - 2$, μ^d is smooth with geometric fibers $\mathbb{P}^{d-g}$.*

Proof.

- Suppose $1 \leq d \leq g$. Note that the fiber of μ^d over $[L] \in W_d(\mathbb{C})$ is $\mathbb{P}H^0(C, L(dz_0))$. Note that μ^d being birational to W_d is equivalent to finding an open subset $U \subseteq W_d$ such that μ^d is an isomorphism over this open subset. We are then motivated to find an open subset $U \subseteq W_d$ whose $\mathbb{C}$-points are those $[L]$ such that $H^0(C, L(dz_0)) = 1$. To do this, we'll do an existence claim and then appeal to upper-semicontinuity.

  Let's start with existence. We claim there exists a degree d line bundle $\mathcal{E} \in \operatorname{Pic}(C)$ such that $H^0(C, \mathcal{E}) = 1$. To do this, we'll use a very geometric incarnation of Riemann-Roch. Since $g(C) \geq 1$, the canonical line bundle ω_C is globally generated, so there is a well-defined canonical map

  $$|\omega_C| : C \to \mathbb{P}^{g-1}.$$

  Note that for any effective divisor $D = \sum_{i=1}^d p_i$ of degree d, we have $H^0(\omega_C(-D))$ is the space of all hyperplanes H in $\mathbb{P}^{g-1}$ such that the pullback $|\omega_C|^*(H)$ contains D. In particular, note that

  $$\dim H^0(\omega_C(-D)) = \dim \mathbb{P}^{g-1} - \dim \operatorname{span}\langle |\omega_C|(D)\rangle,$$

  where $\dim \operatorname{span}\langle |\omega_C|(D)\rangle = \bigcap_{|\omega_C|^*(H) - D \geq 0} H$ is the smallest linear subspace of $\mathbb{P}^{g-1}$ containing the image of D under the canonical map $|\omega_C|$. Then by Riemann-Roch, since $H^0(\omega_C(-D)) \cong H^1(\mathcal{O}_C(D))$, we have

  $$\dim H^0(\mathcal{O}_C(D)) = d - \dim \operatorname{span}\langle |\omega_C|(D)\rangle.$$

  We now utilize this fact. Recall we would like to produce a degree d line bundle $\mathcal{E}$ such that $H^0(C, \mathcal{E}) = 1$. Under the canonical map $|\omega_C| : C \to \mathbb{P}^{g-1}$, since the map is non-degenerate and $d \leq g$, we can pick d points $q_1, \cdots, q_d$ in $\mathbb{P}^{g-1}$ on the image of C which are in general position. They will then span a $d - 1$ linear subspace.

Then pick some point p_i in the preimage of q_i for $1 \leq i \leq d$. Then set $D = \sum_{i=1}^d p_i$ and $\mathcal{E} = \mathcal{O}_C(D)$. Then by geometric Riemann-Roch, we see that $H^0(C, \mathcal{E}) = 1$.

Finally, we finish by appealing to upper-semicontinuity. Again, consider the universal line bundle

$$\mathcal{L}_{\mathrm{univ},d} \in \mathrm{Pic}(C \times \mathrm{Pic}_C^d),$$

and pull it back along the morphism

$$\theta : C \times \mathrm{Pic}_C^0 \xrightarrow{(\mathrm{id}, \times \mathcal{O}_C(dz_0))} C \times \mathrm{Pic}_C^d.$$

Recall we used upper-semicontinuity on $\theta^* \mathcal{L}_{\mathrm{univ},d}$ to define $W_d \subseteq \mathrm{Pic}_C^0$. Now if we restrict to W_d, we use upper-semicontinuity again on $\theta^* \mathcal{L}_{\mathrm{univ},d}|_{W_d}$ to find that the locus whose $\mathbb{C}$-points are line bundles $[L] \in W_d(\mathbb{C})$ such that $H^0(C, L(dz_0)) = 1$ is an open subset U of W_d. Although it could possibly be non-empty, we know that it isn't because we showed existence of a degree d line bundle with $H^0(C, \mathcal{E}) = 1$. Then setting $L = \mathcal{E}(-dz_0)$, we see that U is non-empty.

Then we base change to obtain

$$(\mu^d)^{-1}(U) \to U \subseteq W_d.$$

Since the geometric fiber over a $\mathbb{C}$-point in U is a $\operatorname{Spec} \mathbb{C}$, by [Sta25, Tag 06ND], the map $(\mu^d)^{-1}(U)$ is a closed immersion, and since it is surjective, it is an isomorphism of schemes. This shows that for $1 \leq d \leq g$, $\mathrm{Sym}^d C$ is birational to W_d via μ^d.

- Suppose $1 \leq d < \gamma(C)$, where $\gamma(C)$ is the gonality of C. Let $[L] \in W_d(\mathbb{C})$. So $H^0(C, L(dz_0)) \neq 0$. Suppose $H^0(C, L(dz_0)) \geq 2$. Then we can pick distinct $s_0, s_1 \in H^0(C, L(dz_0))$ which globally generate $L(dz_0)$ which define $C \to \mathbb{P}^1$ where $p \mapsto [s_0(p) : s_1(p)]$... but then we have demonstrated existence of a degree $d < \gamma(C)$ map, contradiction. So we must have $H^0(C, L(dz_0)) = 1$ for all $[L] \in W_d(\mathbb{C})$. In particular, using Theorem 7.5 and [Sta25, Tag 06ND], we have that μ^d is a closed immersion.
- Suppose $d \geq g$. Pick $[L] \in Pic_C^0(\mathbb{C})$. Then by Riemann-Roch,

$$h^0(C, L(dz_0)) - h^1(C, L(dz_0)) = d - g + 1 \geq 1 \implies h^0(C, L(dz_0)) \geq 1,$$

which means that $W_d = \mathrm{Pic}_C^0$ for $d \geq g$ (in particular, our spaces are nice enough here that surjectivity on closed points implies surjectivity everywhere.)
- Suppose $d > 2g - 2$. By the previous statement, $W_d = \mathrm{Pic}_C^0$. Note that for any $[L] \in \mathrm{Pic}_C^0(\mathbb{C})$, we have $H^1(C, L(dz_0)) = H^0(C, \omega_C \otimes L^{-1}(-dz_0)) = 0$ since $d > 2g - 2$, so

$$h^0(C, L(dz_0)) = d - g + 1.$$

In particular, the fiber everywhere over $\mathbb{C}$-points is a copy of $\mathbb{P}^{d-g}$. In this case, all the hypotheses of miracle flatness are satisfied, so by miracle flatness, the map

$\mu^d : \operatorname{Sym}^d C \to \operatorname{Pic}^0_C$ is flat. Since in addition the fibers are all smooth, we have that μ^d is smooth [Sta25, Tag 01V4].

$\square$

Now that we've studied these Abel-Jacobi maps a bit, I want to discuss a special case. Suppose C is a smooth curve of genus $g \geq 2$ and consider

$$\mu^{g-1} : \operatorname{Sym}^{g-1} C \to \operatorname{Pic}^0_C.$$

The image $W_{g-1} \subseteq \operatorname{Pic}^0_C$ is an example of something called a Theta divisor of an abelian variety. Theta divisors are effective divisors associated to a line bundle on an abelian variety which is principally polarized. Let me make these notions more precise.

Let A be any abelian variety (smooth, proper, connected, algebraic group over $\mathbb{C}$), which is necessarily projective. Since A is projective, note Pic^0_A is also an abelian variety, which we define to be the dual abelian variety $A^\vee := \operatorname{Pic}^0_A$ of A. In general, we have a short exact sequence of groups

$$0 \to \operatorname{Pic}^0(A) \to \operatorname{Pic}(A) \to \operatorname{NS}(A) \to 0.$$

The **essential point of our discussion hereafter** is that because A is an abelian variety, and not just any projective variety, the map $\operatorname{Pic}(A) \to \operatorname{NS}(A)$ can be identified with a map $\operatorname{Pic}(A) \to \operatorname{Hom}_{\mathrm{Grp}(\mathrm{Sch}_{\mathbb{C}})}(A, A^\vee)$. In particular, with respect to certain important properties, one can test whether a line bundle $L \in \operatorname{Pic}(A)$ has the aforementioned property via its image $\xi_L \in \operatorname{Hom}_{\mathrm{Grp}(\mathrm{Sch}_{\mathbb{C}})}(A, A^\vee)$.

First, let us construct this group homomorphism $\operatorname{Pic}(A) \to \operatorname{Hom}_{\mathrm{Grp}(\mathrm{Sch}_{\mathbb{C}})}(A, A^\vee)$, the existence of which crucially uses the fact that A is an abelian variety.

Definition 7.8. *Fix $L \in \operatorname{Pic}(A)$. Let $m : A \times A \to A$ denote the multiplication map. We define the Mumford line bundle to be*

$$\Lambda(L) := m^*L \otimes \pi_1^* L^{-1} \otimes \pi_2^* L^{-2} \in \operatorname{Pic}(A \times A).$$

Using the Mumford line bundle, we can define a morphism

$$\xi_L : A \to \operatorname{Pic}_A$$

in the following way: on T-points, ξ_L sends $\phi \in A(T)$ to

$$(\phi, \mathrm{id}_A)^*\Lambda(L) \in \operatorname{Pic}(T \times A), \text{ where } (\phi, \mathrm{id}_A) : T \times A \to A \times A.$$

One sees that this defines a valid morphism of moduli functors and thus of schemes.

Theorem 7.9. *Let A be an abelian $\mathbb{C}$-variety. Let $L \in \operatorname{Pic}(A)$ and $\Lambda(L) \in \operatorname{Pic}(A \times A)$ the Mumford line bundle associated to L. Then*

$$\xi_L : A \to \operatorname{Pic}_A$$

is a group scheme homomorphism which factors as

$$\xi_L : A \to A^\vee := \mathrm{Pic}^0_A \subseteq \mathrm{Pic}_A .$$

Proof. The fact that ξ_L is a group scheme homomorphism and not just a morphism of schemes is due to the famous "Theorem of the Square" [Mil, Theorem of the Square, Theorem 6.7]. The map factorizes through Pic^0_A because A is connected and the identity element $e \in A(\mathbb{C})$ is sent to $\mathcal{O}_A \in \mathrm{Pic}_A(\mathbb{C})$ under ξ_L. □

Note that $\mathrm{Hom}_{\mathrm{Grp}(\mathrm{Sch}_\mathbb{C})}(A, A^\vee)$ has a natural group structure. For any $\xi, \xi' \in \mathrm{Hom}_{\mathrm{Grp}(\mathrm{Sch}_\mathbb{C})}(A, A^\vee)$, we have that $\xi \cdot \xi'$ is defined on T-points via:

$$\xi \cdot \xi' : A(T) \to A^\vee(T), x \mapsto \xi(x)\xi'(x).$$

Since abelian varieties are commutative, $\xi \cdot \xi' \in \mathrm{Hom}_{\mathrm{Grp}(\mathrm{Sch}_\mathbb{C})}(A, A^\vee)$.

Proposition 7.10. *Let A be an abelian $\mathbb{C}$-variety. Then*

$$\mathrm{Pic}(A) \to \mathrm{Hom}_{Grp(Sch_\mathbb{C})}(A, A^\vee)$$

is a homomorphism of groups.

Proof. Suppose $L_1, L_2 \in \mathrm{Pic}(A)$. We'd like to show that $\xi_{L_1} \cdot \xi_{L_2} = \xi_{L_1 \otimes L_2}$ as group scheme homomorphisms

$$A \to A^\vee.$$

But indeed this is true, because on T-points

$$A(T) \to A^\vee(T),$$

for any $\phi \in A(T)$ we have

$$(\xi_{L_1} \cdot \xi_{L_2})(\phi) = (\phi, \mathrm{id}_A)^*\Lambda(L_1) \otimes (\phi, \mathrm{id}_A)^*\Lambda(L_2) \cong (\phi, \mathrm{id}_A)^*\Lambda(L_1 \otimes L_2) = \xi_{L_1 \otimes L_2}(\phi).$$

□

We won't prove the following, but here comes a punchline:

Proposition 7.11. *Let A be an abelian $\mathbb{C}$-variety. Then the homomorphism of groups*

$$\mathrm{Pic}(A) \to \mathrm{Hom}_{Grp(Sch_\mathbb{C})}(A, A^\vee)$$

is identified with the homomorphism of groups

$$\mathrm{Pic}(A) \to NS(A);$$

so there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Pic}^0(A) \arrow[r] & \mathrm{Pic}(A) \arrow[r] \arrow[dr] & NS(A) \arrow[r] \arrow[d, "\cong"] & 0 \\
& & & \mathrm{Hom}_{Grp(Sch_\mathbb{C})}(A, A^\vee) \arrow[r] & 0
\end{tikzcd}$$

In particular, this tells us that $L \in \mathrm{Pic}(A)$ is degree 0 if and only if $\xi_L : A \to A^\vee$ is a zero map. The theme continues for ampleness:

Proposition 7.12. *Let A be an abelian $\mathbb{C}$-variety. A line bundle $L \in \mathrm{Pic}(A)$ is ample if and only if $H^0(A, L) \neq 0$ and*

$$\xi_L \in Hom_{Grp(Sch_{\mathbb{C}})}(A, A^\vee)$$

is an isogeny, i.e.

$$\xi_L : A \to A^\vee$$

is surjective with finite kernel.

Definition 7.13. *Let A be an abelian $\mathbb{C}$-variety. A line bundle $L \in \mathrm{Pic}(A)$ is principally polarized if $\xi_L : A \to A^\vee$ is an isomorphism. A Theta divisor is an effective divisor $D \in |L|$ for principally polarized L.*

Thus, a line bundle $L \in \mathrm{Pic}(A)$ on an abelian variety is principally polarized if it is ample and the map $\xi_L : A \to \mathrm{Pic}^0_A$ induced by its associated Mumford line bundle is an isomorphism. Theta divisors are important; here's one reason.

Theorem 7.14 (Torelli theorem for curves)**.** *Let C be a smooth curve. Then $(\mathrm{Pic}^0_C, \Theta)$ determines the curve C, where Θ is a Theta divisor on Pic^0_C.*

Thus, there are non-isomorphic curves with the same Pic^0_C, but the identifications of Pic^0_C must not be compatible with choice of principal polarization/theta divisor.

Returning to the situation of Abel-Jacobi maps, when $g(C) \geq 2$, we claimed that

$$W_{g-1} \subseteq \mathrm{Pic}^0_C$$

is a theta divisor. So there exists a line bundle $L \in \mathrm{Pic}(\mathrm{Pic}^0_C)$ such that $W_{g-1} \in |L|$ and $\xi_L : \mathrm{Pic}^0_C \to \mathrm{Pic}^0_{\mathrm{Pic}^0_C}$ is an isomorphism.

8. MVC 5/21/25: Vector bundles of higher rank on curves, Ext^1 as coarse moduli space, failure of $\underline{\mathrm{Bun}}_C$

So far we have discussed moduli functors, basic affine GIT, and line bundles on curves. It's time to move towards vector bundles of higher rank on curves.

Let C be a smooth $\mathbb{C}$-curve of genus g. By a vector bundle E of rank $r \geq 1$, we mean either a Zariski-locally $\mathbb{A}^r$-fibration with linear transition functions, or its associated sheaf of sections. We define the degree of E to be $\deg E := \deg \det E \in \mathbb{Z}$. Using this, we can prove a Riemann-Roch theorem for higher rank vector bundles on curves by induction on the rank.

Proposition 8.1 (Riemann-Roch for VB on curves)**.** *Let C be a smooth curve of genus g. Let E be a vector bundle on C of rank $r \geq 1$. Then*

$$\chi(E) = \deg E + r(-g + 1).$$

Proof. By Riemann Roch for line bundles, we know the claim holds when $r = 1$. Assume the claim holds for $r \leq n$. We show it holds for $r = n + 1$.

So suppose E is a rank $n + 1$ vector bundle on C. Since C is projective, it has an ample $\mathcal{O}_C(1)$. Then for a sufficiently large m, we have that $H^0(C, E(m))$ is globally generated. If $H^0(C, E(m)) = \mathbb{C}\langle s_1, \cdots, s_k\rangle$, then consider the global section corresponding to $s_1 + \cdots + s_k$; then we have a short exact sequence

$$0 \to \mathcal{O}_C \to E(m) \to \frac{E(m)}{\mathcal{O}_C} \to 0,$$

and by construction, since $\mathcal{O}_C \to E(m)$ vanishes nowhere and thus has locally constant rank, we have $\frac{E(m)}{\mathcal{O}_C}$ is a vector bundle of rank n. By additivity of Euler characteristic we get

$$\chi(E) = \chi(\mathcal{O}_C(-m)) + \chi(E/\mathcal{O}_C(-m)) = -m - g + 1 + \deg[E/\mathcal{O}_C(-m)] + n(-g + 1)$$

$$-m + \deg[E/\mathcal{O}_C(-m)] + (n + 1)(-g + 1).$$

Note since $\det E = \mathcal{O}_C(-m) \otimes \det[E/\mathcal{O}_C(-m)]$ we have $\deg E = -m + \deg[E/\mathcal{O}_C(-m)]$. Thus we have

$$\chi(E) = \deg E + (n + 1)(-g + 1),$$

as desired. □

So as to preserve the Riemann-Roch formula for coherent sheaves, we define the degree of $F \in Coh(C)$ to be

$$\deg F := \chi(F) - \operatorname{rk} F(-g + 1),$$

where here $\operatorname{rk} F := \dim_{\mathbb{C}(\xi)} F_\xi$, where ξ is the generic point of C. Note this implies that if D is an effective divisor of C, then $\deg \iota_*\mathcal{O}_D = \deg D$, so we get additivity of deg for short exact sequences which include coherent sheaves such as the ideal sheaf short exact sequence of D:

$$0 \to \mathcal{O}_C(-D) \to \mathcal{O}_C \to \iota_*\mathcal{O}_D \to 0.$$

A very useful way to construct higher rank vector bundles is via extensions. We call any short exact sequence of vector bundles

$$0 \to F \to M \to E \to 0$$

an extension of E by F. We say that two extensions of E by F are isomorphic if there exist a diagram

$$\begin{tikzcd} 0 \arrow[r] & F \arrow[r] \arrow[d, "\mathrm{id}"'] & M \arrow[r] \arrow[d, "\cong"] & E \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\ 0 \arrow[r] & F \arrow[r] & M' \arrow[r] & E \arrow[r] & 0 \end{tikzcd}$$

Theorem 8.2. *Let E and F be vector bundles on C. The first derived functor $Ext^1(E, F)$ of the functor* $\mathrm{Hom}_{\mathcal{O}_C}(E, -)$ *is a $\mathbb{C}$-vector space which is the space of all isomorphism classes of extensions of E by F.*

Proof. See subsection 22.1 in the appendix, in particular for the vector space structure on the set of equivalence classes of extensions. □

These Ext groups are quite important as they are related to the infinitesimal structure of the Quot scheme. While scheme-theoretically, $\mathrm{Ext}^1(E, F)$ is simply affine space, one might wonder whether it represents a naturally related moduli functor. One might guess that it represents the moduli functor

$$\underline{\mathrm{Ext}}^1(E, F) : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $\underline{\mathrm{Ext}}^1(E, F)(T) = \mathrm{Ext}^1_{T\times C}(\pi_C^* E, \pi_C^* F)$. This is a natural guess, since any element of $\mathrm{Ext}^1_{T\times C}(\pi_C^* E, \pi_C^* F)$ corresponds to a short exact sequence

$$0 \to \pi_C^* F \to M \to \pi_C^* E \to 0$$

on $T \times C$, and so its restriction to $\{t\} \times C$ is $0 \to F \to M_t \to E \to 0$. It turns out that $\mathrm{Ext}^1(E, F)$ as affine space only represents $\underline{\mathrm{Ext}}^1(E, F)$ in the subcategory of affine schemes. On the entire category of schemes, $\underline{\mathrm{Ext}}^1(E, F)$ fails to be a Zariski sheaf and is not representbale. However, the good news is that its sheafification is represented by $\mathrm{Ext}^1(E, F)$.

Proposition 8.3. *Let E, F be vector bundles on C. Then the affine scheme $Ext^1(E, F)$ is a coarse moduli space for the moduli functor*

$$\underline{Ext}^1(E, F) : (Sch_{\mathbb{C}})^{op} \to Sets,$$

and represents its sheafification.

Proof. Consider what the sheafification of $\underline{\mathrm{Ext}}^1(E, F)$ is: note

$$\underline{\mathrm{Ext}}^1(E, F)(T) = \mathrm{Ext}^1_{T\times C}(\pi_C^* E, \pi_C^* F) \cong H^1(T \times C, \pi_C^*(E^\vee \otimes F)).$$

Then we see that its sheafification on the subcategory of open sets of T is simply $R^1\pi_{T,*}\pi_C^*(E^\vee \otimes F)$. Then its sheafification is the Zariski sheaf

$$\underline{\mathrm{Ext}}^1(E, F)^{\mathrm{sh}} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $T \mapsto H^0(T, R^1\pi_{T,*}\pi_C^*(E^\vee \otimes F))$. Note there is a natural morphism

$$\underline{\mathrm{Ext}}^1(E, F) \to \underline{\mathrm{Ext}}^1(E, F)^{\mathrm{sh}}$$

which on T-points is given by

$$H^1(T \times C, \pi_C^*(E^\vee \otimes F)) \to H^0(T, R^1\pi_{T,*}\pi_C^*(E^\vee \otimes F)).$$

Now note by flat base change

$$H^0(T, R^1\pi_{T,*}\pi_C^*(E^\vee \otimes F)) \cong H^0(T, \pi_T^* R^1\pi_{\operatorname{Spec}\mathbb{C},*}(E^\vee \otimes F))$$

$$\cong H^0(T, \mathcal{O}_T \otimes H^1(C, E^\vee \otimes F)) \cong H^0(T, \mathcal{O}_T) \otimes H^1(C, E^\vee \otimes F)$$

$$H^0(T, \mathcal{O}_T) \otimes \operatorname{Ext}^1(E, F) \cong \operatorname{Ext}^1(E, F)(T).$$

Thus, we see that $\operatorname{Ext}^1(E, F)$ represents the sheafification, and by universal property of the sheafification, it corepresents $\underline{\operatorname{Ext}}^1(E, F)$. Clearly there is a bijection on $\mathbb{C}$-points, so we have it is a coarse moduli space. □

Recall that a Zariski sheaf is representable if and only if it is representable on affines. The moduli functor $\underline{\operatorname{Ext}}^1(E, F)$ is very interesting, as not only is it an important functor, but it gives an example of a non-Zariski sheaf which is still representable on affines.

The representability on affines translates to the following statement: consider the universal family

$$0 \to \pi_C^* F \to \mathcal{U} \to \pi_C^* E \to 0$$

on $\operatorname{Ext}^1(E, F) \times C$. Then since $\operatorname{Ext}^1(E, F)$ represents $\underline{\operatorname{Ext}}^1(E, F)$ on affines $\operatorname{Spec} A$, we have that any element of $\operatorname{Ext}^1_{\operatorname{Spec} A \times C}(E, F)$,

$$0 \to \pi_C^* F \to M \to \pi_C^* E \to 0$$

comes from pulling back the universal family along some morphism $\operatorname{Spec} A \to \operatorname{Ext}^1(E, F)$. The proof of this local universality is given by a Leray spectral sequence calculation.

Now that we've discussed some basics of higher rank vector bundles on curves, we turn to the classification problem. A starting point for inspiration is the case of line bundles on curves. We saw that for any smooth $\mathbb{C}$-curve C (which, crucially, is always projective), we have $\underline{\operatorname{Pic}}_C$ is representable by a quasi-projective Pic_C, with connected components

$$\operatorname{Pic}_C = \bigsqcup \operatorname{Pic}_C^d.$$

Let's try doing an analogous construction for higher rank vector bundles. We define the moduli functor

$$\underline{\operatorname{Bun}}_C : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $T \mapsto \frac{\{E \text{ loc. free. on } T \times C\}}{\sim}$, where we say $E \sim E'$ if there exists some line bundle $L \in \operatorname{Pic}(T)$ such that $E \cong E' \otimes \pi_T^* L$. This equivalence relation seems reasonable, since if $E \sim E'$ then $E_t \cong E'_t$ for every $t \in T$.

The connected components decomposition of Pic_C suggests that we should try to decompose the classification of higher rank vector bundles to classifying vector bundles of fixed rank and degree. So we define the moduli functor

$$\underline{\operatorname{Bun}}_C(r, d) : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

where $T \mapsto \frac{\{E \text{ loc free on } T\times C | \operatorname{rk} E_t = r, \deg E_t = d, \forall t\in T\}}{\sim}$. Note by Proposition 8.1, for $E \in \underline{\mathrm{Bun}}_C(r,d)(T)$ we have

$$\chi(E_t(m)) = \deg E_t + \operatorname{rk}(E_t)\cdot m \cdot \deg \mathcal{O}_C(1) + \operatorname{rk}(E_t)(1-g) = r\cdot \deg \mathcal{O}_C(1)\cdot m + d + r(1-g).$$

So we see that the Hilbert polynomial of a vector bundle on C, with respect to the choice of $\mathcal{O}_C(1)$, is determined by its rank and degree. Then note that since projection $T\times C \to T$ is flat, any vector bundle E on $T\times C$ is T-flat. So the Hilbert polynomial of its fibers are locally constant as a function of $t \in T$, and in particular $\underline{\mathrm{Bun}}_C(r,d) \subset \underline{\mathrm{Bun}}_C$ is both an open and closed subfunctor. So we have

$$\underline{\mathrm{Bun}}_C = \bigsqcup_{(r,d)\in\mathbb{N}^+\times\mathbb{Z}} \underline{\mathrm{Bun}}_C(r,d).$$

Unfortunately, $\underline{\mathrm{Bun}}_C$ fails to be representable, and in general, same goes for $\underline{\mathrm{Bun}}_C(r,d)$.

Firstly, $\underline{\mathrm{Bun}}_C$ fails to even be a Zariski sheaf. Certainly there exist some vector bundles M,N on T such that $M \not\cong N\otimes L$ for any $L\in \operatorname{Pic}(T)$. Then consider $\pi_T^*M, \pi_T^*N \in \underline{\mathrm{Bun}}_C(T)$. Then we cannot have $\pi_T^*M \cong \pi_T^*N\otimes\pi_T^*L$ for some $L \in \operatorname{Pic}(T)$, because since C is projective we have $\mathcal{O}_T \to \pi_{T,*}\mathcal{O}_{T\times C}$ is an isomorphism, so pushing everything forward would imply $M\cong N\otimes L$, contradiction. So π_T^*M and π_T^*N are distinct elements in $\underline{\mathrm{Bun}}_C(T)$. But on a sufficient Zariski cover $\{U_i\}$ of T which locally trivialize both M and N, we see that these distinct elements agree on each $\underline{\mathrm{Bun}}_C(U_i)$.

Here's another bad thing that we don't like: consider the rank 2 vector bundles of degree 0 $\{\mathcal{O}_{\mathbb{P}^1}(-a)\oplus\mathcal{O}_{\mathbb{P}^1}(a)\}$ on $\mathbb{P}^1$. Note that their Castelnuevo-Mumford regularity is unbounded. Then there is no finite type scheme parameterizing them. In particular, $\underline{\mathrm{Bun}}_{\mathbb{P}^1}(2,0)$ could not be a finite-type scheme. A more humble way of seeing this conclusion is that note $H^0(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(-a)\oplus\mathcal{O}_{\mathbb{P}^1}(a)) = a+1$, and so if $\underline{\mathrm{Bun}}_C(2,0)$ were representable, by upper-semicontinuity it could not be a Noetherian scheme and thus could not be finite type.

9. MVC 5/28/25: Semistability for vector bundles on curves

Recall that we tried to generalize the Picard scheme of a curve C, along with its connected components, to moduli functors parameterizing vector bundles of fixed rank and degree on the curve. We ran into some issues, though. Here is another illuminating issue.

Example 9.1 (Jump phenomenon). *Consider $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1)) \cong H^1(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1}(-2)) \cong H^0(\mathbb{P}^1,\mathcal{O}_{\mathbb{P}^1})\cong\mathbb{C}$. Recall that $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1))$ is a coarse moduli space for the moduli functor $\underline{\mathrm{Ext}}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1))$, in particular there is an extension*

$$0\to\pi_{\mathbb{P}^1}^*\mathcal{O}_{\mathbb{P}^1}(-1)\to\mathcal{U}\to\pi_{\mathbb{P}^1}^*\mathcal{O}_{\mathbb{P}^1}(1)\to 0$$

over $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1))\times\mathbb{P}^1$ forming a family over $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1))$ whose fibers over $\mathbb{C}$-points witness the elements of $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1),\mathcal{O}_{\mathbb{P}^1}(-1))$.

Note a priori we know there are two non-isomorphic extensions

$$0 \to \mathcal{O}_{\mathbb{P}^1}(-1) \to \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1) \to \mathcal{O}_{\mathbb{P}^1}(1) \to 0$$

$$0 \to \mathcal{O}_{\mathbb{P}^1}(-1) \to \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1} \to \mathcal{O}_{\mathbb{P}^1}(1) \to 0$$

where the latter is obtained from the Euler exact sequence, and the former is the trivial extension. Note $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1), \mathcal{O}_{\mathbb{P}^1}(-1))$ *is 1-dimensional. This means that all non-trivial extensions will have middle term isomorphic to* $\mathcal{O}_{\mathbb{P}^1}^{\oplus 2}$.

Note both $\mathcal{O}_{\mathbb{P}^1}^{\oplus 2}$ *and* $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$ *belong to* $\underline{Bun}_{\mathbb{P}^1}(2,0)(\mathbb{C})$. *If* $\underline{Bun}_{\mathbb{P}^1}(2,0)(\mathbb{C})$ *were co-represented by some* Z, *then note that this family over* $\mathrm{Ext}^1_{\mathcal{O}_{\mathbb{P}^1}}(\mathcal{O}_{\mathbb{P}^1}(1), \mathcal{O}_{\mathbb{P}^1}(-1)) \cong \mathbb{A}^1$ *is an element of* $\underline{Bun}_{\mathbb{P}^1}(2,0)(\mathbb{A}^1)$ *and so must map to some element* $\mathbb{A}^1 \to Z$ *of* $Z(\mathbb{A}^1)$. *But note that this map is forced to be least constant everywhere outside of the origin. So it is constant everywhere. This implies that if* $\underline{Bun}_{\mathbb{P}^1}(2,0)$ *were co-represented by* Z, *then* $\mathcal{O}_{\mathbb{P}^1}^{\oplus 2}$ *and* $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(1)$ *must map to the same* $\mathbb{C}$*-point of* Z.

So the co-representing space could not be a coarse moduli space. But the bigger issue is, in general, there doesn't seem to be a cohesive way to tell when two vector bundles must map to the same $\mathbb{C}$*-point.*

Example 9.1 motivates us to throw out those line bundles on $\mathbb{P}^1$ which are not balanced. In particular, it turns out that we'll want to do this for any C: instead of considering just vector bundles, we'll want to restrict to those which are semistable. This restriction will be further explained by applying GIT to Quot schemes, which we'll do in later sections. In the end, the moduli functor we'll want to consider is that which parameterizes S-equivalence classes of semistable vector bundles of fixed rank and degree on the curve.

In this section, we'll discuss semistability, Harder-Narasimhan filtration, Jordan-holder filtration and S-equivalence for vector bundles on curves. Everything here is a special case of Section 13, restricted to the case of curves.

Recall from Section 13 that Gieseker (semi)stable or slope (semi)stable sheaves have torsion conditions. For Gieseker, we want the sheaves to be pure, and for slope we want torsion to only exist in codimension at least two. In the case of one-dimensional coherent sheaves on curves C, it having torsion in only codimension at least two is the same as being pure, which is the same as being torsion-free since C is integral. A torsion-free sheaf is locally free away from a codimension at least two locus (see Section 12). This implies pure 1-dimensional coherent sheaves on a curve C are simply vector bundles.

Definition 9.2. *Let* E *be a vector bundle on* C. *Then the slope of* E *is defined as*

$$\mu(E) := \frac{\deg E}{\operatorname{rk} E}$$

where $\deg E := \deg \det(E)$ *and* $\operatorname{rk} E = \dim_{\mathbb{C}(\eta)} E_{\mathbb{C}(\eta)}$ *where* $\eta \in C$ *is the generic point.*

Note that $\mu(E)$ coincides with the general slope definition for coherent sheaves (with torsion only in codimension at least two) given in Section 13. Indeed, the generalized notion of degree restricted to the case of a curve is

$$\alpha_0(E) - \mathrm{rk}(E)\alpha_0(\mathcal{O}_X).$$

By Riemann-Roch 8.1, we have

$$\chi(E(m)) = \deg \mathcal{O}_C(1) \cdot \mathrm{rk}\, E \cdot m + \deg E + \mathrm{rk}(E)(1-g),$$

so $\alpha_0(E) = \deg E + \mathrm{rk}(E)(1-g)$, so indeed $\deg E = \alpha_0(E) - \mathrm{rk}(E)(1-g) = \alpha_0(E) - \mathrm{rk}(E)\alpha_0(\mathcal{O}_X)$.

Definition 9.3. *We say E on C is semistable if for all nontrivial $F \subset E$ where $\mathrm{rk}\, F < \mathrm{rk}\, E$, we have $\mu(F)(\leq)\mu(E)$.*

Note we have dropped the condition that $0 < \mathrm{rk}\, F$ since E is torsion-free and thus can not have any 0-dimensional subsheaf. Also note that essentially by definition, all line bundles on a curves are stable.

Remark 9.4. *The notions of Gieseker (semi)stability and slope (semi)stability on curves coincide. Indeed, $\mu(F)(\leq)\mu(E)$ if and only if $\frac{\alpha_0(F)}{\alpha_1(F)}(\leq)\frac{\alpha_0(E)}{\alpha_1(E)}$, and since C is 1-dimensional, we have $\rho = \frac{\alpha_0}{\alpha_1}$.*

Recall it suffices to check Gieseker (semi)stability of a coherent subsheaf by checking only saturated subsheaves, or equivalently, pure quotients of the same dimension. This means on a curve C, to check that E is (semi)stable, it suffices to check only for subbundles, which we define to be $F \subset E$ such that E/F is locally free.

Proposition 9.5. *Let E be a vector bundle on C. The following are equivalent:*

- *E is (semi)stable.*
- *For every nontrivial proper subbundle $F \subset E$, i.e. $\mathrm{rk}\, F < \mathrm{rk}\, E$ and E/F is locally free, we have $\mu(F)(\leq)\mu(E)$.*
- *For any proper quotient $E \twoheadrightarrow Q$ such that $\mathrm{rk}\, E > \mathrm{rk}\, Q$, we have $\mu(E)(\leq)\mu(Q)$.*
- *For any locally free quotient $E \twoheadrightarrow Q$ with $\mathrm{rk}(E) > \mathrm{rk}(Q)$ we have $\mu(E)(\leq)\mu(Q)$.*

Proof. Special case of Proposition 13.2. □

Now that we've discussed the theory of (semi)stability for vector bundles on curves, here are some concrete things.

Example 9.6 ((Semi)stable vector bundles on projective line)**.** *On $\mathbb{P}^1$, the only stable vector bundles are the line bundles. Clearly any line bundle is stable. Now suppose E is stable. By the Grothendieck-Birkhoff theorem, E will decompose as a direct sum*

$$E \cong \bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(d_i).$$

Suppose $r > 1$ and $d_1 \geq d_2 \geq \cdots \geq d_r$. Then consider the subsheaf $\mathcal{O}_{\mathbb{P}^1}(d_1) \subset E$. The slope of this subsheaf is d_1, while the slope of E is $\frac{\sum d_i}{r} \leq d_1$. This proves the claim.

Furthermore, the only semistable vector bundles on $\mathbb{P}^1$ are the well-balanced vector bundles $\mathcal{O}_{\mathbb{P}^1}(d)^{\oplus r}$. First, we show any well-balanced vector bundle $E \cong \mathcal{O}_{\mathbb{P}^1}(d)^{\oplus r}$ is semistable. If $F \subset E$ is a subsheaf with $0 < \operatorname{rk} F < r$, it is a vector bundle as well, and so admits a decomposition $\bigoplus_{i=1}^{\operatorname{rk} F} \mathcal{O}_{\mathbb{P}^1}(c_i)$. So

$$\bigoplus_{i=1}^{\operatorname{rk} F} \mathcal{O}_{\mathbb{P}^1}(c_i) \subset \bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(d).$$

Note for each c_i we have a morphism $\mathcal{O}_{\mathbb{P}^1}(c_i) \hookrightarrow \bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(d)$, which forces $c_i \leq d$. Then $\mu(F) = \frac{\sum_{i=1}^{\operatorname{rk} F} c_i}{\operatorname{rk} F} \leq d = \mu(E)$. So E is semistable. On the other hand, suppose $E \cong \bigoplus_{i=1}^{r} \mathcal{O}_{\mathbb{P}^1}(d_i)$ is semistable. Suppose the d_i are not all the same. Then WLOG suppose d_1 is maximal. Then $\mathcal{O}_{\mathbb{P}^1}(d_1)$ is a subsheaf of E with slope d_1 which is greater than $\mu(E)$, contradicting the assumed semistability of E. Thus, E must be well-balanced.

Lemma 9.7. *Let E be a (semi)stable vector bundle on C. Let L be a line bundle on C. Then $E \otimes L$ is also (semi)stable.*

Proof. Let F be a nontrivial proper subbundle of $E \otimes L$. Then we have the short exact sequence

$$0 \to F \to E \otimes L \to Q \to 0.$$

Then we obtain the short exact sequence

$$0 \to F \otimes L^\vee \to E \to Q \otimes L^\vee \to 0.$$

Since E is (semi)stable, we have $\mu(F \otimes L^\vee)(\leq)\mu(E)$. But note

$$\mu(F \otimes L^\vee) = \frac{\deg(F \otimes L^\vee)}{\operatorname{rk} F} = \mu(F) - \deg L.$$

Then

$$\mu(F \otimes L^\vee)(\leq)\mu(E) \implies \mu(F)(\leq)\mu(E) + \deg L = \frac{\deg E + \operatorname{rk} E \deg L}{\operatorname{rk} E} = \mu(E \otimes L),$$

thereby showing $E \otimes L$ is (semi)stable. □

Proposition 9.8. *Let E be a semistable vector bundle on C of rank r and degree d. If $\gcd(r, d) = 1$, then E is stable.*

Proof. Suppose $F \subset E$ is a nontrivial proper subsheaf. Then

$$\frac{\deg F}{\operatorname{rk} F} \leq \frac{d}{r} \implies r \cdot \deg F \leq d \cdot \operatorname{rk} F.$$

If we had equality, then since r, d are coprime, we'd have $r \mid \operatorname{rk} F$, but that is impossible since $0 < \operatorname{rk} F < r$. So we must have E is stable. □

We can combine Lemma 9.7 and Proposition 9.8 to prove the following important fact.

Proposition 9.9. *Assume C is a curve of genus $g \geq 1$. Then for every $(r, d) \in \mathbb{N}_+ \times \mathbb{Z}$, there exists a semistable vector bundle of rank r and degree d.*

Proof. We show this by strong induction on the rank. The base case $r = 1$ is clear, since almost by definition, line bundles on curves are stable. Now we show existence of a semistable vector bundle of rank r and degree d, assuming it holds for all vector bundles of rank i where $1 \leq i < r$.

First, we can assume that $d \geq 1$ due to lemma 9.7. Now imagine that (r, d) is an integral point in the first quadrant of cartesian plane. Here, the horizontal axis will be the rank axis, and the vertical axis will be the degree axis. Next, strictly within the interior of the triangle defined by the vertices $\{(0,0), (r, d), (0, d)\}$ plus the line segment $(0, d) \leftrightarrow (r, d)$, let (r_2, d_2) be the integral point that is closest to the line segment $(0, 0) \leftrightarrow (r, d)$. Note that (r_2, d_2) exists since there are a finite number of integral points in the region. Furthermore, if we consider the triangle defined by the vertices $\{(0, 0), (r_2, d_2), (r, d)\}$, note that there are no integral points in the interior or on the sides $(0, 0) \leftrightarrow (r_2, d_2)$ and $(r_2, d_2) \leftrightarrow (r, d)$. Otherwise, this would contradict (r_2, d_2) having minimal distance to the line segment $(0, 0) \leftrightarrow (r, d)$.

Note $0 < r_2 < r$ and $0 < d_2 \leq d$, and $\frac{d_2}{r_2} > \frac{d}{r}$. Let $r_1 = r - r_2$ and $d_1 = d - d_2$. So $0 < r_1 < r$ and $0 \leq d_1 < d$. We have the inequalities

$$\frac{d_1}{r_1} < \frac{d}{r} < \frac{d_2}{r_2}.$$

Now let E_1 and E_2 be semistable vector bundles of rank and degree (r_1, d_1) and (r_2, d_2), respectively, which we know existence by our inductive hypothesis. By construction, we know that $\gcd(r_2, d_2) = 1$. So E_2 is in fact stable, by Proposition 9.8, which is an important detail that will come to play near the end of the proof.

We have

$$\dim_{\mathbb{C}} \operatorname{Ext}^1(E_2, E_1) = \dim_{\mathbb{C}} H^0(C, E_1^\vee \otimes E_2 \otimes \omega_C) \geq \deg(E_1^\vee \otimes E_2) + r_1 r_2(2g - 2) + r_1 r_2(1 - g)$$

$$= \deg(E_1^\vee) \operatorname{rk}(E_2) + \operatorname{rk}(E_1^\vee) \deg(E_2) + r_1 r_2(g - 1) = -d_1 r_2 + r_1 d_2 + r_1 r_2(g - 1) > 0$$

since $\frac{d_2}{r_2} > \frac{d_1}{r_1}$ and $g \geq 1$. Then let

$$0 \longrightarrow E_1 \longrightarrow V \longrightarrow E_2 \longrightarrow 0$$

be a nontrivial extension. Let $F \subset V$ be the maximal destablising subsheaf of V.

I would like to show that $F = V$. Let's assume for the sake of contradiction that $F \neq V$. Then by construction of the maximal destablising subsheaf, we must have $\mu(F) > \mu(V)$. Now consider the subsheaves $F \cap E_1 \subset E_1$ and $\frac{F}{F \cap E_1} \subset E_2$. There are 3 cases to consider:

(1) $\frac{F}{F \cap E_1} = 0$, i.e. $F \subset E_1$.
(2) Neither $F \cap E_1$ nor $\frac{F}{F \cap E_1}$ are equal to 0.
(3) $F \cap E_1 = 0$.

We'll show that all 3 cases are impossible, and conclude by contradiction then that $F = V$ and thus V is semistable. First, note that $0 \leq \mathrm{rk}(F \cap E_1) \leq \min\{r_1, \mathrm{rk}\, F\}$ and if $F \cap E_1 \neq 0$ then $\mu(F \cap E_1) \leq \frac{d_1}{r_1}$. Next, note that $0 \leq \mathrm{rk}(\frac{F}{F \cap E_1}) \leq \min\{r_2, \mathrm{rk}\, F\}$, and if $\frac{F}{F\cap E_1} \neq 0$, then $\mu(\frac{F}{F\cap E_1}) \leq \frac{d_2}{r_2}$.

If $\frac{F}{F\cap E_1} = 0$, then $F \subset E_1$. But then $\mu(F) \leq \mu(E_1)$, while $\mu(F) > \frac{d}{r} > \mu(E_1)$. So this cannot happen.

If neither $F \cap E_1$ nor $\frac{F}{F\cap E_1}$ are zero, then since F is semistable we have

$$\mu(F \cap E_1) \leq \frac{d_1}{r_1} < \frac{d}{r} < \mu(F) \leq \mu(\frac{F}{F \cap E_1}) \leq \frac{d_2}{r_2}.$$

These slope inequalities, plus by how $\mathrm{rk}(F \cap E_1)$ and $\mathrm{rk}(\frac{F}{F\cap E_1})$ are bounded, and since $\mathrm{rk}\, F = \mathrm{rk}(F \cap E_1) + \mathrm{rk}(\frac{F}{F\cap E_1})$, imply that $(\mathrm{rk}\, F, \deg F)$ lies on the triangle given by vertices $\{(0,0), (r_2, d_2), (r, d)\}$, except it cannot line on side $(0,0) \leftrightarrow (r, d)$. Since there are no integral points in the interior of this triangle or on the sides $(0,0) \leftrightarrow (r_2, d_2)$, $(r_2, d_2) \leftrightarrow (r, d)$, besides the vertices, we must have $(\mathrm{rk}\, F, \deg F) = (r_2, d_2)$. But this forces $\mathrm{rk}(F \cap E_1) = 0$, which means $F \cap E_1 = 0$, so this case of neither $F \cap E_1$ nor $\frac{F}{F\cap E_1}$ being zero cannot happen.

Then the last case is that $F \cap E_1 = 0$. But by assumption we have the inequalities

$$\frac{d}{r} < \mu(F) \leq \frac{d_2}{r_2},$$

and $0 < \mathrm{rk}\, F \leq r_2$. Again by arguing about the integral points within and on the triangle with vertices $\{(0,0), (r_2, d_2), (r, d)\}$, we find $(\mathrm{rk}\, F, \deg F) = (r_2, d_2)$. This means that $F \hookrightarrow E_2$ but note also that $\mu(F) = \mu(E_2)$. Recall E_2 is stable since $\gcd(r_2, d_2) = 1$. Then we must have $F \cong E_2$. But this means that the short exact sequence $0 \to E_1 \to V \to E_2 \to 0$ splits, contradicting our assumption that it is a nontrivial extension. Altogether this establishes that V must be semistable. □

Combining Proposition 9.9 and Proposition 9.8 implies the following corollary.

Corollary 9.10. *Let C be a curve of genus $g \geq 2$. Then for any $(r, d) \in \mathbb{N}_+ \times \mathbb{Z}$ where $\gcd(r, d) = 1$, there exists a stable vector bundle of rank r and degree d.*

And with a little bit more work, one can deduce the following.

Remark 9.11. [Alp25, Theorem 8.2.5] *There exists a stable vector bundle of rank r and degree d for any (r, d).*

Now that we've said something concrete, we move towards defining the desired moduli functor. The objects that our moduli functor will parameterize not semistable sheaves, but equivalence classes of these. The equivalence relation will be given by S-equivalence. So let's mention the Harder-Narasimhan filtration and Jordan-Holder filtration.

Proposition 9.12. *Let E be a vector bundle on C. There exists a unique filtration of vector bundles*

$$0 = E_0 \subset E_1 \subset \cdots \subset E_\ell = E$$

such that $\frac{E_i}{E_{i-1}}$ is semistable and

$$\mu(E_1) > \mu(\frac{E_2}{E_1}) > \cdots > \mu(\frac{E}{E_{\ell-1}}).$$

Proof. Special case of Theorem 13.8. □

We also have the Jordan-Holder filtration.

Proposition 9.13. *Let E be a semistable vector bundle on C. Then E admits an increasing filtration (not necessarily unique)*

$$0 = E_0 \subset E_1 \subset \cdots \subset E_\ell$$

such that each E_i is a semistable vector bundle with $\mu(E_i) = \mu(E)$ and $\frac{E_i}{E_{i-1}}$ is a stable vector bundle with $\mu(\frac{E_i}{E_{i-1}}) = \mu(E)$. The direct sum of stable vector bundles, called the associated graded of E,

$$gr(E) = \bigoplus_{i=1}^{\ell} \frac{E_i}{E_{i-1}}$$

is unique.

Proof. Special case of Proposition 13.9 and Proposition 13.10. □

We say that two vector bundles E and E', necessarily of the same rank and degree, are S-equivalent if $\mathrm{gr}(E) \cong \mathrm{gr}(E')$. We say a vector bundle is polystable if it is a direct sum of stable vector bundles all of the same slope.

Before we finally define the moduli functor, let me assert the following about $\mathrm{Bun}^\mu(C) \subseteq Coh(C)$, the subcategory of semistable vector bundles on C with slope μ.

Proposition 9.14. *The full subcategory $Bun^\mu(C) \subseteq Coh(C)$ of semistable vector bundles on C with slope μ is an abelian subcategory closed under extensions. Furthermore, it is*

- *Noetherian, meaning ascending filtrations by subobjects stabilise as the rank increases but is bounded,*
- *and Artinian, meaning descending filtrations by subobjects stabilise as the rank decreases but is bounded.*

Proof. The noetherian and artinian claims follow because C is Noetherian and we are dealing with coherent sheaves [Sta25, Tag 01Y7].

The main content is that the full subcategory $\mathrm{Bun}^\mu(C)$ is closed under kernels and cokernels, which does not hold in general for locally free sheaves (recall this failure is why, in the first place, we enlarge the category of vector bundles to to coherent sheaves).

Let's first show closure under extensions. Given a short exact sequence

$$0 \to F \to E \to G \to 0$$

where $F, G \in \mathrm{Bun}^{\mu}(C)$, we claim $E \in \mathrm{Bun}^{\mu}(C)$. Note E has no torsion section, otherwise one of F or G would have a torsion section. Being a torsion free sheaf on a 1-dimensional integral scheme, we have E is a vector bundle. Furthermore, $\deg F + \deg G = \deg E$ and $\operatorname{rk} F + \operatorname{rk} G = \operatorname{rk} E$. So

$$\mu(E) = \frac{\operatorname{rk} F \cdot \mu + \operatorname{rk} G \cdot \mu}{\operatorname{rk} E} = \mu.$$

Now suppose $E' \subset E$ is a nontrivial subsheaf with $\operatorname{rk} E' < \operatorname{rk} E$, so necessarily $0 < \operatorname{rk} E'$. Then note we have a short exact sequence

$$0 \to F \cap E' \to E' \to \frac{E'}{F \cap E'} \to 0.$$

If the quotient is 0, then $E' = F$, then $\mu(E') \leq \mu(E) = \mu$. If $F \cap E' = 0$, then $E' \cong \frac{E'}{F \cap E'} \hookrightarrow G$ and since G is semistable we find $\mu(E') \leq \mu$. Finally, if neither of these are true, then we have E' is an extension of vector bundles such that $\mu(F \cap E') \leq \mu$ and $\frac{E'}{F \cap E'} \leq \mu$. Then

$$\mu(E') = \frac{\operatorname{rk}(F \cap E')\mu(F \cap E') + \operatorname{rk}(\frac{E'}{F \cap E'})\mu(\frac{E'}{F \cap E'})}{\operatorname{rk}(E')} \leq \mu.$$

So indeed, the extension $E \in \mathrm{Bun}^{\mu}_{C}$. This shows that the subcategory is closed under extensions.

Now we show it is closed under kernels and cokernels. Suppose $F \to G$ is a morphism between semistable vector bundles of slope μ. Consider the kernel

$$0 \to K \to F \to G.$$

Assume it's nontrivial, so K is a vector bundle of positive rank. Then $\mu(K) \leq \mu$ since F is semistable. Note $F/K \hookrightarrow G$ which implies $\mu(F/K) \leq \mu$. But F is semistable, so $\mu(F/K) \geq \mu$. This implies $\mu(F/K) = \mu$, and note

$$\mu(F) = \frac{\operatorname{rk}(K)\mu(K) + \operatorname{rk}(F/K)\mu(F/K)}{\operatorname{rk} F} = \frac{\operatorname{rk} K \mu(K) + \operatorname{rk}(F/K)\mu}{\operatorname{rk} F} = \mu \implies \mu(K) = \mu.$$

So $\mu(K) = \mu$ and if $F' \subset K$ nontrivial then $\mu(F') \leq \mu$ by semistability of F. This implies K is semistable of slope μ. So we have closure under kernels.

Next we have closure under cokernels. Consider again $\theta : F \to G$. If it is a nontrivial morphism, let $I = \mathrm{im}\ \theta \subset G$. Note we must have $\mu(I) = \mu$ since by semistability of F and G we have $\mu = \mu(F) \leq \mu(I) \leq \mu(G) = \mu$. Note I is saturated in G, because we could filter G in terms of its Jordan-Holder filtration, and the morphism θ would need to define a nontrivial map from F to one of the stable quotient summands of $\mathrm{gr}(G)$. But by Proposition 13.3, this map would need to be a surjection. We see then that I needs to be one of the Jordan-Holder filtrations of G, and thus must be saturated. This is because the quotient G/F will then be expressible by a series of extensions by vector bundles, and so itself will be a vector bundle. Furthermore, $\mu(G/F) \le \mu(G)$ by semistability of G. But there is a short exact sequence

$$0 \to I \to G \to G/F \to 0,$$

and since I and G both have slope μ, we must have $\mu(G/F) = \mu$. Thus, G/F is a vector bundle of slope μ. Furthermore, it is semistable as $G/F \subset G$ and using the semistability of G. This shows closure under cokernels. □

Definition 9.15. *The moduli functor of semistable vector bundles of rank r and degree d is*

$$\underline{Bun}_C^{ss}(r,d) : (Sch_{\mathbb{C}})^{op} \to Sets$$

where

$$T \mapsto \frac{\{E \text{ loc. free on } T \times C \text{ and } E_t \text{ semistable rank } r \text{ and degree } d\}}{\pi_T^* \operatorname{Pic}(T)}.$$

What we'll hope for is that this moduli functor is co-represented. Note that $\underline{\mathrm{Bun}}_C^{ss}(r,d)$ admitting a coarse moduli space in general is still quite difficult. At least in the genus 0 case, i.e. $\mathbb{P}^1$, every polystable vector bundle is a well-balanced vector bundle, so isomorphism classes of semistable vector bundles and S-equivalence classes coincide.

But for genus $g \ge 1$, we cannot hope for a coarse moduli space in general. For example, $\operatorname{Ext}^1(\mathcal{O}_C, \mathcal{O}_C)$ has dimension g, and if one takes an affine line in this space corresponding to an extension where the middle term is semistable, then over this line we get nontrivial semistable vector bundles, but the central fiber over the origin is $\mathcal{O}_C \oplus \mathcal{O}_C$. So the existence of semistable but not polystable vector bundles unfortunately still gives rise to a jump phenomena that we don't like.

One positive thing though is that semistable vector bundles of fixed rank and degree on C form a bounded set.

Proposition 9.16. *Let E be a semistable vector bundle on C of rank r. Suppose $\mu(E) > 2g-2$. Then $H^1(C,E) = 0$.*

Proof. Note $H^1(C,E) \cong H^0(C, E^\vee \otimes \omega_C)$. Note $H^0(C, E^\vee \otimes \omega_C)$ is the space of morphisms $\mathcal{O}_C \to E^\vee \otimes \omega_C$, which is the space of morphisms $E \to \omega_C$. Note E is semistable and ω_C is semistable (also stable, but doesn't matter for the punchline here). But $\mu(E) > \mu(C)$. This implies the only morphism is the 0 map. So $H^1(C,E) = 0$. □

Proposition 9.17. *Consider the set $\underline{Bun}_C^{ss}(r,d)(\mathbb{C})$ of all semistable vector bundles of degree r and rank d. Let λ be any integer such that $\frac{d+r\lambda}{r} > 2g-1$. Then there exists a vector space V such that for every $E \in \underline{Bun}_C^{ss}(r,d)(\mathbb{C})$, we have a surjection*

$$V \otimes L^\vee \twoheadrightarrow E.$$

where L is a line bundle on C of degree λ.

Proof. Let λ be as asserted in the statement. Let L be a line bundle on C of degree λ. Then note for every $E \in \underline{\mathrm{Bun}}_C^{ss}(r,d)(\mathbb{C})$, we have $E \otimes L$ is a line bundle of degree $d + r\lambda$. Thus $\mu(E \otimes L) > 2g-1$. This implies that $H^1(C, E \otimes L) = 0$. In particular, note all the $E \otimes L$ share the same Hilbert polynomial P and since $H^1(C, E \otimes L) = 0$, we have $\dim_{\mathbb{C}} H^0(C, E \otimes L) = P(0)$. So we let $H^0(C, E \otimes L) = V$, which is a vector space of dimension $P(0)$ and thus isomorphic across all E.

Note that for any $p \in C(\mathbb{C})$, if we take the short exact sequence

$$0 \to \mathcal{O}_C(p) \otimes (E \otimes L) \to E \otimes L \to (E \otimes L)(p) \to 0,$$

we find that $H^1(\mathcal{O}_C(p) \otimes (E \otimes L)) = 0$ since $\mu(\mathcal{O}_C(p) \otimes (E \otimes L)) > 2g-2$. Its vanishing forces a surjection $H^0(C, E \otimes L) \to H^0(C, (E \otimes L)(p))$. Since this is true for every $p \in C(\mathbb{C})$, this implies that the global sections $H^0(C, E \otimes L)$ globally generates the vector bundle $E \otimes L$. Thus, we have the desired surjection

$$V \otimes \mathcal{O}_C \twoheadrightarrow E \otimes L,$$

which implies we have the desired surjection

$$V \otimes L^\vee \twoheadrightarrow E$$

for every E. □

Proposition 9.17 shows that we can exhibit every semistable vector bundle of fixed rank and degree on C as the quotient of the same coherent sheaf. This implies that $\underline{\mathrm{Bun}}_C^{ss}(r,d)(\mathbb{C})$ is a bounded set, in the sense of [HL, Lemma 1.7.6].

10. MVC 6/4/25: Co-representability of moduli functor of semistable vector bundles part 1

In the previous section, we showed that semistable vector bundles of fixed rank and degree on a curve C all arise as quotients from the same vector bundle. We will use this to relate the sheafification

$$\underline{\mathrm{Bun}}_C^{ss}(r,d)^{sh}$$

of the moduli functor of semistable vector bundles of rank r and degree d on C to the universal categorical quotient of a certain group action on a Quot scheme.

First, let E be a semistable vector bundle of rank r and degree d on a curve C of genus g. Fix an ample line bundle $\mathcal{O}_C(p)$, for some $p \in C(\mathbb{C})$. Let $n \in \mathbb{Z}_{\geq 0}$ so that

$$\frac{d}{r} + n > 2g - 1.$$

Then for every such E, we have a quotient map

$$V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E,$$

induced by an isomorphism $V \to H^0(C, E \otimes \mathcal{O}_C(np))$. Note $\dim_{\mathbb{C}} V = d + rn + r(-g+1)$, since $\chi(E \otimes \mathcal{O}_C(np)) = h^0(C, E \otimes \mathcal{O}_C(np)) = d + rn + r(-g+1)$. All of this follows from Proposition 9.17. Since we've exhibited each semistable vector bundle of rank r and degree d as a quotient, this motivates us to take a look at the corresponding Quot scheme.

So consider the Quot scheme

$$Q := \mathrm{Quot}_C(V \otimes \mathcal{O}_C(-np), P(m)),$$

where $P(m) = rm + d + r(-g+1)$. Now consider the universal quotient

$$V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ}}$$

of coherent sheaves on $Q \times C$. Recall that the locus of $t \in Q$ such that $E_{\mathrm{univ},t}$ is a vector bundle and semistable is open by Proposition 16.2. Furthermore, if we tensor the universal surjection by $\pi_C^* \mathcal{O}_C(np) \in \mathrm{Pic}(Q \times C)$ and take the kernel of the resulting surjection, we obtain the short exact sequence

$$0 \to K \to V \otimes \mathcal{O}_{Q \times C} \to E_{\mathrm{univ}} \otimes \pi_C^* \mathcal{O}_C(np) \to 0.$$

Since everything here is flat we have $K_t = \ker[V \otimes \mathcal{O}_C \twoheadrightarrow E_{\mathrm{univ},t} \otimes \mathcal{O}_C(np)]$ by [HL, Lemma 2.1.4, Page 33]. Since K is Q-flat, by upper-semicontinuity we have that the locus of $t \in Q$ such that $H^0(C, K_t) = 0$ is open. Thus there is an open locus $U \subseteq Q$ such that for $t \in U$ we have $V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ},t}$ is a surjection such that we have an injection

$$0 \to V \otimes \mathcal{O}_U \to \pi_{U,*}[E_{\mathrm{univ}}|_U \otimes \pi_C^* \mathcal{O}_C(np)]$$

which is injective on fibers. Note that $R^i \pi_{U,*}[E_{\mathrm{univ}}|_U \otimes \pi_C^* \mathcal{O}_C(np)] = 0$ for all $i > 0$, so $\pi_{U,*}[E_{\mathrm{univ}}|_U \otimes \pi_C^* \mathcal{O}_C(np)]$ is a vector bundle. Thus the above injection is a morphism of vector bundles of locally constant rank, therefore the cokernel is a vector bundle. But note that the injection of vector bundles is an injection of fibers, and the vector bundles have the same rank. Thus it is actually an isomorphism of vector bundles. Thus, we define $R^{ss} := U$ to be the open loci of $t \in R^{ss}$ such that $E_{\mathrm{univ},t}$ is semistable and the pushforward

$$V \otimes \mathcal{O}_{R^{ss}} \to \pi_{R^{ss},*}[E_{\mathrm{univ}}|_{R^{ss}} \otimes \pi_C^* \mathcal{O}_C(np)]$$

of $V \otimes \mathcal{O}_{R^{ss} \times C} \twoheadrightarrow E_{\mathrm{univ}}|_{R^{ss}} \otimes \pi_C^* \mathcal{O}_C(np)$, is an isomorphism.

In summary, $R^{ss} \subset Q$ is the open loci of $t \in Q$ such that the restriction $V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ},t}$ of the universal surjection is such that the induced map $V \to H^0(C, E_{\mathrm{univ},t} \otimes \mathcal{O}_C(np))$ is an isomorphism. Note that this induced map $V \to H^0(C, E_{\mathrm{univ},t} \otimes \mathcal{O}_C(np))$ recovers $V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ},t}$. Relatively speaking then, R^{ss} comes equipped with its own universal surjection

$$q_{R^{ss}} : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ}}|_{R^{ss}}$$

such that tensoring by $\pi_C^* \mathcal{O}_C(np)$ and pushing forward by $\pi_{R^{ss}}$, we obtain an isomorphism

$$V \otimes \mathcal{O}_{R^{ss}} \to \pi_{R^{ss},*}[E_{\mathrm{univ}}|_{R^{ss}} \otimes \pi_C^* \mathcal{O}_C(np)].$$

Note that this isomorphism recovers $q_{R^{ss}}$, as post-composing the pullback of this isomorphism with the canonical push-pull map, the composition

$$V \otimes \mathcal{O}_{R^{ss}\times C} \longrightarrow \pi_{R^{ss}}^{*}\pi_{R^{ss},*}[E_{\mathrm{univ}}|_{R^{ss}} \otimes \pi_C^{*}\mathcal{O}_C(np)] \longrightarrow E_{\mathrm{univ}}|_{R^{ss}} \otimes \pi_C^{*}\mathcal{O}_C(np)$$

equals $q_{R^{ss}}$.

Proposition 10.1. *The functor of points of the quasi-projective $\mathbb{C}$-variety R^{ss} may be expressed as*

$$T \mapsto \{[q : V\otimes\pi_C^{*}\mathcal{O}_C(-np) \twoheadrightarrow F] \in Q(T) \mid F_t \textit{ semistable } \forall t \in T; q \textit{ induced by iso } V\otimes\mathcal{O}_T \cong \pi_{T,*}[F\otimes\pi_C^{*}\mathcal{O}_C(np)]\}.$$

Proof. First, we show that this functor is a subfunctor of the functor of points of R^{ss}. Suppose we have $[q : V \otimes \pi_C^{*}\mathcal{O}_C(-np) \twoheadrightarrow F] \in Q(T)$ satisfyig the conditions of the moduli functor. Note that the condition that q be induced by $V \otimes \mathcal{O}_T \cong \pi_{T,*}[F \otimes \pi_C^{*}\mathcal{O}_C(np)]$ is redundant wording; the necessary condition here is simply that the pushforward of

$$V \otimes \mathcal{O}_{T\times C} \xrightarrow{q\otimes\mathrm{id}} F \otimes \pi_C^{*}\mathcal{O}_C(np)$$

along $\pi_{T,*}$ is an isomorphism

$$V \otimes \mathcal{O}_T \to \pi_{T,*}[F \otimes \pi_C^{*}\mathcal{O}_C(np)],$$

and this isomorphism automatically recovers q. We've added the redundant wording to emphasize this fact that the isomorphism recovers q.

Now suppose we are given the data of such a q. Note q must come from pulling back the universal surjection over $Q \times C$ along some $T \to Q$. Let K denote the kernel

$$0 \to K \to V \otimes \mathcal{O}_{T\times C} \to F \otimes \pi_C^{*}\mathcal{O}_C(np) \to 0.$$

Note F_t is semistable for every $t \in T$, and since we have the isomorphism

$$V \otimes \mathcal{O}_T \to \pi_{T,*}[F \otimes \pi_C^{*}\mathcal{O}_C(np)],$$

we have $\pi_{T,*}(K) = 0$. Then $H^0(C, K_t) = 0$ for every $t \in T$. Altogether, this implies that the morphism $T \to Q$ must factor through R^{ss}.

Conversely, given any morphism $T \to R^{ss}$, pulling back the universal surjection over R^{ss} gives exactly a $[q : V \otimes \pi_C^{*}\mathcal{O}_C(-np) \twoheadrightarrow F] \in Q(T)$ such that F_t is semistable for every $t \in T$ which is induced by an isomorphism $V \otimes \mathcal{O}_T \to \pi_{T,*}[F \otimes \pi_C^{*}\mathcal{O}_C(np)]$. $\square$

Note that the immediately aforementioned F must be locally free on $T \times C$, since each F_t is a vector bundle. Then we have a morphism of moduli functors

$$R^{ss} \to \underline{\mathrm{Bun}}_C^{ss}(r, d)$$

where $R^{ss}(T) \to \underline{\mathrm{Bun}}_C^{ss}(r, d)(T)$ sends

$$[V \otimes \pi_C^{*}\mathcal{O}_C(-np) \twoheadrightarrow F] \mapsto [F].$$

Note that R^{ss} includes not only the information of $\underline{\mathrm{Bun}}_C^{ss}(r,d)$, but also choices of bases for global sections. An element $[V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow F] \in R^{ss}(T)$ is not just the information of F parameterizing semistable vector bundles over T, but also a choice of isomorphism $V \otimes \mathcal{O}_T \to \pi_{T,*}(F \otimes \pi_C^* \mathcal{O}_C(np))$. At the level of $\mathbb{C}$-points, an element $[V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E] \in R^{ss}(\mathbb{C})$ is not just the datum of a semistable vector bundle E, but also a choice of isomorphism $V \to H^0(C, E \otimes \mathcal{O}_C(np))$. This suggests that it may be possible to define a GL_V action on R^{ss} so that the morphism $R^{ss} \to \underline{\mathrm{Bun}}_C^{ss}(r,d)$ is GL_V-equivariant.

Indeed this is possible. Recall that the T-points of GL_V are $\mathrm{Aut}_T(V \otimes \mathcal{O}_T)$. Then we define the group action

$$\mathrm{GL}_V \times Q \to Q$$

to be so that on T-points the map $\mathrm{GL}_V(T) \times Q(T) \to Q(T)$ is given by

$$(\theta, [q : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow F]) \mapsto q \circ (\pi_T^*(\theta) \otimes \mathrm{id}_{\pi_C^* \mathcal{O}_C(-np)}) : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow F.$$

What this map does is given the data $(\theta, [q : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow F])$, note $\theta \in \mathrm{Aut}_T(V \otimes \mathcal{O}_T)$ may also be thought of as parameterizing linear isomorphisms of V with respect to $t \in T$. So if one thinks of $[q : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow F]$ as parameterizing surjections $V \otimes \mathcal{O}_C(-np) \twoheadrightarrow F_t$, then our action sends this parameterization to the surjection of coherent sheaves parameterizing

$$V \otimes \mathcal{O}_C(-np) \xrightarrow{\theta_t \otimes \mathrm{id}} V \otimes \mathcal{O}_C(-np) \twoheadrightarrow F.$$

Note the GL_V action we've defined respects the equivalence relation on $Q(T)$, and satisfies the definition of a group action. Furthermore, we see that R^{ss} is GL_V-invariant, since F remains the same, and if q is induced by an isomorphism

$$V \otimes \mathcal{O}_T \to \pi_{T,*}(F \otimes \pi_C^* \mathcal{O}_C(np)),$$

then $q \circ (\pi_T^*(\theta) \otimes \mathrm{id}_{\pi_C^* \mathcal{O}_C(-np)})$ is induced by the composition

$$V \otimes \mathcal{O}_T \xrightarrow{\theta} V \otimes \mathcal{O}_T \to \pi_{T,*}(F \otimes \pi_C^* \mathcal{O}_C(np)).$$

So there is an induced action

$$\mathrm{GL}_V \times R^{ss} \to R^{ss},$$

so that the morphism of moduli functors

$$R^{ss} \to \underline{\mathrm{Bun}}_C^{ss}(r,d)$$

is GL_V-equivariant. Then there is an induced morphism of moduli functors

$$[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}} \to \underline{\mathrm{Bun}}_C^{ss}(r,d).$$

Theorem 10.2. *The morphism of moduli functors*

$$[\frac{R^{ss}}{\mathrm{GL}_V}]^{pre} \to \underline{Bun}_C^{ss}(r,d)$$

sheafifies to an isomorphism

$$[\frac{R^{ss}}{\mathrm{GL}_V}]^{sh} \to (\underline{Bun}_C^{ss}(r,d))^{sh}$$

of Zariski sheaves.

Proof. We first show surjection. Suppose we have an element of $(\underline{\mathrm{Bun}}_C^{ss}(r,d))^{sh}(T)$. Then such an element may be thought of as the following datum: an open Zariski cover $\{U_i\}$ of T and $E_i \in \underline{\mathrm{Bun}}_C^{ss}(r,d)(U_i)$ such that

$$E_i \cong E_j \otimes \pi^*_{U_i \cap U_j} R_{ij}$$

over $U_i \cap U_j$, for some $R_{ij} \in \mathrm{Pic}(U_i \cap U_j)$. Note that for each i, we have $R^k \pi_{U_i,*}(E_i \otimes \pi_C^* \mathcal{O}_C(np)) = 0$ for all $k > 0$. So $\pi_{U_i,*}(E_i \otimes \pi_C^*\mathcal{O}_C(np))$ is a vector bundle of rank $\dim V$. Then we can choose the U_i so that such pushforwards are all trivial vector bundles over U_i, isomorphic to $V \otimes \mathcal{O}_{U_i}$.

Applying $\pi_{U_i \cap U_j,*}$ to the isomorphism

$$E_i \otimes \pi_C^*\mathcal{O}_C(np) \cong E_j \otimes \pi_C^*\mathcal{O}_C(np) \otimes \pi^*_{U_i\cap U_j} R_{ij},$$

on the overlap $U_i \cap U_j$, we obtain

$$\pi_{U_i\cap U_j,*}(E_i \otimes \pi_C^*\mathcal{O}_C(np)) \cong \pi_{U_i \cap U_j,*}(E_j \otimes \pi_C^*\mathcal{O}_C(np)) \otimes R_{ij},$$

where the right hand side follows by the projection formula. This forces $R_{ij} \cong \mathcal{O}_{ij}$, since we have shrunk the U_i enough so that the pushforwards are both isomorphic to $V \otimes \mathcal{O}_{U_i \cap U_j}$.

Now choose isomorphisms $V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(E_i \otimes \pi_C^*\mathcal{O}_C(np))$ and $V \otimes \mathcal{O}_{U_j} \to \pi_{U_j,*}(E_j \otimes \pi_C^*\mathcal{O}_C(np))$. Then post-composing the pullbacks of these isomorphisms with the canonical push-pull morphisms,

$$\pi^*_{U_i}\pi_{U_i,*}(E_i \otimes \pi_C^*\mathcal{O}_C(np)) \twoheadrightarrow E_i \otimes \pi_C^*\mathcal{O}_C(np)$$

and

$$\pi^*_{U_j}\pi_{U_j,*}(E_j \otimes \pi_C^*\mathcal{O}_C(np)) \twoheadrightarrow E_j \otimes \pi_C^*\mathcal{O}_C(np),$$

then twisting everything down by $\pi_C^*\mathcal{O}_C(-np)$, we obtain surjections

$$q_i : V \otimes \pi_C^*\mathcal{O}_C(-np) \twoheadrightarrow E_i,$$

over $U_i \times C$, and

$$q_j : V \otimes \pi_C^*\mathcal{O}_C(-np) \twoheadrightarrow E_j,$$

over $U_j \times C$. These surjections q_i and q_j define elements of $R^{ss}(U_i)$ and $R^{ss}(U_j)$ which map to $E_i \in \underline{\mathrm{Bun}}_C^{ss}(r,d)(U_i)$ and $E_j \in \underline{\mathrm{Bun}}_C^{ss}(r,d)(U_j)$, respectively. Furthermore, we claim that the restrictions of q_i and q_j over $U_i \cap U_j \times C$ are in the same $\mathrm{GL}_V(U_i \cap U_j)$ orbit class. To see why, recall that over $U_i \cap U_j \times C$ we have

$$E_i \cong Ej \implies E_i \otimes \pi_C^* \mathcal{O}_C(np) \cong E_j \otimes \pi_C^* \mathcal{O}_C(np)$$
$$\implies \pi_{U_i \cap U_j,*}(E_i \otimes \pi_C^* \mathcal{O}_C(np)) \cong \pi_{U_i \cap U_j,*}(E_j \otimes \pi_C^* \mathcal{O}_C(np)).$$

Then the restrictions of the isomorphisms we chose to induce q_i and q_j to $U_i \cap U_j$ may be used to define

$$\theta_{ij} : V \otimes \mathcal{O}_{U_i \cap U_j} \to \pi_{U_i \cap U_j,*}(E_i \otimes \pi_C^* \mathcal{O}_C(np)) \cong \pi_{U_i \cap U_j,*}(E_j \otimes \pi_C^* \mathcal{O}_C(np)) \to V \otimes \mathcal{O}_{U_i \cap U_j},$$

so that $\theta_{ij} \in \mathrm{GL}_V(U_i \cap U_j)$. Then we have the following commutative diagram

$$\begin{tikzcd}
V \otimes \mathcal{O}_{U_i \cap U_j \times C} \arrow[r] & {[\pi_{U_i \cap U_j}^* \pi_{U_i \cap U_j,*}(E_j \otimes \pi_C^* \mathcal{O}_C(np))]} \arrow[r, two heads] & E_j \otimes \pi_C^* \mathcal{O}_C(np) \\
V \otimes \mathcal{O}_{U_i \cap U_j \times C} \arrow[r] \arrow[u, "\pi_{U_i \cap U_j}^*(\theta_{ij})"] & {[\pi_{U_i \cap U_j}^* \pi_{U_i \cap U_j,*}(E_i \otimes \pi_C^* \mathcal{O}_C(np))]} \arrow[r, two heads] \arrow[u] & E_i \otimes \pi_C^* \mathcal{O}_C(np) \arrow[u]
\end{tikzcd}$$

Twisting everything down by $\pi_C^* \mathcal{O}_C(-np)$, we obtain the diagram

$$\begin{tikzcd}
V \otimes \pi_C^* \mathcal{O}_C(-np) \arrow[r, two heads, "q_j"] & E_j \\
V \otimes \pi_C^* \mathcal{O}_C(-np) \arrow[r, two heads, "q_i"'] \arrow[u, "\pi_{U_i \cap U_j}^*(\theta_{ij}) \otimes \mathrm{id}_{\pi_C^* \mathcal{O}_C(-np)}"] & E_i \arrow[u]
\end{tikzcd}$$

This diagram shows that the action of θ_{ij} on q_j, namely the composition $q_j \circ \pi_{U_i \cap U_j}^*(\theta_{ij}) \otimes \mathrm{id}_{\pi_C^* \mathcal{O}_C(-np)}$, belongs to the same equivalence class of q_i where the equivalence relation we're referring to here is one inherited from the Quot functor. Altogether, this shows that the restrictions of q_i and q_j over $U_i \cap U_j \times C$ are in the same $\mathrm{GL}_V(U_i \cap U_j)$ orbit class, which finishes our proof of surjectivity.

To show injectivity, it suffices to show the following: suppose $[q_M : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow M_i] \in R^{ss}(U_i)$ such that q_M is induced by an isomorphism

$$V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(M_i \otimes \pi_C^* \mathcal{O}_C(np))$$

and suppose $[q_N : V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow N_i] \in R^{ss}(U_i)$ such that q_N is induced by an isomorphism

$$V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(N_i \otimes \pi_C^* \mathcal{O}_C(np)).$$

Furthermore, suppose that $[M_i] = [N_i] \in \underline{\mathrm{Bun}}_C^{ss}(r,d)(U_i)$, i.e. $M_i \cong N_i \otimes \pi_{U_i}^* R_i$ for some $R_i \in \mathrm{Pic}(U_i)$. But we can shrink all the U_i, i.e. take a finer cover, so that $R_i \cong \mathcal{O}_{U_i}$, so $M_i \cong N_i$. We would like to show then that q_M and q_N must belong in the same $\mathrm{GL}_V(U_i)$ orbit class. If we show this, then the proof of injectivity follows.

Since $M_i \cong N_i$, then $M_i \otimes \pi_C^* \mathcal{O}_C(np) \cong N_i \otimes \pi_C^* \mathcal{O}_C(np)$, so

$$\pi_{U_i,*}(M_i \otimes \pi_C^* \mathcal{O}_C(np)) \cong \pi_{U_i,*}(N_i \otimes \pi_C^* \mathcal{O}_C(np)).$$

Then using the isomorphisms inducing q_M and q_N, we can define

$$\theta_i : V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(M_i \otimes \pi_C^* \mathcal{O}_C(np)) \cong \pi_{U_i,*}(N_i \otimes \pi_C^* \mathcal{O}_C(np)) \to V \otimes \mathcal{O}_{U_i},$$

so that $\theta_i \in \mathrm{GL}_V(U_i)$. We see that we have a commuting diagram

$$\begin{tikzcd}
V \otimes \mathcal{O}_{U_i} \arrow[r] & \pi_{U_i}^* \pi_{U_i,*}[N_i \otimes \pi_C^* \mathcal{O}_C(np)] \arrow[r, two heads] & N_i \otimes \pi_C^* \mathcal{O}_C(np) \\
V \otimes \mathcal{O}_{U_i} \arrow[r] \arrow[u, "\pi_{U_i}^*(\theta_i)"] & \pi_{U_i}^* \pi_{U_i,*}[M_i \otimes \pi_C^* \mathcal{O}_C(np)] \arrow[r, two heads] \arrow[u] & M_i \otimes \pi_C^* \mathcal{O}_C(np) \arrow[u]
\end{tikzcd}$$

and twisting down by $\pi_C^* \mathcal{O}_C(-np)$, we obtain a commuting diagram

$$\begin{tikzcd}
V \otimes \pi_C^* \mathcal{O}_C(-np) \arrow[r, two heads, "q_N"] & N_i \\
V \otimes \pi_C^* \mathcal{O}_C(-np) \arrow[r, two heads, "q_M"'] \arrow[u, "\pi_{U_i}^*(\theta_i) \otimes \mathrm{id}_{\pi_C^* \mathcal{O}_C(-np)}"] & M_i \arrow[u]
\end{tikzcd}$$

This shows that the action of θ_i on q_N is in the same Quot functor equivalence class as q_M, thereby showing that q_N and q_M lie in the same $\mathrm{GL}_V(U_i)$ orbit. Then if we consider two elements in $[\frac{R^{ss}}{\mathrm{GL}_V}]^{sh}(T)$ which map to the same element of $(\underline{\mathrm{Bun}}_C^{ss}(r,d))^{sh}(T)$, then it is straightforward to check that what we have shown implies injectivity. $\square$

Why do we like Theorem 10.2? Because by GIT, the action of GL_V on R^{ss} will admit a universal categorical quotient, and thus by Theorem 10.2, it will co-represent the moduli functor $\underline{\mathrm{Bun}}_C^{ss}(r,d)$.

Remark 10.3. *Note that in order to obtain Theorem 10.2, it was crucial that we imposed the isomorphism on global sections condition that is a part of the definition of R^{ss}. Here's an example of why: consider $\underline{Bun}_{\mathbb{P}^1}^{ss}(1,2)$. Note the slope here is 2, which is greater than $2g-1 = -1$ in this situation; V here has dimension 3. The only $\mathbb{C}$-point is $\mathcal{O}_{\mathbb{P}^1}(2)$, which is also the only $\mathbb{C}$-point of the sheafification. But note if we defined $V \to H^0(\mathbb{P}^1_{[X_0:X_1]}, \mathcal{O}_{\mathbb{P}^1}(2))$ so that its image is 2-dimensional sub-vector space $\mathbb{C}\langle X_0^2, X_1^2\rangle$, then we still get a surjection $V \otimes \mathcal{O}_{\mathbb{P}^1} \twoheadrightarrow \mathcal{O}_{\mathbb{P}^1}(2)$, but it will not be GL_3 equivalent to a surjection induced by an isomorphism $V \cong H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2))$. So in the construction of R^{ss} in this case, if we did not require the isomorphism condition, then immediately we see that Theorem 10.2 would no longer hold, since then we would not even have a bijection on $\mathbb{C}$-points.*

Now considering the universal surjection $V \otimes \pi_C^* \mathcal{O}_C(-np) \twoheadrightarrow E_{\mathrm{univ}}|_{R^{ss}}$ over R^{ss}, we define

$$R^s \subseteq R^{ss}$$

to be the open loci of $t \in R^{ss}$ such that E_t is further required to be stable, which is open by Proposition 16.2. Then note that the morphism $R^{ss} \to \underline{\mathrm{Bun}}_C^{ss}(r,d)$ restricts to a morphism

$$[\frac{R^s}{\mathrm{GL}_V}] \to \underline{\mathrm{Bun}}_C^s(r,d),$$

which sheafifies to an isomorphism

$$(\frac{R^s}{\mathrm{GL}_V})^{sh} \cong (\underline{\mathrm{Bun}}_C^s(r,d))^{sh}$$

by the same argument given in Theorem 10.2. One might expect that in order to show that $(\underline{\mathrm{Bun}}_C^s(r,d))^{sh}$ is co-represented by some scheme, we would find some projective GIT setup with respect to the action of GL_V on the Quot scheme Q. But in fact, it turns out the right thing to do is to consider instead the induced action of SL_V on Q. We explain why, and sketch the proof of Theorem 10.4 in the general case of semistable coherent sheaves on a projective variety, in section 18.

Theorem 10.4. *Applying the projective GIT construction to the induced action of* SL_V *on* $Q := Quot_C(V \otimes \mathcal{O}_C(-np), rm + d + r(-g+1))$*, denote the semistable and stable locus of* Q *as* Q^{ss} *and* Q^s*. Recall that there is a universal categorical quotient*

$$\pi : Q^{ss} \to Q /\!\!/ \mathrm{GL}_V =: U_C(r,d),$$

which restricts to an actual geometric quotient $\pi : Q^s \to \pi(Q^s) =: U_C^s(r,d)$*. Then* $Q^{ss} = R^{ss}$ *and* $Q^s = R^s$*.*

Theorem 10.4 identifies R^{ss} and R^s with the semistable and stable loci. This implies that $\underline{\mathrm{Bun}}_C^{ss}(r,d)$ is co-represented by the projective GIT quotient $U_C(r,d) := Q /\!\!/ \mathrm{GL}_V$, and $\underline{\mathrm{Bun}}_C^s(r,d)$ admits coarse moduli space $U_C^s(r,d) := \pi(Q^s) \subseteq Q /\!\!/ \mathrm{GL}_V$.

Since $U_C(r,d)$ only co-represents $\underline{\mathrm{Bun}}_C^{ss}(r,d)$ we should ask what its $\mathbb{C}$-points. It turns out that its $\mathbb{C}$-points are not the semistable vector bundles, but exactly the S-equivalence classes of them. In order to show this, we first need the following lemma which we can use to show that semistable vector bundles having the same associated gradeds is a closed condition.

Lemma 10.5. *Let* T *be an irreducible affine* $\mathbb{C}$*-variety and let* $F \in \underline{Bun}_C^{ss}(r,d)(T)$*. Suppose that for an open* $T_0 \subseteq T$ *we have for all* $t \in T_0$

$$gr(F_t) \cong E_1 \oplus \cdots \oplus E_n,$$

where E_i *are stable vector bundles of slope* $\frac{d}{r}$*. Then actually for all* $t \in T$*, we have* $gr(F_t) \cong \bigoplus_{i=1}^n E_i$*.*

Proof. We proceed by induction on n. First consider when $n = 1$. So for $t \in T_0$, we have $F_t \cong E_1$. Consider the locus $T_1 := \{t \in T \mid \mathrm{Hom}_{\mathcal{O}_C}(E_1, F_t) \neq 0\}$. If we consider the vector bundle $\pi_C^* E_1^\vee \otimes F$, then by upper-semicontinuity, we have that the locus of t such that $H^0(C, (\pi_C^* E_1^\vee \otimes F)|_t) = H^0(C, E_1^\vee \otimes F_t) = \mathrm{Hom}_{\mathcal{O}_C}(E_1, F_t) \neq 0$ is closed by upper-semicontinuity, and this locus is exactly T_1. So $T_1 \subseteq T$ is closed, but furthermore $T_1 \supseteq T_0$. Since T is irreducible, this implies that $T_1 = T$. So for all $t \in T$, we have $\mathrm{Hom}_{\mathcal{O}_C}(E_1, F_t) \neq 0$. But then for every t, there exists a nontrivial homomorphism $E_1 \to F_t$, but E_1 is stable of the same slope as F_t which is a priori semistable. This forces the homomorphism to be an injection

$E_1 \hookrightarrow F_t$. But note that E_1 and F_t have the same Hilbert polynomial. This implies that we must have $E_1 \cong F_t$, which completes our base case.

Now suppose the claim holds for n. We show it holds for $n+1$. So for all $t \in T_0$, we have $\mathrm{gr}(F_t) \cong \bigoplus_{j=1}^{n+1} E_j$. For $1 \leq i \leq n+1$, define $T_i = \{t \in T \mid \mathrm{Hom}_{\mathcal{O}_C}(E_i, F_t) \neq 0\}$. Then note that

$$T = (T \setminus T_0) \cup T_1 \cup \cdots \cup T_{n+1}.$$

We have this covering since $T_1 \cup \cdots \cup T_{n+1}$ cover T_0. Since T is irreducible and each of these T_i are closed, we must have $T_i = T$ for some i. Then for all $t \in T$, we have there exists some nontrivial $E_i \to F_t$. Again, since E_i is stable of the same slope as F_t, these morphisms must be injections $E_i \hookrightarrow F_t$. Now if we consider the family $\pi_C^* E_i^\vee \otimes F$ on $T \times C$, parameterized over T, we can pushforward by $\pi_{T,*}$ to obtain the quasi-coherent sheaf $\pi_{T,*}(\pi_C^* E_i^\vee \otimes F)$, and note that the stalk of $\pi_{T,*}(\pi_C^* E_i^\vee \otimes F)$ is nonzero for every $t \in T$ since $\mathrm{Hom}_{\mathcal{O}_C}(E_i, F_t) \neq 0$.

Suppose $T_0 \neq T$. Then pick some closed point $z \in T \setminus T_0$. Since T is affine and $\pi_{T,*}(\pi_C^* E_i^\vee \otimes F)$ is quasi-coherent and its stalk at every point is nonzero, we can pick some global section $s \in H^0(T, \pi_{T,*}(\pi_C^* E_i^\vee \otimes F))$ such that $s(z) \neq 0$. Let U denote an affine open where s is invertible and note U contains z. Then s gives a family of injections $E_i \hookrightarrow F_t$ for $t \in U$. Then over U, we get a short exact sequence

$$0 \to \pi_C^* E_i \xrightarrow{s} F \to Q \to 0$$

where Q is defined to be the quotient. Note by irreducibility, $U \cap T_0 \neq \emptyset$, and for $t \in U \cap T_0$ we have $\mathrm{gr}(F_t) \cong \bigoplus_{j=1}^{n+1} E_j$. The short exact sequence then implies that $\mathrm{gr}(Q_t) \cong \bigoplus_{j \neq i} E_j$ for $t \in U \cap T_0$. Then by the inductive hypothesis, we have $\mathrm{gr}(Q_t) \cong \bigoplus_{j \neq i} E_j$ for all $t \in U$. The short exact sequence then implies that $\mathrm{gr}(F_t) \cong \bigoplus_{j=1}^{n+1} E_j$ for all $t \in U$, including $z \in T \setminus T_0$. This implies that actually we must have $T_0 = T$. □

By lemma 10.5, we see then that semistable vector bundles in a flat family having the same associated gradeds is actually a closed condition, as we can cover the parameter base by affine opens.

Proposition 10.6. *The $\mathbb{C}$-points of $U_C(r,d)$ are S-equivalence classes of semistable vector bundles.*

Proof. By Theorem 10.4, since $U_c(r,d) := R^{ss} /\!\!/ \mathrm{GL}_V$, every $\mathbb{C}$-point of $U_c(r,d)$ will correspond to some semistable vector bundle. The task at hand here is to show $[E] = [E'] \in U_C(r,d)(\mathbb{C})$ if and only if $\mathrm{gr}(E) \cong \mathrm{gr}(E')$.

First, suppose $\mathrm{gr}(E) \cong \mathrm{gr}(E') \cong \bigoplus_{j=1}^{n} E_j$. Note it suffices to show that $[E] = [\bigoplus_{j=1}^{n} E_j] \in U_C(r,d)(\mathbb{C})$. If $n = 1$, this is trivially true. Let's take a look at when $n = 2$. This is something we've seen before. If $\mathrm{gr}(E) = E_1 \oplus E_2$, then there is some WLOG short exact sequence

$$0 \to E_1 \to E \to E_2 \to 0,$$

so we can look at the nonzero vector space $\mathrm{Ext}^1(E_2, E_1)$ and take the affine line $\mathbb{A}^1_t$ corresponding to the above short exact sequence. This will give us a flat family of vector bundles over $\mathbb{A}^1$, such that the nonzero fibers give E and the central fiber gives $E_1 \oplus E_2$. For reasons we've discussed before, this jumping behavior forces $[E] = [E_1 \oplus E_2] \in U_C(r, d)(\mathbb{C})$. Now consider the case of when $n = 3$, so $\mathrm{gr}(E) = E_1 \oplus E_2 \oplus E_3$. Suppose we have a Jordan-Holder filtration

$$0 \subset E_1 \subset E' \subset E$$

such that $E'/E_1 \cong E_2$ and $E/E' \cong E_3$. So we have short exact sequences

$$0 \to E_1 \to E' \to E_2 \to 0$$

$$0 \to E' \to E \to E_3 \to 0.$$

Again, taking the affine line in $\mathrm{Ext}^1(E_3, E')$ corresponding to the latter short exact sequence gives us a flat family of vector bundles parameterized over $\mathbb{A}^1_t$ whose nonzero fiber is E and whose central fiber is $E' \oplus E_3$. This implies that $[E] = [E' \oplus E_3] \in U_C(r, d)(\mathbb{C})$. Taking the affine line in $\mathrm{Ext}^1(E_2, E_1)$ corresponding to the former short exact sequence gives us a flat family χ, a vector bundle over $\mathbb{A}^1 \times C$, parameterized over $\mathbb{A}^1_t$ such that $\chi_t \cong E'$ but $\chi_0 \cong E_1 \oplus E_2$. Then we can consider the flat family $\chi \oplus \pi_C^* E_3$, so that $(\chi \oplus \pi_C^* E_3)_t \cong E' \oplus E_3$ for $t \neq 0$, but $(\chi \oplus \pi_C^* E_3)_0 \cong E_1 \oplus E_2 \oplus E_3$. This family then implies that

$$[E_1 \oplus E_2 \oplus E_3] = [E' \oplus E_3] = [E] \in U_C(r, d)(\mathbb{C}).$$

The proof of this direction of the claim holds for higher n by an analogous argument.

Finally, suppose that $[E] = [E'] \in U_C(r, d)$. If we pick some lifts $q : V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E$ and $q' : V \otimes \mathcal{O}_C(-np) \twoheadrightarrow E'$, then by GIT, $[E] = [E']$ implies

$$\overline{\mathrm{GL}_V \cdot q} \cap \overline{\mathrm{GL}_V \cdot q'} \neq \emptyset.$$

Suppose z is in the intersection, and take an affine open U of z. Then $U \cap \overline{\mathrm{GL}_V \cdot q} \neq 0$ and $U \cap \overline{\mathrm{GL}_V \cdot q'}$ are nonempty closed affine subschemes of U.

Note that there is an open loci of $t \in U \cap \overline{\mathrm{GL}_V \cdot q}$ such that the associated graded is simply $\mathrm{gr}(E)$. Then by lemma 10.5, we have for every $t \in U \cap \overline{\mathrm{GL}_V \cdot q}$, the associated gradeds are all isomorphic to $\mathrm{gr}(E)$, including over z. By the same argument applied to $U \cap \overline{\mathrm{GL}_V \cdot q'}$, we have the the associated graded over z is also isomorphic to $\mathrm{gr}(E')$. This implies $\mathrm{gr}(E) \cong \mathrm{gr}(E')$, as desired. □

As a corollary, the $\mathbb{C}$-points of $U_C^s(r, d)$ are exactly the stable vector bundles of rank r and degree d on C.

11. MVC 2/6/25: Higgs bundles

Definition 11.1. *A Higgs bundle on C over $\mathbb{C}$ is a pair (E, ϕ) where E is a vector bundle on C and $\phi \in \mathrm{Hom}(E, E \otimes \omega_C)$. Here ϕ is called the Higgs field.*

A morphism of Higgs bundles $\alpha : (E, \phi) \to (F, \psi)$ is a morphism of $\mathcal{O}_C$-modules $\alpha : E \to F$ such that the diagram

$$\begin{tikzcd} E \arrow[r, "\phi"] \arrow[d, "\alpha"'] & E \otimes \omega_C \arrow[d, "\alpha\otimes\mathrm{id}"] \\ F \arrow[r] & F \otimes \omega_C \end{tikzcd}$$

Definition 11.2. *We say (E, ϕ) is slope-semistable if for all ϕ-invariant subsheaves $F \subseteq E$ we have $\mu(F) \leq \mu(E)$. We say it is stable if it is strict.*

Definition 11.3. *Given $(r, d) \in \mathbb{N}\times\mathbb{Z}$, we consider the moduli functor of semistable Higgs bundles of fixed rank r andd egree d on the C to be:*

$$\underline{Higgs}^{ss}_C(r, d) : (Sch_{\mathbb{C}})^{op} \to Sets$$

where $T \mapsto \{(\mathcal{E}, \Phi) \mid \mathcal{E}$ v.b. on $C; \Phi : \mathcal{E} \to \mathcal{E} \otimes \pi_C^* \omega_C$ s.t. $(\mathcal{E}, \Phi)|_{\{t\}\times C}$ ss rank r deg $d\}/\sim$, where the equivalent relation is that $(\mathcal{E}, \Phi) \sim (\mathcal{E}, \Phi')$ if and only if there exists $N \in \mathrm{Pic}(T)$ such that $(\mathcal{E}', \Phi') \cong (\mathcal{E} \otimes \pi_T^* N, \Phi \otimes id_{\pi_T^* N})$.

We also define the subfunctor $\underline{Higgs}^{s}_C(r, d) \subseteq \underline{Higgs}^{ss}_C(r, d)$ parameterizing only stable Higgs bundles.

There are examples where (E, ϕ) is a stable pair, but the underlying E is not slope-stable.

Example 11.4. *Let C be smooth curve of genus $g \geq 2$. Let $\theta \in \mathrm{Pic}^{g-1}(C)$ such that $\theta^2 \cong \omega_C$. Consider the vector bundle $E := \theta \oplus \theta^{-1}$. This is unstable as a bundle. But you can put a Higgs field on it so that it becomes stable as a pair. A Higgs field on E is an element $\phi \in \mathrm{Hom}((\theta \oplus \theta^{-1}), (\theta \oplus \theta^{-1}) \otimes \omega_C)$. So a Higgs field on E is an element of the form $\begin{pmatrix} 0 & s \\ 1 & 0 \end{pmatrix}$ where $s \in H^0(\omega_C^2)$. One can see that (E, ϕ_s) is stable for all $s \in H^0(\omega_C^2)$.*

As in the case of vector bundles, we have a Harder-Narasimhan filtration for Higgs bundles. And also a Jordan-Holder filtration for semistable Higgs bundles. This leads to the defintion of the associated graded:

$$\mathrm{gr}(E, \phi) := \oplus_{i=1}^k (A_i, \phi_i),$$

where (A_i, ϕ_i) are stable Higgs bundles.

Theorem 11.5. *Let C be a smooth curve over $\mathbb{C}$ of genus $g \geq 2$, and $(r, d) \in \mathbb{N} \times \mathbb{Z}$. There exists a normal integral quasi-projetive scheme $M_C(r, d)$ together with a natural transformation*

$$\beta_i : \underline{Higgs}^{ss}_C(r, d) \to M_C(r, d)$$

such that

(1) *The pair $(M_C(r, d), \beta)$ co-represents the functor $\underline{Higgs}^{ss}_C(r, d)$;*

(2) *The closed points of* $M_C(r,d)$ *are the equivalence classes of semistable Higgs bundles under the equivalence relation given by* $(E_1,\phi_1) \sim (E_2,\phi_2) \iff gr(E_1,\phi_1) \cong gr(E_2,\phi_2)$.

(3) *There exists a smooth open subset* $M_C^s(r,d) \subseteq M_C(r,d)$ *such that* $(M_C^s(r,d),\beta)$ *is a coarse moduli space for* $\underline{Higgs}_C^s(r,d)$.

(4) *When* $\gcd(r,d) = 1$*, then* $(M_C^s(r,d),\beta)$ *is a fine moduli space for* $\underline{Higgs}_C^{ss}(r,d) = \underline{Higgs}_C^s(r,d)$.

If E itself is stable, then (E,ϕ) is a stable Higgs bundle for any Higgs field $\phi \in \mathrm{Hom}(E, E\otimes \omega_C)$. And note $\mathrm{Hom}(E, E\otimes\omega_C) = Ext^1(E,E) = T_E^* U_C^s(r,d)$. This shows that at least on closed points we have a map

$$T^*U_C^s(r,d)(\mathbb{C}) \to M_C^s(r,d)(\mathbb{C}).$$

Theorem 11.6. *There is an open immersion*

$$T^*U_C^s(r,d) \hookrightarrow M_C^s(r,d).$$

In particular, $\dim M_C^s(r,d) = r^2(2g-2)+2$.

Proof. We'll sketch the proof here. Let us assume that $\gcd(r,d)=1$, and $U := U_C^s(r,d)$ and $M := M_C^s(r,d)$ are smooth fine moduli space. Let $\mathcal{E} \in \underline{\mathrm{Bun}}_C^s(r,d)(U)$ be a universal family. Then we have isomorphisms of $\mathcal{O}_U$ modules:

$$T_U \cong R^1\pi_{U,*}\mathcal{E}\backslash(\mathcal{E}), T_U^* \cong \pi_{U,*}(\mathcal{E}\backslash(\mathcal{E}) \otimes \pi_C^*\omega_C),$$

where the latter is by relative Serre duality. In order to produce a morphism $T^*U \to M$, it is enough to produce a family of Higgs bundles parameterized by T^*U i.e. an element in $\underline{\mathrm{Higgs}}_C^S(r,d)(T^*U)$.

Consider the projection $C\times T^*U \xrightarrow{\mathrm{id}_C\times T} C\times U \xrightarrow{\pi_U} U$. Consider the vector bundle $(id_C\times\pi)^*\mathcal{E}$ on $C\times T^*U$. We want a Higgs field on it, i.e. an element of

$$H^0(C\times T^*U, \mathcal{E}\backslash((\mathrm{id}_C\times\pi)^*\mathcal{E}\otimes\pi_C^*\omega_C) \cong H^0(C\times T^*U, (\mathrm{id}_C\times\pi)^*(\mathcal{E}\backslash(\mathcal{E})\otimes\pi_C^*\omega_C)).$$

$$\cong H^0(C\times U, (\mathrm{id}_C\times\pi)_*\mathcal{O}_{C\times T^*U}\otimes(\mathcal{E}\backslash(\mathcal{E})\otimes\pi_C^*\omega_C)).$$

$$\cong H^0(C\times U, \pi_U^*\pi_*\mathcal{O}_{T^*U}\otimes(\mathcal{E}\backslash(\mathcal{E})\otimes\pi_C^*\omega_C)) \cong H^0(U, \pi_*\mathcal{O}_{T^*U}\otimes\pi_{U,*}(\mathcal{E}\backslash(\mathcal{E})\otimes\pi_C^*\omega_C))$$

$$\cong H^0(U, \mathrm{Sym}^* T_U\otimes T_U^*).$$

Here the maps we are using fit in the diagram

$$\begin{tikzcd} C\times T^*U \arrow[r] \arrow[d] & T^*U \arrow[d, "\pi"] \\ C\times U \arrow[r, "\pi_U"'] & U \end{tikzcd}$$

Canonical section in $H^0(U, \mathrm{Sym}^* T_U\otimes T_U^*)$, check that it defines a morphism $T^*U \to M$ which on closed points is given by the one above.

Now to show it is an open immersion. Injectivity on closed points follows from the fact that we're dealing with stable sheaves. In order to conclude we want to show it is etale. Suffices to show that we can fill in the following diagram:

$$\begin{tikzcd} \operatorname{Spec}\mathbb{C} \arrow[r] \arrow[d] & T^*U \arrow[d] \\ \operatorname{Spec}\mathbb{C}[\epsilon] \arrow[r] \arrow[ur, dashed, "\exists!"] & U \end{tikzcd}$$

The data given by such a diagram are: a closed point $(E,\phi) \in T^*U(\mathbb{C})$ i.e. a vector bundle E stable on C and an element $\phi \in \operatorname{Ext}^1(E,E)^\vee \cong \operatorname{Hom}(E, E\otimes\omega_C)$. A family $(E_\epsilon,\phi_\epsilon) \in \underline{\mathrm{Higgs}}^s_C(r,d)(\operatorname{Spec}\mathbb{C}[\epsilon])$ such that $(E_\epsilon,\phi_\epsilon)|_C \cong (E,\phi)$. And $E_\epsilon \in \underline{\mathrm{Bun}}^s_C(r,d)(\operatorname{Spec}\mathbb{C}[\epsilon]) = U(\operatorname{Spec}\mathbb{C}[\epsilon])$. Classifying morphism $f_{E_\epsilon} : \operatorname{Spec}\mathbb{C}[\epsilon] \to U$ such that $E_\epsilon \cong (f^*_{E_\epsilon} \times id_C)^*\mathcal{E}$.

Note the T-points of cotangent bundle $T^*U(T) = \{(g,s) \mid g : T \to U, s \in H^0(T, g^*T^*U))\}$

We want to show that $(E_\epsilon,\phi_\epsilon)$ defines a $\operatorname{Spec}\mathbb{C}[\epsilon]$ valued point of T^*U. We have $\operatorname{Hom}(E_\epsilon, E_\epsilon\otimes\pi_C^*\omega_C) \cong H^0(C[\epsilon], \mathcal{E}\backslash(E_\epsilon)\otimes\pi_C^*\omega_C)$, where $C[\epsilon] = \mathbb{C}[\epsilon]\times C$. So this equals

$$H^0(C[\epsilon], (\mathrm{id}_C\times f_{E_\epsilon})^*(\mathcal{E}\backslash(E)\otimes\pi_C^*\omega_C)) = H^0(\operatorname{Spec}\mathbb{C}[\epsilon], f^*_{E_\epsilon} f_{U,*}(\mathcal{E}\backslash(\mathcal{E})\otimes\pi_C^*\omega_C)).$$

See open immersion from this. □

Example 11.7. *A rank one Higgs bundle on C is (L,ϕ), but this is just a line bundle with a global section of ω_C. Independent of L. So this tells you that*

$$M_C(1,d) \cong \operatorname{Pic}^d(C)\times H^0(C,\omega_C) \cong T^*\operatorname{Pic}^d(C).$$

There's something to mention here called a spectral correspondence. The toy model here is Higgs bundles on a point. This is pairs (V,ϕ) where V is a finite dimensional vector space and $\phi \in End(V)$. This underlies a $\mathbb{C}[\lambda]$-module structure on V, where the action of λ is given by ϕ. There is a ring map here $\mathbb{C}[\lambda] \to End(V)$. This is a torsion module, supported at the eigenvalues of ϕ. And if you want to be more precise, we have a free resolution of V as a $\mathbb{C}[\lambda]$-module:

$$V\otimes\mathbb{C}[\lambda] \xrightarrow{\phi\otimes id - id_V\otimes\lambda} V\otimes\mathbb{C}[\lambda] \to V \to 0.$$

The fiber at each eigenvalue $\lambda_0 \in \mathbb{C}$ of V is given by $V\otimes\frac{\mathbb{C}[\lambda]}{(\lambda-\lambda_0)} \cong \operatorname{coker}(\phi - \lambda_0 id)$, which is the λ_0 eigenspace of the dual map. Vice versa, given a finitely generated torsion $\mathbb{C}[\lambda]$-module M, the action of λ defines an endomorphism of the underlying finite dimensional vector space M. More geometrically, seeing M as a torsion sheaf on $\mathbb{A}^1_\lambda$, we have that $\pi_*M = (V,\phi)$ where $\phi \in End(V)$, where $\pi : \mathbb{A}^1_\lambda \to \operatorname{Spec}\mathbb{C}$.

Example 11.8. *Take the structure sheaf of fat point. Have $\mathbb{C}[t]/t^2$ as a $\mathbb{C}[\lambda]$-module. Then look at $\mathbb{C}[\lambda]/\lambda^2 \cong_{\mathbb{C}} \mathbb{C}\langle 1\rangle \oplus \mathbb{C}\langle\lambda\rangle$. The associated Higgs bundle is $(\mathbb{C}^2, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix})$.*

12. GMS: Coherent sheaves, torsion-free, reflexive, purity

Suppose X is a $\mathbb{C}$-variety (integral and finite type, can also just replace X with integral Noetherian schemes). The local models of coherent sheaves on X are sections of vector bundles and ideal sheaves of $\mathbb{C}$-subvarieties of X. Another simple but useful way of motivating coherent sheaves is the following: there is a functor from the category of holomorphic vector bundles on X (really we mean on the analytic space that is comprised of the $\mathbb{C}$-rational points) to locally free sheaves on X. The issue however is that not every morphism of locally free sheaves on X comes from a morphism of vector bundles. Locally, such a morphism of locally free sheaves is a matrix of local functions. But this matrix need not have locally constant rank. However, if one locally trivializes a morphism of vector bundles, the morphism is required to be locally constant rank, so that the quotient is again a vector bundle.

Then the category of coherent sheaves on X is the smallest enlargement of the category of locally free sheaves, so that every morphism has a cokernel. Note also that any kernel of a morphism of locally free sheaves will be coherent since we are working in a Noetherian situation.

Fix $\mathcal{F} \in Coh(X)$. An important aspect of understanding coherent sheaves is the notion of torsion and torsion-freeness. This is related to the notion of the support of a section.

Suppose $s \in \mathcal{F}$ is a torsion section. Then if $\mathcal{F}|_{\mathrm{Spec}(A)} = \widetilde{M}$, then viewing s as an element of M, there is an element $f \in A$ such that $f \cdot s = 0$. In other words, the support of s is contained by the vanishing loci $V(f) = \mathrm{Spec}(A/f) \subsetneq \mathrm{Spec}(A)$. We can also go in the reverse direction. Suppose $\mathrm{Spec}(A/I) \subsetneq \mathrm{Spec}(A)$ is a closed subscheme, and the support of $s \in M$ is contained by $\mathrm{Spec}(A/I)$. Let $f \in \sqrt{I}$. Then f^n vanishes on this closed subscheme for some $n \geq 1$. Note $\mathrm{Spec}(A/f^n) \supseteq \mathrm{Spec}(A/I)$. Then we claim $f^n \cdot s = 0$. Why? Because $f^n \cdot s$ vanishes over $\mathrm{Spec}(A/f^n)$, and it also vanishes over $\mathrm{Spec}(A) \setminus \mathrm{Spec}(A/I)$.

Thus, torsion of a section is intimately related to being supported on a proper subvariety.

Definition 12.1. *Let $\mathcal{F} \in Coh(X)$. Then we define $\mathcal{F}$ to be a torsion sheaf if $\mathcal{F}_p$ is a torsion $\mathcal{O}_{X,p}$-module for every $p \in X$.*

Proposition 12.2. *Suppose $\mathcal{F} \in Coh(X)$. Then $\mathcal{F}$ is a torsion coherent sheaf if and only if $\mathcal{F}$ is supported on a proper subvariety of X.*

Proof. Forward direction: let $\mathrm{Spec}(A) \subseteq X$ be an open and $\mathcal{F}|_{\mathrm{Spec}(A)} \cong \widetilde{M}$. Then we can pick generators $s_1, \cdots, s_n$ for M. We have $M \otimes_A k(A) = 0$, so there will exist elements $f_1, \cdots, f_n \in A$ such that $f_i \cdot s_i = 0$. Then letting $f = \prod_{i=1}^n f_i$, we find $f \cdot s_i = 0$ for $1 \leq i \leq n$, so the support of M is contained by $\mathrm{Spec}(A/f)$. □

Equivalently, $\mathcal{F}$ is torsion if and only if $\mathcal{F}_\eta = 0$ where η is the generic point of X. So the stalk at the generic point tests whether $\mathcal{F}$ is torsion. Suppose $\mathcal{F}_\eta \neq 0$. Locally, this means

$M \otimes_A K(A) \neq 0$ where $k(A)$ is the field of fractions of A. Then $M \otimes_A K(A)$ kills off torsion, and will be a $k(A)$-vector space of dimension r. Note if we tensor the map $A^r \to M$ by $- \otimes_A K(A)$, we obtain $K(A)^r \to K(A)^r$. This implies $A^r \to M$ is an injection. Then we obtain a short exact sequence

$$0 \to A^r \to M \to T \to 0,$$

and tensoring by $- \otimes_A K(A)$, we see that T is torsion. Since T is torsion, it is supported on a proper subvariety. This local computation then demonstrates that any coherent $\mathcal{F}$ such that $\mathcal{F}_\eta \neq 0$ (i.e. it is not torsion) will be locally free on some open subset of X.

Definition 12.3. *Let X be a complete nonsingular $\mathbb{C}$-scheme. Let $\mathcal{F} \in Coh(X)$. We say $\mathcal{F}$ is torsion-free if $\mathcal{F}_p$ is a torsion-free $\mathcal{O}_{X,p}$-module for every $p \in X$. Equivalently, if the multiplication map $- \cdot s : \mathcal{F}_p \to \mathcal{F}_p$ is injective for every $s \in \mathcal{O}_{X,p} \setminus \{0\}$.*

Proposition 12.4. *Let (R, m) be a regular local Noetherian ring and let M be a finitely generated R-module. Let M_{tors} denote the sub R-module of torsion elements. TFAE:*

(1) *$M_{tors} = 0$, i.e. M is torsion free.*
(2) *$M \hookrightarrow R^n$ for some n*
(3) *$M \hookrightarrow R^n$ where $n = \mathrm{rk}(M) := \dim_{K(R)} M \otimes_R K(R)$.*

Proof. Clearly (3) $\implies$ (2) $\implies$ (1). So it remains to prove (1) $\implies$ (3). Since M is torsion free, we have an injection

$$M \hookrightarrow M \otimes_R M(R) \cong K(R)^n.$$

Let $e_1, \cdots, e_n$ be a basis for $K(R)^n$, and let $g_1, \cdots, g_t$ be generators for M as a R-module. Then express the image of the g_i under the embedding in terms of the e_i:

$$g_k = \sum_{i=1}^{n} c_{ki} e_i.$$

These $c_{ki} \in K(R)$. Then there exist an element $r \in R$ which clears the denominators of all c_{ki}. Since the endomorphism of M given by multiplication by r is injective, the composition

$$M \to^{\cdot r} M \hookrightarrow M \otimes_R K(R)$$

gives us an injection $M \hookrightarrow R^n$. □

Proposition 12.5. *Let R, M have the assumed hypotheses. Then*

$$M_{tors} = \ker(M \to M^{\vee\vee}).$$

Proof. First we show

$$M_{\text{tors}} \subseteq \ker(M \to M^{\vee\vee}).$$

Let $x \in M_{\text{tors}}$, so that there exist $s \in R$ such that $s \cdot x = 0$. Then x maps to the element in $M^{\vee\vee}$ which R-linearly evaluates $\phi \in M^\vee$ via $\phi \mapsto \phi(x)$. But note $s \cdot m = 0$, which means the image of $s \cdot m$ is zero. Thus, $\phi(s \cdot m) = s \cdot \phi(m) = 0$ for every $\phi \in M^\vee$. But $s, \phi(m) \in R$, which is a regular local ring and thus an integral domain, so this forces $\phi(m) = 0$. Thus, $m \in \ker(M \to M^{\vee\vee})$.

To see that we actually have equality, first consider the following. Suppose $x \in M_{\text{tors}}$ as before. Note $M^\vee := \text{Hom}_R(M, R)$. For any $\phi \in M^\vee$, we have $\phi(s \cdot x) = 0 \implies s \cdot \phi(x) = 0 \implies \phi(x) = 0$. This means that $\text{Hom}_R(M, R) \cong \text{Hom}_R(M/M_{\text{tors}}, R)$. This implies that we have equality $M_{\text{tors}} = \ker(M \to M^{\vee\vee})$ if and only if

$$M/M_{\text{tors}} \hookrightarrow (M/M_{\text{tors}})^{\vee\vee}$$

is an injection. So assume that M is torsion free. Then $M \hookrightarrow R^n$ for some n. The following diagram

$$\begin{tikzcd} M \arrow[r] \arrow[d, hook] & M^{\vee\vee} \arrow[d, hook] \\ R^n \arrow[r, "="'] & (R^n)^{\vee\vee} \cong R^n \end{tikzcd}$$

implies that indeed injectivity holds. So we have the desired equality. □

Theorem 12.6. *Let X be a nonsingular $\mathbb{C}$-variety. Let $\mathcal{F} \in Coh(X)$ be a torsion-free sheaf. Then $\mathcal{F}$ is locally free outside of a closed subset of codimension ≥ 2.*

Recall that every coherent sheaf which is not torsion will be locally free on some open subset. Being torsion-free guarantees that the locus on which the coherent sheaf is not locally free is smaller than codimension 1. In particular, any torsion-free sheaf on a curve is locally free everywhere.

Definition 12.7. *Let R, M have the hypotheses as before. We say that M is reflexive if*

$$M \to M^{\vee\vee}$$

is an isomorphism.

Proposition 12.8. *TFAE:*

(1) *M is reflexive*
(2) *There exists a finite R-module N such that $M \cong N^\vee$*
(3) *M is torsion free and for all ideals I of R such that $\operatorname{codim} V(I) \geq 2$, the natural map $H^0(\operatorname{Spec} R, \tilde{M}) \to H^0(\operatorname{Spec} R \setminus V(I), \tilde{M}|_{\operatorname{Spec}(R) \setminus V(I)})$ is a natural isomorphism.*

Definition 12.9. *Let X be a nonsingular $\mathbb{C}$-variety. Then $\mathcal{F} \in Coh(X)$ is reflexive if*

$$\mathcal{F} \to \mathcal{F}^{\vee\vee}$$

is an isomorphism.

Theorem 12.10. *Suppose $\mathcal{F}$ is a reflexive sheaf. Then $\mathcal{F}$ is locally free outside of a closed subset of codimension ≥ 3.*

In particular, any reflexive sheaf on a surface is locally free everywhere.

Corollary 12.11. *Let X be a nonsingular $\mathbb{C}$-variety. A rank 1 reflexive sheaf $\mathcal{F}$ is locally free everywhere.*

Proof. Choose some $\mathrm{Spec}(A)$ and $\tilde{M} \cong \mathcal{F}|_{\mathrm{Spec}(A)}$. We have there is some $I \subseteq A$ such that $\mathrm{codim}\, V(I) \geq 2$ and $\tilde{M}|_{\mathrm{Spec}(A)\setminus V(I)}$ is locally free, so is a line bundle. Note $\mathrm{Pic}(\mathrm{Spec}(A)) = \mathrm{Pic}(\mathrm{Spec}(A) \setminus V(I)) = 0$ by Hartog's, so $\tilde{M}|_{\mathrm{Spec}(A)\setminus V(I)} \cong \mathcal{O}_{\mathrm{Spec}(A)\setminus V(I)}$. Since M is reflexive, we have $M = A$. □

Proposition 12.12. *Let X be a smooth $\mathbb{C}$-variety and $\mathcal{F} \in Coh(X)$. Then $\mathcal{F}^\vee$ is reflexive.*

Proof. First, we see the stalks $\mathcal{F}_p^\vee \cong \mathrm{Hom}_{\mathcal{O}_{X,p}}(\mathcal{F}_p, \mathcal{O}_{X,p})$ are torsion-free $\mathcal{O}_{X,p}$-modules. So $\mathcal{F}^\vee$ is torsion-free. Then

$$\mathcal{F}^\vee \hookrightarrow (\mathcal{F}^\vee)^{\vee\vee}$$

is an injection. We claim that it is actually an isomorphism. First, let

$$\mathcal{E}^1 \to \mathcal{E}^0 \to \mathcal{F} \to 0$$

be the first few terms of a (finite) resolution of $\mathcal{F}$ by locally free sheaves. Then we can dualize to obtain

$$0 \to \mathcal{F}^\vee \to (\mathcal{E}^0)^\vee \to N \to 0,$$

where N is the dual of $\ker(\mathcal{E}^0 \to \mathcal{F})$. Note N is torsion free and $(\mathcal{E}^0)^\vee$ is reflexive since it is locally free. Thus, by [Sta25, Lemma 0EB8], $\mathcal{F}^\vee$ is reflexive. □

Corollary 12.13. *Let $\mathcal{F}$ be a coherent sheaf on a smooth surface X. Then $\mathcal{F}^\vee$ is locally free.*

From now on, let X be a Noetherian scheme. Note for $\mathcal{F} \in Coh(X)$, the support $\mathrm{Supp}(\mathcal{F}) \subset X$ is closed.

Definition 12.14. *Let X be a Noetherian scheme and $\mathcal{F} \in Coh(X)$. We say $\mathcal{F}$ is pure of dimension d if for every coherent subsheaf $E \subset F$ we have $\dim E = d$.*

Purity generalizes torsion-free. If X is an integral scheme, $\mathcal{F} \in Coh(X)$ is torsion-free if and only if $\mathcal{F}$ is pure of dimension $\dim(X)$. Purity in general for a coherent sheaf on an integral scheme is torsion-freeness on the support.

13. GMS: Semistability, Harder-Narasimhan filtration, Jordan-Holder filtration

Recall the following motivation: if X is a complete nonsingular projective $\mathbb{C}$-variety, then using its exponential exact sequence, we can non-canonically identify the set $\mathrm{Pic}^{c_1}(X)$ of line bundles on X with fixed Chern class with $\frac{H^1(X,\mathcal{O}_X)}{H^1(X,\mathbb{Z})}$. In particular, this is a torus and is the fine moduli space with the universal family of line bundles of X with Chern class

c_1. We want to reproduce such a moduli space for higher rank vector bundles of some fixed Chern class information. However, in order to get a separated finite type scheme structure on such a moduli space, we have to restrict to semistable bundles. Then, in order to compactify the moduli space, we have to include semistable coherent sheaves. Thus, in this section, let me discuss the notions of semistability. We'll introduce in this section Gieseker semi-stability and μ-semistability, how they're related, and the Harder-Narasimhan filtration. We begin with Gieseker semi-stability.

Let X be a Noetherian projective scheme. Choose a polarization $H \in |\mathcal{O}_X(1)|$. Semistability is always defined with respect to a very ample divisor, the choice of which is important – a semistable coherent sheaf may no longer be semistable with respect to another choice of H. The Hilbert polynomial of $E \in Coh(X)$ may be written as

$$P(E, m) = \sum_{i=0}^{\dim E} \alpha_i(E) \frac{m^i}{i!},$$

where $\dim E$ is the dimension of the support of E, $\alpha_i(E) \in \mathbb{Z}$ for every i, and $\alpha_{\dim E} > 0$ [HL, Lemma 1.2.1]. We define the reduced Hilbert polynomial of E to be

$$\rho(E) := \frac{P(E, m)}{\alpha_{\dim E}(E)}.$$

Note there is a natural lexicographic order on polynomials via comparison of coefficients.

Definition 13.1. *Let $E \in Coh(X)$. Then E is Geiseker (semi)stable if E is pure and for any nonzero proper $F \subset E$,*

$$\rho(F)(\le)\rho(E).$$

Alternatively, we could equivalently say E is semistable if for any nonzero proper $F \subset E$ we have $\alpha_d(E)P(F)(\le)\alpha_d(F)P(E)$. This implicitly forces E to be pure, because if $\alpha_d(F) = 0$, then we must have $P(F) = 0$, so F is the zero sheaf. Here are some useful ways to think about Gieseker (semi)stability.

Proposition 13.2. *Let $E \in Coh(X)$ be pure of dimension d. TFAE:*

1. *E is Gieseker (semi)stable*
2. *For every proper saturated subsheaf $F \subset E$, we have $\rho(F)(\le)\rho(E)$*
3. *For every proper quotient $E \to Q$ such that $\alpha_d(Q) > 0$, we have $\rho(E)(\le)\rho(Q)$.*
4. *For every proper quotient $E \to Q$ where Q is pure of dimension d, we have $\rho(E)(\le)\rho(Q)$.*

Proof. Before we go on, let me say what a saturated subsheaf is: $F \subset E$ is saturated if E/F is pure of dimension $d = \dim E$ or 0. For any $F \subset E$, its saturation F^{sat} is the kernel

$$E \to E/F \to \frac{E/F}{T_{d-1}(E/F)},$$

i.e. the saturation F^{sat} is the minimal coherent subsheaf of E containing F such that its quotient is pure of dimension $d = \dim E$, or it is zero. Something to note of the saturation is that $\alpha_d(F) = \alpha_d(F^{\text{sat}})$. This can be seen from the fact that $\alpha_d(F) \le \alpha_d(E)$. If we have equality, then this is obvious. But if $\alpha_d(F) < \alpha_d(E)$, then $\alpha_d(E/F) = \alpha_d(E) - \alpha_d(F) = \alpha_d(\frac{E/F}{T_{d-1}(E/F)}) = \alpha_d(E) - \alpha_d(F^{\text{sat}})$.

Now let's prove the proposition. It's clear that $1 \implies 2$ and $3 \implies 4$. Note given $0 \to F \to E \to Q \to 0$, we have $\alpha_d(F) + \alpha_d(Q) = \alpha_d(E)$ and $P(F) + P(Q) = P(E)$, so

$$\alpha_d(F)[\rho(F) - \rho(E)] = \alpha_d(Q)[\rho(E) - \rho(Q)].$$

Then clearly $4 \implies 2$.

Next, $2 \implies 1$ because given nontrivial proper $F \subset E$, we have $F \subset F^{\text{sat}} \subset E$ and if $F^{\text{sat}} = E$ then $\alpha_d(F) = \alpha_d(E)$ so $\rho(F) < \rho(E)$. Otherwise, $\rho(F^{\text{sat}})(\le)\rho(E)$ by assumption, and $P(F) \le P(F^{\text{sat}})$ but $\alpha_d(F) = \alpha_d(F^{\text{sat}})$ so $\rho(F) \le \rho(F^{\text{sat}})$ so $\rho(F)(\le)\rho(E)$, as desired.

Finally $1 \implies 3$ by using the above equation. □

Note that if $F, G \in Coh(X)$ are pure sheaves, then if we wanted nontrivial $F \to G$, we need $\dim F \le \dim G$. At least in the case of $\dim F = \dim G$ we can say something in regards to (semi)stability.

Proposition 13.3. *Let $F, G \in Coh(X)$ be semistable of dimension d. Let $\theta : F \to G$ be a morphism.*

1. *If $\rho(F) > \rho(G)$, then θ must be the zero map.*
2. *If $\rho(F) = \rho(G)$, and F is stable then θ must be injective. If G is stable, then θ must be surjective.*
3. *If $\rho(F) = \rho(G)$ and $\alpha_d(F) = \alpha_d(G)$, then if either F or G is stable, then θ is an isomorphism.*

Proof.

1. Suppose θ was not the zero map. Then there would be a nontrivial cokernel of $\ker\theta \to F$, and $F/\ker\theta \hookrightarrow G$ but F being semistable means $\rho(F/\ker\theta) \ge \rho(F) > \rho(G)$, contradicting G being semistable.
2. If F is stable, then if θ had a kernel then $F/\ker\theta \hookrightarrow G$ and $\rho(F/\ker\theta) > \rho(F) = \rho(G)$, contradicting semistability of G. So θ must be injective. If G is stable, then again $F/\ker\theta \hookrightarrow G$ but couldn't be equal to G, and $\rho(F/\ker\theta) \ge \rho(F) = \rho(G)$, contradicting stability of G.
3. In this case, $P(F) = P(G)$. If F is stable, then we have an injection $F \hookrightarrow G$ but equality of Hilbert polynomials imply $G/F = 0 \implies F \cong G$. If G is stable, we have surjection $F \to G$ but equality of Hilbert polynomial implies $\ker[F \to G] = 0 \implies F \cong G$.

□

Now we discuss μ-(semi)stability or slope (semi)stability.

Definition 13.4. *Let X be a Noetherian projective scheme. Let $E \in Coh(X)$ have dimension* $\dim E = \dim X$. *We define*

$$\mathrm{rk}(E) := \frac{\alpha_{\dim X}(E)}{\alpha_{\dim X}(\mathcal{O}_X)}$$

and

$$\deg E := \alpha_{\dim X-1}(E) - \mathrm{rk}(E)\alpha_{\dim X-1}(\mathcal{O}_X).$$

The slope of E is defined to be

$$\mu(E) := \frac{\deg E}{\mathrm{rk}\, E}.$$

In general, these quantities are rational numbers, including $\mathrm{rk}\, E$. When X is integral, then $\mathrm{rk}\, E$ equals $\mathrm{rk}\, E_\eta$, where η is the generic point of X, and thus is a positive integer. When X is a smooth projective variety, then the Hirzebruch-Riemann-Roch formula implies $\deg E = (c_1(E).H^{\dim X-1})$, where H is the very ample divisor that the Hilbert polynomial is written in terms of. One sees from this that changing the amble divisor H with, for example, nH, alters $\mu(E)$.

Definition 13.5. *Let X be a Noetherian projective scheme and let $E \in Coh(X)$ such that* $\dim E = \dim X$. *Then E is μ-(semi)stable if $T_{\dim X-2}(E) = T_{\dim X-1}(E)$ and for $F \subset E$ with* $0 < \mathrm{rk}\, F < \mathrm{rk}\, E$ *we have*

$$\mu(F)(\leq)\mu(E).$$

So a μ-(semi)stable sheaf is a coherent sheaf of maximal dimension whose torsion lives in codimension at least two, satisfying the slope inequality. Equivalently, a coherent sheaf of maximal dimension is μ-(semi)stable if for all $F \subset E$ with $\mathrm{rk}\, F < \mathrm{rk}\, E$ we have $\mathrm{rk}\, E \deg F (\leq) \mathrm{rk}\, F \deg E$. This forces the torsion living in codimension 2 aspect because if $\mathrm{rk}\, F = 0$, then we must have $\deg F = 0$. But $\deg F = 0$ means $\alpha_{\dim X-1}(F) = 0$.

Proposition 13.6. *Let X be a Noetherian projective scheme. Let $E \in Coh(X)$ be pure of maximal dimension. Then*

$$\mu - \textit{stable} \implies \textit{Gieseker stable} \implies \textit{Gieseker semistable} \implies \mu - \textit{semistable}.$$

Proof. Suppose $E \in Coh(X)$ is μ-stable. Then for $F \subset E$ with $\mathrm{rk}\, F < \mathrm{rk}\, E$ we have

$$\mathrm{rk}\, E \deg F < \mathrm{rk}\, F \deg E$$

$$\implies \frac{\alpha_{\dim X}(E)}{\alpha_{\dim X}(\mathcal{O}_X)}[\alpha_{\dim X-1}(F) - \frac{\alpha_{\dim X}(F)}{\alpha_{\dim X}(\mathcal{O}_X)}\alpha_{\dim X-1}(\mathcal{O}_X)] < \frac{\alpha_{\dim X}(F)}{\alpha_{\dim X}(\mathcal{O}_X)}[\alpha_{\dim X-1}(E) - \frac{\alpha_{\dim X}(E)}{\alpha_{\dim X}(\mathcal{O}_X)}\alpha_{\dim X-1}(\mathcal{O}_X)]$$

$$\implies \alpha_{\dim X}(E)\alpha_{\dim X-1}(F) < \alpha_{\dim X}(F)\alpha_{\dim X-1}(E) \implies \frac{\alpha_{\dim X-1}(F)}{\alpha_{\dim X}(F)} < \frac{\alpha_{\dim X-1}(E)}{\alpha_{\dim X}(E)}.$$

Then for $F \subset E$, if $\alpha_d(F) = \alpha_d(E)$ then $P(F) < P(E) \implies \rho(F) < \rho(E)$. Otherwise, $\alpha_d(F) < \alpha_d(E) \implies \mathrm{rk}\, F < \mathrm{rk}\, E$ so using the above inequality we see $\rho(F) < \rho(E)$. So μ-stable implies Gieseker stable.

The middle implication is obvious. For the last implication, suppose $F \subset E$ such that $\operatorname{rk} F < \operatorname{rk} E$, so $\alpha_{\dim X}(F) < \alpha_{\dim X}(E)$. Since E is pure, we cannot have $\alpha_{\dim X}(F) = 0$. We want to verify that

$$\mu(F) \leq \mu(E),$$

which is equivalent to showing that $\frac{\alpha_{\dim X-1}(F)}{\alpha_{\dim X}(F)} \leq \frac{\alpha_{\dim X-1}(E)}{\alpha_{\dim X}(E)}$. But since $\rho(F) \leq \rho(E)$ by assumption, this must be true. □

Remark 13.7. *μ-(semi)stability is calculationally more tractable. One reason is at least because it only involves the first two coefficients of the Hilbert polynomial. This makes calculating slope easier. Also for example, the dualizing a slope semistable sheaf is again slope semistable.*

Recall Grothendieck's vector bundle theorem: every vector bundle on $\mathbb{P}^n$ splits into a direct sum of line bundles. For example, we might have some vector bundle that splits into the direct sum

$$\mathcal{O}_{\mathbb{P}^n}(5) \oplus \mathcal{O}_{\mathbb{P}^n}(5) \oplus \mathcal{O}_{\mathbb{P}^n}(4) \oplus \mathcal{O}_{\mathbb{P}^n}(-1) \oplus \mathcal{O}_{\mathbb{P}^n}(-1) \oplus \mathcal{O}_{\mathbb{P}^n}(-1).$$

There is a natural filtration here, as with all such line bundle decompositinos, so that each graded quotient is semistabl:

$$0 \subset \mathcal{O}_{\mathbb{P}^n}(5)^{\oplus 2} \subset \mathcal{O}_{\mathbb{P}^n}(5)^{\oplus 2} \oplus \mathcal{O}_{\mathbb{P}^n}(4) \subset \mathcal{O}_{\mathbb{P}^n}(5)^{\oplus 2} \oplus \mathcal{O}_{\mathbb{P}^n}(4) \oplus \mathcal{O}_{\mathbb{P}^n}(-1)^{\oplus 3}$$

While vector bundles in general do not split, the Harder-Narasimhan filtration generalizes this filtration for pure coherent sheaves over any Noetherian projective scheme.

Theorem 13.8 (Harder-Narasimhan filtration)**.** *Let X be a Noetherian projective scheme. Let $E \in Coh(X)$ be pure of dimension d. Then there exists a filtration*

$$0 = HN_0(E) \subset HN_1(E) \subset \cdots \subset HN_\ell(E) = E$$

such that $\rho_i(E) = \rho(\frac{HN_i(E)}{HN_{i-1}(E)})$ are Gieseker semistable of dimension d and

$$\rho_{\max} := \rho_1 > \cdots > \rho_\ell =: \rho_{\min}.$$

The filtration exists uniquely, and is called the Harder-Narasimhan filtration.

To prove this, one needs the fact for every pure sheaf E, there exists $F \subset E$ such that for $G \subset E$ we have $\rho(F) \geq \rho(G)$ and if $\rho(F) = \rho(G)$ then $F \supset G$. This is called a maximal destablizing subsheaf, and is semistable and unique.

Every pure sheaf has a maximal destablizing subsheaf [HL, Lemma 1.3.5]. The way I like to think about this is: for example, every non-unit element in a ring is contained in a maximal ideal. Such a fundamental fact requires Zorn's lemma. Similarly, that every pure sheaf contains a maximal destablizing susbheaf is such a fundamental fact, that it is really a consequence of Zorn's lemma (and that X is Noetherian scheme [Sta25, Tag 01Y7]).

Let's show existence of the Harder-Narasimhan filtration. Let $E \in Coh(X)$ be pure of dimension d. Let $E_1 \subset E$ be its maximal destablizing subsheaf, which is semistable. Note if $\alpha_d(E_1) = \alpha_d(E)$, then combined with the fact that $\rho(E_1) \geq \rho(E)$ and $P(E_1) \leq P(E)$ we find $E_1 = E$. So in this case, we are done. Otherwise, $\alpha_d(E_1) < \alpha_d(E)$ and $\alpha_d(E/E_1) = \alpha_d(E) - \alpha_d(E_1)$. So the quantity α_d gives a decreasing positive sequence of integers until we land on a quotient which is semistable. So we can assume that E/E_1 admits a filtration

$$0 = F_0 \subset F_1 \subset \cdots \subset F_\ell = E/E_1,$$

such that $\rho(F_i/F_{i-1}) > \rho(F_{i+1}/F_i)$. Then letting E_{i+1} be the preimage of F_i under $E \to E/E_1$, we get a filtration

$$0 \subset E_1 \subset E_2 \subset \cdots \subset E_{\ell+1} = E.$$

For $j \geq 2$ we see $E_j/E_{j-1} = F_{j-1}/F_{j-2}$ which is semistable of dimension d and $\rho_2 > \cdots > \rho_{\ell+1}$ where $\rho_j = \rho(E_j/E_{j-1})$. So it remains to show $\rho(E_1) > \rho(E_2/E_1)$. But by definition of E_1 being maximal destablizing and E_2 containing E_1, we must have $\rho(E_2) < \rho(E_1)$. Then looking at $0 \to E_1 \to E_2 \to E_2/E_1 \to 0$ and the induced equation

$$\alpha_d(E_1)[\rho(E_1) - \rho(E_2)] = \alpha_d(E_2/E_1)[\rho(E_2) - \rho(E_2/E_1)],$$

we have $\rho(E_2) > \rho(E_2/E_1) \implies \rho(E_1) > \rho(E_2/E_1)$.

Finally we show uniqueness. Suppose we have two filtrations

$$0 \subset E_1 \subset \cdots \subset E_\ell = E \text{ and } 0 \subset E'_1 \subset \cdots \subset E'_t = E.$$

WLOG we can assume $\rho_{\max,E} \geq \rho_{\max,E'}$. Let j be the smallest index such that $E_1 \subset E'_j$. So $E_1 \hookrightarrow E'_j \to E'_j/E'_{j-1}$ is a nontrivial map. If $j \geq 2$, then $\rho(E_1) > \rho(E'_j/E'_{j-1})$, which makes it a zero map. So we must have $j = 1$, so $E_1 \hookrightarrow E'_1$. But then $\rho(E_1) \leq \rho(E'_1)$. So $\rho(E_1) = \rho(E'_1)$ so by symmetry $E'_1 \subset E_1 \implies E_1 = E'_1$. Then we can take E/E_1 and apply the same argument to E/E_1 and show that the filtrations agree at every term. Thus, the Harder-Narasimhan filtration exists uniquely.

Finally, we end this section with the Jordan-Holder filtration. Recall that for a vector bundle over $\mathbb{P}^n$, its Harder-Narasimhan filtration is such that each semistable graded bundle is a direct sum of isomorphic line bundles. In fact any semistable vector bundle over $\mathbb{P}^n$ is a direct sum of isomorphic line bundles. This means one can filter any semistable vector bundle over $\mathbb{P}^n$ so that the graded pieces are stable line bundles. The Jordan-Holder filtration generalizes this.

Proposition 13.9 (Jordan-Holder filtration)**.** *Let X be a Noetherian projective scheme and $F \in Coh(X)$ semistable of dimension d. Then there exists a filtration*

$$0 \subset F_1 \subset \cdots \subset F_\ell = F$$

such that for each $1 \leq i \leq \ell$, $\rho(F_i) = \rho(F)$ (and thus the F_i are semistable of dimension d) and each $\frac{F_i}{F_{i-1}}$ is stable with $\rho(\frac{F_i}{F_{i-1}}) = \rho(F)$.

Proof. Suppose F is such that there exist no nontrivial proper $F_1 \subset F$ with $\rho(F_1) = \rho(F)$. This implies F is stable, so we're done. Then suppose there does exist nontrivial proper $F_1 \subset F$ such that $\rho(F_1) = \rho(F)$. Clearly F_1 is also semistable of dimension d. Suppose there are no nontrivial F_2 such that $F_1 \subset F_2 \subset F$ and $\rho(F_2) = \rho(F)$. Then we claim F/F_1 is stable with $\rho(F/F_1) = \rho(F)$. Indeed, suppose we have $E \subset F/F_1$. Then there is a lift $F_1 \subset \widetilde{E} \subset F$ such that $\widetilde{E}/F_1 = E$. Then using $0 \to F_1 \to \widetilde{E} \to E \to 0$, we find

$$\alpha_d(F_1)[\rho(F_1) - \rho(\widetilde{E})] = \alpha_d(E)[\rho(\widetilde{E}) - \rho(E)].$$

Since $\rho(\widetilde{E}) < \rho(F_1) = \rho(F)$, we have $\rho(\widetilde{E}) > \rho(E)$. So $\rho(E) < \rho(F) = \rho(F/F_1)$. So we see F/F_1 is semistable. And note $\rho(F/F_1) = \rho(F)$ because using $0 \to F_1 \to F \to F/F_1 \to 0$, we find $\rho(F_1) = \rho(F)$ implies $\alpha_d(F/F_1) > 0$ since $P(F_1) < P(F)$ but $\rho(F_1) = \rho(F)$, so $\alpha_d(F/F_1) > 0$ and $\rho(F_1) = \rho(F) \implies \rho(F) = \rho(F/F_1)$.

Altogether, the above implies that we can find a filtration

$$\cdots \subset F_{i-1} \subset F_i \subset \cdots \subset F$$

such that each $\rho(F_i) = \rho(F)$ and F_i/F_{i-1} is stable with $\rho(F_i/F_{i-1}) = \rho(F)$. However, since X is a Noetherian scheme, we have that actually this chain must be finite [Sta25, Tag 01Y7], which completes our proof. □

Jordan-Holder filtrations are not unique. For example, one could take the direct sum of two non-isomorphic stable line bundles of the same ρ, and one would have two different filtrations. However, the graded pieces are unique.

Proposition 13.10. *Let X be a Noetherian projective scheme. Let $F \in Coh(X)$ be semistable of dimension d. Let*

$$0 \subset F_1 \subset \cdots \subset F_\ell = F$$

be a Jordan-Holder filtration of F. Then $\bigoplus_{i=1}^{\ell} \frac{F_i}{F_{i-1}}$ is independent of filtration.

Proof. Suppose

$$0 \subset F_1 \subset \cdots \subset F_\ell = F \text{ and } 0 \subset F_1' \subset \cdots \subset F_t' = F$$

are two Jordan-Holder filtrations of F. Let's suppose that the quantity ℓ is minimal among all Jordan-Holder filtrations of F. We induct on these minimal lengths.

We'd like to prove that

$$\bigoplus_{i=1}^{\ell} \frac{F_i}{F_{i-1}} \cong \bigoplus_{j=1}^{t} \frac{F_j'}{F_{j-1}'}.$$

Let j be minimal such that $F_1 \subset F_k'$. Then there is a nontrivial morphism

$$0 \to F_1 \to F_k' \to F_k'/F_{k-1}' \to 0.$$

Since F'_k/F'_{k-1} and F_1 are both stable, this implies that $F_1 \cong F'_k/F'_{k-1}$. Then consider F/F_1. It admits the Jordan-Holder filtration

$$0 \subset \frac{F_2}{F_1} \subset \cdots \subset \frac{F_\ell}{F_1} = \frac{F}{F_1}.$$

On the other hand, we can obtain another Jordan-Holder filtration of F/F_1 from F'_* via the following way. Note we have a short exact sequence

$$0 \to F'_{k-1} \to \frac{F}{F_1} \to \frac{F}{F'_k} \to 0.$$

So we claim that

$$0 \subset F'_1 \subset \cdots \subset F'_{k-1} \subset E_{k+1} \subset \cdots \subset E_t = \frac{F}{F_1},$$

where $E_j = \ker[\frac{F}{F_1} \to \frac{F}{F'_j}]$, is a Jordan-Holder filtration. By inductive hypothesis, this implies that

$$\bigoplus_{i=2}^{\ell} \frac{F_i}{F_{i-1}} \cong \bigoplus_{j=1, j\neq k}^{t} \frac{F'_j}{F'_{j-1}}$$

are isomorphic, and since $F_1 \cong F'_k/F'_{k-1}$, we obtain the claim. □

14. GMS: Quot scheme and its representability

In this section we will define the moduli functor for Quot schemes and sketch representability. This is an important construction which generalizes Grassmannians and Hilbert schemes, and will be crucial for constructing moduli spaces of sheaves.

Let's warm ourselves up towards defining the Quot functor. The most basic example is the Grassmannian. Recall the Grassmannian functor is the moduli functor

$$\mathrm{Gr}(r, V) : \mathrm{Sch}_{\mathbb{C}}^{\mathrm{op}} \to \mathrm{Sets}$$

where $T \mapsto \{\mathcal{E} \text{ rank r on T} \mid V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{E}\}/\sim$, where here $\mathcal{E} \sim \mathcal{E}'$ if there exists an isomorphism $\mathcal{E} \cong \mathcal{E}'$ such that its composition $V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{E}$ equals $V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{E}'$. In essence, the Grassmannian functor sends T to rank r locally free quotient classes of $V^* \otimes \mathcal{O}_T$. But note further that $V^* \otimes \mathcal{O}_T$ can be thought of as the pullback of the coherent sheaf on $\operatorname{Spec} \mathbb{C}$ given by the vector space V^* along $T \to \operatorname{Spec} \mathbb{C}$. Then we can generalize $\mathrm{Gr}(r, V)$ to the following moduli functor.

Proposition 14.1. *Let $\mathcal{V} \in Coh(S)$ for some Noetherian scheme S. We define*

$$\mathrm{Gr}_S(\mathcal{V}, r) : (Sch/S)^{op} \to Sets$$

where $[f : T \to S] \in Sch/S$ maps to $\{\mathcal{E} \text{ rank } r \text{ on } T \mid f^*\mathcal{V} \twoheadrightarrow \mathcal{E}\}/\sim$. Then $\mathrm{Gr}_S(\mathcal{V}, r)$ is representable by some element of Sch/S.

Proof. Suppose $S = \bigcup U_i = \bigcup \operatorname{Spec} A_i$. Then consider the moduli functors $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r)$. Suppose we show that $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r)$ is representable by $[\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r) \to U_i] \in \mathrm{Sch}/U_i$ (note we are abusing notation here slightly). Then consider the elements

$$\mathrm{Gr}_S(\mathcal{V}, r)([T \to S]) = \{\mathcal{E} \text{ rank r on T } \mid f^*\mathcal{V} \twoheadrightarrow \mathcal{E}\}/\sim .$$

Note $T \to S$ is glued together by $T_i \to U_i$, where $T_i = U_i \times_S T$, and note T_i is an open of T. And $f^*\mathcal{V} \twoheadrightarrow \mathcal{E}$ is also glued together by $(f^*\mathcal{V})|_{T_i} = f^*(\mathcal{V}|_{U_i}) \twoheadrightarrow \mathcal{E}|_{T_i}$. Each $f^*(\mathcal{V}|_{U_i}) \twoheadrightarrow \mathcal{E}|_{T_i}$ defines an element of $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r)([T_i \to U_i])$ and thus corresponds to a morphism

$$\begin{tikzcd} T_i \arrow[r] \arrow[d] & \mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r) \arrow[dl] \\ U_i & \end{tikzcd}$$

Since $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r) \to U_i$ is the universal family, this morphism is unique, and thus should glue together. Thus, the $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r) \to U_i$ should glue together to a S-scheme $\mathrm{Gr}_S(\mathcal{V}, r) \to S$ such that the family $f^*\mathcal{V} \twoheadrightarrow \mathcal{E}$ corresponds uniquely to a S-morphism $T \to \mathrm{Gr}_S(\mathcal{V}, r)$. This $\mathrm{Gr}_S(\mathcal{V}, r) \to S$ should then represent our functor.

Thus, it suffices to prove that each $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r) : (\mathrm{Sch}/U_i)^{\mathrm{op}} \to \mathrm{Sets}$ is representable. Note we can assume that $\mathcal{V}|_{U_i} \cong \widetilde{M}$ for some A_i-module M. Since M is coherent, there is an exact sequence

$$A_i^{\oplus n'} \xrightarrow{\phi} A_i^{\oplus n} \xrightarrow{\psi} M \to 0.$$

Now note that

$$\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i}, r)([T \xrightarrow{f} U_i]) = \{\mathcal{E} \text{ rank r on T} \mid f^*\widetilde{M} \twoheadrightarrow \mathcal{E}\} \hookrightarrow \mathrm{Gr}_{U_i}(\mathcal{O}_{A_i}^{\oplus n}, r)[T \xrightarrow{f} U_i] = \{\mathcal{E} \text{ rank r on T} \mid \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{E}\}$$

because given $f^*\tilde{M} \twoheadrightarrow \mathcal{E}$, we obtain the surjection $\mathcal{O}_T^{\oplus n} \xrightarrow{f^*\psi} f^*\widetilde{M} \twoheadrightarrow \mathcal{E}$. Furthermore, note that $\mathrm{Gr}_{U_i}(\mathcal{O}_{A_i}^{\oplus n}, r) = \mathrm{Gr}_{\operatorname{Spec} A_i}(\mathcal{O}_{A_i}^{\oplus n}, r) = \operatorname{Spec} A_i \times \mathrm{Gr}(k^{\oplus n}, r)$ which can be deduced immediately on the level of points. Then we claim that

$$\mathrm{Gr}_{\operatorname{Spec} A_i}(\widetilde{M}, r)$$

is representable by a closed subscheme of $\mathrm{Gr}_{\operatorname{Spec} A_i}(\mathcal{O}_{A_i}^{\oplus n}, r) = \operatorname{Spec} A_i \times \mathrm{Gr}(k^{\oplus n}, r)$.

Now, when does an element $\mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{E}$ factor as $\mathcal{O}_T^{\oplus n} \xrightarrow{f^*\psi} f^*\widetilde{M} \twoheadrightarrow \mathcal{E}$? Exactly when the composition

$$\mathcal{O}_T^{\oplus n'} \xrightarrow{f^*\phi} \mathcal{O}_T^{\oplus n} \to \mathcal{E}$$

is zero. Note that any element $[\mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{E}] \in \mathrm{Gr}_{U_i}(\mathcal{O}_{A_i}^{\oplus n}, r)([T \to U_i])$ comes from pulling back, along a unique morphism

$$\begin{tikzcd} T \arrow[rr, "\theta"] \arrow[dr, "f"'] & & \mathrm{Gr}_{U_i}(\mathcal{O}_{A_i}^{\oplus n}, r) \arrow[dl, "\pi"] \\ & U_i & \end{tikzcd}$$

the universal element

$$[\mathcal{O}^{\oplus n}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}] \in \mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)[\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r) \to U_i] \leftrightarrow \mathrm{id} \in \mathrm{Hom}_{\mathrm{Sch}/U_i}(\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r), \mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)).$$

In other words,

$$\theta^*[\mathcal{O}^{\oplus n}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}] = [\mathcal{O}^{\oplus n}_T \twoheadrightarrow \mathcal{E}] \in \mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)[T \xrightarrow{f} U_i].$$

Note pulling back $\mathcal{O}^{\oplus n'}_{A_i} \xrightarrow{\phi} \mathcal{O}^{\oplus n}_{A_i}$ along π yields

$$\mathcal{O}^{\oplus n'}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \to \mathcal{O}^{\oplus n}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)},$$

and pulling back again along θ then equals pulling back by f from the start, yielding

$$\mathcal{O}^{\oplus n'}_T \xrightarrow{f^*\phi} \mathcal{O}^{\oplus n}_T.$$

We see then that

$$\mathcal{O}^{\oplus n'}_T \xrightarrow{f^*\phi} \mathcal{O}^{\oplus n}_T \to \mathcal{E}$$

is zero when θ maps T to the locus where

$$\mathcal{O}^{\oplus n'}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \to \mathcal{O}^{\oplus n}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}$$

is zero. This occurs along the closed subscheme of $\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)$ whose ideal sheaf is given by the elements of $\mathcal{O}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)}$ occuring in the $n' \times n$ matrix defining the morphism $\mathcal{O}^{\oplus n'}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)} \to \mathcal{O}^{\oplus n}_{\mathrm{Gr}_{U_i}(\mathcal{O}^{\oplus n}_{A_i},r)}$. This closed subscheme exactly represents the moduli functor $\mathrm{Gr}_{U_i}(\mathcal{V}|_{U_i},r) = \mathrm{Gr}_{\mathrm{Spec}\,A_i}(\widetilde{M},r)$. □

We now make our final generalization to the Quot scheme. Let's be clear about how we are generalizing. When we constructed the Grassmannian moduli functor, we began with a diagram

$$\begin{array}{c} \mathrm{Spec}\,\mathbb{C} \\ \big\downarrow \mathrm{id} \\ \mathrm{Spec}\,\mathbb{C} \end{array}$$

and a locally free sheaf on $\mathrm{Spec}\,\mathbb{C}$, namely a vector space V^*. Then for all $T \to \mathrm{Spec}\,\mathbb{C}$,

$$\mathrm{Gr}(V,r)([T \to \mathrm{Spec}\,\mathbb{C}]) = \{\mathcal{E} \text{ locally free rank r } \mid V^* \otimes \mathcal{O}_T \twoheadrightarrow \mathcal{E}\}/\sim$$

Next, in Proposition 14.1 we generalized this construction by replacing $\mathrm{Spec}\,\mathbb{C}$ with a Noetherian scheme S,

$$\begin{array}{c} S \\ \big\downarrow \mathrm{id} \\ S \end{array}$$

and replaced $V^* \in Coh(\mathrm{Spec}\,\mathbb{C})$ with $\mathcal{V} \in Coh(S)$. Then for all $[T \to S] \in \mathrm{Sch}/S$,

$$\mathrm{Gr}_S(\mathcal{V},r)([T \xrightarrow{f} S]) = \{\mathcal{E} \text{ locally free rank r on T} \mid f^*\mathcal{V} \twoheadrightarrow \mathcal{E}\}.$$

Here is our final generalization. We replace $S \xrightarrow{\mathrm{id}} S$ with projective S-morphism

$$\begin{array}{c} X \\ \big\downarrow \pi \\ S \end{array}$$

where S is Noetherian. Let $\mathcal{O}_X(1)$ be a π-ample line bundle. We replace $\mathcal{V} \in Coh(S)$ with $\mathcal{H} \in Coh(X)$. Fix a polynomial $P \in \mathbb{Q}[z]$. Then for all $T \xrightarrow{f} S$

$\mathrm{Quot}_{X/S}(\mathcal{H}, P)([T \xrightarrow{f} S]) := \{\mathcal{F} \in Coh(T \times_S X)$ T-flat of Hilbert polynomial P $\mid \mathcal{H}_T \twoheadrightarrow \mathcal{F}\}/\sim$

where $\mathcal{H}_T$ is the pullback of $\mathcal{H}$ along $T \times_S X \to X$.

Thus, in defining the Quot scheme, we've made two simultaneous generalizations from Proposition 14.1. The first is we've replaced $S \xrightarrow{\mathrm{id}} S$ with a projective morphism of Noetherian schemes $X \xrightarrow{\pi} S$. The second is that, instead of considering just locally free quotients of $\mathcal{H}_T$, we consider coherent quotients which form flat families with constant Hilbert polynomial.

Theorem 14.2. *Let $X \to S$ be a projective morphism of Noetherian schemes. Let $\mathcal{O}_X(1)$ be a π-ample line bundle, $\mathcal{H} \in Coh(X)$, and $P \in \mathbb{Q}[z]$. The moduli functor*

$$Quot_{X/S}(\mathcal{H}, P) : (Sch/S)^{op} \to Sets$$

is representable by a projective S-scheme.

Proof. **Step 1: representability in most basic case.** We first do the special case that $X = \mathbb{P}^n$ and $S = \operatorname{Spec} \mathbb{C}$. So we have some $\mathcal{H} \in Coh(X)$ and polynomial $P \in \mathbb{Q}[z]$. By definition, we have for all $T \in (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}}$

$$\begin{array}{ccc} T \times \mathbb{P}^n & \xrightarrow{\pi_{\mathbb{P}^n}} & \mathbb{P}^n \\ {\scriptstyle \pi_T}\big\downarrow & & \big\downarrow \\ T & \longrightarrow & \operatorname{Spec} \mathbb{C} \end{array} \quad \mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{H}, P)(T) = \left\{\pi_{\mathbb{P}^n}^* \mathcal{H} \twoheadrightarrow \mathcal{F} \;\middle|\; \begin{array}{c} \mathcal{F} \in Coh(T \times \mathbb{P}^n) \text{ flat over T} \\ \\ \text{and } P(\mathcal{F}_t) = P, \forall t \in T(\mathbb{C}) \end{array} \right\}/\sim$$

Note that each element $[\pi_{\mathbb{P}^n}^* \mathcal{H} \twoheadrightarrow \mathcal{F}]$ is parameterizing a family of short exact sequences

$$0 \to \mathcal{K}_t \to \mathcal{H} \to \mathcal{F}_t \to 0$$

for every $t \in T$, where $\mathcal{K}_t$ is the restriction of $\mathcal{K} = \ker[\pi_{\mathbb{P}^n}^* \mathcal{H} \twoheadrightarrow \mathcal{F}]$ to $\{\operatorname{Spec} \mathbb{C}(t)\} \times \mathbb{P}^n$. Note that this is true because $\pi_{\mathbb{P}^n}^* \mathcal{H}$ and $\mathcal{F}$ are flat over T [HL, Lemma 2.1.4]; in general, restriction need not preserve short exact sequences as tensor product is only right exact.

Now, if we fix T and an element $[\pi_{\mathbb{P}^n}^* \mathcal{H} \twoheadrightarrow \mathcal{F}]$, then by definition the set (or more precisely, the set of isomorphism classes of) $\{\mathcal{F}_t\}_{t \in T(\mathbb{C})}$ is a bounded collection of coherent sheaves. But even better, we have that actually the set of isomorphism classes of all $\mathcal{F}_t$, where $t \in T(\mathbb{C})$ *for any* T, is a bounded collection of sheaves. Equivalently, a bounded collection of sheaves is one where the number of distinct Hilbert polynomials are finite and there is a uniform upper bound on the Castelnuevo-Mumford regularities [HL, Section 1.7, Lemma 1.7.6] . We already knew that $\mathcal{F}_t$, for any $t \in T(\mathbb{C})$ and any T, has Hilbert polynomial P by definition of the Quot functor. So the corollary of actual interest here, of what is essentially by definition of the Quot functor, is that there is a uniform bound on the Castelnuevo-Mumford regularities of $\mathcal{F}_t$ for all $t \in T(\mathbb{C})$ across all schemes T.

Then using this uniform bound on regularities, and using the construction one might expect in going from $(iii) \to (i)$ in [HL, Lemma 1.7.6] to get uniform bound on the regularities of the $\mathcal{K}_t's$, there exists an m, independent of choice of $T \in \mathrm{Sch}_{\mathbb{C}}$, such that for all $m' \geq m$, we have

$$0 \to \mathcal{K}_t(m') \to \mathcal{H}(m') \to \mathcal{F}_t(m') \to 0$$

is a short exact sequence such that for $i > 0$, we have

$$H^i(\{t\} \times \mathbb{P}^n, \mathcal{K}_t(m')) = H^i(\{t\} \times \mathbb{P}^n, \mathcal{H}(m')) = H^i(\{t\} \times \mathbb{P}^n, \mathcal{F}_t(m')) = 0,$$

for any $t \in T(\mathbb{C})$ across all $T \in \mathrm{Sch}_{\mathbb{C}}$. Globally, this means that for all $m' \geq m$, we have a short exact sequence

$$0 \to \mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m') \to \pi_{\mathbb{P}^n}^*\mathcal{H} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m') \to \mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m') \to 0$$

such that for $i > 0$,

$$R^i\pi_{T,*}(\mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')) = R^i\pi_{T,*}(\pi_{\mathbb{P}^n}^*\mathcal{H} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')) = R^i\pi_{T,*}(\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')) = 0,$$

which means there is a short exact sequence

$$0 \to \pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')] \to \pi_{T,*}[\pi_{\mathbb{P}^n}^*\mathcal{H} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')] \to \pi_{T,*}[\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')] \to 0.$$

Note that $\pi_{T,*}[\pi_{\mathbb{P}^n}^*\mathcal{H} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')] \cong \mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m'))$. Furthermore, note that $\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')$ is locally free of rank $\dim H^0(\{t\} \times \mathbb{P}^n, \mathcal{F}_t(m')) = P(m') = \chi(\mathcal{F}_t(m'))$, since it is T-flat and all the higher direct images vanish. So it is a rank $P(m')$ bundle on T. Thus, we have obtained an element

$$[\mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m')) \twoheadrightarrow \pi_{T,*}[\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')]] \in \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m'))^\vee, P(m')).$$

In particular, for every $m' \geq m$, we have defined a map

$$\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P) \to \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m'))^\vee, P(m')).$$

We claim we can just focus on the map

$$\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P) \to \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m)).$$

Note m is greater than or equal to the Castelnuevo-Mumford regularity of each of the $\mathcal{K}_t$, for $t \in T(\mathbb{C})$ across all $T \in \mathrm{Sch}_{\mathbb{C}}$. This implies we have a surjection [HL, Section 1.7, Lemma 1.7.2]

$$H^0(\{t\} \times \mathbb{P}^n, \mathcal{K}_t(m)) \otimes H^0(\{t\} \times \mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m' - m)) \twoheadrightarrow H^0(\{t\} \times \mathbb{P}^n, \mathcal{K}_t(m')).$$

This means

$$\pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)] \otimes \pi_{T,*}\mathcal{O}_{T\times\mathbb{P}^n}(m'-m) \twoheadrightarrow \pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m')],$$

so the short exact sequence

$$0 \to \pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)] \to \pi_{T,*}[\pi_{\mathbb{P}^n}^*\mathcal{H} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)] \to \pi_{T,*}[\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)] \to 0$$

For this reason, we concentrate solely on the map

$$\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P) \to \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m)),$$

which we claim is an injection. Let's verify this for some fixed $T \in (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}}$. Suppose $[\pi_{\mathbb{P}^n}^*\mathcal{H} \twoheadrightarrow \mathcal{F}_1]$ and $[\pi_{\mathbb{P}^n}^*\mathcal{H} \twoheadrightarrow \mathcal{F}_2]$ are two elements, such that

$$\mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow \pi_{T,*}[\mathcal{F}_1 \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)]$$

and

$$\mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow \pi_{T,*}[\mathcal{F}_2 \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)]$$

are isomorphic elements of $\mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m)), P(m))(T)$. Then there is a global identification of vector bundles which identifies maps of fibers

$$H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow H^0(\{t\} \times \mathbb{P}^n, \mathcal{F}_{1,t}(m))$$

$$H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow H^0(\{t\} \times \mathbb{P}^n, \mathcal{F}_{2,t}(m))$$

over $t \in T(\mathbb{C})$. Using surjectivity of multiplication by $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m'-m))$, the aforementioned identifications of surjections extend to identifications of surjections of graded $\mathbb{C}[x_0, \cdots, x_n] = \bigoplus_{k\geq 0} H^0(\mathcal{O}_{\mathbb{P}^n}(k))$ modules:

$$\Gamma_{*,\geq m}\mathcal{H} = \bigoplus_{m'\geq m} H^0(\mathbb{P}^n, \mathcal{H}(m')) \twoheadrightarrow \bigoplus_{m'\geq m} H^0(\{t\} \times \mathbb{P}^n, \mathcal{F}_{1,t}(m')) = \Gamma_{*,\geq m}\mathcal{F}_{1,t}$$

$$\Gamma_{*,\geq m}\mathcal{H} = \bigoplus_{m'\geq m} H^0(\mathbb{P}^n, \mathcal{H}(m')) \twoheadrightarrow \bigoplus_{m'\geq m} H^0(\{t\} \times \mathbb{P}^n, \mathcal{F}_{1,t}(m')) = \Gamma_{*,\geq m}\mathcal{F}_{2,t}.$$

Noting that in general a graded S-module $\Gamma_*\mathcal{G}$, where S is a graded rings, produces the same quasicoherent sheaf over $\mathrm{Proj} S$ as $\Gamma_{*,\geq m}\mathcal{G}$ for any m, we find that the aforementioned identifications then identify the surjection

$$\mathcal{H} \twoheadrightarrow \mathcal{F}_{1,t}$$

with the surjection

$$\mathcal{H} \twoheadrightarrow \mathcal{F}_{2,t}.$$

But since these identifications are being parameterized globally for all $t \in T(\mathbb{C})$, we have identifications of the surjections

$$\pi_{\mathbb{P}^n}^*\mathcal{H} \twoheadrightarrow \mathcal{F}_1$$

and

$$\pi_{\mathbb{P}^n}^*\mathcal{H} \twoheadrightarrow \mathcal{F}_2.$$

Thus,

$$\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P) \to \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m)) =: G$$

is an injection of moduli functors. Finally, we claim that $\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P)$ is represented by a locally closed subscheme of the Grassmannian. Consider the dualized tautological sequence

$$0 \to \mathcal{S} \to G \times H^0(\mathbb{P}^n, \mathcal{H}(m)) \to \mathcal{Q} \to 0.$$

This parameterizes injections of vector spaces

$$0 \to \mathcal{S}_t \to H^0(\mathbb{P}^n, \mathcal{H}(m))$$

for $t \in G(\mathbb{C})$. Using multiplication by $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m'-m))$, these injections induce injections

$$0 \to S_t \cdot \mathbb{C}[x_0, \cdots, x_n] \to \bigoplus_{m' \geq m} H^0(\mathbb{P}^n, \mathcal{H}(m'))$$

for every $t \in G(\mathbb{C})$; in particular, the injections are parameterized, so this corresponds to some injection

$$0 \to S \cdot \widetilde{\mathbb{C}[x_0, \cdots}, x_n] \to \pi_{\mathbb{P}^n}^* \mathcal{H}$$

of coherent sheaves on $G \times \mathbb{P}^n$. Then we take a flattening stratification of the quotient $\pi_{\mathbb{P}^n}^* \mathcal{H} / S \cdot \widetilde{\mathbb{C}[x_0, \cdots}, x_n]$ and take the locus over G where the quotient is a flat-family parameterizing coherent sheaves of Hilbert polynomial P. This is a locally closed subscheme of $\mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m))$, and we claim it represents $\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P)$.

Given some $[\pi_{\mathbb{P}^n}^* \mathcal{H} \twoheadrightarrow \mathcal{F}] \in \mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P)(T)$, we have $\mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow \pi_{T,*}[\mathcal{F} \otimes \mathcal{O}_{T \times \mathbb{P}^n}(m)]$ comes from pulling back $\mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m)) \times H^0(\mathbb{P}^n, \mathcal{H}(m)) \twoheadrightarrow \mathcal{Q}$ along some morphism $T \xrightarrow{\theta} \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m))$. In particular, the kernel $0 \to \pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T \times \mathbb{P}^n}(m)] \to \mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m))$ comes from pulling back $0 \to \mathcal{S} \to G \times H^0(\mathbb{P}^n, \mathcal{H}(m))$ along this morphism $T \xrightarrow{\theta} \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m))$.

In particular, for $t \in T(\mathbb{C})$, we have that

$$0 \to H^0(\{t\} \times \mathbb{P}^n, \mathcal{K}_t(m)) \to H^0(\mathbb{P}^n, \mathcal{H}(m))$$

is identified with

$$0 \to \mathcal{S}_{\theta(t)} \to H^0(\mathbb{P}^n, \mathcal{H}(m)),$$

so that

$$0 \to H^0(\{t\} \times \mathbb{P}^n, \mathcal{K}_t(m)) \cdot \mathbb{C}[x_0, \cdots, x_n] \to \bigoplus_{m' \geq m} H^0(\mathbb{P}^n, \mathcal{H}(m'))$$

is identified with

$$0 \to S_{\theta(t)} \cdot \mathbb{C}[x_0, \cdots, x_n] \to \bigoplus_{m' \geq m} H^0(\mathbb{P}^n, \mathcal{H}(m'))$$

and since these identifications happen globally, we see that

$$0 \to \pi_{T,*}[\mathcal{K} \otimes \mathcal{O}_{T \times \mathbb{P}^n}(m)] \to \mathcal{O}_T \otimes H^0(\mathbb{P}^n, \mathcal{H}(m))$$

comes from pulling back

$$0 \to S \cdot \mathbb{C}[\widetilde{x_0, \cdots}, x_n] \to \pi_{\mathbb{P}^n}^* \mathcal{H}$$

along $T \xrightarrow{\theta} \mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{H}(m))^\vee, P(m))$. In particular, since for every $t \in T(\mathbb{C})$, we have the fiber of $\pi_{T,*}[\mathcal{F} \otimes \mathcal{O}_{T\times\mathbb{P}^n}(m)]$ over t has Hilbert polynomial P, we see that θ must factor through the desired locus. This completes the proof of representability of $\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P)$.

Step 2: projectivity of most basic case. Next, one shows that $\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{H}, P)$ is projective; since by construction it is quasi-projective it remains to show it is proper. One uses the valuative criterion for properness.

Step 3: reduction of general case. Now let $X \to S$ be an arbitrary projective morphism. We want to show that $\mathrm{Quot}_{X/S}(\mathcal{H}, P)$ is representable by a projective S-scheme, where $\mathcal{H} \in Coh(X)$ and $P \in \mathbb{Q}[z]$. Since $X \to S$ is projective, consider the diagram

$$\begin{tikzcd} X \arrow[r, hook, "\iota"] \arrow[d] & \mathbb{P}^n \times S \arrow[dl] \\ S & \end{tikzcd}$$

Note $\iota_*\mathcal{H}$ remains coherent on $\mathbb{P}^n \times S$. We claim that

$$\mathrm{Quot}_{X/S}(\mathcal{H}, P) \cong \mathrm{Quot}_{\mathbb{P}^n_S/S}(\iota_*\mathcal{H}, P).$$

Let $T \to S$ be some morphism. Then consider the diagram

$$\begin{tikzcd} T \times_S X \arrow[r, "\pi_X"] \arrow[d, hook, "\iota'"'] & X \arrow[d, hook, "\iota"] \\ T \times \mathbb{P}^n \arrow[r, "\pi_{\mathbb{P}^n\times S}"'] \arrow[d, "\pi_T"'] & \mathbb{P}^n \times S \arrow[d] \\ T \arrow[r] & S \end{tikzcd}$$

Suppose we have some element $[\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{F}] \in \mathrm{Quot}_{X/S}(\mathcal{H}, P)([T \to S])$. We claim that

$$[\iota'_*(\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{F})] \in \mathrm{Quot}_{\mathbb{P}^n_S/S}(\iota_*\mathcal{H}, P)([T \to S]).$$

First of all, note that since $\iota : X \to \mathbb{P}^n \times S$ is a closed embedding and thus an affine morphism,

$$\iota'_*\pi_X^*\mathcal{H} \cong \pi_{\mathbb{P}^n\times S}^*\iota_*\mathcal{H}$$

by [Sta25, Tag 02KE]. Thus we see that

$$\iota'_*(\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{F}) = \pi_{\mathbb{P}^n\times S}^*\iota_*\mathcal{H} \to \iota'_*\mathcal{F}.$$

Now we claim that $\pi_{\mathbb{P}^n\times S}^*\iota_*\mathcal{H} \to \iota'_*\mathcal{F}$ remains a surjection of coherent sheaves on $T \times \mathbb{P}^n$. This is because ι' is a closed embedding and thus an affine morphism, and higher direct images of pushforward of affine morphisms always vanish (this is a relative version of the fact that higher cohomologies of coherent sheaves on affines always vanish), so pushforward by affine morphisms is always exact.... thus we have the surjection. And $\iota'_*\mathcal{F}$ is coherent on $T \times \mathbb{P}^n$. Thus,

$$[\iota'_*(\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{F})] = [\pi^*_{\mathbb{P}^n \times S}\iota_*\mathcal{H} \to \iota'_*\mathcal{F}] \in \mathrm{Quot}_{\mathbb{P}^n_S/S}(\iota_*\mathcal{H}, P)([T \to S]),$$

so we have defined a valid morphism of Quot functors. Clearly it is injective. Now we claim that it is surjective. Suppose we have a surjection of coherent sheaves

$$\pi^*_{\mathbb{P}^n \times S}\iota_*\mathcal{H} \twoheadrightarrow \mathcal{G}$$

on $T \times \mathbb{P}^n$. Note that the fact this is a surjection means that the support of $\mathcal{G} \in Coh(T \times \mathbb{P}^n)$ is actually contained in $T \times_S X$. Then note that $\mathcal{G} \to \iota'_*(\iota')^*\mathcal{G}$ is an isomorphism. Locally, if ι' is given as $\mathrm{Spec}(A/I) \to \mathrm{Spec}\, A$, then $\mathcal{G}$ is an A-module M whose support is contained in $\mathrm{Spec}(A/I)$. Since everything is Noetherian here, I is finitely generated, so in particular every $x \in I$ we have $x \cdot M = 0$. So the fact that $\mathcal{G} \to \iota'_*(\iota')^*\mathcal{G}$ reflects the fact that locally M is isomorphic to $M \otimes A/I$ as A-modules when for every $i \in I$, we have $i \cdot M = 0$. Furthermore, note that $\iota'_*\pi_X^*\mathcal{H} = \pi^*_{\mathbb{P}^n \times S}\iota_*\mathcal{H}$ implies that $(\iota')^*\pi^*_{\mathbb{P}^n \times S}\iota_*\mathcal{H} \cong \pi_X^*\mathcal{H}$. All of this is to say that

$$(\iota')^*[\pi^*_{\mathbb{P}^n \times S}\iota_*\mathcal{H} \twoheadrightarrow \mathcal{G}] \cong \pi_X^*\mathcal{H} \twoheadrightarrow (\iota')^*\mathcal{G}.$$

such that applying $(\iota')_*$ gives back what we started with. This shows that

$$\mathrm{Quot}_{X/S}(\mathcal{H}, P) \cong \mathrm{Quot}_{\mathbb{P}^n \times S/S}(\iota_*\mathcal{H}, P).$$

Step 4: Final step. It remains to show that $\mathrm{Quot}_{\mathbb{P}^n \times S/S}(\mathcal{H}, P)$ is representable by a projective S-scheme, for any $\mathcal{H} \in Coh(\mathbb{P}^n \times S)$. Recall that since $\mathcal{H}$ is coherent, there exists an exact sequence

$$\mathcal{O}_{\mathbb{P}^n \times S}(-m')^{\oplus n'} \to \mathcal{O}_{\mathbb{P}^n \times S}(-m)^{\oplus n} \to \mathcal{H} \to 0,$$

on $\mathbb{P}^n_S$, and note that for any $T \to S$, applying $\pi^*_{\mathbb{P}^n_S}$, the sequence

$$\mathcal{O}_{T \times \mathbb{P}^n}(-m')^{\oplus n'} \to \mathcal{O}_{T \times \mathbb{P}^n}(-m)^{\oplus n} \to \pi^*_{\mathbb{P}^n_S}\mathcal{H} \to 0$$

remains exact. Now, any surjection $\pi^*_{\mathbb{P}^n_T}\mathcal{H} \twoheadrightarrow \mathcal{F}$ over $T \times \mathbb{P}^n$ induces a surjection

$$\mathcal{O}^{\oplus n}_{T \times \mathbb{P}^n} \twoheadrightarrow \mathcal{F}.$$

And any surjection $\mathcal{O}^{\oplus n}_{T \times \mathbb{P}^n} \xrightarrow{\theta} \mathcal{F}$ factors through a surjection $\pi^*_{\mathbb{P}^n_T}\mathcal{H} \twoheadrightarrow \mathcal{F}$ if and only if the composition

$$\mathcal{O}_{T \times \mathbb{P}^n}(-m')^{\oplus n'} \to \mathcal{O}_{T \times \mathbb{P}^n}(-m)^{\oplus n} \xrightarrow{\theta} \mathcal{F}$$

is 0. Then we have an injection of moduli functors

$$\mathrm{Quot}_{\mathbb{P}^n \times S/S}(\mathcal{H}, P) \hookrightarrow S \times \mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P).$$

If we consider the universal element

$$\mathrm{id} \in \mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P)(\mathrm{Quot}_{\mathbb{P}^n/\mathrm{Spec}\,\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P))$$

which with respect to the diagram

$$\begin{tikzcd}
\mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P) \times S \times \mathbb{P}^n \arrow[r, "\phi"] \arrow[d] & \mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P) \times \mathbb{P}^n \arrow[r, "\pi_{\mathbb{P}^n}"] \arrow[d, "\pi_Q"'] & \mathbb{P}^n \arrow[d] \\
\mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P) \times S \arrow[r] & \mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P) \arrow[r] & \operatorname{Spec}\mathbb{C}
\end{tikzcd}$$

looks like the surjection of coherent sheaves $\pi_{\mathbb{P}^n}^*\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n} \twoheadrightarrow \mathcal{F}_{\mathrm{univ}}$ over $\mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P) \times \mathbb{P}^n$, then $\mathrm{Quot}_{\mathbb{P}^n \times S/S}(\mathcal{H}, P)$ is represented by the closed subscheme of $S \times \mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P)$ that is the locus where the morphism

$$\phi^*\mathcal{O}_{\mathbb{P}^n_S}(-m')^{\oplus n'} \to \phi^*\mathcal{O}_{\mathbb{P}^n_S}(-m)^{\oplus n} \cong \phi^*\pi_{\mathbb{P}^n}^*\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n} \twoheadrightarrow \phi^*\mathcal{F}_{\mathrm{univ}}$$

is zero, i.e. the locus of $t \in S \times \mathrm{Quot}_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}(-m)^{\oplus n}, P)$ where

$$\mathcal{O}_{\mathrm{Quot}\times S\times\mathbb{P}^n,t}(-m')^{\oplus n'} \to \mathcal{O}_{\mathrm{Quot}\times S\times\mathbb{P}^n,t}(-m)^{\oplus n} \to \phi^*\mathcal{F}_{\mathrm{univ}}|_t$$

is zero. □

Question 14.3. *Note that for $Quot_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{H}, P)$ to be nonempty, there has to be a compatible choice of $P \in \mathbb{Q}[z]$. The choice of P actually comes from considering the short exact sequence of coherent sheaves on* $\mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)), P(m))$ *associated to the tautological sequence on the Grassmannian (see proof of Step 1 of Theorem 14.2). In particular, there are only finitely many flattening stratification loci for the quotient coherent sheaf on* $\mathrm{Gr}(H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)), P(m)) \times \mathbb{P}^n$ [HL, Theorem 2.1.5]. *So if one fixes the data of $\mathcal{H} \in Coh(\mathbb{P}^n)$, which determines m, one needs to be careful about the Hilbert polynomial P. A question is: how difficult is to determine a complete list of all those Hilbert polynomials?*

Example 14.4. *The Quot scheme*

$$Quot_{\mathbb{P}^n/\operatorname{Spec}\mathbb{C}}(\mathcal{O}_{\mathbb{P}^n}, P)$$

is the fine moduli space for closed subschemes $X \subset \mathbb{P}^n$ with Hilbert polynomial P, i.e. $\chi(\mathcal{O}_X(m)) = P(m)$. These are examples of Hilbert schemes.

Example 14.5. *The Hilbert scheme of n points on a projective variety is given by $Quot_{X/\mathbb{C}}(\mathcal{O}_X, n)$. Note when X is a smooth curve C, we have*

$$Quot_{C/\mathbb{C}}(\mathcal{O}_C, n) \cong \mathrm{Sym}^n C.$$

Finally, we saw in the process of Theorem 14.2 that the Quot scheme is projective relative to its base. In fact, we can nicely describe a line bundle on the Quot scheme ample relative to the base.

Proposition 14.6. *Let $X \xrightarrow{f} S$ be a projective morphism of Noetherian schemes, $\mathcal{H} \in Coh(X)$, and $P \in \mathbb{Q}[z]$. Over the Quot scheme $Quot_{X/S}(\mathcal{H}, P)$, consider the universal quotient*

$$\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{Q},$$

of coherent sheaves on $Quot_{X/S}(\mathcal{H}, P) \times_S X$*. Then for sufficiently large* m*, we have*

$$\det \pi_{Quot_{X/S}(\mathcal{H},P),*} \mathcal{Q}(m)$$

is an S*-very-ample line bundle on* $Quot_{X/S}(\mathcal{H}, P)$*.*

Proof. Recall from the proof of Theorem 14.2 that for the universal quotient $\pi_X^* \mathcal{H} \twoheadrightarrow \mathcal{Q}$, whose kernel we denote as $\mathcal{K}$, there exists m such that for all $m' \geq m$, we have that

$$0 \to \mathcal{K}_z \to \mathcal{H}_z \to \mathcal{Q}_z \to 0$$

is a short exact sequence of coherent sheaves on X_t for every $z \in \mathrm{Quot}_{X/S}(\mathcal{H}, P)(\mathbb{C})$ lying in the fiber over $t \in S(\mathbb{C})$, here z will correspond to the $\mathbb{C}$-point $[\mathcal{H} \twoheadrightarrow \mathcal{Q}_z]$, and such that for all $i > 0$,

$$R^i \pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P),*} \mathcal{K}(m') = R^i \pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P),*} \pi_X^* \mathcal{H}(m') = R^i \pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P),*} \mathcal{Q}(m') = 0.$$

In particular, using pushforward along $\pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P)}$, we saw from the proof of Theorem 14.2 that there is an injection from the Quot scheme

$$\mathrm{Quot}_{X/S}(\mathcal{H}, P) \xrightarrow{\phi} \mathrm{Gr}_S(f_* \mathcal{H}(m), P(m))$$

to the relative Grassmannian, so that if $S^\vee$ denotes the dual of the tautological bundle on $\mathrm{Gr}_S(f_* \mathcal{H}(m), P(m))$, we see that

$$\phi^* S^\vee \cong \pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P),*} \mathcal{Q}(m).$$

Furthermore, there is an embedding from the relative Grassmannian to relative projective space:

$$\mathrm{Gr}_S(f_* \mathcal{H}(m), P(m)) \xrightarrow{\psi} \mathbb{P}_S\Big(\bigwedge^{P(m)} f_* \mathcal{H}(m) \Big)$$

so that the pullback of the $\mathcal{O}(1)$ on $\mathbb{P}_S(\bigwedge^{P(m)} f_* \mathcal{H}(m))$, which is S-very-ample, is $\det S^\vee$. This implies that

$$\phi^* \psi^* \mathcal{O}(1) \cong \det \pi_{\mathrm{Quot}_{X/S}(\mathcal{H},P),*} \mathcal{Q}(m),$$

which implies that the latter is S-very-ample, as desired. □

15. GMS: Tangent space of Quot scheme and Fogarty's result

Let's start with some brief reminders about Zariski tangent spaces. Let X be a k-scheme. Let $z \in X$ be any point, not necessarily k-rational. We define the Zariski tangent space of X at z to be the $k(z)$-vector space $(m_z/m_z^2)^\vee := \mathrm{Hom}_{k(z)}(m_z/m_z^2, k(z))$. This is because in differential geometry, tangent vectors at a point should be thought of as directions in which one takes directional derivatives of functions vanishing at that point. These directional derivatives evaluate such functions to the field, and the act of taking a directional derivative is by definition to apply a derivation. Derivations only see the linear part of functions vanishing at the point, hence why we consider m_z/m_z^2. Zariski tangent spaces of affine subvarieties are exactly as we'd want them to be.

Proposition 15.1. *Let $X \subseteq \mathbb{A}^n$ be an affine k-subvariety, so $X = \operatorname{Spec} \frac{\mathbb{C}[x_1,\cdots,x_n]}{\langle f_1,\cdots,f_t\rangle}$. Let*

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_k}{\partial x_1} & \cdots & \frac{\partial f_t}{\partial x_n} \end{pmatrix}$$

denote the Jacobian of $k^n \xrightarrow{(f_1,\cdots,f_t)} k^t$. Let $p \in X(k)$ be a k-rational point. Then the Zariski tangent space of X at p is identified with $\ker J|_p$.

Proposition 15.2. *Let X be a k-scheme. Let $z \in X$ be some point and let $k(z)$ denote its function field $\mathcal{O}_{X,z}/m_{x,z}$. Then there is a one-to-one correspondence between the Zariski tangent space of X at z*

$$\operatorname{Hom}_{k(z)}(\frac{m_{X,z}}{m_{X,z}^2}, k(z))$$

and morphisms $\operatorname{Spec} \frac{k(z)[\epsilon]}{\epsilon^2} \to X$ whose image set-wise is z.

Proof. Suppose $\operatorname{Spec} \frac{k(z)[\epsilon]}{\epsilon^2} \to X$ is a morphism whose image set-wise is z. Such a morphism always factors through

$$\begin{tikzcd} \operatorname{Spec} \frac{k(z)[\epsilon]}{\epsilon^2} \arrow[rr] \arrow[dr] & & X \\ & \operatorname{Spec} \mathcal{O}_{X,z} \arrow[ur] & \end{tikzcd}$$

so that the data of a morphism $\operatorname{Spec} \frac{k(z)[\epsilon]}{\epsilon^2} \to X$ whose image set-wise is z is equivalent to specifying a morphism of k-schemes

$$\operatorname{Spec} \frac{k(z)[\epsilon]}{\epsilon^2} \to \operatorname{Spec} \mathcal{O}_{X,z},$$

or a map

$$\mathcal{O}_{X,z} \to \frac{k(z)[\epsilon]}{\epsilon^2}.$$

Since $\mathcal{O}_{X,z}/m_{x,z} =: k(z)$, such a map is equivalent to the data of a $k(z)$-linear ring map

$$m_{X,z} \to \frac{(\epsilon)}{\epsilon^2},$$

and since $m_{X,z}$ is finitely generated $k(z)$-module, this is equivalent to the data of a $k(z)$-linear ring map

$$\frac{m_{X,z}}{m_{X,z}^2} \to \frac{(\epsilon)}{\epsilon^2},$$

which is equivalent to the data of an element of

$$\operatorname{Hom}_{k(z)}(\frac{m_{X,z}}{m_{X,z}^2}, k(z))$$

□

In particular, when one has a moduli functor $\mathcal{M} : (\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Sets}$, one can define the Zariski tangent space of $\mathcal{M}$ at $z \in \mathcal{M}(\operatorname{Spec} k)$ to be

$$T_z\mathcal{M} := \{v \in \mathcal{M}(\operatorname{Spec} \frac{k[\epsilon]}{\epsilon^2}) \mid v \mapsto z \text{ under } \mathcal{M}(\operatorname{Spec} \frac{k[\epsilon]}{\epsilon^2}) \to \mathcal{M}(k)\}.$$

We now describe Zariski tangent spaces of Quot schemes at $\mathbb{C}$-rational points. We first consider the most general non-relative case.

Theorem 15.3. *Let X be a projective $\mathbb{C}$-variety and $\mathcal{H} \in Coh(X)$. Fix a polynomial $P \in \mathbb{Q}[z]$. Let $[\mathcal{H} \twoheadrightarrow \mathcal{F}] \in Quot_{X/\mathbb{C}}(\mathcal{H}, P)(\mathbb{C})$ be a $\mathbb{C}$-rational point, so that*

$$0 \to \mathcal{G} \to \mathcal{H} \to \mathcal{F} \to 0$$

is a short exact sequence of coherent sheaves on X. Then

$$T_{[\mathcal{H}\twoheadrightarrow\mathcal{F}]}\, Quot_{X/\mathbb{C}}(\mathcal{H}, P) \cong \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F}).$$

Proof. Consider the diagram

$$\begin{tikzcd} X \arrow[r, "\iota"] \arrow[d] & X \times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2} \arrow[r, "\pi_X"] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} \mathbb{C} \arrow[r] & \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2} \arrow[r] & \operatorname{Spec} \mathbb{C} \end{tikzcd}$$

We have that $T_{[\mathcal{H}\twoheadrightarrow\mathcal{F}]}\mathrm{Quot}_{X/\mathbb{C}}(\mathcal{H}, P)$ is identified with isomorphism classes of surjections $\pi_X^*\mathcal{H} \twoheadrightarrow \mathcal{Q}$ where $\mathcal{Q} \in Coh(X \times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2})$ is $\operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$-flat and $\mathcal{Q}_0 \cong \mathcal{F}$, such that such surjections pull back to the surjection $\mathcal{H} \twoheadrightarrow \mathcal{F}$.

Note that a coherent sheaf $Q \in Coh(X \times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2})$ is $\operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$-flat if and only if

$$0 \to Q \otimes (\epsilon) \to Q \to Q_{\mathbb{C}} \to 0.$$

where we define $Q_{\mathbb{C}} := Q \otimes_{\mathcal{O}_{X\times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}}} \mathcal{O}_X$. Locally, this short exact sequence is describe in the following way: if M is a finitely generated $A \otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$-module, then the short exact sequence translates to the short exact sequence of $A \otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$-modules,

$$0 \to M \otimes_{A\otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}} \epsilon A \to M \to M \otimes_{A\otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}} A \to 0,$$

where we have the pushout diagram

$$\begin{tikzcd} M \arrow[r] & M \otimes_{A\otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}} A \\ A \otimes_{\mathbb{C}} \frac{\mathbb{C}[\epsilon]}{\epsilon^2} \arrow[u] \arrow[r, "\epsilon\mapsto 0"'] & A \arrow[u] \end{tikzcd}$$

Note that $\mathcal{Q}_{\mathbb{C}} \otimes (\epsilon) = \mathcal{Q} \otimes (\epsilon)$, since $\mathcal{Q} \otimes (\epsilon) \otimes (\epsilon) = 0$. Now, with respect to our $\mathbb{C}$-point, we always have the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & \mathcal{G} \otimes (\epsilon) \arrow[d] \arrow[dr, "\alpha"] & & \iota_* \mathcal{G} \arrow[d] & \\
0 \arrow[r] & \mathcal{H} \otimes (\epsilon) \arrow[r] \arrow[d] & \pi_X^* \mathcal{H} \arrow[r] \arrow[dr, "\beta"'] & \iota_* \mathcal{H} \arrow[r] \arrow[d] & 0 \\
 & \mathcal{F} \otimes (\epsilon) \arrow[d] & & \iota_* \mathcal{F} \arrow[d] & \\
 & 0 & & 0 &
\end{tikzcd}$$

and, in particular, note the morphisms α and β. Since $\beta \circ \alpha = 0$, we can define

$$\widehat{\mathcal{F}} := \frac{\ker \beta}{\operatorname{im} \alpha} \in \operatorname{Coh}(X \times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}),$$

which fits into the short exact sequence

$$0 \to \mathcal{F} \otimes (\epsilon) \to \widehat{\mathcal{F}} \to \iota_* \mathcal{G} \to 0. \tag{1}$$

Remember that we have this short exact sequence a priori – it is merely data associated to our $\mathbb{C}$-point $[\pi_X^* \mathcal{H} \twoheadrightarrow \mathcal{F}]$. We claim that tangent vectors of the $\operatorname{Quot}_{X/\mathbb{C}}(\mathcal{H}, P)$ at $[\pi_X^* \mathcal{H} \twoheadrightarrow \mathcal{F}]$ bijectively correspond with splittings of short exact sequence 1.

Indeed, given any tangent vector

$$[\pi_X^* \mathcal{H} \twoheadrightarrow \mathcal{Q}] \in T_{[\mathcal{H} \twoheadrightarrow \mathcal{F}]} \operatorname{Quot}_{X/\mathbb{C}}(\mathcal{H}, P),$$

since both $\pi_X^* \mathcal{H}$ and $\mathcal{Q}$ are $\operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$-flat, we can construct the diagram of coherent sheaves on $X \times \operatorname{Spec} \frac{\mathbb{C}[\epsilon]}{\epsilon^2}$:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{G} \otimes (\epsilon) \arrow[r] \arrow[d] & \pi_X^* \mathcal{G} \arrow[r] \arrow[d] & \iota_* \mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{H} \otimes (\epsilon) \arrow[r] \arrow[d] & \pi_X^* \mathcal{H} \arrow[r] \arrow[d] & \iota_* \mathcal{H} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F} \otimes (\epsilon) \arrow[r] \arrow[d] & \mathcal{Q} \arrow[r] \arrow[d] & \iota_* \mathcal{F} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

With respect to short exact sequence 1, note that the injection $\iota_*\mathcal{G} \cong \frac{\pi_X^*\mathcal{G}}{\mathcal{G}\otimes(\epsilon)} \hookrightarrow \widehat{\mathcal{F}}$ provides a splitting:

$$0 \to \iota_*\mathcal{G} \to \widehat{\mathcal{F}} \to \iota_*\mathcal{G} \to 0.$$

Note in an abelian category, a left splitting is equivalent to a right splitting which is equivalent to $\widehat{\mathcal{F}}$ being a direct sum. So we see that our tangent vector corresponded to a splitting of short exact sequence 1. We can define the inverse of this map. Suppose we have a splitting of short exact sequence 1:

$$0 \to \mathcal{S} \to \widehat{\mathcal{F}} \to \iota_*\mathcal{G} \to 0,$$

where $\mathcal{S} \cong \iota_*\mathcal{G}$. Then define $\widehat{\mathcal{S}}$ to be the preimage of $\mathcal{S} \subset \widehat{\mathcal{F}} = \frac{\ker\beta}{\operatorname{im}\alpha}$ under the map $\ker\beta \to \frac{\ker\beta}{\operatorname{im}\alpha}$. Then this defines for us a diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{G}\otimes(\epsilon) \arrow[r] \arrow[d] & \widehat{\mathcal{S}} \arrow[r] \arrow[d] & \iota_*\mathcal{G} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{H}\otimes(\epsilon) \arrow[r] \arrow[d] & \pi_X^*\mathcal{H} \arrow[r] \arrow[d] & \iota_*\mathcal{H} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{F}\otimes(\epsilon) \arrow[r] \arrow[d] & \frac{\pi_X^*\mathcal{H}}{\widehat{\mathcal{S}}} \arrow[r] \arrow[d] & \iota_*\mathcal{F} \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

so from the splitting we obtain a tangent vector

$$[\pi_X^*\mathcal{H} \twoheadrightarrow \frac{\pi_X^*\mathcal{H}}{\widehat{\mathcal{S}}}] \in T_{[\mathcal{H}\twoheadrightarrow\mathcal{F}]}\mathrm{Quot}_{X/\mathbb{C}}(\mathcal{H}, P) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F}).$$

Since the maps we've constructed are inverses of each other, we have a bijective correspondence. Furthermore, note that all the possibile splittings of

$$0 \to \mathcal{F}\otimes(\epsilon) \to \widehat{\mathcal{F}} \to \iota_*\mathcal{G} \to 0,$$

is a torsor of

$$\mathrm{Hom}_{\mathcal{O}_{X\times\mathrm{Spec}\,\frac{\mathbb{C}[\epsilon]}{\epsilon^2}}}(\iota_*\mathcal{G}, \mathcal{F}\otimes(\epsilon)) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F})$$

so we have the identification

$$T_{[\mathcal{H}\twoheadrightarrow\mathcal{F}]}\mathrm{Quot}_{X/\mathbb{C}}(\mathcal{H}, P) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F}).$$

$\square$

Corollary 15.4 (Grassmannian Zariski tangent space and tangent bundle)**.** *Consider the Grassmannian* $\mathrm{Gr}(V, r)$*. Consider the* $\mathbb{C}$*-point corresponding to an* r*-dimensional subvector space* $W \subseteq V$*, or equivalently* $V^* \twoheadrightarrow W^*$*. Then*

$$T_{[V^* \twoheadrightarrow W]} \mathrm{Gr}(V, R) \cong \mathrm{Hom}_{\mathcal{O}_{\mathrm{Spec}\,\mathbb{C}}}((V/W)^*, W^*) \cong \mathrm{Hom}_{\mathbb{C}}(W, V/W).$$

In particular, if we consider the tautological short exact sequence on the $G := \mathrm{Gr}(V, r)$*:*

$$0 \to \mathcal{S} \to \mathcal{O}_G \otimes V \to \mathcal{Q} \to 0,$$

which is the dual of

$$0 \to \mathcal{Q}^\vee \to \mathcal{O}_G \otimes V^* \to \mathcal{S}^\vee \to 0,$$

where $\mathcal{S}^\vee$ *is the universal bundle on* $\mathrm{Gr}(V, r)$ *(with respec to how we've described its functor of points, if we dualized the functor of points description then* $\mathcal{S}$ *would be universal), then the tangent bundle is the* Hom*-sheaf* $T_G \cong \mathrm{Hom}(\mathcal{S}, \mathcal{Q})$*.*

The following requires a lot more deformation theory to prove.

Theorem 15.5. [HL, Proposition 2.2.8] *Let* X *be a projective* $\mathbb{C}$*-variety and* $\mathcal{H} \in Coh(X)$*. Fix a polynomial* $P \in \mathbb{Q}[z]$*. Let* $[q : \mathcal{H} \twoheadrightarrow F] \in Quot_{X/\mathbb{C}}(\mathcal{H}, P)(\mathbb{C})$ *be a* $\mathbb{C}$*-rational point. Let* $K := \ker q$*. Then*

$$\dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_X}(K, F) \geq \dim_{[q]} Quot_{X/\mathbb{C}}(\mathcal{H}, P) \geq \dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_X}(K, F) - \dim_{\mathbb{C}} Ext^1(K, F).$$

If equality holds in the latter inequality, then $Quot_{X/\mathbb{C}}(\mathcal{H}, P)$ *is a local complete intersection near* $[q]$*.*

If $Ext^1(K, F) = 0$*, then* $Quot_{X/\mathbb{C}}(\mathcal{H}, P)$ *is smooth at* $[q]$*.*

Remark 15.6. *Suppose* X *is a projective variety. Then*

$$Quot_{X/\mathbb{C}}(\mathcal{O}_X, P)$$

is the fine moduli space parameterizing closed subschemes of X *with Hilbert polynomial* P*. In particular, if we consider the closed point* $[\mathcal{O}_X \twoheadrightarrow \iota_* \mathcal{O}_Z] \in Quot_{X/\mathbb{C}}(\mathcal{O}_X, P)(\mathbb{C})$ *where* $Z \subseteq X$ *is some closed subscheme (with the appropriate Hilbert polynomial* P*), then we have the canonical isomorphism*

$$T_{[Z]} Quot_{X/\mathbb{C}}(\mathcal{O}_X, P) \cong \mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z).$$

But note that $\mathrm{Hom}_{\mathcal{O}_X}(I_Z, \mathcal{O}_Z) \cong \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z)$*. Then if* X *and* Z *are both smooth, we have that*

$$T_{[Z]} Quot_{X/\mathbb{C}}(\mathcal{O}_X, P) \cong \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{I}_Z/\mathcal{I}_Z^2, \mathcal{O}_Z) \cong H^0(Z, \mathcal{N}_{Z/X}).$$

Before we move on to discussing the Zariski tangent space at a closed point of the Quot scheme in the most general setting, let's discuss Fogarty's result as an example. Recall that given a projective $\mathbb{C}$-scheme X, the Hilbert scheme of n points on X is

$$X^{[n]} := \mathrm{Quot}_{X/\mathbb{C}}(\mathcal{O}_X, n).$$

Lemma 15.7. [GF05, Lemma 3.1] *Let X be a projective scheme. Suppose X is connected. Then $X^{[n]}$ is geometrically connected (connected as a scheme and has $\mathbb{C}$-rational point).*

Proposition 15.8. *Let X be a connected smooth projective curve or surface. Then $X^{[n]}$ is smooth.*

Proof. Note that $X^{[n]}$ is generically smooth, so to show smoothness it suffices to show that all the tangent spaces at each $\mathbb{C}$-point have the same dimensions.

Suppose first that X is a smooth projective curve C. Let $\iota : Z \hookrightarrow X$ be a closed subscheme of length n, i.e. Z is an effective sum of points whose degree is n. Note that when Z is smooth, we have

$$T_{[Z]}\mathrm{Quot}_{C/\mathbb{C}}(\mathcal{O}_C, n) = H^0(Z, \mathcal{N}_{Z/C}) = n,$$

which means that for any $[Z] \in \mathrm{Quot}_{C/\mathbb{C}}(\mathcal{O}_C, n)(\mathbb{C})$ we have $\dim_{\mathbb{C}} T_{[Z]}\mathrm{Quot}_{C/\mathbb{C}}(\mathcal{O}_C, n) \geq n$, since $X^{[n]}$ is connected. Thus, to show smoothness of $X^{[n]}$, it suffices to show that for any $[Z]$ we have $\dim_{\mathbb{C}} T_{[Z]}\mathrm{Quot}_{C/\mathbb{C}}(\mathcal{O}_C, n) = \dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \leq n$. Note the short exact sequence

$$0 \to \mathcal{I}_Z \to \mathcal{O}_C \to \iota_*\mathcal{O}_Z \to 0$$

induces a long exact sequence

$$0 \to \mathrm{Hom}_{\mathcal{O}_C}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_C, \iota_*\mathcal{O}_Z) \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \to \cdots$$

Note that $\mathrm{Hom}_{\mathcal{O}_C}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \cong \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_C, \iota_*\mathcal{O}_Z)$. Then we have an injection

$$0 \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z)$$

Note that by Serre duality, which states that over a smooth projective variety we have $\mathrm{Ext}^i(\mathcal{F}, \mathcal{G}) \cong \mathrm{Ext}^{\dim X - i}(\mathcal{G}, \mathcal{F} \otimes \omega_X)^\vee$, we find that

$$\begin{aligned} \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) &\cong \mathrm{Ext}^0(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z \otimes \omega_C)^\vee \cong \mathrm{Hom}_{\mathcal{O}_C}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z \otimes \omega_C)^\vee \\ &\cong \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{O}_Z, \omega_C|_Z)^\vee \cong \mathbb{C}^n. \end{aligned}$$

So $\dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \leq n$, as desired.

Now suppose X is a smooth projective surface. Let $\iota : Z \hookrightarrow X$ be a closed zero-dimensional subscheme of length n. Note that when Z is smooth, we have

$$\dim_{\mathbb{C}} T_{[Z]}X^{[n]} = h^0(Z, \mathcal{N}_{Z/X}) = 2n.$$

Noting that $X^{[n]}$ is generically smooth and connected, this implies that for any $[Z] \in X^{[n]}(\mathbb{C})$ we have $\dim_{\mathbb{C}} T_{[Z]}X^{[n]} \geq 2n$. Then it suffices to show that for all $[Z]$ we have

$$\dim_{\mathbb{C}} T_{[Z]}X^{[n]} = \dim_{\mathbb{C}} H^0(\mathcal{I}_Z, \mathcal{O}_Z) \leq 2n.$$

Again, we can consider the long exact sequence

$$0 \to \mathrm{Hom}_{\mathcal{O}_C}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_C, \iota_*\mathcal{O}_Z) \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \to \cdots$$

Note that $\mathrm{Hom}_{\mathcal{O}_C}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \cong \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{O}_C, \iota_*\mathcal{O}_Z) \cong \mathbb{C}^n$. Thus, we have the injection

$$0 \to \mathrm{Hom}_{\mathcal{O}_C}(\mathcal{I}_Z, \iota_*\mathcal{O}_Z) \to \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z).$$

It would be convenient if we could show that $\dim_{\mathbb{C}} \mathrm{Ext}^1(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = n$. However, we cannot deduce this as simply as we did in the case that X was a curve by applying Serre duality. Instead, consider the quantity

$$\chi(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = \sum_{i=0}^{2} \dim_{\mathbb{C}} \mathrm{Ext}^i(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z).$$

By Serre duality,

$$\mathrm{Ext}^2(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) \cong \mathrm{Ext}^0(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z \otimes \omega_X)^\vee \cong \mathrm{Hom}_{\mathcal{O}_X}(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z \otimes \omega_X)^\vee$$
$$\cong \mathrm{Hom}_{\mathcal{O}_Z}(\mathcal{O}_Z, \omega_X|_Z)^\vee \cong \mathbb{C}^n.$$

Then it suffices to show that $\chi(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = 0$. Indeed, we can pick a finite resolution of $\iota_*\mathcal{O}_Z$ by locally free sheaves:

$$0 \to \mathcal{E}_\ell \to \cdots \to \mathcal{E}_0 \to \iota_*\mathcal{O}_Z \to 0,$$

which the Euler characteristic is bilinear with respect to:

$$\chi(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = \sum_{j=0}^{\ell} (-1)^j \chi(\mathcal{E}_j, \iota_*\mathcal{O}_Z).$$

Note that for each $0 \le j \le \ell$, we have

$$\chi(\mathcal{E}_j, \iota_*\mathcal{O}_Z) = \sum_{i=0}^{2} (-1)^i \dim_{\mathbb{C}} \mathrm{Ext}^i(\mathcal{E}_j, \iota_*\mathcal{O}_Z) = \sum_{i=0}^{2} (-1)^i \dim_{\mathbb{C}} \mathrm{Ext}^i(\mathcal{O}_X, \mathcal{E}_j^\vee \otimes \iota_*\mathcal{O}_Z)$$
$$= \sum_{i=0}^{2} (-1)^i h^i(X, \mathcal{E}_j^\vee \otimes \iota_*\mathcal{O}_Z) = h^0(X, \mathcal{E}_j^\vee \otimes \iota_*\mathcal{O}_Z) = n \cdot \mathrm{rk}\, \mathcal{E}_j$$

Then

$$\chi(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = \sum_{j=0}^{\ell} (-1)^j n \cdot \mathrm{rk}\, \mathcal{E}_j.$$

Recall how the rank was defined in Definition 13.4. Then we see that

$$\chi(\iota_*\mathcal{O}_Z, \iota_*\mathcal{O}_Z) = n \cdot \mathrm{rk}(\iota_*\mathcal{O}_Z) = 0.$$

□

Remark 15.9. *In the process of proving Fogarty's result for surfaces (Proposition15.8), we utilized the Euler characteristic for a pair of coherent sheaves:*

$$\chi(E, F) = \sum (-1)^i \dim_{\mathbb{C}} \mathit{Ext}^i(E, F).$$

It is bilinear with respect to the pairing, and generalizes the usual Euler characteristic for a coherent sheaf:

$$\chi(\mathcal{O}_X, F) := \sum (-1)^i \dim_{\mathbb{C}} \mathit{Ext}^i(\mathcal{O}_X, F) = \sum (-1)^i h^i(X, F) =: \chi(F),$$

This is useful for solving for higher Ext groups; a common technique is to combine its bilinearity with the fact that one can find a finite resolution of any coherent sheaf by locally free sheaves, which we performed in Proposition 15.8.

Remark 15.10 (Hilbert-Chow morphism)**.** *In the case of curves, one has that* $C^{[n]} \cong \mathrm{Sym}^n C$*. In the case of surfaces* S*, the symmetric product* $\mathrm{Sym}^n S$ *can be singular in general. It is a nontrivial fact that there is a morphism called the Hilbert-Chow morphism*

$$\pi : S^{[n]} \to \mathrm{Sym}^n S$$

which exibits $S^{[n]}$ *as a resolution of singularities of* $\mathrm{Sym}^n S$*. In particular, it is a crepant resolution, which simply means that* $\omega_{S^{[n]}} \cong \pi^* \omega_{\mathrm{Sym}^n S}$ *(so for example, blowing up a plane at a point is not a crepant resolution).*

Finally, we provide some description of Zariski tangent spaces of $\mathbb{C}$-points of Quot schemes in the relative setting.

Theorem 15.11. *Let* $X \to S$ *be a projective morphism of* $\mathbb{C}$*-schemes. Let* $\mathcal{H} \in \mathit{Coh}(X)$ *and fix a polynomial* $P \in \mathbb{Q}[z]$*. Consider the Quot scheme*

$$\pi : \mathit{Quot}_{X/S}(\mathcal{H}, P) \to S.$$

Let $[\mathcal{H} \twoheadrightarrow \mathcal{F}]$ *be some* $\mathbb{C}$*-point over* $t \in S(\mathbb{C})$*, where* $\mathcal{H} \twoheadrightarrow \mathcal{F}$ *is a surjection of coherent sheaves over* X_t*. Then the following is an exact sequence:*

$$0 \to \mathrm{Hom}_{\mathcal{O}_{X_t}}(\mathcal{G}, \mathcal{F}) \to T_{[\mathcal{H} \twoheadrightarrow \mathcal{F}]} \mathit{Quot}_{X/S}(\mathcal{H}, P) \xrightarrow{d\pi} T_t S \to \mathit{Ext}^1(\mathcal{G}, \mathcal{F})$$

At the very least, the injection is clear, since the scheme-theoretic fiber of $\mathrm{Quot}_{X/S}(\mathcal{H}, P) \to S$ over t is $\mathrm{Quot}_{X_t/\mathbb{C}}(\mathcal{H}, P)$, and $T_t S$ is a natural target for the quotient space of this injection.

16. GMS: Openness of purity and semistability, relative Harder-Narasimhan filtration

Throughout this section, if $F \in \mathit{Coh}(X)$ where X projective, with respect to a choice of ample line bundle $\mathcal{O}_X(1)$, we let $\alpha_i(F)$ denote the i-th integral coefficient of the Hilbert polynomial of F, in the sense of Section 13. If F has dimension d, then we define $\widehat{\mu}(F) := \frac{\alpha_{d-1}(F)}{\alpha_d(F)}$ and $\rho(F) := \frac{P(F)}{\alpha_d(F)}$.

In this section, as an application of Quot schemes, we prove the following two fundamental facts:

(1) purity and Gieseker (semi)stability are open properties for flat families;

(2) flat families over an integral base admit a relative Harder-Narasimhan filtration over a dense open locus.

A crucial lemma that we will use to prove both applications is the following.

Lemma 16.1. *Suppose $X \xrightarrow{f} S$ is a projective morphism of Noetherian schemes, and fix $\mathcal{O}_X(1)$ f-very-ample. Suppose $\mathcal{F} \in Coh(X)$ is a S-flat family of Hilbert polynomial $P \in \mathbb{Q}[z]$. Fix some constant c. Then*

$$S = \{P' \in \mathbb{Q}[z] \mid \exists t \in S(\mathbb{C}) \text{ s.t. } \mathcal{F}_t \twoheadrightarrow \mathcal{Q} \text{ pure dim. } d \text{ }, P(\mathcal{Q}) = P' \text{ and } \widehat{\mu}(\mathcal{Q}) \leq c\}$$

is a finite set.

Proof. This is an application of [HL, Grothendieck's Lemma 1.7.9, Page 29] and [HL, Lemma 1.7.6, Page 28]. ◻

Proposition 16.2. *Let $X \xrightarrow{f} S$ be a projective morphism of Noetherian schemes. Let $\mathcal{O}_X(1)$ be a f-very-ample line bundle and let $\mathcal{F} \in Coh(X)$ be S-flat. Then the purity and Gieseker (semi)stability of the $\mathcal{F}_t$, where $t \in S(\mathbb{C})$, is open on S.*

Proof. Since $\mathcal{F}$ is S-flat, the Hilbert polynomials of $\mathcal{F}_t$ are locally constant as a function of $s \in S$. Then it suffices to prove the claim assuming that every $\mathcal{F}_t$ has the same fixed Hilbert polynomial P of degree d.

Let us first show openness of purity. We begin by examining what allows for failure of purity of a $\mathcal{F}_t$. If $\mathcal{F}_t$ were not pure, then there would exist $K \subset \mathcal{F}_t$ with $\alpha_d(K) = 0$. In particular, we can take the saturation K_{sat} of K with respect to $\mathcal{F}_t$, so that we have $0 \to K_{sat} \to \mathcal{F}_t \to \mathcal{Q} \to 0$, with $\mathcal{Q}$ pure of dimension d and $\alpha_d(\mathcal{Q}) = \alpha_d(\mathcal{F}_t)$. In other words, $\mathcal{F}_t$ fails to be pure if and only if there exists a pure d-dimensional quotient $\mathcal{Q}$ such that $\widehat{\mu}(\mathcal{Q}) \leq \widehat{\mu}(\mathcal{F}_t)$ and $\alpha_d(\mathcal{Q}) = \alpha_d(\mathcal{F}_t)$.

Then define the sets

$$A = \{P' \mid \exists t \in S(\mathbb{C}) \text{ s.t. } \exists \mathcal{F}_t \twoheadrightarrow \mathcal{Q} \text{ pure dim. d with } P(Q) = P' \text{ and } \widehat{\mu}(P') = \widehat{\mu}(\mathcal{Q}) \leq \widehat{\mu}(\mathcal{F}_t)\},$$

$$A_{\text{pure}} = \{P' \in A \mid \deg(P' - P) < d\},$$

and let $V_{P'} \subseteq S$ denote the image of the map

$$\text{Quot}_{X/S}(\mathcal{F}, P') \to S.$$

Note this is a projective morphism, so $V_{P'} \subseteq S$ is closed. In particular, note that A is finite by lemma 16.1, so A_{pure} is finite. Then $\bigcup_{P' \in A_{\text{pure}}} V_{P'} \subseteq S$ is a closed subset. Then we see that $\mathcal{F}_t$ is pure for $t \in S(\mathbb{C})$ if and only if $t \in S \setminus \bigcup_{P' \in A_{\text{pure}}} V_{P'}$, which completes the argument.

Now we show openess of Gieseker (semi)stability. We begin by examining what allows for failure of Gieseker semistability of a $\mathcal{F}_t$. For one, $\mathcal{F}_t$ could fail to be pure. If, however, $\mathcal{F}_t$ were pure, it could still fail to be Gieseker semistable if there exists a pure d-dimensional quotient $\mathcal{F}_t \twoheadrightarrow \mathcal{Q}$ with $\rho(Q) < \rho(E)$ by Proposition 13.2. Note that since $\rho(Q) < \rho(E)$, we have $\widehat{\mu}(Q) \le \widehat{\mu}(E)$. Then we define the subset

$$A_{\mathrm{gss}} = \{P' \in A \mid \rho(P') < \rho(P)\}.$$

Note A_{gss} is finite since A is finite. Then we see that $\mathcal{F}_t$ is Gieseker semistable if and only if $t \in S \setminus \bigcup_{P' \in A_{\mathrm{pure}} \cup A_{\mathrm{gss}}} V_{P'}$. This completes the claim. If we replace A_{gss} with

$$A_{\mathrm{gs}} = \{P' \in A \mid \rho(P') \le \rho(P)\},$$

then the same argument demonstrates the claim for Gieseker stability. □

We can also show openness of μ-(semi)stability, which is not written up in [HL].

Proposition 16.3. *Let $X \xrightarrow{f} S$ be a projective morphism of Noetherian schemes. Let $\mathcal{O}_X(1)$ be f-very-ample, and let $\mathcal{F} \in Coh(X)$ be S-flat of Hilbert polynomial P of degree d. Suppose further that X_t has dimension d for general $t \in S$. Then the loci of $t \in S(\mathbb{C})$ such that $\mathcal{F}_t$ is μ-(semi)stable is open.*

Proof. Let's focus first on μ-semistability. Recall that $E \in Coh(A)$ of dimension d for some projective $(A, \mathcal{O}_A(1))$ variety of dimension d is μ-semistable if and only if $T_{d-2}(E) = T_{d-1}(E) = 0$ and for all subsheaves K such that $0 < \operatorname{rk} K < \operatorname{rk} E$ one has $\mu(K) \le \mu(E)$.

First, we show that the loci of $t \in S(\mathbb{C})$ such that $T_{d-2}(\mathcal{F}_t) = T_{d-1}(\mathcal{F}_t) = 0$ is open. In other words, we want to show that torsion only existing in codimension at least 2 is an open property.

First, recall that $T_{d-2}(\mathcal{F}_t) = T_{d-1}(\mathcal{F}_t) = 0$ fails to hold if there exists a subsheaf $K \subset \mathcal{F}_t$ such that $\alpha_d(K) = 0$ but $\alpha_{d-1}(K) \neq 0$. If we take the quotient $0 \to K \to \mathcal{F}_t \to Q \to 0$, then failure is equivalent to there existing a surjection $\mathcal{F}_t \twoheadrightarrow Q$ with $\alpha_d(\mathcal{F}_t) = \alpha_d(Q)$ but $\alpha_{d-1}(\mathcal{F}_t) > \alpha_{d-1}(Q)$. Note if Q is not pure of dimension d, then we can quotient out $T_{d-1}(Q)$, to get a surjection $\mathcal{F}_t \twoheadrightarrow Q \twoheadrightarrow \frac{Q}{T_{d-1}(Q)}$, and note $\alpha_d(\mathcal{F}_t) = \alpha_d(Q) = \alpha_d(\frac{Q}{T_{d-1}(Q)})$ and $\alpha_{d-1}(\mathcal{F}_t) > \alpha_{d-1}(Q) \ge \alpha_{d-1}(\frac{Q}{T_{d-1}(Q)})$. Thus, $T_{d-2}(\mathcal{F}_t) = T_{d-1}(\mathcal{F}_t) = 0$ fails to hold if and only if there exists a pure dimension d quotient $\mathcal{F}_t \twoheadrightarrow Q$ such that $\alpha_d(\mathcal{F}_t) = \alpha_d(Q)$ and $\alpha_{d-1}(\mathcal{F}_t) > \alpha_{d-1}(Q)$.

Then define the sets

$$A = \{P' \mid \exists t \in S(\mathbb{C}) \text{ s.t. } \exists \mathcal{F}_t \twoheadrightarrow \mathcal{Q} \text{ pure dim. d with } P(Q) = P' \text{ and } \widehat{\mu}(P') = \widehat{\mu}(\mathcal{Q}) \le \widehat{\mu}(\mathcal{F}_t)\},$$

$$A_{\mathrm{torsCodim1}} = \{P' \in A \mid \alpha_d(P') = \alpha_d(P) \text{ and } \alpha_{d-1}(P') < \alpha_{d-1}(P)\}.$$

Note A is finite by lemma 16.1, so $A_{\mathrm{torsCodim1}}$ is finite. For every $P' \in A_{\mathrm{torsCodim1}}$, consider the image $V_{P'} \subseteq S$ of the map

$$\mathrm{Quot}_{X/S}(\mathcal{F}, P') \to S.$$

Since this morphism is projective, $V_{P'}$ is a closed subset of S. Then we see that $\mathcal{F}_t$ satisfies $T_{d-2}(\mathcal{F}_t) = T_{d-1}(\mathcal{F}_t)$ if and only if $t \in S \setminus \bigcup_{P' \in A_{\mathrm{torsCodim1}}} V_{P'}$. So the torsion condition for μ-(semi)stability is open.

Now let's study the other way in which $\mathcal{F}_t$ could fail to be μ-semistable. In this other way for failure, there would exist some subsheaf $K \subset \mathcal{F}_t$ with $0 < \operatorname{rk} K < \operatorname{rk} \mathcal{F}_t$ such that $\mu(K) > \mu(\mathcal{F}_t)$. Equivalently, $\mathcal{F}_t$ fails in this way if there exists $K \subset \mathcal{F}_t$ such that $0 < \alpha_d(K) < \alpha_d(\mathcal{F}_t)$ and $\widehat{\mu}(K) > \widehat{\mu}(\mathcal{F}_t)$.

Let's think of how to express this in terms of quotients. If we take $0 \to K \to \mathcal{F}_t \to Q \to 0$, then note $\alpha_d(K) + \alpha_d(Q) = \alpha_d(\mathcal{F}_t)$ and $\alpha_{d-1}(K) + \alpha_{d-1}(Q) = \alpha_{d-1}(\mathcal{F}_t)$. And in particular, $P(K) + P(Q) = P(\mathcal{F}_t)$. Note we have $0 < \alpha_d(K) < \alpha_d(\mathcal{F}_t)$ if and only if $0 < \alpha_d(Q) < \alpha_d(\mathcal{F}_t)$. Stating $\widehat{\mu}(K) > \widehat{\mu}(\mathcal{F}_t)$ ends up being somewhat tautological. But at least note that $\widehat{\mu}(K) > \widehat{\mu}(\mathcal{F}_t)$ implies that $\rho(K) > \rho(\mathcal{F}_t)$ which implies that $\rho(\mathcal{F}_t) > \rho(Q)$ so $\widehat{\mu}(F) \geq \widehat{\mu}(Q)$. Also note that for Q if we take further quotient $Q/T_{d-1}(Q)$, and correspondingly take the saturation K_{sat} of K in $\mathcal{F}_t$, then we see that $\alpha_d(K_{\mathrm{sat}}) = \alpha_d(K)$ but $\alpha_{d-1}(K_{\mathrm{sat}}) \geq \alpha_{d-1}(K)$, which means that $\widehat{\mu}(K_{\mathrm{sat}}) \geq \widehat{\mu}(K) > \widehat{\mu}(\mathcal{F}_t)$.

All in all this means we can define the subset

$$A_{\mu-\mathrm{ss}} = \{P' \in A \mid 0 < \alpha_d(P') < \alpha_d(P) \text{ and } \widehat{\mu}(P - P') > \widehat{\mu}(P)\} \subseteq A.$$

Then for $t \in S(\mathbb{C})$, we have $\mathcal{F}_t$ is μ-semistable if and only if $t \in S \setminus \bigcup_{P' \in A_{\mathrm{torsCodim1}} \cup A_{\mu-\mathrm{ss}}}$ and X_t has dimension d. So we see μ-semistability is an open condition. To see this for μ-stability, simply replace $A_{\mu-\mathrm{ss}}$ with the subset

$$A_{\mu-\mathrm{s}} = \{P' \in A \mid 0 < \alpha_d(P') < \alpha_d(P) \text{ and } \widehat{\mu}(P - P') \geq \widehat{\mu}(P)\} \subseteq A.$$

□

Proposition 16.4. *Let $X \xrightarrow{f} S$ be a projective morphism of Noetherian scheme, where S is integral. Fix a f-very-ample line bundle $\mathcal{O}_X(1)$. Let $\mathcal{F} \in Coh(X)$ be an S-flat family of coherent sheaves on the fibers of f, all with the same fixed Hilbert polynomial P of degree d. Then there is a projective birational morphism $g : T \to S$ of integral $\mathbb{C}$-schemes*

$$\begin{array}{ccc} T \times_S X & \xrightarrow{g_X} & X \\ \downarrow & & \downarrow f \\ T & \xrightarrow{g} & S \end{array}$$

and a filtration

$$0 \subset HN_1(g_X^*\mathcal{F}) \subset \cdots \subset HN_\ell(g_X^*\mathcal{F}) = g_X^*\mathcal{F}$$

such that

- *each $\frac{HN_i(g_X^*\mathcal{F})}{HN_{i-1}(g_X^*\mathcal{F})}$ is T-flat and*
- *there is a dense open subscheme $U \subset T$ such that for every $t \in U(\mathbb{C})$, one has $HN_i(g_X^*\mathcal{F})_t = HN_i(\mathcal{F}_{g(t)})$.*

Furthermore, $g : T \to S$ is universal in the sense that for any other projective birational $g' : T' \to S$ satisfying the above properties, g' factors as some $T' \to T \xrightarrow{g} S$.

Proof. Here's our strategy.

(1) First, we show that for any such $\mathcal{F}$, there exists a projective birational morphism $g_1 : T_1 \to S$ of integral $\mathbb{C}$-schemes such that there is a quotient map $(g_1)_X^*\mathcal{F} \twoheadrightarrow \mathcal{Q}_1$ of coherent sheaves on $T_1 \times_S X$ such that $\mathcal{Q}_1$ is T_1-flat and there is a dense open $U \subset T_1$ such that for all $t \in U(\mathbb{C})$, we have that $Q_{1,t} \cong (g_1)_X^*[\frac{\mathrm{HN}_{\max}(\mathcal{F}_{g_1(t)})}{\mathrm{HN}_{\max-1}(\mathcal{F}_{g_1(t)})}]$.

(2) Second, we show that such a $g_1 : T_1 \to S$ is universal in the following sense: for any other $g_1' : T_1' \to S$ admitting a quotient $(g_1')_X^*\mathcal{F} \twoheadrightarrow \mathcal{Q}_1'$ such that $\mathcal{Q}_1'$ is T_1' flat and such that there exists a dense open subscheme $U \subset T_1'$ such that for all $t \in U(\mathbb{C})$ we have that $Q'_{1,t} \cong (g_1')_X^*[\frac{\mathrm{HN}_{\max}(\mathcal{F}_{g_1'(t)})}{\mathrm{HN}_{\max-1}(\mathcal{F}_{g_1'(t)})}]$, (*deep breath*) the map $g_1' : T_1' \to S$ factors through as some $T_1' \to T_1 \xrightarrow{g_1} S$.

If we show the above two things then we are done. Because then we can iterate the procedure, i.e. once we obtain $g_1 : T_1 \to S$, and we can perform the same procedure on $(\ker[(g_1)_X^*\mathcal{F} \twoheadrightarrow \mathcal{Q}_1], T_1 \times_S X \to T_1)$, replacing $(\mathcal{F}, X \to S)$. Note that Q_1 is T_1-flat and since $\mathcal{F}$ is S-flat then $(g_1)_X^*\mathcal{F}$ is T_1-flat, so $\ker[(g_1)_X^*\mathcal{F} \twoheadrightarrow \mathcal{Q}_1]$ is T_1-flat. The procedure must terminate since the Harder-Narasimhan filtration for some isomorphic copy of $\mathcal{F}_t$ in the eventual desired open locus is finite. Then we will define $g : T \to S$ to be the composition

$$g : T_\ell \xrightarrow{g_\ell} T_{\ell-1} \to \cdots \to T_1 \to S,$$

which will have the desired properties. Note that in order to say that the global filtration restricted to $\mathbb{C}$-points of the eventual open locus gives the Harder-Narasimhan filtration of the restriction of $g_X^*\mathcal{F}$, we are using [HL, Lemma 2.1.4], so we see that $\mathcal{F}$ being S-flat is a crucial assumption.

Let's first show item (1). Consider the set

$$A = \{P' \mid \exists t \in S(\mathbb{C}) \text{ s.t. } \exists \mathcal{F}_t \twoheadrightarrow \mathcal{Q} \text{ pure dim. d with } P(Q) = P' \text{ and } \widehat{\mu}(P') = \widehat{\mu}(\mathcal{Q}) \le \widehat{\mu}(\mathcal{F}_t)\},$$

Note that A is finite by lemma 16.1. For each $P' \in A$, let $V_{P'} \subseteq S$ denote the image of the map $\mathrm{Quot}_{X/S}(\mathcal{F}, P')$; since it is the image of a projective morphism, $V_{P'}$ is closed. Furthermore, note that $\bigcup_{P' \in A} V_{P'} = S$. This is because for any $t \in S(\mathbb{C})$, we always have the surjection $\mathcal{F}_t \twoheadrightarrow \frac{\mathcal{F}_t}{T_{d-1}(\mathcal{F}_t)}$ and $P(\frac{\mathcal{F}_t}{T_{d-1}(\mathcal{F}_t)}) \in A$. This is because $\frac{\mathcal{F}_t}{T_{d-1}(\mathcal{F}_t)}$ is pure of dimension d and since $\alpha_d(\frac{\mathcal{F}_t}{T_{d-1}(\mathcal{F}_t)}) = \alpha_d(\mathcal{F}_t)$, we find that $\widehat{\mu}(\frac{\mathcal{F}_t}{T_{d-1}(\mathcal{F}_t)}) \le \widehat{\mu}(\mathcal{F}_t)$.

Here comes the first reason why we want S to be integral: since S is integral it is irreducible, which implies that there exists at least one $P' \in A$ such that $V_{P'} = S$. Define

$$A_{\mathrm{surj}} = \{P' \in A \mid V_{P'} = S\}.$$

Define the following order relation $\preceq$ on A_{surj}: we say that $P_1' \preceq P_2'$ if and only if $\rho(P_1') < \rho(P_2')$ or $P_1' > P_2'$ when $\rho(P_1') = \rho(P_2')$. This order relation $\preceq$ totally orders A_{surj}, and thus there exists an absolute minimal element which we call $P_- \in A_{\mathrm{surj}}$.

Finally, define the set

$$A_{\mathrm{rej}} = \{P' \in A \setminus A_{\mathrm{surj}} \mid \rho(P') < \rho(P_-)\}.$$

Then consider $U = S \setminus \bigcup_{P' \in A_{\text{rej}}} V_{P'}$. Note that $\text{Quot}_{X/S}(\mathcal{F}, P_-)$ surjects onto S. So for every $t \in S(\mathbb{C})$, we have there exist some $\mathcal{F}_t \twoheadrightarrow \mathcal{Q}_t$ where $P(\mathcal{Q}_t) = P_-$.

We claim that $[\mathcal{F}_t \twoheadrightarrow \mathcal{Q}_t] \cong [\mathcal{F}_t \twoheadrightarrow \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}]$ if and only if $t \in U$. First we show the forward direction. We show it by contradiction. Suppose we have the isomorphism but $t \notin U$. This implies $\mathcal{F}_t$ admits a quotient $\mathcal{F}_t \twoheadrightarrow \mathcal{Q}'_t$ where $P(\mathcal{Q}'_t) \in A_{\text{rej}}$. Among the $P' \in A_{\text{rej}}$ such that $t \in V_{P'}$, let $\mathcal{F}_t \twoheadrightarrow \mathcal{Q}'_t$ be so that $\rho(\mathcal{Q}'_t)$ is minimal. Then $\mathcal{Q}'_t$ is forced to be pure and semistable, otherwise the minimality of $\rho(Q'_t)$ would be violated. Then we have found a semistable quotient $\mathcal{Q}'_t$ of $\mathcal{F}_t$ such that $\rho(\mathcal{Q}'_t) < \rho(\mathcal{Q}_t) = \rho_{\min}(\mathcal{F}_t)$, contradicting the uniqueness of the Harder-Narasimhan filtration of $\mathcal{F}_t$. Thus, we must have $t \in U$. On the other hand, if $t \in U$, then we claim that $[\mathcal{F}_t \twoheadrightarrow \mathcal{Q}_t] \cong [\mathcal{F}_t \twoheadrightarrow \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}]$. Indeed, firstly $\mathcal{Q}_t$ must be pure, otherwise we could take a further quotient

$$\mathcal{F}_t \twoheadrightarrow \mathcal{Q}_t \twoheadrightarrow \frac{\mathcal{Q}_t}{T_{d-1}(\mathcal{Q}_t)},$$

and we have that $\alpha_d(\mathcal{Q}_t) = \alpha_d(\frac{\mathcal{Q}_t}{T_{d-1}(\mathcal{Q}_t)})$ while $P(\mathcal{Q}_t) > P(\frac{\mathcal{Q}_t}{T_{d-1}(\mathcal{Q}_t)})$, so $\rho(Q_t) > \rho(\frac{\mathcal{Q}_t}{T_{d-1}(\mathcal{Q}_t)})$. Then $P(\frac{\mathcal{Q}_t}{T_{d-1}(\mathcal{Q}_t)}) \in A$ and either it is in A_{surj} or A_{rej}. It cannot be in A_{surj} by minimality of P_-, so it would have to be in A_{rej} but this contradicts our assumption that $t \in U$. Thus, we must have that $\mathcal{Q}_t$ is pure. Furthermore, it is semistable, because for any proper pure d-dimensional quotient $\mathcal{Q}_t \twoheadrightarrow \mathcal{Q}^o_t$, we can compose to a surjection $\mathcal{F}_t \twoheadrightarrow \mathcal{Q}^o_t$. If $\rho(\mathcal{Q}^o_t) < \rho(Q_t) = \rho(P_-)$, then by minimality of P_- we'd have $P(\mathcal{Q}^o_t) \in A_{\text{rej}}$ but this would contradict t being in U. Thus, Q_t must be semistable. Now that we've shown Q_t is Gieseker semistable, we state the conclusion of why $[\mathcal{F}_t \twoheadrightarrow \mathcal{Q}_t] \cong [\mathcal{F}_t \twoheadrightarrow \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}]$. If we did not have this isomorphism, then we'd be forced to have $\rho(Q_t) > \rho(\frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)})$ by uniqueness of the Harder-Narasimhan filtration. But then $\mathcal{F}_t$ would admit a semistable quotient with smaller reduced Hilbert polynomial than $\rho(Q_t) = \rho(P_-)$, which would violate our assumption that $t \in U$.

Then with respect to the morphism $\pi : \text{Quot}_{X/S}(\mathcal{F}, P_-) \to S$, we restrict to

$$\pi^{-1}(U) \to U.$$

By uniqueness of Harder-Narasimhan filtration, this map is bijective. Note the map is also proper. Furthermore, for any $[\mathcal{F}_t \twoheadrightarrow \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}] \in \pi^{-1}(U)(\mathbb{C})$, we have that

$$T_{[\mathcal{F}_t \twoheadrightarrow \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}]}\text{Quot}_{X/S}(\mathcal{F}, P_-) \cong \text{Hom}_{\mathcal{O}_{X_t}}(\text{HN}_{\max-1}(\mathcal{F}_t), \frac{\text{HN}_{\max}(\mathcal{F}_t)}{\text{HN}_{\max-1}(\mathcal{F}_t)}) = 0,$$

which implies that the relative cotangent sheaf $\Omega_{\pi^{-1}(U)/U}$ is 0. Then the map $\pi^{-1}(U) \to U$ is unramified, bijective, and proper. Here comes the second reason for our integrality of the base assumption: since the target U is integral and thus reduced, this implies $\pi^{-1}(U) \to U$ is an isomorphism. Finally, let T_1 denote the closure of $\pi^{-1}(U)$ in $\text{Quot}_{X/S}(\mathcal{F}, P_-)$ with reduced scheme structure. Then $g_1 : T_1 \to S$ is a projective birational morphism of integral schemes satisfying item (1).

Now we show item (2), universality. Suppose $g_1' : T_1' \to S$ is any other morphism with the obvious properties. Suppose that over open $U' \subset S$ we have $(g_1')^{-1}(U') \to U'$ is an isomorphism such that the surjection $(g_1')_X^* \mathcal{F} \twoheadrightarrow \mathcal{Q}_1'$ is such that for $t \in (g_1')^{-1}(U')(\mathbb{C})$ we have $Q_{1,t}' \cong (g_1')_X^* [\frac{\mathrm{HN}_{\max}(\mathcal{F}_{g_1'(t)})}{\mathrm{HN}_{\max -1}(\mathcal{F}_{g_1'(t)})}]$. Note by universality of the Quot scheme,

$$\begin{tikzcd} & \mathrm{Quot}_{X/S}(\mathcal{F}, P_-) \arrow[d, "\pi"] \\ T_1' \arrow[ur, "\theta"] \arrow[r, "g_1'"'] & S \end{tikzcd}$$

g_1' factors through some $\theta : T_1' \to \mathrm{Quot}_{X/S}(\mathcal{F}, P_-)$. The map θ restricts to a morphism

$$(g_1')^{-1}(U' \cap U) \to \pi^{-1}(U)$$

over S, and further restricts to an isomorphism $(g_1')^{-1}(U' \cap U) \cong \pi^{-1}(U \cap U')$ over $U' \cap U \subseteq S$. Since $g_1' : T_1' \to S$ is projective and T_1' is integral, note that the image of θ is an integral closed subscheme of $\mathrm{Quot}_{X/S}(\mathcal{F}, P_-)$ which contains the open subscheme $U \subseteq T_1$. In particular, the image of θ must then contain T_1. In fact, the image must be T_1. Otherwise, $\mathrm{im}\ \theta = (\mathrm{im}\ \theta \cap [\mathrm{Quot}_{X/S}(\mathcal{F}, P_-) \setminus U]) \cup T_1$, contradicting the irreducibility of $\mathrm{im}\ \theta$. Thus, $g_1' : T_1' \to S$ factors through $g_1 : T_1 \to S$.

□

Remark 16.5. *There is probably an analogous statement of Proposition 16.4 in terms of μ-(semi)stability.*

17. GMS: Hilbert-Mumford criterion

In section 10 we showed that the sheafification of the moduli functor $\underline{\mathrm{Bun}}_C^{ss}(r,d)$ is isomorphic to the sheafification of a quotient moduli functor obtained by the group action of GL_V on a certain GL_V-invariant open locus of a certain Quot scheme. We state this more generally in the higher-dimensional case of semistable sheaves on projective $(X, \mathcal{O}_X(1))$ later on. The eventual goal is to show that a scheme co-represents the moduli functor $\underline{\mathrm{Bun}}_C^{ss}(r,d)$ and its natural generalization to higher-dimesional $(X, \mathcal{O}_X(1))$. In order to do this, we will actually restrict to an action of SL_V on a loci R^{ss} of a Quot scheme parameterizing certain semistable sheaves equipped with global section data. While we did not show it in section 10 (nor did we show yet why we need to restrict to a SL_V action), this R^{ss} corresponds to the GIT semistable locus with respect to the SL_V action. With respect to this section, the point of saying all of this is that projective GIT is an important procedure for constructing the co-representing moduli spaces of Gieseker semistable sheaves, and we would like to clarify how R^{ss} corresponds to the Gieseker semistable locus, and in turn, motivate the definition of Gieseker semistability. This section then is about understanding semistable points in the projective GIT situation better, and motivating a criterion for determining such points.

In section 4 we gave the basics of GIT. Recall the setup. Let G be a reductive algebraic group over $\mathbb{C}$ (or any algebraically closed field of characteristic 0) acting a projective $\mathbb{C}$-variety X. Suppose $X \to \mathbb{P}(V)$ is a G-linear equivariant embedding with respect to a G-linearisation $G \to \mathrm{GL}_V$. Then recall that we take the cone $\tilde{X} = \operatorname{Spec} S \subseteq V$ where $S = \bigoplus_{m=0}^{\infty} H^0(X, \mathcal{O}_X(m))$ and G preserves the grading of S. Then we take the G-invariant functions S^G and then by projective GIT, there is a universal categorical quotient with all the properties of Proposition 4.3 (but in the non-affine case)

$$X^{ss} \to \mathrm{Proj} S^G =: X//G,$$

where the semistable locus X^{ss} is comprised of all $p \in X$ such that there exists positive-degree homogeneous $f \in S_r^G$ such that $f(p) \neq 0$. This quotient morphism restricts to an actual geometric quotient

$$X^s \to \mathrm{im}\ (X^s)$$

where $p \in X^s$ if and only if $p \in X^{ss}$ and the $\dim G_p = 0$ where $G_p \subset G$ is the stabilizer of p and the orbit Gp is closed in X^{ss}.

In section 4.3, we saw several examples where we could determine X^{ss} because we could compute S^G. But in practice, it can be extremely difficult to compute S^G. So we'll want to have some alternative descriptions of X^{ss} and X^s. Note that $X^{ss} = X \cap \mathbb{P}(V)^{ss}$ and $X^s = X \cap \mathbb{P}(V)^s$. So it suffices to provide these alternative descriptions just in the situation of $\mathbb{P}(V)$ and some morphism $G \to \mathrm{GL}_V$ where G is reductive. The first is called the topological Hilbert-Mumford criterion.

Proposition 17.1 (Topological Hilbert-Mumford criterion)**.** [Hos23, Proposition 3.1] *Let $x = [v] \in \mathbb{P}(V)$. Then*

- *x is semistable if and only if $0 \notin \overline{G \cdot v}$.*
- *x is stable if and only if $\dim G_v = 0$ and $G \cdot v$ closed in V.*

Proof. We prove only the first statement. Suppose x is semistable. Then there exist some positive degree homogeneous $f \in V_+^G$ such that $f(x) \neq 0$. Since f is G-invariant, we have $f|_{G \cdot v}$ is some non-zero constant function. But if $0 \in \overline{G \cdot v}$, we will have a discontinuous jump since $f(0) = 0$. So we must have $0 \notin \overline{G \cdot v}$.

Now suppose $0 \notin \overline{G \cdot v}$. Then 0 and $\overline{G \cdot v}$ are disjoint G-invariant closed subschemes of V. Then by affine GIT, they collapse to different closed points z_1 and z_2, respectively, of $\operatorname{Spec} V^G$. But then there is a G-invariant function $f \in V^G$ such that $f(z_2) \neq 0 \implies f(v) \neq 0$. We can write f as a sum of its homogeneous parts, and thus we will find some homogeneous positive degree f_+ such that $f_+(v) \neq 0$, as desired.

Not that the statements holds for any choice of lift $v \in V$ of $x = [v] \in \mathbb{P}(V)$. ◻

The particular case of $G = \mathbb{G}_m$ admits a nice enough description that we can describe semistable points with respect to a $\mathbb{G}_m$ action by some numerical criterion. The special fact about $\mathbb{G}_m$ is that for any[7] $\mathbb{G}_m \to \mathrm{GL}_V$, we have a finite decomposition $V = \bigoplus_{d\in\mathbb{Z}} V_d$ such that $t \in \mathbb{G}_m$ acts on $v \in V_d$ via $t \cdot v = t^d v$. For $x = [v] \in \mathbb{P}(V)$, we define

$$\mathrm{wt}_{\mathbb{G}_m}(x) := \mathrm{wt}_{\mathbb{G}_m}(v) = \{d \mid v = \sum v_d \text{ where } 0 \neq v_d \in V_d\} \subset \mathbb{Z}.$$

Proposition 17.2. *Let $x = [v] \in \mathbb{P}(V)$. Suppose we have a linear $\mathbb{G}_m \to \mathrm{GL}_V$ action. Then*

- *x is semistable if and only if $0 \in conv(wt_{\mathbb{G}_m}(x))$.*
- *x is stable if and only if $0 \in int(conv(wt_{\mathbb{G}_m}(x)))$.*

Here conv denotes taking the $\mathbb{R}$-convex hull of the given vertices.

Proof.

- We know that x is semistable if and only if $0 \notin \overline{\mathbb{G}_m \cdot v}$, by Proposition 17.1,. What are the limit points of $\mathbb{G}_m \cdot v$? Choose a lift $v \in V$ of x. If $v = \sum_d v_d$, then the limit points are
$$\lim_{t\to 0} t \cdot v = \lim_{t\to 0} \sum t^d v_d \text{ and } \lim_{t\to\infty} t \cdot v = \lim_{t\to\infty} \sum t^d v_d.$$
Then 0 not being a limit point means that
$$\lim_{t\to 0} \sum t^d v_d \neq 0 \text{ and } \lim_{t\to\infty} \sum t^d v_d \neq 0.$$
This is true if and only if the d cannot be all positive or all negative, which is equivalent to saying that $0 \in \mathrm{conv}(\mathrm{wt}_{\mathbb{G}_m}(x))$. This shows the claim.
- Recall x is stable if and only if x is semistable, $\dim(\mathbb{G}_m)_x = 0$, and $\mathbb{G}_m \cdot x$ is closed in $\mathbb{P}(V)^{ss}$. Note that the limit points of $\mathbb{G}_m \cdot x$ are the following: if $v = \sum_d v_d inV$ is a lift of x, then we have
$$\lim_{t\to 0} t \cdot x = [\lim_{t\to 0} \sum_d \frac{t^d v_d}{t^{\min d}}] = [v_{\min d}]$$
$$\lim_{t\to\infty} t \cdot x = [\lim_{t\to\infty} \sum_d \frac{t^d v_d}{t^{\max d}}] = [v_{\max d}].$$
Now suppose x is stable. Then it is semistable, so $0 \in \mathrm{conv}(\mathrm{wt}_{\mathbb{G}_m}(x))$. Among the d appearing in $v = \sum_d v_d$, consider $\min d$ and $\max d$. We have so far that $\min d \leq 0$ and $0 \leq \max d$. Suppose $\min d = 0$. This implies $\lim_{t\to 0} t \cdot x = [v_0]$. But if x is stable, then we must have $[v_0] \in \mathbb{G}_m \cdot x$. So there exist t so that $\sum_d t^d v_d = \lambda v_0$. But this only occurs if $\max d = 0$. But then $x = [v_0]$, which is a fixed point of $\mathbb{G}_m$, so $(\mathbb{G}_m)_x = \mathbb{G}_m$, which is not 0 dimensional, contradicting x being stable. So $\min d \neq 0$, so $\min d < 0$. The same argument shows that $\max d \neq 0$, so $0 < \max d$. Thus, x being stable implies $0 \in \mathrm{int}(\mathrm{conv}(\mathrm{wt}_{\mathbb{G}_m}(x)))$.

  On the other hand, suppose $0 \in \mathrm{int}(\mathrm{conv}(\mathrm{wt}_{\mathbb{G}_m}(x)))$. Then clearly x is semistable. Since $\min d < 0 < \max d$, if we consider setting $[t \cdot v] = [v]$, then only finitely many $t \in \mathbb{G}_m$ can do this, so $\dim(\mathbb{G}_m)_x = 0$. Finally, as we computed above, we have

[7]This generalizes to any morphism from a torus T to GL_V [Spe].

$\lim_{t\to 0} t\cdot x = [v_{\min}]$ and $\lim_{t\to\infty} t\cdot x = [v_{\max}]$. But by the criterion for semistability, we see that neither of these limit points can be in the stable locus. Thus, $\mathbb{G}_m \cdot x$ is closed in $\mathbb{P}(V)$. Thus, x is stable.

□

So we like $\mathbb{G}_m$ actions because the representation theory of $\mathbb{G}_m$ makes their semistable loci amendable to a numerical criterion. But there's another reason. Returning to the general case of reductive G, if $x \in \mathbb{P}(V)$ is semistable with respect to G, then x is semistable with respect to any subgroup $H \subseteq G$. In particular, any $\mathbb{G}_m \subseteq H$. The Hilbert-Mumford criterion may be thought of as a converse; that the semistability of a point with respect to G can be tested against $\mathbb{G}_m$'s.

Definition 17.3. *Let G be a reductive group. We call any nontrivial homomorphism $\lambda : \mathbb{G}_m \to G$ a 1 parameter subgroup (1-PS).*

Definition 17.4. *Let G be a reductive group admitting a G-linearisation $G \to \mathrm{GL}_V$. Let $\lambda : \mathbb{G}_m \to G$ be a 1-PS. Define the Hilbert-Mumford weight of $x = [v] \in \mathbb{P}(V)$ with respect to λ to be*

$$\mu(x,\lambda) := -\min\{d \mid \sum_d v_d \text{ where } v_d \neq 0\},$$

where $V = \bigoplus V_d$ is the decomposition with respect to $\lambda(t)$.

Note that $\mu(x,\lambda)$ is the unique integer such that

$$\lim_{t\to 0} t^{\mu(x,\lambda)}\lambda(t)\cdot x \neq 0 \text{ and exists .}$$

This is because $\lim_{t\to 0} t^{\mu(x,\lambda)}\lambda(t)\cdot x = \lim_{t\to 0}\sum \frac{t^d v_d}{t^{\min d}} = v_{\min d}$. If $\mu(x,\lambda)$ were any smaller, then the limit wouldn't exist. Any larger, and the limit would be 0. Here are some other facts.

Proposition 17.5. *Let $\lambda : \mathbb{G}_m \to G$ be a 1-PS and $G \to \mathrm{GL}_V$ a G-linearisation. Then*

- $\mu(x,\lambda^n) = n\mu(x,\lambda)$
- *Denoting $x_0 := \lim_{t\to 0}\lambda(t)\cdot x \in \mathbb{P}(V)$, we have $\mu(x,\lambda) = \mu(x_0,\lambda)$.*
- $\mu(gx, g\lambda g^{-1}) = \mu(x,\lambda)$ *for all $g \in G$.*

Proof.

- Note $\lambda^n(t) := \lambda(t)^n$. So if we have the decomposition $V = \bigoplus V_d$ with respect to the λ, then the decomposition with respect to λ^n is $V = \bigoplus V_{nd}$. Then given $x = [v] \in \mathbb{P}(V)$, if $v = \sum_d v_d$ with respect to λ, then $v = \sum_d v_{nd}$ with respect to λ^n. This shows $\mu(x,\lambda^n) = n\mu(x,\lambda)$.
- Note $x_0 = \lim_{t\to 0}\lambda(t)\cdot x = [v_{\min}]$. So $\mu(x,\lambda) = \mu(x_0,\lambda)$.
- Have $\lim_{t\to 0} t^n g\lambda(t)g^{-1}\cdot gv = \lim_{t\to 0} t^n g\lambda(t)\cdot v$, so we see that the n making this limit nonzero and exist is $-d$ where d is the minimum index of $g\lambda(t)\cdot v$ which is the minimal index for $\lambda(t)\cdot v$ which is $-\mu(x,\lambda)$. This implies the claim.

□

Now here's the Hilbert-Mumford criterion for $\mathbb{G}_m$ actions, i.e. semistability with respect to a $\mathbb{G}$m actions is completed determined by Hilbert-Mumford weights of 1-parameter subgroups.

Proposition 17.6. *Let $\mathbb{G}_m \to \mathrm{GL}_V$ be a linearisation. Let $\lambda : \mathbb{G}_m \to \mathbb{G}_m$, where $t \mapsto t$, be a 1-PS (the identity 1-PS). Let $x = [v] \in \mathbb{P}(V)$.*

- *x is semistable if and only if $\mu(x, \lambda) \geq 0$ and $\mu(x, \lambda^{-1}) \geq 0$.*
- *x is stable if and only if $\mu(x, \lambda) > 0$ and $\mu(x, \lambda^{-1}) > 0$.*

Proof.

- Suppose $v = \sum_d v_d$ with respect to the decomposition $V = \bigoplus_d V_d$ given by the $\mathbb{G}_m$ action. Recall x is semistable if and only if $0 \notin \overline{\mathbb{G}_m \cdot v}$, if and only if
$$\lim_{t\to 0} \lambda(t) \cdot v \neq 0 \text{ and } \lim_{t\to\infty} \lambda(t) \cdot v \neq 0,$$
if and only if $\mu(x, \lambda) \geq 0$ and $\mu(x, \lambda^{-1}) \geq 0$, respectively.
- Recall x is stable if and only if $0 \in \mathrm{int}(\mathrm{conv}(\mathrm{wt}_{\mathbb{G}_m}(x)))$. So $\min d < 0 < \max d$. Then this is exactly saying that $\mu(x, \lambda) = -\min d > 0$ and $\mu(x, \lambda^{-1}) = \max d > 0$.

$\square$

Note that any 1-PS $\lambda : \mathbb{G}_m \to \mathbb{G}_m$ of a linearisation $\mathbb{G}_m \to \mathrm{GL}_V$ is given by $\operatorname{Spec} \mathbb{C}[t^{\pm 1}] \to \operatorname{Spec} \mathbb{C}[t^{\pm 1}]$ where $t \mapsto t^a$ for some a. But this implies that any 1-PS is a power of either λ or λ^{-1} in the above proposition. Since $\mu(x, \lambda^n) = n\mu(x, \lambda)$, we see that what we have shown implies that whether $x \in \mathbb{P}(V)$ is (semi)stable with respect to a $\mathbb{G}_m$ action can be tested against all the 1-parameter subgroups. This special case of a $\mathbb{G}_m$ action generalizes to all reductive G.

Theorem 17.7 (Hilbert-Mumford Criterion)**.** *Let G be a reductive group, and $G \to \mathrm{GL}_V$ a linearisation. Let $x \in \mathbb{P}(V)$. Then x is semistable if and only if $\mu(x, \lambda) \geq 0$ for all non-trivial 1-PS $\lambda : \mathbb{G}_m \to G$; x is stable if and only if the inequalities are strict.*

Theorem 17.7 takes some work to prove, so we'll only state it here. Usually one has to invoke the Cartan-Iwahori decomposition of a reductive group G and prove the following:

Theorem 17.8. *Let G be a reductive group acting linearly on V. Suppose $v \in V$ such that $0 \in \overline{G \cdot v}$. Then there exist a 1-PS $\lambda : \mathbb{G}_m \to G$ such that $\lim_{t\to 0} \lambda(t) \cdot v = 0$.*

Remark 17.9. *When starting with reductive group G acting on projective X, the Hilbert-Mumford weight $\mu(x, \lambda)$ for some $x \in X \subset \mathbb{P}(V)$ and $\lambda : \mathbb{G}_m \to G$ depends on the linearisation $G \to \mathrm{GL}_V$.*

18. GMS: Construction of moduli space of semistable sheaves

Let X be a connected projective $\mathbb{C}$-scheme (or any algebraically closed field of characteristic 0) and choose an ample line bundle $\mathcal{O}_X(1)$. Fix a polynomial $P \in \mathbb{Q}[z]$. Then we define the moduli functor of semistable coherent sheaves on X with Hilbert polynomial P to be

$$\underline{\mathrm{Bun}}^{ss}_{X,P} : (\mathrm{Sch}_{\mathbb{C}})^{\mathrm{op}} \to \mathrm{Sets}$$

such that $\underline{\mathrm{Bun}}^{ss}_{X,P}(T) = \{F \in \mathrm{Coh}(T \times X)\ \text{T-flat}\ \mid \forall t : F_t \text{ semistable}; P(F) = P\}/\pi_T^* \operatorname{Pic}(T)$.

Definition 18.1. *The moduli space of semistable coherent sheaves with Hilbert polynomial P on $(X, \mathcal{O}_X(1))$ is defined to be the scheme which co-represents the moduli functor $\underline{Bun}^{ss}_{X,P}$.*

18.1. **Setting up the GL_V action on a Quot scheme.** In this section, we finally sketch the construction of a scheme co-representing $\underline{\mathrm{Bun}}^{ss}_{X,P}$. In section 10 we did the beginning of the construction in the case of curves, but this section will subsume what we did as a special case. First we need the following fact.

Theorem 18.2. *The set of semistable coherent sheaves with Hilbert polynomial P on $(X, \mathcal{O}_X(1))$ is bounded.*

Theorem 18.2 is proved in Chapter 3 of [HL] (specifically [HL, Theorem 3.3.7]). While we showed this statement quite easily in the case of curves in Proposition 9.17, we will blackbox this general fact. In particular, by [HL, Lemma 1.7.6], there exists $m \in \mathbb{Z}^+$ such that for any semistable coherent sheaf F with Hilbert polynomial P on $(X, \mathcal{O}_X(1))$, we have F is globally generated by $H^0(X, F \otimes \mathcal{O}_X(m))$ and $H^i(X, F \otimes \mathcal{O}_X(m)) = 0$ for all $i > 0$. Then if V is a vector space of dimension $P(m)$, by choosing an isomorphism $V \to H^0(X, F \otimes \mathcal{O}_X(m))$, for every $F \in \underline{\mathrm{Bun}}^{ss}_{X,P}(\mathbb{C})$ we have a surjection

$$V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F.$$

Then if we consider the Quot scheme $Q = \mathrm{Quot}_X(V \otimes \mathcal{O}_X(-m), P)$, every data of semistable F and an identification $V \to H^0(X, F \otimes \mathcal{O}_X(m))$ appears as a $\mathbb{C}$-point of Q. Consider the universal quotient $V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}$ on $Q \times X$. Define K to be the kernel

$$0 \to K \to V \otimes \mathcal{O}_{Q \times X} \to \mathcal{E}_{\mathrm{univ}} \otimes \pi_X^* \mathcal{O}_X(m) \to 0.$$

Then define $R^{ss} \subseteq Q$ to be the loci of $t \in Q$ where

(1) $H^0(X, K_t) = 0$
(2) $E_{\mathrm{univ},t}$ is pure of dimension $\deg P$ and is Gieseker semistable.

Then R^{ss} is open by upper-semicontinuity and Theorem 16.2. Further define $R^s \subseteq R^{ss}$ to be the loci where $E_{\mathrm{univ},t}$ is Gieseker stable, which is again open by Theorem 16.2. Note that for every $t \in R^s$, we have $H^i(X, \mathcal{E}_{\mathrm{univ},t} \otimes \mathcal{O}_X(m)) = 0$ for all $i > 0$. Then the induced map $V \to H^0(X, \mathcal{E}_{\mathrm{univ},t} \otimes \mathcal{O}_X(m))$ (which itself induces back the map of sheaves) which is an injection is actually an isomorphism.

Proposition 18.3. *Let $T \in (Sch_{\mathbb{C}})^{op}$. The T-points of R^{ss} are*

$$\{[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(T) \mid \forall t : F_t \text{ semistable }, \pi_{T,*}(q \otimes id_{\pi_X^* \mathcal{O}_X(m)}) : V \otimes \mathcal{O}_T \cong \pi_{T,*}[F \otimes \pi_X^* \mathcal{O}_X(m)]\}.$$

The T-points of R^s are similar, except one requires F_t to be stable for every $t \in T$.

Proof. Suppose we have $[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(T)$ satisfyig the conditions of the moduli functor. Note that the induced map

$$\pi_{T,*}(q \otimes \mathrm{id}_{\pi_X^* \mathcal{O}_X(m)}) : V \otimes \mathcal{O}_T \cong \pi_{T,*}[F \otimes \pi_X^* \mathcal{O}_X(m)],$$

which is an isomorphism by assumption, recovers q. Note q must come from pulling back the universal surjection over $Q \times X$ along some $T \to Q$. Let K denote the kernel

$$0 \to K \to V \otimes \mathcal{O}_{T \times X} \xrightarrow{q \otimes \mathrm{id}_{\pi_X^* \mathcal{O}_X(m)}} F \otimes \pi_X^* \mathcal{O}_X(m) \to 0.$$

Since $\pi_{T,*}(q \otimes \mathrm{id}_{\pi_X^* \mathcal{O}_X(m)})$ is an isomorphism, we have $\pi_{T,*}(K) = 0$. So $H^0(C, K_t) = 0$ for every $t \in T$. Then since F_t is semistable for every $t \in T$, altogether this implies that the morphism $T \to Q$ must factor through R^{ss}.

Conversely, given any morphism $T \to R^{ss}$, pulling back the universal surjection over R^{ss} gives exactly a $[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(T)$ such that F_t is semistable for every $t \in T$ which is induced by an isomorphism $V \otimes \mathcal{O}_T \to \pi_{T,*}[F \otimes \pi_X^* \mathcal{O}_X(m)]$. The proof of the description of the T-points of R^s is essentially the same. □

Now there is a morphism of moduli functors

$$R^{ss} \to \underline{\mathrm{Bun}}_{X,P}^{ss}$$

which on T-points sends

$$[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in R^{ss}(T) \mapsto F \in \underline{\mathrm{Bun}}_{X,P}^{ss}(T).$$

Now we define a GL_V action on Q so that R^{ss} is GL_V-invariant and the above morphism is GL_V-invariant. The action $\mathrm{GL}_V \times Q \to Q$ is given on T-points by the following map $\mathrm{GL}_V(T) \times Q(T) \to Q(T)$:

$$(\theta, [q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F]) \mapsto [q \circ (\pi_T^* \theta \otimes \mathrm{id}_{\pi_X^* \mathcal{O}_X(-m)}) : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F].$$

One can check that this is indeed a group action, and respects the equivalence relation on $Q(T)$. Furthermore, we see that R^{ss} is GL_V-invariant, and the morphism $R^{ss} \to \underline{\mathrm{Bun}}_{X,P}^{ss}$ is GL_V-invariant. This implies that we have a morphism of moduli functors

$$[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}} \to \underline{\mathrm{Bun}}_{X,P}^{ss}.$$

Theorem 18.4. *The morphism of moduli functors $[\frac{R^{ss}}{\mathrm{GL}_V}]^{pre} \to \underline{Bun}_{X,P}^{ss}$ sheafifies to an isomorphism of Zariski sheaves*

$$[\frac{R^{ss}}{\mathrm{GL}_V}]^{sh} \to (\underline{Bun}_{X,P}^{ss})^{sh}.$$

Proof. We first show surjection. Suppose we have an element of $(\underline{\mathrm{Bun}}_{X,P}^{ss})^{sh}(T)$. Then such an element may be thought of as the following datum: an open Zariski cover $\{U_i\}$ of T and $E_i \in \underline{\mathrm{Bun}}_{X,P}^{ss}(U_i)$ such that

$$E_i \cong E_j \otimes \pi_{U_i \cap U_j}^* R_{ij}$$

over $U_i \cap U_j$, for some $R_{ij} \in \operatorname{Pic}(U_i \cap U_j)$. Note that for each i, we have $R^k\pi_{U_i,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m)) = 0$ for all $k > 0$. So $\pi_{U_i,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m))$ is a vector bundle of rank $\dim V$. Then we can choose the U_i so that such pushforwards are all trivial vector bundles over U_i, isomorphic to $V \otimes \mathcal{O}_{U_i}$.

Applying $\pi_{U_i\cap U_j,*}$ to the isomorphism

$$E_i \otimes \pi_X^*\mathcal{O}_X(m) \cong E_j \otimes \pi_X^*\mathcal{O}_X(m) \otimes \pi_{U_i\cap U_j}^*R_{ij},$$

on the overlap $U_i \cap U_j$, we obtain

$$\pi_{U_i\cap U_j,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m)) \cong \pi_{U_i\cap U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m)) \otimes R_{ij},$$

where the right hand side follows by the projection formula. This forces $R_{ij} \cong \mathcal{O}_{ij}$, since we have shrunk the U_i enough so that the pushforwards are both isomorphic to $V \otimes \mathcal{O}_{U_i\cap U_j}$.

Now choose isomorphisms $V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m))$ and $V \otimes \mathcal{O}_{U_j} \to \pi_{U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m))$. Then post-composing the pullbacks of these isomorphisms with the canonical push-pull morphisms,

$$\pi_{U_i}^*\pi_{U_i,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m)) \twoheadrightarrow E_i \otimes \pi_X^*\mathcal{O}_X(m)$$

and

$$\pi_{U_j}^*\pi_{U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m)) \twoheadrightarrow E_j \otimes \pi_X^*\mathcal{O}_X(m),$$

then twisting everything down by $\pi_X^*\mathcal{O}_X(-m)$, we obtain surjections

$$q_i : V \otimes \pi_X^*\mathcal{O}_X(-m) \twoheadrightarrow E_i,$$

over $U_i \times X$, and

$$q_j : V \otimes \pi_X^*\mathcal{O}_X(-m) \twoheadrightarrow E_j,$$

over $U_j \times X$. These surjections q_i and q_j define elements of $R^{ss}(U_i)$ and $R^{ss}(U_j)$ which map to $E_i \in \underline{\mathrm{Bun}}_{X,P}^{ss}(U_i)$ and $E_j \in \underline{\mathrm{Bun}}_{X,P}^{ss}(U_j)$, respectively. Furthermore, we claim that the restrictions of q_i and q_j over $U_i \cap U_j \times X$ are in the same $\mathrm{GL}_V(U_i \cap U_j)$ orbit class. To see why, recall that over $U_i \cap U_j \times X$ we have

$$E_i \cong Ej \implies E_i \otimes \pi_X^*\mathcal{O}_X(m) \cong E_j \otimes \pi_X^*\mathcal{O}_X(m)$$
$$\implies \pi_{U_i\cap U_j,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m)) \cong \pi_{U_i\cap U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m)).$$

Then the restrictions of the isomorphisms we chose to induce q_i and q_j to $U_i \cap U_j$ may be used to define

$$\theta_{ij} : V \otimes \mathcal{O}_{U_i\cap U_j} \to \pi_{U_i\cap U_j,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m)) \cong \pi_{U_i\cap U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m)) \to V \otimes \mathcal{O}_{U_i\cap U_j},$$

so that $\theta_{ij} \in \mathrm{GL}_V(U_i \cap U_j)$. Then we have the following commutative diagram

$$\begin{tikzcd}
V \otimes \mathcal{O}_{U_i\cap U_j\times X} \arrow[r] & {[\pi_{U_i\cap U_j}^*\pi_{U_i\cap U_j,*}(E_j \otimes \pi_X^*\mathcal{O}_X(m))]} \arrow[r, two heads] & E_j \otimes \pi_X^*\mathcal{O}_X(m) \\
V \otimes \mathcal{O}_{U_i\cap U_j\times X} \arrow[r] \arrow[u, "\pi_{U_i\cap U_j}^*(\theta_{ij})"] & {[\pi_{U_i\cap U_j}^*\pi_{U_i\cap U_j,*}(E_i \otimes \pi_X^*\mathcal{O}_X(m))]} \arrow[r, two heads] \arrow[u] & E_i \otimes \pi_X^*\mathcal{O}_X(m) \arrow[u]
\end{tikzcd}$$

Twisting everything down by $\pi_X^*\mathcal{O}_X(-m)$, we obtain the diagram

$$\begin{tikzcd}
V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, two heads, "q_j"] & E_j \\
V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, two heads, "q_i"'] \arrow[u, "\pi^*_{U_i\cap U_j}(\theta_{ij})\otimes \mathrm{id}_{\pi_X^*\mathcal{O}_X(-m)}"] & E_i \arrow[u]
\end{tikzcd}$$

This diagram shows that the action of θ_{ij} on q_j, namely the composition $q_j \circ \pi^*_{U_i\cap U_j}(\theta_{ij}) \otimes \mathrm{id}_{\pi_X^*\mathcal{O}_X(-m)}$, belongs to the same equivalence class of q_i where the equivalence relation we're referring to here is one inherited from the Quot functor. Altogether, this shows that the restrictions of q_i and q_j over $U_i \cap U_j \times X$ are in the same $\mathrm{GL}_V(U_i \cap U_j)$ orbit class, which finishes our proof of surjectivity.

To show injectivity, it suffices to show the following: suppose $[q_M : V \otimes \pi_X^*\mathcal{O}_X(-m) \twoheadrightarrow M_i] \in R^{ss}(U_i)$ such that q_M is induced by an isomorphism

$$V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(M_i \otimes \pi_X^*\mathcal{O}_X(m))$$

and suppose $[q_N : V \otimes \pi_X^*\mathcal{O}_X(-m) \twoheadrightarrow N_i] \in R^{ss}(U_i)$ such that q_N is induced by an isomorphism

$$V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(N_i \otimes \pi_X^*\mathcal{O}_X(m)).$$

Furthermore, suppose that $[M_i] = [N_i] \in \underline{\mathrm{Bun}}^{ss}_{X,P}(U_i)$, i.e. $M_i \cong N_i \otimes \pi_{U_i}^* R_i$ for some $R_i \in \mathrm{Pic}(U_i)$. But we can shrink all the U_i, i.e. take a finer cover, so that $R_i \cong \mathcal{O}_{U_i}$, so $M_i \cong N_i$. We would like to show then that q_M and q_N must belong in the same $\mathrm{GL}_V(U_i)$ orbit class. If we show this, then the proof of injectivity follows.

Since $M_i \cong N_i$, then $M_i \otimes \pi_X^*\mathcal{O}_X(m) \cong N_i \otimes \pi_X^*\mathcal{O}_X(m)$, so

$$\pi_{U_i,*}(M_i \otimes \pi_X^*\mathcal{O}_X(m)) \cong \pi_{U_i,*}(N_i \otimes \pi_X^*\mathcal{O}_X(m)).$$

Then using the isomorphisms inducing q_M and q_N, we can define

$$\theta_i : V \otimes \mathcal{O}_{U_i} \to \pi_{U_i,*}(M_i \otimes \pi_X^*\mathcal{O}_X(m)) \cong \pi_{U_i,*}(N_i \otimes \pi_X^*\mathcal{O}_X(m)) \to V \otimes \mathcal{O}_{U_i},$$

so that $\theta_i \in \mathrm{GL}_V(U_i)$. We see that we have a commuting diagram

$$\begin{tikzcd}
V \otimes \mathcal{O}_{U_i} \arrow[r] & \pi_{U_i}^*\pi_{U_i,*}[N_i \otimes \pi_X^*\mathcal{O}_X(m)] \arrow[r, two heads] & N_i \otimes \pi_X^*\mathcal{O}_C(m) \\
V \otimes \mathcal{O}_{U_i} \arrow[r] \arrow[u, "\pi_{U_i}^*(\theta_i)"] & \pi_{U_i}^*\pi_{U_i,*}[M_i \otimes \pi_X^*\mathcal{O}_X(m)] \arrow[r, two heads] \arrow[u] & M_i \otimes \pi_X^*\mathcal{O}_X(m) \arrow[u]
\end{tikzcd}$$

and twisting down by $\pi_X^*\mathcal{O}_X(-m)$, we obtain a commuting diagram

$$\begin{tikzcd}
V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, two heads, "q_N"] & N_i \\
V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, two heads, "q_M"'] \arrow[u, "\pi^*_{U_i}(\theta_i)\otimes \mathrm{id}_{\pi_X^*\mathcal{O}_X(-m)}"] & M_i \arrow[u]
\end{tikzcd}$$

This shows that the action of θ_i on q_N is in the same Quot functor equivalence class as q_M, thereby showing that q_N and q_M lie in the same $\mathrm{GL}_V(U_i)$ orbit. Then if we consider two elements in $[\frac{R^{ss}}{\mathrm{GL}_V}]^{sh}(T)$ which map to the same element of $(\underline{\mathrm{Bun}}^{ss}_{X,P})^{sh}(T)$, then it is straightforward to check that what we have shown implies injectivity.

□

18.2. **Descending to SL_V action.** This is the point in the construction which we stopped at in the special case of semistable vector bundles on curves, so what follows will be new. By Theorem 18.4, any scheme co-representing $[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{sh}}$ also co-represents $(\underline{\mathrm{Bun}}^{ss}_{X,P})^{\mathrm{sh}}$. The hope was to find a scheme co-representing $[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{sh}}$ by doing a projective GIT construction with respect to the action of GL_V on Q, so that the semistable locus coincides exactly with R^{ss} and the stable locus coincides exactly with R^s. It turns out, however, that this is not the right action – disregarding whatever linearization of the GL_V action we pick, the stabilizer of any $\mathbb{C}$-point in $Q(\mathbb{C})$ will be positive-dimensional. We will see why in the process of the following proposition.

Proposition 18.5. *Let $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(\mathbb{C})$ be a $\mathbb{C}$-point induced by an isomorphism $V \to H^0(X, F \otimes \mathcal{O}_X(m))$. Let $Stab(F,q)$ denote the stabilizer of $[q]$. Then there is an isomorphism of group schemes*

$$Stab(F,q) \cong \mathrm{Aut}(F).$$

Proof. The T-points of the group scheme $\mathrm{Aut}(F)$ are

$$\mathrm{Aut}(F)(T) = \{\phi \in \mathrm{Hom}_{\mathcal{O}_{T\times X}}(\pi_X^* F, \pi_X^* F) \mid \forall t : \phi_t \text{ isomorphism } \}.$$

See subsection 22.2 in the appendix to see why. Note that $Stab(F,q)$ is the pullback of the following diagram of schemes

$$\begin{tikzcd} Stab(F,q) \arrow[r] \arrow[d] & \operatorname{Spec}\mathbb{C} \arrow[d] \\ \mathrm{GL}_V \times \operatorname{Spec}\mathbb{C} \arrow[r] & Q \end{tikzcd}$$

So the T-points of $Stab(F,q)$ are $\theta \in \mathrm{GL}_V(T)$ such that

$$\begin{tikzcd} V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, "\pi_T^*(\theta)\otimes \mathrm{id}"] \arrow[dr, "\pi^* q"'] & V \otimes \pi_X^*\mathcal{O}_X(-m) \arrow[r, two heads, "\pi^* q"] & \pi_X^* F \\ & \pi_X^* F \arrow[ur, dashed, "\exists \psi_T(\theta)"'] & \end{tikzcd}$$

$[\pi^* q : V \otimes \pi_X^*\mathcal{O}_X(-m) \to \pi_X^* F] \in Q(T)$, which is the pullback of $[q] \in Q(\operatorname{Spec}\mathbb{C})$ along $T \to \operatorname{Spec}\mathbb{C}$, is equivalent to the T-point $\theta \cdot [\pi^* q] \in Q(T)$.

Thinking of $\theta \in \mathrm{GL}_V(T)$ as a family of $\theta_t \in \mathrm{GL}(V)$ parameterized by $t \in T$ clarifies that $\theta \in Stab(F,q)$ if and only if for each $t \in T$ we have $\ker(\pi_q^*|_t) = \ker q \subset V \otimes \mathcal{O}_{\{t\}\times X}(-m)$ is invariant under the induced $\theta_t : V \otimes \mathcal{O}_{\{t\}\times X}(-m) \to V \otimes \mathcal{O}_{\{t\}\times X}(-m)$. In particular, we see that $\mathbb{G}_m \subset Stab(F,q)$, as $\lambda \in \mathbb{G}_m(T)$ corresponds to $\theta_t := \lambda_t \cdot \mathrm{id}_V$, which certainly preserves the kernel. Note this holds true for any $[q] \in Q(\mathbb{C})$; we haven't used anywhere here so far the assumption we made on q in the proposition statement.

Next we define a morphism of moduli functors $Stab(F,q) \to \mathrm{Aut}(F)$ via on T-points the map

$$\psi_T : Stab(F,q)(T) \to \mathrm{Aut}(F)(T),$$

where $\theta \mapsto \psi_T(\theta)$. Let's show surjectivity. Suppose we have $\Phi \in \mathrm{Aut}(F)(T)$. So $\Phi : \pi_X^* F \to \pi_X^* F$ is an isomorphism, and can be thought of as a family of isomorphisms $\Phi_t : F \to F$ for every $t \in T$. For each $t \in T$, we can define θ_t to be the unique map making the diagram

$$\begin{tikzcd} V \arrow[r, "\theta_t"] \arrow[d] & V \arrow[d] \\ H^0(\{t\} \times X, F \otimes \mathcal{O}_X(m)) & H^0(\{t\} \times X, F \otimes \mathcal{O}_X(m)) \arrow[l, "\Phi_t^*"] \end{tikzcd}$$

commute. Note θ_t exists and is unique because the vertical arrows are isomorphisms by assumption. Then these θ_t define $\theta \in Stab(F,q)(T)$ such that $\psi_T(\theta) = \Phi$. Injectivity follows again because of this construction procedure and because the vertical maps in the above diagrams are isomorphisms by assumption. □

In the middle of the proof of the previous proposition, we saw that for every $[q] \in Q(\mathbb{C})$, we have $Stab(F,[q])$ contains a copy of $\mathbb{G}_m$. Usually at this point what happens is that one wants to the mod out $\mathbb{G}_m \subset \mathrm{GL}_V$ and descend the action of GL_V on Q to a PGL_V action. Then in order to get a nice linearization, one lifts the action to SL_V, since $\mathrm{SL}_V \to \mathrm{PGL}_V$ is a finite and a surjection etale locally. This is described very briefly in the foundational paper [Sim94, Theorem 1.21]. Instead, I will opt to just prove the following directly. Note there is a natural SL_V action on Q since $\mathrm{SL}_V \subset \mathrm{GL}_V$ is a subgroup scheme, so there is an induced map

$$[\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}} \to [\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}} \to \underline{\mathrm{Bun}}^{ss}_{X,P}.$$

Proposition 18.6. *The morphism of moduli functors $[\frac{R^{ss}}{\mathrm{SL}_V}]^{pre} \to [\frac{R^{ss}}{\mathrm{GL}_V}]^{pre}$ sheafifies to a morphism of Zariski sheaves*

$$[\frac{R^{ss}}{\mathrm{SL}_V}]^{sh} \to [\frac{R^{ss}}{\mathrm{GL}_V}]^{sh}$$

which is etale locally an isomorphism, i.e. the morphism of moduli functors etale sheafifies to an isomorphism of etale sheaves.

Proof. To show the statement, it suffices to show the following:

- (injection etale locally) for any $T \in \mathrm{Sch}_{\mathbb{C}}$, if $\alpha, \beta \in [\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}}(T)$ map to the same element in $[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}}(T)$, then there exists some surjective etale $T' \to T$ such that $\alpha|_{T'} = \beta|_{T'}$.

– (surjection etale locally) for any $T \in \mathrm{Sch}_{\mathbb{C}}$, if $\beta \in [\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}}(T)$, there exists some etale surjective $T' \to T$ such that some $\alpha \in [\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}}(T')$ maps to $\beta|_{T'}$.

Surjection etale locally is easy to see. If $\beta \in [\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}}(T)$, then it is the $\mathrm{GL}_V(T)$ orbit class of some $[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in R^{ss}(T)$. Then we can simple take $T' = T$, and note $[q] \in [\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}}(T)$ maps to $[q] = [\beta] \in [\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}}(T)$.

Next we examine injection etale locally. Suppose $[q : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F] \in [\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}}(T)$ and $[q' : V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow F'] \in [\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{pre}}(T)$ such that they map to the same element in $[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{pre}}(T)$, i.e. there exist some $\theta \in \mathrm{GL}_V(T)$ such that there is a commuting diagram

$$\begin{tikzcd} V \otimes \pi_X^* \mathcal{O}_X(-m) \arrow[r, "q"] & F \arrow[d, "\cong"] \\ V \otimes \pi_X^* \mathcal{O}_X(-m) \arrow[u, "\pi_T^*(\theta)\otimes \mathrm{id}"] \arrow[r, "q'"'] & F' \end{tikzcd}$$

Then we want to show that there is some surjective etale $T' \to T$ such that $[q]|_{T'}, [q]'|_{T'} \in R^{ss}(T')$ are in the same $\mathrm{SL}_V(T')$ orbit. To see why this is true and how we might construct $T' \to T$, let's first do some wishful thinking: suppose we wanted to show $[q], [q]' \in R^{ss}(T)$ were in the same $\mathrm{SL}_V(T)$ orbit. What would we do? Well given $\theta \in \mathrm{GL}_V(T)$, one can think of θ as parameterizing $\theta_t \in GL(V)$ as a function of $t \in T$. Then we get an element $\det\theta \in \mathbb{G}_m(T)$ where $(\det\theta)_t := \det\theta_t$. So inverting, we get an element

$$\frac{1}{\det\theta} \in \mathbb{G}_m(T).$$

Here's the wishful thinking: if $\frac{1}{(\det\theta)^{\frac{1}{\dim V}}} \in \mathbb{G}_m(T)$ were well-defined, then the following diagram would show that

$$\begin{tikzcd} V \otimes \pi_X^* \mathcal{O}_X(-m) \arrow[dr, "q"] & \\ V \otimes \pi_X^* \mathcal{O}_X(-m) \arrow[u, "\pi_T^*(\frac{1}{(\det\theta)^{\frac{1}{\dim V}}}\cdot \mathrm{id})\otimes \mathrm{id}"] \arrow[r, "q"'] & F \arrow[d, "\cong"] \\ V \otimes \pi_X^* \mathcal{O}_X(-m) \arrow[u, "\pi_T^*(\theta)\otimes \mathrm{id}"] \arrow[r, "q'"'] & F' \end{tikzcd}$$

$[q]$ and $[q]'$ are in the same $\mathrm{SL}_V(T)$ orbit class, since $\frac{1}{(\det\theta)^{\frac{1}{\dim V}}} \cdot \mathrm{id} \circ \theta \in \mathrm{SL}_V(T)$. But $\frac{1}{(\det\theta)^{\frac{1}{\dim V}}} \in \mathbb{G}_m(T)$ is not necessarily well-defined. For example, if one considers the map $\mathbb{G}_m \to \mathrm{GL}_2$ given by $t \mapsto \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix}$, then we cannot take the square root of the determinant map. One needs an etale squaring map in this case.

So the resolution is this: take the fiber product

$$\begin{tikzcd} T_{\dim V} \arrow[r, "p_{\dim V}"] \arrow[d, "\frac{1}{(\det\theta)^{\frac{1}{\dim V}}}"'] & T \arrow[d, "\frac{1}{\det\theta}"] \\ \mathbb{G}_m \arrow[r, "t\mapsto t^{\dim V}"'] & \mathbb{G}_m \end{tikzcd}$$

Since the $\dim V$-th power map $\mathbb{G}_m \to \mathbb{G}_m$ is surjective and etale, we see $T_{\dim V} \to T$ is surjective etale. And we have

$$\frac{1}{(\det\theta)^{\frac{1}{\dim V}}} \in \mathbb{G}_m(T_{\dim V}).$$

Then the diagram

$$\begin{tikzcd} V\otimes \pi_X^*\mathcal{O}_X(-m) \arrow[drr, "p^*_{\dim V}(q)=q|_{T'}"] & & \\ V\otimes \pi_X^*\mathcal{O}_X(-m) \arrow[u, "\pi^*_{T_{\dim V}}(\frac{1}{(\det\theta)^{\frac{1}{\dim V}}}\cdot\mathrm{id})\otimes\mathrm{id}"] \arrow[rr, "p^*_{\dim V}(q)=q|_{T'}"'] & & p^*_{\dim V}(F) \arrow[d, "\cong"] \\ V\otimes \pi_X^*\mathcal{O}_X(-m) \arrow[u, "\pi^*_{T_{\dim V}}(\theta)\otimes\mathrm{id}"] \arrow[rr, "p^*_{\dim V}(q')=q|_{T'}"'] & & p^*_{\dim V}(F) \end{tikzcd}$$

shows that $[q]|_{T'}$ and $[q']|_{T'} \in R^{ss}(T')$ are in the same $\mathrm{SL}_V(T')$ orbit, where $T' \to T$ is surjective etale. $\square$

The reason why we like Proposition 18.6 is because then a scheme X co-represents $[\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{sh}}$ if and only if it co-represents $[\frac{R^{ss}}{\mathrm{GL}_V}]^{\mathrm{sh}}$. This follows from the more general fact that if $M_1 \to M_2$ is a morphism of moduli functors which etale sheafifies to an isomorphism of etale sheaves, then if $M_1 \to X$ is a morphism where X is a scheme, then there is a unique morphism $M_2 \to X$ such that $M_1 \to X$ factors through $M_1 \to M_2 \to X$. This is beacuse of the universal property of etale sheafifications, and the fact that schemes are also etale sheaves. More precisely, representable functors satisfy etale descent[8]

Then if we construct a GIT setup for the SL_V action on Q such that the semistable locus is exactly R^{ss} and the stable locus is exactly R^s, then by GIT the universal categorical quotient of the semistable locus will be a scheme co-representing $[\frac{R^{ss}}{\mathrm{SL}_V}]^{\mathrm{sh}}$ and thus co-representing $\underline{\mathrm{Bun}}^{ss}_{X,P}$ by Proposition 18.6 and Proposition 18.4. It's easy to see that analogs of the aforementioned two propositions hold for $\underline{\mathrm{Bun}}^{s}_{X,P}$, so that in the GIT setup the universal categorical quotient of R^{ss} will restrict to a geometric quotient of R^s which will then be a coarse moduli space for $\underline{\mathrm{Bun}}^{s}_{X,P}$.

[8]Actually, representable functors satisfy fpqc descent which implies etale descent. Some references are mentioned in this thread: `https://math.stackexchange.com/questions/4326243/why-do-schemes-give-sheaves-on-etale-site`

18.3. **The GIT setup for SL_V action on $\overline{R} := \overline{R^{ss}}$.** Define $\overline{R} \subseteq Q$ to be the Zariski closure of R^{ss} in Q. Note $\overline{R}$ is SL_V invariant since R^{ss} is SL_V-invariant. We will now construct a GIT setup for the action of SL_V on $\overline{R}$. Once we construct the GIT setup, we will want to verify that GIT-(semi)stable loci $\overline{R}^{ss}$ and $\overline{R}^{s}$ coincide exactly with R^{ss} and R^s.

To construct the GIT setup for SL_V acting on $\overline{R}$, we will embed $\overline{R}$ into a projective space using an embedding of Q into projective space given by the very ample line bundle described in Proposition 14.6. Let's describe concretely how this embedding of Q goes on the level of $\mathbb{C}$-points.

Let $V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}$ denote the universal surjection parameterized by Q. Consider the short exact sequence

$$0 \to K \to V \otimes \mathcal{O}_{Q\times X} \to \mathcal{E}_{\mathrm{univ}} \otimes \pi_X^* \mathcal{O}_X(m) \to 0.$$

Then for sufficiently large $\ell > 0$, we have

$$\forall i > 0 : R^i\pi_{Q,*}[K \otimes \pi_X^*\mathcal{O}_X(\ell)] = R^i\pi_{Q,*}[V \otimes \pi_X^*\mathcal{O}_X(\ell)] = 0$$

Note we already know $R^i\pi_{Q,*}[\mathcal{E}_{\mathrm{univ}} \otimes \pi_X^*\mathcal{O}_X(n)]$ for all $n \geq m$ and $i > 0$. Then fixing ℓ, pushing forward onto Q gives us a short exact sequence of vector bundles

$$0 \to \pi_{Q,*}[K \otimes \pi_X^*\mathcal{O}_X(\ell)] \to V \otimes H^0(X, \mathcal{O}_X(\ell)) \to \pi_{Q,*}[\mathcal{E}_{\mathrm{univ}} \otimes \pi_X^*\mathcal{O}_X(m+\ell)] \to 0.$$

This manuever provides the embedding

$$Q \hookrightarrow \mathrm{Gr}([V \otimes H^0(X, \mathcal{O}_X(\ell))]^\vee, P(m+\ell)).$$

In other words, for every $t \in Q(\mathbb{C})$, the point $t = [V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \mathcal{E}_{\mathrm{univ},t}]$ is mapped to the element

$$[V\otimes H^0(X, \mathcal{O}_X(\ell)) \twoheadrightarrow H^0(X, \mathcal{E}_{\mathrm{univ},t}\otimes\mathcal{O}_X(m+\ell))] = [H^0(X, \mathcal{E}_{\mathrm{univ},t}\otimes\mathcal{O}_X(m+\ell))^\vee \hookrightarrow [V\otimes H^0(X, \mathcal{O}_X(\ell))]^\vee],$$

in $\mathrm{Gr}([V\otimes H^0(X, \mathcal{O}_X(\ell))]^\vee, P(m+\ell))(\mathbb{C})$. Then wedging to the $P(m+\ell)$ degree, in summary we find that the very ample line bundle in Proposition 14.6 provides a projective embedding of our Quot scheme which can be described as:

$$[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(\mathbb{C})$$

maps to

$$[\bigwedge^{P(m+\ell)} H^0(X, F(m+\ell))^\vee] \in \mathbb{P}(\bigwedge^{P(m+\ell)} [V \otimes H^0(X, \mathcal{O}_X(\ell))]^\vee).$$

This embedding also corresponds a natural linearization to the action of SL_V on Q. In particular, $\theta \in \mathrm{SL}_V(\mathbb{C})$ acts on $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F]$ by

$$[q] \mapsto [\theta \cdot q] = [V \otimes \mathcal{O}_X(-m) \xrightarrow{\theta\otimes\mathrm{id}} V \otimes \mathcal{O}_X(-m) \xrightarrow{q} F] \in Q(\mathbb{C}).$$

In terms of the embedding, the image of $[\theta \cdot q]$ in the projective embedding is the line in $\bigwedge^{P(m+\ell)}[V\otimes H^0(X,\mathcal{O}_X(\ell))]^\vee$ which is the image of the line spanned by $\bigwedge^{P(m+\ell)} H^0(X,F(m+\ell))^\vee$, under the linear isomorphism

$$\bigwedge^{P(m+\ell)}[V\otimes H^0(X,\mathcal{O}_X(\ell))]^\vee \to \bigwedge^{P(m+\ell)}[V\otimes H^0(X,\mathcal{O}_X(\ell))]^\vee$$

induced by $V \xrightarrow{\theta} V$. In other words, our embedding of Q naturally offers the following linearization of the action of SL_V on Q: a map

$$\mathrm{SL}_V \longrightarrow \mathrm{GL}_{\bigwedge^{P(m+\ell)}[V\otimes H^0(X,\mathcal{O}_X(\ell))]^\vee},$$

which at least on $\mathbb{C}$-points sends $\theta \in \mathrm{SL}_V(\mathbb{C})$ sends θ to the linear automorphism induced by the following procedure:

(1) Start with $V \xrightarrow{\theta} V$ to get $V\otimes\mathcal{O}_X(-m) \xrightarrow{\theta\otimes\mathrm{id}} V\otimes\mathcal{O}_X(-m)$.
(2) Twist and take global sections: $V\otimes H^0(X,\mathcal{O}_X(\ell)) \xrightarrow{\theta\otimes\mathrm{id}} V\otimes H^0(X,\mathcal{O}_X(\ell))$
(3) Dualize the map, and then wedge by $\bigwedge^{P(m+\ell)}$.

Now we restrict this linearization setup to $\overline{R}$. Finally, we would like to show the following:

Theorem 18.7. *Let $\overline{R}^{ss}$ and $\overline{R}^{s}$ denote the semistable and stable loci with respect to the aforementioned GIT setup for the action of SL_V on $\overline{R}$. Then $\overline{R}^{ss} = R^{ss}$ and $\overline{R}^{s} = R^{s}$.*

Let's first focus on sketching why $\overline{R}^{ss} = \overline{R}$. Computing the homogeneous coordinate ring of the affine cone $\widetilde{\overline{R}} \subset \bigwedge^{P(m+\ell)}[V\otimes H^0(X,\mathcal{O}_X(\ell))]^\vee$ is terribly difficult. Luckily we have the Hilbert-Mumford criterion to get a handle on $\overline{R}^{ss}$. In particular, $[q] \in \overline{R}^{ss}$ if and only if $\mu([q],\lambda) \geq 0$ for all 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$. Note that $-\mu([q],\lambda)$ is the weight which $\lambda(t)$ acts on $\lim_{t\to 0}\lambda(t)\cdot[q]$. So let's compute $\lim_{t\to 0}\lambda(t)\cdot[q]$.

18.4. Applying the Hilbert-Mumford criterion to the GIT setup.

Proposition 18.8. *Let $[q : V\otimes\mathcal{O}_X(-m) \twoheadrightarrow F] \in \overline{R}(\mathbb{C})$.* [9] *Let $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$ be any 1-PS, which is given by a decomposition $V = \bigoplus_d V_d$. Note all but finitely many V_d are 0. Let $V_{\leq n} = \bigoplus_{d\leq n} V_d$ and let $F_{\leq n} \subset F$ denote the image of the restriction of q:*

$$V_{\leq n}\otimes\mathcal{O}_X(-m) \to F.$$

Define $F_d = \frac{F_{\leq d}}{F_{\leq d-1}}$. Then we have surjections

$$q_d : V_d\otimes\mathcal{O}_X(-m) \twoheadrightarrow F_d.$$

We claim that

$$[\lim_{t\to 0}\lambda(t)\cdot q] = [\bigoplus_d q_d] \in \overline{R}(\mathbb{C}).$$

[9] At this point of the discussion the restriction to $\overline{R}$ is actually not necessary, so the proposition also holds for $\mathbb{C}$-points of Q.

Proof. In order to show

$$[\lim_{t\to 0}\lambda(t)\cdot q] = [\bigoplus_d q_d] \in \overline{R}(\mathbb{C}),$$

it suffices to construct a flat family of surjections over $\mathbb{A}^1\times X$ such that over $0\neq t\in\mathbb{A}^1$, the surjection is identified with $\lambda(t)\cdot[q]$, while over 0, the surjection is identified with $\bigoplus_d q_d$.

A coherent sheaf on $\mathbb{A}^1\times X$ whose restriction to $\{0\}\times X$ gives $\bigoplus_d F_d$ is

$$\bigoplus_d F_{\leq d}\cdot T^d,$$

where T^d should be thought of as a 'placeholder,' in the sense that the action of $\mathbb{C}[t]$ on this module is that t acts on $F_{\leq d}\cdot T^d$ by inclusion in the copy of $F_{\leq d+1}$ marked by $F_{\leq d+1}\cdot T^{d+1}$. So the action of t is shifting, which makes clear why its restriction to $\{0\}\times X$ is $\bigoplus_d F_d$.

Now consider the coherent sheaf $V\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T]$ on $\mathbb{A}^1\times X$. This can be described in the following pieces: we have copies of $V\otimes\mathcal{O}_X(-m) = \bigoplus_d V_d\otimes\mathcal{O}_X(-m)$ marked by powers T^k, and the action of $\mathbb{C}[T]$ shifts and scales.Then define the following surjection of coherent sheaves on $\mathbb{A}^1\times X$:

$$\theta : V\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T]\to\bigoplus_d F_{\leq d}\cdot T^d,$$

where θ restricted to $V_d\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}$ is the map

$$V_d\otimes\mathcal{O}_X(-m)\xrightarrow{q\cdot T^d} F_{\leq d}\cdot T^d\subseteq\bigoplus_d F_{\leq d}\cdot T^d.$$

The rest of the map θ is generated by these restrictions of θ on $V_d\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}$ using the $\mathbb{C}[t]$ action. Note that the restriction θ_0 of this map θ to $\{0\}\times X$ is exactly

$$\theta_0 = \bigoplus_d q_d.$$

The structure of θ is also described by the following diagram:

$$\begin{tikzcd}
\bigoplus_d F_{\leq d}\cdot T^d \arrow[r] & \bigoplus_d F_{\leq d}\cdot T^d \\
V\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T] \arrow[u, "\theta"] \arrow[r] & \bigoplus_d V_{\leq d}\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T] \arrow[u]
\end{tikzcd}$$

This diagram can be extended to the following:

$$\begin{tikzcd}
\bigoplus_d F_{\leq d}\cdot T^d \arrow[r] & \bigoplus_d F_{\leq d}\cdot T^d \arrow[r] & F\cdot T^{-N}\mathbb{C}[T] \\
V\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T] \arrow[u, "\theta"] \arrow[r] & \bigoplus_d V_{\leq d}\otimes\mathcal{O}_X(-m)\cdot\mathbb{C}[T] \arrow[u] \arrow[r] & V\otimes\mathcal{O}_X(-m)\cdot T^{-N}\mathbb{C}[T] \arrow[u, "q\otimes T^{-N}\mathbb{C}[T]"']
\end{tikzcd}$$

where $-N$ is the largest index so that $F_{\leq d}$ and $V_{\leq d}$ for all $d < -N$. Then we can restrict this diagram to $(\mathbb{A}^1 \setminus \{0\}) \times X$, which means inverting T, to obtain the diagram of coherent sheaves

$$\begin{tikzcd}
F \cdot \mathbb{C}[T^{\pm 1}] \arrow[rr] & & F \cdot \mathbb{C}[T^{\pm 1}] \\
V \otimes \mathcal{O}_X(-m) \cdot \mathbb{C}[T^{\pm 1}] \arrow[u, "\theta"] \arrow[rr] & & V \otimes \mathcal{O}_X(-m) \cdot \mathbb{C}[T^{\pm 1}] \arrow[u, "q \otimes \mathbb{C}[T^{\pm 1}]"']
\end{tikzcd}$$

Note that the horizontal arrows are isomorphisms. The bottom horizontal arrow acts on $V_d \otimes \mathcal{O}_X(-m)$ by $v \mapsto T^d \cdot v$. The top horizontal arrow is equality. Then if we restrict this diagram to $\{t\} \times X$, we see that θ_t is equal to $\lambda(t) \cdot q$. □

So fix any 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$ given by some decomposition $V = \bigoplus_d V_d$. Then for $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in \overline{R}(\mathbb{C})$, we have

$$[\lim_{t \to 0]} \lambda(t) \cdot q] = [\bigoplus_d q_d] \in \overline{R}(\mathbb{C})$$

by Proposition 18.8. Now we compute the weight of the $\lambda(t)$ action on the line in $\bigwedge^{P(m+\ell)} [V \otimes H^0(X, \mathcal{O}_X(\ell))]^\vee$ corresponding to the image of $[\bigoplus_d q_d]$ under the embedding of $\overline{R}$ into projective space that we previously discussed. Starting with

$$\bigoplus_d q_d : \bigoplus V_d \otimes \mathcal{O}_X(-m) \to \bigoplus_d F_d,$$

we twist and take cohomology to obtain

$$\bigoplus V_d \otimes H^0(X, \mathcal{O}_X(\ell)) \to \bigoplus H^0(X, F_d(m+\ell)),$$

then we dualize and wedge to get the line

$$\mathbb{C} \cdot [\bigwedge^{P(m+\ell)} H^0(X, F_d(m+\ell))^\vee] \subset \bigwedge^{P(m+\ell)} V_d^\vee \otimes H^0(X, \mathcal{O}_X(\ell))^\vee.$$

Then since each $H^0(X, F_d(m+\ell))^\vee \subset V_d^\vee \otimes H^0(X, \mathcal{O}_X(\ell))^\vee$, we see that

$$\lambda(t) \cdot \mathbb{C} \cdot [\bigwedge^{P(m+\ell)} H^0(X, F_d(m+\ell))^\vee] = t^{\sum_d d \cdot P(F_d, m+\ell)} \cdot \mathbb{C} \cdot [\bigwedge^{P(m+\ell)} H^0(X, F_d(m+\ell))^\vee].$$

In summary, we have shown the following proposition.

Proposition 18.9. *With respect to our chosen embedding and linearization for the action of* SL_V *on* $\overline{R}$*, for any 1-PS* $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$ *given by* $V = \bigoplus_d V_d$ *and* $[q] \in \overline{R}(\mathbb{C})$*, we have*

$$\mu([q], \lambda) = \mu([\oplus_d q_d], \lambda) = -\sum_d d \cdot P(F_d, m+\ell).$$

Note $\dim_{\mathbb{C}} H^0(X, F_d(m+\ell)) = P(F_d, m+\ell)$.[10]

[10] Again, note that everything here still holds for Q and it doesn't actually matter here that we are restricting to $\overline{R}$.

Now in order for a point $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in \overline{R}^{ss}$, we must have that for all 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$

$$\mu([q], \lambda) = -\sum_d d \cdot P(F_d, m + \ell) \geq 0.$$

Since SL_V is the moduli space for matrices of determinant 1, we have that for any decomposition $V = \bigoplus_d V_d$ corresponding to a $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$

$$\sum_d d \cdot \dim V_d = 0.$$

Using this, we find that

$$\mu([q], \lambda) = -\sum_d d \cdot P(F_d, m+\ell) = \frac{1}{\dim V}[\sum_d d \cdot \dim(V_d) \cdot P(F, m+\ell) - d \cdot \dim(V) \cdot P(F_d, m+\ell)]$$

$$= \frac{1}{\dim V}[\sum_d \dim V \cdot P(F_{\leq d}, m+\ell) - \sum_d \dim V_{\leq d} \cdot P(F, \ell)]$$

Remark 18.10. *Note that a 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$ can never have a decomposition of the form $V = V_n$ for some $n \in \mathbb{Z}$. This is because $V = V_0$ is the trivial decomposition, and $V = V_n$ for $n \neq 0$ is not possible since we must have $\det \lambda(t) = 1$. So any non-trivial 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$ must be associated to a decomposition $V = \bigoplus_d V_d$ such that at least two indices are nonzero.*

This remark is small but important – it clarifies why in the following lemma we can test (semi)stability of a point against only non-trivial proper linear subspaces.

Lemma 18.11. *A closed point $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \overline{R}$ is a (semi)stable point if and only if for all non-trivial proper linear subspaces $V' \subset V$ and $F' := q(V' \otimes \mathcal{O}_X(-m)) \subset F$ we have*

$$\dim V \cdot P(F', m+\ell)(\geq) \dim V' \cdot P(F, m+\ell).$$

Proof. For any $V' \subset V$ define

$$\theta(V') := \dim V \cdot P(F', m+\ell) - \dim V' \cdot P(F, m+\ell).$$

Now $[q]$ is a semistable point if and only if $\mu([q], \lambda) \geq 0$ for all 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$. This means for all decompositions $V = \bigoplus_d V_d$, we have $[q]$ (semi)stable point if and only if

$$\sum_d \theta(V_{\leq d})(\geq)0.$$

So suppose $\theta(V')(\geq)0$ for all proper $V' \subset V$. Then $\theta(V_{\leq d})(\geq)0$ for each d, so we see $[q]$ is a (semi)stable point. This proves the reverse direction.

We prove the forward direction by contrapositive. Suppose for some proper $V' \subset V$ we have $\theta(V')(<)0$. Then if V^+ is a complementary subspace so $V = V' + V^+$, we can construct a decomposition $V_{-\dim V'} = V'$ and $V_{\dim V"} = V^+$. This corresponds uniquely to a 1-PS $\lambda : \mathbb{G}_m \to \mathrm{SL}_V$, and with respect to λ we have

$$\mu([q], \lambda) = \frac{1}{\dim V} \sum_d \theta(V_{\leq d}) = \frac{1}{\dim V} \theta(V_{-\dim V'})(<)0.$$

Then $[q]$ is not (semi)stable. □

Lemma 18.12. *For sufficiently large ℓ, $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in \overline{R}$ is (semi)stable if and only if for all nontrivial $F' \subset F$ and $V' := H^0(X, F(m)) \cap ((q \otimes id_{\mathcal{O}_X(-m)})_*)^{-1}[H^0(X, F'(m))]$ we have*

$$\dim V \cdot P(F')(\geq) \dim V' \cdot P(F).$$

Furthermore, for sufficiently large ℓ, the necessary ℓ is uniform across all $[q] \in \overline{R}(\mathbb{C})$ (or $Q(\mathbb{C})$).

Proof. Just to clarify, the notation $V' := H^0(X, F(m)) \cap ((q \otimes \mathrm{id}_{\mathcal{O}_X(-m)})_*)^{-1}[H^0(X, F'(m))]$ here means we take $q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F$ and consider the induced map

$$V \to H^0(X, F(m)),$$

and define V' to be the preimage of $H^0(X, F'(m)) \subset H^0(X, F(m))$ along this map.

First, suppose $[q]$ is (semi)stable. Then by the previous lemma, we have for all nontrivial proper $V' \subset V$ that

$$\dim V \cdot P(F', m + \ell)(\geq) \dim V' \cdot P(F, m + \ell).$$

Next we claim that the set $\{P(F') \mid \text{non-trivial proper } V' \subset V\}$ is actually finite. For each $0 < r < \dim V$, we can consider $G_r := \mathrm{Gr}(V, r)$ and if we consider the tautalogous rank r bundle

$$0 \to S \to V \otimes G_r$$

and letting $\pi_X : G_r \times X \to X$ denote the canonical projection, we have $\pi_{G_r}^* S \otimes \pi_X^* \mathcal{O}_X(-m) \hookrightarrow V \otimes \pi_X^* \mathcal{O}_X(-m)$. Pulling back $q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F$, we obtain surjection $V \otimes \pi_X^* \mathcal{O}_X(-m) \to \pi_X^* F$ of coherent sheaves on $G_r \times X$, and so we can consider the image of the injection

$$\pi_{G_r}^* S \otimes \pi_X^* \mathcal{O}_X(-m) \hookrightarrow V \otimes \pi_X^* \mathcal{O}_X(-m) \xrightarrow{\pi_X^* q} \pi_X^* F$$

on $G_r \times X$. Then since there are finitely many $0 < r < \dim V$ and by implication of a bounded set of sheaves, we find $\{P(F')\}$ is indeed finite. In fact, we can show that $\{P(F')\}$ is finite even when considering altogether all $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in Q(\mathbb{C})$. Instead of just considering $\pi_X : G_r \times X \to X$, we can consider $G_r \times Q \times X$, and letting $\pi_{Q \times X} : G_r \times Q \times X \to Q \times X$ denote the canonical projection, we can pull back the universal surjection $V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow \mathcal{E}_{\mathrm{univ}}$ along $\pi_{Q \times X}$ and consider the image of the injection

$$\pi_{G_r}^* S \otimes \pi_X^* \mathcal{O}_X(-m) \hookrightarrow V \otimes \pi_X^* \mathcal{O}_X(-m) \twoheadrightarrow \pi_{Q \times X}^* \mathcal{E}_{\mathrm{univ}}$$

of coherent sheaves on $G_r \times Q \times X$. So the fiber of this image over $([V]', [q]) \in G_r(\mathbb{C}) \times Q(\mathbb{C})$ is the image sheaf induced by $V' \subset V$ under the map q. By definition of bounded set of sheaves, we find a stronger result: that $\{P(F')\}$ is finite across all q and nontrivial proper $V' \subset V$.

Then for ℓ sufficiently large, since there are only finitely many $P(F')$, we have

$$\dim V \cdot P(F', m+\ell)(\geq) \dim V' \cdot P(F, m+\ell) \implies \dim V \cdot P(F')(\geq) \dim V' \cdot P(F).$$

Now suppose $F' \subset F$ and let V' be as defined in the proposition statement. Then the image of V' which we denote as F^+ will be contained by F'. So $P(F^+)(\leq) P(F')$. Then

$$\dim V \cdot P(F^+)(\geq) \dim V' \cdot P(F) \implies \dim V \cdot P(F')(\geq) \dim V' \cdot P(F)$$

as desired.

Now we prove the reverse direction. Suppose for all $F' \subset F$ and V' as defined in the proposition statement we have

$$\dim V \cdot P(F')(\geq) \dim V' \cdot P(F).$$

Now let $V' \subset V$ be non-trivial proper linear subspace, and let $F^+ \subset F$ denote the corresponding image sheaf. Let V^+ be as defined in the proposition statement with respect to F^+. Then $V' \subset V^+$. Then by assumption we have

$$\dim V \cdot P(F^+)(\geq) \dim V^+ \cdot P(F)(\geq) \dim V' \cdot P(F).$$

Then since ℓ is sufficiently large, this implies

$$\dim V \cdot P(F^+)(\geq) \dim V' \cdot P(F)$$

as desired. □

Proposition 18.13. *For $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in \overline{R}$ to be a GIT semistable point, it is necessary that the induced*

$$V \to H^0(X, F(m))$$

is an injection, and that no submodule $F' \subset F$ of dimension $\leq d-1$ has a global section.

Proof. Suppose $[q]$ were a GIT semistable point. Let ℓ be sufficiently large, in the sense of the previous lemma. Assume for the sake of contradiction that $V \to H^0(X, F(m))$ were not an injection. Then there would exist $V' \subset V$ mapping to the 0 sheaf. This contradicts the inequality

$$\dim V \cdot P(F', m+\ell) \geq \dim V' \cdot P(F, m+\ell),$$

which is supposed to hold. So it is necessary that $V \to H^0(X, F(m))$ is an injection.

On the other hand, suppose $F' \subset F$ is a torsion subsheaf but has a global section. Let V' be the preimage of $H^0(X, F'(m))$ in the map $V \to H^0(X, F(m))$. Then $[q]$ being semistable implies

$$\dim V \cdot P(F') \geq \dim V' \cdot P(F).$$

But since F' is torsion, we have $\deg P(F') < \deg P(F)$, contradiction. $\square$

18.5. **Concluding section of construction; showing GIT-(semi)stable $\iff$ Gieseker (semi)stable.** Finally, we are in a position to prove that

$$\overline{R}^{ss} = R^{ss} \text{ and } \overline{R}^{s} = R^{s}.$$

Before we begin we need to make a comment about the quantities m and ℓ. We want m to be sufficiently large so that lemma 22.7 is satisfied, and such that any Gieseker semistable sheaf with reduced Hilbert polynomial equal to ρ and whose leading coefficient α_d is less than or equal to $\alpha_d(P)$ is also m-regular. Changing m affects changes the specific Quot scheme $Q := \mathrm{Quot}(V \otimes \mathcal{O}_X(-m), P)$ and the dimension of V. Furthermore, we want ℓ to be sufficiently large so that lemma 18.12 can be invoked uniformly for any $[q] \in \overline{R}(\mathbb{C})$. Changing ℓ (and also changing m) changes the Grassmannian that Q embeds in, and thus the projective space that Q embeds in.

First suppose that $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in R^{ss}(\mathbb{C})$. So the induced $V \to H^0(X, F(m))$ is an isomorphism and F is Gieseker semistable. Let $F' \subset F$ be a nontrivial proper subsheaf. Since F is pure, we know $0 < \alpha_d(F') \leq \alpha_d(F)$. So either $0 < \alpha_d(F') < \alpha_d(F)$ or $\alpha_d(F') = \alpha_d(F)$.

Suppose $\alpha_d(F') = \alpha_d(F)$. Note a priori $\dim V' = h^0(X, F'(m)) \leq h^0(X, F(m)) = \dim V$. But if $F' \neq F$, this must be a strict inequality since $V \to H^0(X, F(m))$ is an isomorphism and $F(m)$ is globally generated by its global sections; otherwise if we had equality, then the image of $V \otimes \mathcal{O}_X(-m) \to F$ would be F', not F. Thus, $\dim V \cdot \alpha_d(F') = \dim V \cdot \alpha_d(F) > \dim V' \cdot \alpha_d(F)$ which implies

$$\dim V \cdot P(F') > \dim V' \cdot P(F).$$

Note this is the inequality used in lemma 18.12. So far so good.

This leaves us to check the inequality in lemma 18.12 for $F' \subset F$ such that $0 < \alpha_d(F') < \alpha_d(F)$. By lemma 16.1 and by how m was chosen, we have

$$h^0(X, F'(m))(\leq)\alpha_d(F') \cdot \rho(m).$$

In the case that $[q] \in R^s$, we have this is a strict inequality. Thus,

$$\dim V \cdot \alpha_d(F') = \alpha_d(F) \cdot \rho(m) \cdot \alpha_d(F') > h^0(X, F'(m)) \cdot \alpha_d(F) = \dim V' \cdot \alpha_d(F)$$

$$\implies \dim V \cdot P(F) > \dim V' \cdot P(F).$$

Hence, by lemma 18.12, we have $[q] \in R^s \implies [q] \in \overline{R}^s$. Now suppose $[q] \in R^{ss} \setminus R^s$, so F is strictly semistable. Then for $F' \subset F$ where $0 < \alpha_d(F') < \alpha_d(F)$, if we have $h^0(X, F'(m)) < \alpha_d(F') \cdot \rho(m)$ then, as we just argued, $\dim V \cdot P(F) > \dim V' \cdot P(F)$. But if $h^0(X, F'(m)) = \alpha_d(F') \cdot \rho(m)$ note this means $\rho(F') = \rho(F)$ by lemma 22.7. Then by how m was chosen, we have F' is also m-regular. Then

$$\dim V' \cdot P(F) = \alpha_d(F') \cdot \rho(m) \cdot \alpha_d(F) \cdot \rho = \alpha_d(F) \cdot \rho(m) \cdot \alpha_d(F') \cdot \rho = \dim V \cdot P(F').$$

Then we conclude, by lemma 18.12, that $[q] \in R^{ss} \setminus R^{s} \implies [q] \in \overline{R}^{ss} \setminus \overline{R}^{s}$.

Now we show the reverse direction. Note it suffices to show that $[q] \in \overline{R}^{ss} \implies R^{ss}$. Why? Suppose we indeed showed this. Then if $[q] \in \overline{R}^{s}$, then we must have $[q] \in R^{s}$. Otherwise, $[q] \in R^{ss} \setminus R^{s}$, but we showed in the forward direction that this implies $[q] \in \overline{R}^{ss} \setminus \overline{R}^{s}$, contradicting $[q]$ being in $\overline{R}^{s}$. So let us now verify that

$$[q] \in \overline{R}^{ss} \implies [q] \in R^{ss}.$$

Since $[q]$ is a GIT semistable point, lemma 18.12 implies we have for all nontrivial proper $F' \subset F$ that

$$\dim V \cdot P(F') \geq \dim V' \cdot P(F),$$

where V' is the preimage of $H^0(X, F'(m))$ along the map $V \to H^0(X, F(m))$, which a priori is not necessarily an isomorphism. Taking leading coefficients, we obtain the following inequality:

$$\alpha_d(F) \cdot \rho(m) \cdot \alpha_d(F') \geq \dim V' \cdot \alpha_d(F) \implies \rho(m) \cdot \alpha_d(F') \geq \dim V'.$$

Since $[q] \in \overline{R} := \overline{R^{ss}}$, F can be deformed into a semistable sheaf and thus a pure sheaf. Then by lemma 22.8 in the appendix, there exists a morphism

$$\phi : F \to E$$

such that $\ker \phi = T_{d-1}(F)$ and E is pure with $P(F) = P(E)$. Since $[q] \in \overline{R}^{ss}$ is a GIT semistable point, the induced map

$$V \to H^0(X, F(m)) \to H^0(X, E(m))$$

is injective because $V \to H^0(X, F(m))$ is injective by proposition 18.13 and $H^0(X, F(m)) \to H^0(X, E(m))$ is injective by the following argument: there is an exact sequence

$$0 \to H^0(X, T_{d-1}(F)(m)) \to H^0(X, F(m)) \to H^0(X, E(m)),$$

and $H^0(X, T_{d-1}(F)(m)) = 0$ by proposition 18.13.

We now show that E must be Gieseker semistable. Let

$$E \to E^+$$

be any quotient sheaf with $0 < \alpha_d(E^+) < \alpha_d(E)$ and define $F' = \ker[F \to E \to E^+]$ and V' to be the corresponding preimage of $H^0(X, F'(m))$ under the map $V \to H^0(X, F(m))$. Note F' will surject onto $\ker[E \to E^+]$ by $\phi|_{F'}$, so $\alpha_d(F') \geq \alpha_d(\ker[E \to E^+]) > 0$. Since $F' \subset F$ is a nontrivial proper subsheaf, we have $\rho(m) \cdot \alpha_d(F') \geq \dim V'$. Thus,

$$h^0(X, E^+(m)) \geq h^0(X, F(m)) - h^0(X, F'(m)) \geq \dim V - \dim V' \geq \alpha_d(F)\rho(m) - \alpha_d(F')\rho(m) = \alpha_d(E^+)\rho(m).$$

Then by lemma 22.7, E is Gieseker semistable.

In particular, note E is semistable with Hilbert polynomial P. Thus, V and $H^0(X, E(m))$ have the same dimension, but we showed earlier that the map $V \to H^0(X, E(m))$ is an injection; thus it is actually an isomorphism and V generates E. Furthermore, the map

$q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow E$ factors through F. Then $V \to H^0(X, E(m))$ generating E forces $\phi : F \to E$ to be surjective. Since $P(F) = P(E)$, this implies $\phi : F \to E$ is actually an isomorphism. Hence F is semistable and $V \to H^0(X, F(m))$ is an isomorphism, which implies $[q] \in R^{ss}(\mathbb{C})$.

In summary, we have shown the following: we took the locus $R^{ss} \subseteq Q := \mathrm{Quot}(V \otimes \mathcal{O}_X(-m), P)$ and considered the SL_V action on $\overline{R} := \overline{R^{ss}}$. We linearized this SL_V action on $\overline{R}$, and used the Hilbert-Mumford criterion to show that the GIT (semi)stable loci of this GIT setup are exactly the loci R^{ss} and R^s. Then by the theory of GIT, the induced morphism

$$R^{ss} \to \overline{R} /\!\!/ \mathrm{SL}_V$$

is morphism satisfying all the properties of Theorem 4.3 generalized to the non-affine setting, and the restriction of this morphism to R^s defines a geometric quotient. By Proposition 18.6, we have that $\overline{R} /\!\!/ \mathrm{SL}_V$ co-represents the moduli functor $\underline{\mathrm{Bun}}^{ss}_{X,P}$. We let $M^{ss}_{X,\mathcal{O}_X(1),P} := \overline{R} /\!\!/ \mathrm{SL}_V$ denote the moduli space of semistable sheaves with Hilbert polynomial P on $(X, \mathcal{O}_X(1))$. We let $M^{s}_{X,\mathcal{O}_X(1),P}$ denote the moduli space of stable sheaves with Hilbert polynomial P on $(X, \mathcal{O}_X(1))$, which is a coarse moduli space for the action of SL_V restricted to R^s, and is a coarse moduli space for the moduli functor $\underline{\mathrm{Bun}}^{s}_{X,P}$. To wrap up this section, we show the following about the $\mathbb{C}$-points of $M^{ss}_{X,\mathcal{O}_X(1),P}$ and $M^{s}_{X,\mathcal{O}_X(1),P}$.

Proposition 18.14. *The $\mathbb{C}$-points of $M^{ss}_{X,\mathcal{O}_X(1),P}$ are exactly the S-equivalence classes of Gieseker semistable sheaves on $(X, \mathcal{O}_X(1))$ with Hilbert polynomial P.*

Proof. By GIT, the $\mathbb{C}$-points of $M^{ss}_{X,\mathcal{O}_X(1),P}$ come from $\mathrm{SL}_V(\mathbb{C})$-orbits of points in $R^{ss}(\mathbb{C})$, so we know that every $\mathbb{C}$-point of $M^{ss}_{X,\mathcal{O}_X(1),P}$ is represented by some $[q : V \otimes \mathcal{O}_X(-m) \to F] \in R^{ss}(\mathbb{C})$.

We would like to show that $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] = [q' : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F'] \in$ We would like to show that $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] = [q' : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F'] \in (\mathbb{C})$ if and only if $\mathrm{gr}(F) \cong \mathrm{gr}(F')$.

We first show the reverse direction. We'd like to show that $\mathrm{gr}(F) \cong \mathrm{gr}(F')$ implies that $[q] = [q'] \in M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$. Note it suffices to show that there is some $[\overline{q} : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \bigoplus Q_i]$, where $\bigoplus Q_i \cong \mathrm{gr}(F)$ is polystable, such that $[q] = [\overline{q}] \in M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$. To show this, consider the Jordan-Holder filtration

$$0 \subset JH_1(F) \subset \cdots \subset JH_\ell(F) = F$$

of F. For $1 \leq i \leq \ell$, define $Q_i := \frac{JH_i(F)}{JH_{i-1}(F)}$. Define $V_{\leq t}$ to be the preimage of $H^0(X, JH_i(F)(m)) \subseteq H^0(X, F(m))$ under the isomorphism $V \to H^0(X, F(m))$. Note that by how m was chosen in obtaining $M^{ss}_{X,\mathcal{O}_X(1),P}$ (see first paragraph of the concluding argument), since each $JH_i(F)$ is semistable with reduced Hilbert polynomial equal to ρ, then each $JH_i(F)$ is m-regular, so $V_{\leq t} \to H^0(X, JH_i(F)(m))$ is an isomorphism which generates $JH_i(F)(m)$. Hence, we have surjections $V_{\leq t} \otimes \mathcal{O}_X(-m) \twoheadrightarrow JH_t(F)$. Picking subspaces $V_t \subset V$ to identify with $\frac{V_{\leq t}}{V_{\leq t-1}}$, we obtain surjections $V_t \otimes \mathcal{O}_X(-m) \twoheadrightarrow Q_t$. This defines for us an element

$$[\overline{q} : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \bigoplus_i Q_i] \in R^{ss}(\mathbb{C}),$$

and note that there exists a 1-PS $\lambda : \mathbb{G}_m \to \mathrm{GL}_V$ so that we can do the same construction as in Proposition 18.8 to show that

$$\lim_{t \to 0} \lambda(t) \cdot [q] = [\overline{q}].$$

A priori, we see this implies that $[\overline{q}]$ is in the closure of the $\mathrm{GL}_V(\mathbb{C})$ orbit of $[q]$. But we claim it is also in the closure of the $\mathrm{SL}_V(\mathbb{C})$ orbit of $[q]$. Indeed, in the construction we obtain a flat family of sheaves parameterized by $\mathbb{A}^1$ such that for $0 \neq t \in \mathbb{A}^1$, we have that the restriction is $\lambda(t) \cdot [q]$, while for $t = 0$ the restriction is identified with $\overline{q}$. Note

$$[\lambda(t) \cdot q] = [\frac{1}{(\det \lambda(t))^{\frac{1}{\dim V}}} \cdot \lambda(t) \cdot q] \in R^{ss}(\mathbb{C}),$$

so we see that $[\overline{q}]$ is indeed in the closure of the $\mathrm{SL}_V(\mathbb{C})$ orbit of $[q]$ (even though λ does not necessarily factor through SL_V). Then since $[\overline{q}] \in \overline{\mathrm{SL}_V(\mathbb{C}) \cdot [q]}$, we have

$$\overline{\mathrm{SL}_V(\mathbb{C}) \cdot [\overline{q}]} \cap \overline{\mathrm{SL}_V(\mathbb{C}) \cdot [q]} \neq \emptyset,$$

and so by GIT theory on the map $R^{ss} \to M^{ss}_{X,\mathcal{O}_X(1),P}$, we must then have $[\overline{q}] = [q] \in M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$. This completes the proof of the reverse direction.

Now we prove the forward direction: if $[q] = [q'] \in M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$, then $\mathrm{gr}(F) \cong \mathrm{gr}(F')$. Not every element of $M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$ can be represented by some $[\overline{q} : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \bigoplus Q_i]$ where $\bigoplus Q_i$ is polystable. Thus, it suffices to show that if $[q] = [\overline{q}] \in M^{ss}_{X,\mathcal{O}_X(1),P}(\mathbb{C})$, then $\mathrm{gr}(F) \cong \bigoplus Q_i$.

First we show that $\mathrm{SL}_V(\mathbb{C}) \cdot [\overline{q}]$ is always closed. Suppose that we have some point $[\gamma : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow E] \in \overline{\mathrm{SL}_V(\mathbb{C}) \cdot [\overline{q}]}$. Then there exists some smooth curve C and a flat family $\mathcal{E}$ of coherent sheaves on X parameterized by C such that $\mathcal{E}_t \cong \bigoplus_i Q_i$ for $t \neq 0$ and $\mathcal{E}_0 \cong E$. Let us rewrite the polystable sheaf $\bigoplus_i Q_i$ as $\bigoplus F_i^{n_i}$, where now we are accounting for multiplicites. Note that $n_i = \dim_{\mathbb{C}} \mathrm{Hom}(F_i, \bigoplus Q_i)$. This is because any homomorphism from F_i to the polystable sheaf must be injective as F_i is stable and has the same reduced Hilbert polynomial as the polystable sheaf. Then the homomorphism is defined by F_i to all the stable F_j summands. But if the map is nontrivial, then $F_i \cong F_j$. So any homomorphism $F_i \to \bigoplus Q_i$ must actually be a homomorphism given by $F_i \to F_i^{n_i}$, and note the endomorphisms of a stable sheaf are exactly $\mathbb{C}$ by [HL, Corollary 1.2.8] since we are working over an algebraically closed base field. Now note that $\mathcal{E} \in \mathrm{Coh}(C \times X)$ and consider $\pi_X^* F_i \in \mathrm{Coh}(C \times X)$. Taking sheaf-hom we obtain a family of sheaves $\mathcal{H}(\pi_X^* F_i, \mathcal{E})$ which is C-flat. This implies the function

$$t \in C \mapsto \dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_X}(F_i, \mathcal{E}_t)$$

is upper-semicontinuous function and $t \mapsto n_i$ for all $t \neq 0$, which implies $\dim_{\mathbb{C}} \operatorname{Hom}_{\mathcal{O}_X}(F_i, \mathcal{E}_0) = \dim_{\mathbb{C}} \operatorname{Hom}_{\mathcal{O}_X}(F_i, E) = n_i^E \geq n_i$. Note for any nonzero $\phi \in \operatorname{Hom}_{\mathcal{O}_X}(F_i, \mathcal{E}_0)$, we have $\phi : F_i \to \mathcal{E}_0$ is injective since F_i is stable and $\mathcal{E}_0$ is semistable of the same slopes. So ϕ is an isomorphism of F_i onto its image. Note that ϕ is simply multiplication by a nonzero scalar up to some identification of the image with F_i.

Then we claim the following: suppose $\phi, \psi \in \operatorname{Hom}_{\mathcal{O}_X}(F_i, \mathcal{E}_0)$ are linearly independent in the $\mathbb{C}$-vector space. So $\phi(F_i), \psi(F_i) \subseteq \mathcal{E}_0$. We claim that $\phi(F_i) \cap \psi(F_i) = 0$. Assume for the sake of contradiction the intersection is some nonzero subsheaf. Then since ϕ and ψ are simply scalar multiplications modulo some identification of $\phi(F_i)$ and $\psi(F_i)$ with F_i, there exists a nonzero linear combination $\lambda_\phi \phi + \lambda_\psi \psi : F_i \to \mathcal{E}_0$ which maps all of $\phi(F_i) \cap \psi(F_i)$ to zero. Then since $\lambda_\phi \phi + \lambda_\psi \psi : F_i \to \mathcal{E}_0$ is simply given by multiplication by a scalar up modulo some identification of the image with F_i, this implies it is multiplication by zero. So $\lambda_\phi \phi + \lambda_\psi \psi = 0$. But since ϕ and ψ are linearly independent, we must have $\lambda_\phi = \lambda_\psi = 0$, contradiction. So we must have $\phi(F_i) \cap \psi(F_i) = 0$.

Altogether, this implies we have an injection

$$F_i \otimes \operatorname{Hom}_{\mathcal{O}_X}(F_i, \mathcal{E}_0) \cong F_i^{\oplus n_i^E} \hookrightarrow \mathcal{E}_0.$$

Given distinct i and j, note that the images of $F_i^{\oplus n_i^E}$ and $F_j^{\oplus n_j^E}$ have 0 intersection in $\mathcal{E}_0$. Then we have an injection

$$\bigoplus_i F_i^{\oplus n_i^E} \hookrightarrow \mathcal{E}_0.$$

Noting $n_i^E \geq n_i$, since $E \cong \mathcal{E}_0$ has the same Hilbert polynomial as $\bigoplus_i F_i^{\oplus n_i}$, we must have then that $n_i^E = n_i$ and $\bigoplus F_i^{\oplus n_i} \cong E$.

Then suppose $[q] = [\overline{q}] \in M^{ss}_{X, \mathcal{O}_X(1), P}(\mathbb{C})$. Then

$$\overline{\mathrm{SL}_V(\mathbb{C}) \cdot [q]} \cap \overline{\mathrm{SL}_V(\mathbb{C}) \cdot [\overline{q}]} = \overline{\mathrm{SL}_V(\mathbb{C}) \cdot [q]} \cap \mathrm{SL}_V(\mathbb{C}) \cdot [\overline{q}] \neq \emptyset.$$

But note there exist some $[V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \operatorname{gr}(F)]$ which is in $\overline{\mathrm{SL}_V(\mathbb{C}) \cdot [q]}$. So if $\operatorname{gr}(F) \not\cong \bigoplus Q_i \cong \bigoplus F_i^{n_i}$, then by intersection of all these closures of orbits, we'd have $[V \otimes \mathcal{O}_X(-m) \twoheadrightarrow \operatorname{gr}(F)]$ and $[\overline{q}]$ give the same point in $M^{ss}_{X, \mathcal{O}_X(1), P}(\mathbb{C})$. But this is a contradiction: since we just showed that the $\mathrm{SL}_V(\mathbb{C})$ orbits of surjections of polystable sheaves are closed, their orbits must be distinct and GIT tells us that they cannot map to the same $\mathbb{C}$-point. Thus, we must have $\operatorname{gr}(F) \cong \bigoplus_i Q_i$. □

Corollary 18.15. *The $\mathbb{C}$-points of $M^s_{X, \mathcal{O}_X(1), P}$ are exactly the stable sheaves, so $M^s_{X, \mathcal{O}_X(1), P}$ is a coarse moduli space for the moduli functor $\underline{Bun}^s_{X,P}$.*

19. Luna's etale slice theorem, tangent space of moduli of stable sheaves

The purpose of this section is to get some feeling for Luna's etale slice theorem and how it allows for some local study of GIT quotients. In the end we'll be able to say something about our newly constructed $M^s_{X, \mathcal{O}_X(1), P}$.

Theorem 19.1. *Let G be a reductive group acting on X and suppose $X \to X /\!\!/ G$ is a good quotient. Suppose $z \in X(\mathbb{C})$ is a closed point with closed G-orbit, so its stabilizer G_z is reductive as well.*

Then there exists a G_z-invariant locally closed subscheme $S \subset X$, passing through z, such that there exists the following pullback diagram

$$\begin{tikzcd} (S \times G) /\!\!/ G_z \arrow[r] \arrow[d] & X \arrow[d] \\ S /\!\!/ G_z \arrow[r] & X /\!\!/ G \end{tikzcd}$$

where $S /\!\!/ G_z \to X /\!\!/ G$ is etale and $(S \times G) /\!\!/ G_z \to X$ is a G-equivariant etale morphism, descending from $S \times G \to X$.

Let's do a (somewhat trivial) example to get a feeling for what's going on here. Consider the scaling action of $\mathbb{G}_m$ on $X = \mathbb{A}^2 \setminus \{0\}$. In this case, $X /\!\!/ G \cong \mathbb{P}^1$. Suppose we pick the point $z \in \mathbb{A}^2 \setminus \{0\}$ corresponding to $(1, 1)$. Note its stabilizer is trivial. now any locally closed $S \subset X$ passing through z will be a G_z-invariant subscheme, but actually not just any S will work. Let's first pick a suitable S, and then afterwards we'll pick a bad S.

Let S be the closed subscheme of $\mathbb{A}^2 \setminus \{0\}$ defined by $y = -x + 1$. So $S \cong \mathbb{A}^1$. Then we claim we have a commutative diagram

$$\begin{tikzcd} \mathbb{A}^1 \times \mathbb{G}_m \arrow[r] \arrow[d] & \mathbb{A}^2 \setminus \{0\} \arrow[d] \\ \mathbb{A}^1 \arrow[r] & \mathbb{P}^1 \end{tikzcd}$$

Indeed, one can think of the map $\mathbb{A}^1 \times \mathbb{G}_m \to \mathbb{A}^2 \setminus \{0\}$ as producing all the points which lie on lines through the origin intersecting $y = -x + 1$. So its image is $\mathbb{A}^2 \setminus \{0\}$ minus the loci $\{y = -x\}$. The map $\mathbb{A}^1 \to \mathbb{P}^1$ is an embedding, missing only the point in $\mathbb{P}^1$ corresponding to $\{y = -x\}$. Thus, the S we have chosen here satisfies Luna's etale slice theorem.

Even though the stabilizer of z is trivial, there are unsuitable S. For example, let S be the line given by $y = x$. This passes through z and is invariant under the stabilizer as the stabilizer is trivial. But the map $S /\!\!/ G_z \cong \mathbb{A}^1 \setminus \{0\} \to X /\!\!/ G$ will not even be etale. It will be $\mathbb{A}^1 \setminus \{0\}$ collapsing to the point in $\mathbb{P}^1$ corresponding to that line.

In summary, one should think of this S passing through z as somehow being a transverse slice to the G-orbit of z. One should think of $(S \times G) /\!\!/ G_z$ as accumulating the G-orbits of this transverse slice, and quotienting by G_z. The fact that the maps $S /\!\!/ G_z \to X /\!\!/ G$ and $(S \times G) /\!\!/ G_z \to X$ are etale means that the local behavior of $X \to X /\!\!/ G$ is captured by $(S \times G) /\!\!/ G_z \to S /\!\!/ G_z$. We can apply Luna's etale slice theorem to the geometric quotient $R^s \to M^s_{X, \mathcal{O}_X(1), P}$ to conclude the following:

Proposition 19.2. *The geometric quotient $R^s \to M^s_{X, \mathcal{O}_X(1), P}$ is a principal* PGL_V *bundle.*

Proof. Pick any $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in R^s(\mathbb{C})$. Note since $[q]$ is a GIT stable point, its SL_V-orbit is closed. Then consider the stabilizer $\mathrm{SL}_{V,[q]}$ of $[q]$, which is the pullback of the diagram

$$\begin{tikzcd} \mathrm{SL}_{V,[q]} \arrow[r] \arrow[d] & \operatorname{Spec}\mathbb{C} \arrow[d] \\ \mathrm{SL}_V \times \{q\} \arrow[r] & Q \end{tikzcd}$$

Note this diagram factors as

$$\begin{tikzcd} \mathrm{SL}_{V,[q]} \arrow[r] \arrow[d] & \mathrm{GL}_{V,[q]} \arrow[r] \arrow[d] & \operatorname{Spec}\mathbb{C} \arrow[d] \\ \mathrm{SL}_V \times \{q\} \arrow[r] & \mathrm{GL}_V \times \{q\} \arrow[r] & Q \end{tikzcd}$$

We showed in Proposition 18.5 that $\mathrm{GL}_{V,[q]} \cong \mathrm{Aut}(F)$. Since F is stable and we are working over an algebraically closed field, we have $\mathrm{Aut}(F) \cong \mathbb{C}^\times$ by [HL, Corollary 1.2.8]. In particular, $\mathrm{SL}_{V,[q]}$ is the closed subscheme

$$\mathrm{SL}_V \cap \mathrm{GL}_{V,[q]} \subseteq \mathrm{GL}_V .$$

This implies that $\mathrm{SL}_{V,[q]}$ is the closed subscheme given by the following union of closed points of $\mathrm{GL}_V(\mathbb{C})$

$$\{\mu \cdot \mathrm{id}_V \mid \mu^{\dim V} = 1, \mu \in \mathbb{C}^\times\},$$

i.e. the stabilizer of $[q]$ with respect to the SL_V action is the scalar matrices given by the $\dim(V)$-th roots of unity.

Luna's etale slice theorem tells us that there exists locally closed $S \subset R^s$ passing through $[q]$ which is $\mathrm{SL}_{V,[q]}$ invariant such that we have the pullback diagram

$$\begin{tikzcd} (S \times \mathrm{SL}_V) /\!\!/ \mathrm{SL}_{V,[q]} \arrow[r] \arrow[d] & R^s \arrow[d] \\ S /\!\!/ \mathrm{SL}_{V,[q]} \arrow[r] & M^s_{X,\mathcal{O}_X(1),P} \end{tikzcd}$$

where the top horizontal morphism is SL_V-equivariant and etale, and the bottom horizontal map is etale. But note that the action of $\mathrm{SL}_{V,[q]} \subset \mathbb{G}_m$ on S is trivial. In fact, we showed in the discussion surrounding Proposition 18.5 that the action of $\mathbb{G}_m \subset \mathrm{GL}_V$ on Q is trivial. Note that there is a short exact sequence [11]

$$1 \to \mathrm{SL}_{V,[q]} \to \mathrm{SL}_V \to \mathrm{PGL}_V \to 1,$$

[11] The morphism $\mathrm{SL}_V \to \mathrm{PGL}_V$ is only surjective etale locally.

so the diagram provided to us by Luna's etale slice theorem is really the diagram

$$\begin{tikzcd} S \times \mathrm{PGL}_V \arrow[r] \arrow[d] & R^s \arrow[d] \\ S \arrow[r] & M^s_{X,\mathcal{O}_X(1),P} \end{tikzcd}$$

By definition, this implies $R^s \to M^s_{X,\mathcal{O}_X(1),P}$ is a principal PGL_V bundle. □

Recall that for $Q = \mathrm{Quot}_{X,\mathcal{H},P}$, we have the for $[q : \mathcal{H} \to F] \in Q(\mathbb{C})$ we have the identification

$$T_{[q]}Q \cong \mathrm{Hom}_{\mathcal{O}_X}(K, F)$$

where $K = \ker q$, and we have an inequality

$$\dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_X}(K, F) \geq \dim_{[q]} Q \leq \dim_{\mathbb{C}} \mathrm{Hom}_{\mathcal{O}_X}(K, F) - \dim_{\mathbb{C}} \mathrm{Ext}^1(K, F).$$

If $\mathrm{Ext}^1(K, F) = 0$ then Q is smooth at $[q]$ [HL, Proposition 2.2.8].

Proposition 19.3. *Let $[F] \in M^s_{X,\mathcal{O}_X(1),P}(\mathbb{C})$ be a stable sheaf. Then*

$$T_{[F]}M^s_{X,\mathcal{O}_X(1),P} \cong \mathit{Ext}^1(F, F).$$

Furthermore, if $\mathit{Ext}^2(F, F) = 0$, then $M^s_{X,\mathcal{O}_X(1),P}$ is smooth at $[F]$.

Proving this dimension bound on $M^s_{X,\mathcal{O}_X(1),P}$ rigorously requires deformation theory [HL, Theorem 4.5.1]. I will only sketch the general idea of why this proposition is true.

Recall that since $R^s \to M^s_{X,\mathcal{O}_X(1),P}$ is a principal PGL_V bundle, then we can find locally closed S passing through $[F]$ such that we have the diagram

$$\begin{tikzcd} S \times \mathrm{PGL}_V \arrow[r] \arrow[d] & R^s \arrow[d] \\ S \arrow[r] & M^s_{X,\mathcal{O}_X(1),P} \end{tikzcd}$$

where the bottom horizontal map is etale and the top horizontal map is equivariant and etale. Since etale morphisms induce isomorphisms on tangent spaces, it suffices to make our identification with respect to $S \times \mathrm{PGL}_V \to S$. In particular, we'd like to make the identification

$$T_{[F]}S \cong \mathrm{Ext}^1(F, F).$$

Considering a lift $[q : V \otimes \mathcal{O}_X(-m) \twoheadrightarrow F] \in (S \times \mathrm{PGL}_V)(\mathbb{C})$, we have $T_{[q]} \cong \mathrm{Hom}_{\mathcal{O}_X}(K, F)$. If we look at the short exact sequence $0 \to K \to V \otimes \mathcal{O}_X(-m) \to F \to 0$, we can obtain the long exact sequence

$$0 \to \mathrm{Hom}_{\mathcal{O}_X}(F, F) \to \mathrm{Hom}_{\mathcal{O}_X}(V \otimes \mathcal{O}_X(-m), F) \to \mathrm{Hom}_{\mathcal{O}_X}(K, F) \to \mathrm{Ext}^1(F, F) \to \mathrm{Ext}^1(V \otimes \mathcal{O}_X(-m), F)$$

$$\to \mathrm{Ext}^1(K, F) \to \mathrm{Ext}^2(F, F) \to \cdots$$

Note $Ext^i(V\otimes\mathcal{O}_X(-m),F) \cong V\otimes H^1(X,F(m)) = 0$ for $i > 0$ since m was chosen so that F is m-regular. So $\mathrm{Ext}^1(K,F) \cong \mathrm{Ext}^2(F,F)$. So we see that $\mathrm{Ext}^2(F,F) = 0$ implies $\mathrm{Ext}^1(K,F) = 0$, so R^s is smooth at the lift of $[F]$. This somehow implies that S is smooth at $[F]$, and by the etale map then $M^s_{X,\mathcal{O}_X(1),P}$ is smooth at $[F]$.

Now concentrating on

$$0 \to \mathrm{Hom}_{\mathcal{O}_X}(F,F) \to \mathrm{Hom}_{\mathcal{O}_X}(V\otimes\mathcal{O}_X(-m),F) \xrightarrow{\phi} \mathrm{Hom}_{\mathcal{O}_X}(K,F) \to \mathrm{Ext}^1(F,F) \to 0.$$

Note that $\mathrm{Hom}_{\mathcal{O}_X}(F,F) \cong \mathbb{C} \cong \dim_{\mathbb{C}} \mathrm{Stab}_{\mathrm{GL}_V,[F]}$, by Proposition 18.5, and $\mathrm{Hom}_{\mathcal{O}_X}(V\otimes\mathcal{O}_X(-m),F) \cong V\otimes V$, which has dimension $(\dim V)^2$. And $\dim(V)^2-1$ is the dimension of PGL_V. Note $\dim \mathrm{PGL}_V + \dim S = \dim T_{[q]}R^s$. So we see at least that $\dim_{\mathbb{C}} \mathrm{Ext}^1(K,F) = \dim T_{[F]}S$. Indeed, $\mathrm{Hom}_{\mathcal{O}_X}(K,F)$ is the tangent space $T_{[q]}(S\times \mathrm{PGL}_V)$, $\mathrm{Hom}_{\mathcal{O}_X}(F,F)$ is the tangent space of the stabilizer of $[q]$ in GL_V, and the image of ϕ is the tangent space of the GL_V orbit of $[q]$ at $[q]$, which implies the claim.

There is still much more to say about the local study of $M^{ss}_{X,\mathcal{O}_X(1),P}$. A good next reference is [Las94], where Laszlo uses Luna's etale slice theorem to do a local study of $M^{ss}_{X,\mathcal{O}_X(1),P}$ in the case that X is a smooth projective connected curve.

20. Globally generated vector bundles and determinantal conditions, Serre correspondence on surfaces

Now that we have detailed the construction of $M^{ss}_{X,\mathcal{O}_X(1),P}$, it would be nice to give some examples. So far we know what the story looks like for $\mathbb{P}^1$, and for curves of genus $g \geq 1$ we showed there exists a semistable vector bundle of rank r and degree d for every r and d, by Proposition 9.9. We'd like to also say something in the case of higher dimensional varieties. In this section, we will use the Serre correspondence to prove that for sufficiently large c_2 there exists stable rank 2 vector bundles on any nonsingular projective surface, relative to a fixed line bundle.

First we motivate the Serre correspondence and our eventual construction of stable rank 2 vector bundles on smooth projective surfaces via a discussion on globally generated vector bundles and determinantal varieties. Consider the following claim:

Proposition 20.1. *Let E be a globally generated vector bundle of rank r on a smooth $\mathbb{C}$-variety X. Let h denote the dimension of the $\mathbb{C}$-vector space $H := H^0(X,E)$. Then we have a surjection*

$$H\otimes\mathcal{O}_X \twoheadrightarrow E.$$

Fix an integer $0 < v \leq h$ and $0 \leq s \leq \min\{v,r\}$. Let $V \subset H$ be a v-dimensional subspace. Let $X_s \subset X$ denote the locus with respect to V, over which the the image of the morphism

$$V\otimes\mathcal{O}_X \hookrightarrow H\otimes\mathcal{O}_X \twoheadrightarrow E$$

has rank $\leq s$ on fibers of E. For general v-dimensional subspace $V \subset H$, the locus X_s is empty or has codimension $(v-s)(r-s)$. The singular locus of X_s is exactly X_{s-1}.

So $X_s \subset X$ is the locus of

$$\{p \in X \mid \langle v(p)\rangle_{\forall v \in V} \subset E_p \text{ has rank } \le s\}.$$

The claim of Proposition 20.1 fundamentally boils down to the following fact about determinantal varieties: consider the vector space of linear maps

$$\mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C}^r).$$

Then the locus

$$V_s = \{\phi \in \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C}^r) \mid \mathrm{rk}\,\phi \le s\}$$

is a determinantal variety, described as the subvariety of $\mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C}^r)$ which is the vanishing of all $(s+1) \times (s+1)$ minors. It has codimension $(v-s)(r-s)$. It's singular locus is exactly V_{s-1}.

Lemma 20.2. *V_s is a normal variety of codimension $(v-s)(r-s)$ whose singular locus is exactly V_{s-1}.*

Proof. Details of the proof can be found in [HGCA, II.1 - II.3]. □

Let's see how Proposition 20.1 comes from Lemma 20.2. First, note for any v-dimensional subspace $V \hookrightarrow H$, there exists a surjection

$$\iota^* : \mathrm{Hom}_{\mathbb{C}}(H, W) \twoheadrightarrow \mathrm{Hom}_{\mathbb{C}}(V, W).$$

Note that $(\iota^*)^{-1}(V_s)$ is either empty or has codimension $(v-s)(r-s)$. Next, note the surjection

$$H \otimes \mathcal{O}_X \twoheadrightarrow E$$

corresponds to a morphism

$$\phi : X \longrightarrow \mathrm{Gr}(H, \mathbb{C}^r)$$

from X to the Grassmannian parameterizing r-dimensional quotients $H \twoheadrightarrow \mathbb{C}^r$. Then

$$X_s = \phi^{-1}(\{[\theta : H \twoheadrightarrow \mathbb{C}^r] \mid \mathrm{rk}\ \mathrm{im}(V \hookrightarrow H \xrightarrow{\theta} \mathbb{C}^r) \le s\}).$$

Let $G_s := \{[\theta : H \twoheadrightarrow \mathbb{C}^r] \mid \mathrm{rk}\ \mathrm{im}(V \hookrightarrow H \xrightarrow{\theta} \mathbb{C}^r) \le s\} \subset \mathrm{Gr}(H, \mathbb{C}^r)$. So $X_s = \phi^{-1}(G_s)$. If we consider the bundle projection

$$\pi : U \longrightarrow \mathrm{Gr}(H, \mathbb{C}^r)$$

where $U \subset \mathrm{Hom}_{\mathbb{C}}(H, \mathbb{C}^r)$ is the open subspace of surjective linear maps $H \twoheadrightarrow \mathbb{C}^r$, then we see that

$$G_s = \pi(U \cap (\iota_*)^{-1}(V_s)).$$

Since $(\iota^*)^{-1}(V_s)$ is either empty or has codimension $(v-s)(r-s)$, and its singular locus is exactly $(\iota^*)^{-1}(V_{s-1})$, then by construction $G_s \subset \mathrm{Gr}(H, \mathbb{C}^r)$ is either empty or has codimension $(v-s)(r-s)$ and its singular locus is exactly G_{s-1}.

Next, note that GL_H has a transitive action on $\mathrm{Gr}(H, \mathbb{C}^r)$. Then we can apply Kleiman's transversality theorem to the diagram

$$\begin{tikzcd} & G_s \arrow[d] \\ X \arrow[r, "\phi"'] & \mathrm{Gr}(H, \mathbb{C}^r) \end{tikzcd}$$

noting that

$$X_s = X \times_{\mathrm{Gr}(H,\mathbb{C}^r)} G_s.$$

By Kleiman's transversality theorem, for general $\theta \in \mathrm{GL}(H)$,

$$X \times_{\mathrm{Gr}(H,\mathbb{C}^r)} [\theta \cdot G_s]$$

is either empty or has codimension $(v-s)(r-s)$. This corresponds exactly to saying that for general v-dimensional $V \subset H$, we have X_s is empty or has codimension $(v-s)(r-s)$. Furthermore, by the second part of Kleiman's transversality theorem, we have $X_s \setminus X_{s-1}$ is smooth.

Example 20.3. *Let X be a smooth variety of dimension d. Suppose E is a globally generated vector bundle of rank $r > d$. Then let $v = r - d$ and $s = r - d - 1$. Then for a general $(r-d)$ dimensinal subspace $V \subset H^0(X, E)$ if we consider the morphism*

$$V \otimes \mathcal{O}_X \to E$$

then X_{r-d-1} is either empty or has codimension $(v-s)(r-d) = d+1$. This implies X_{r-d-1} is empty. But X_{r-d-1} is exactly the locus where $V \otimes \mathcal{O}_X \to E$ fails to be injective on fibers. This implies for general $(r-d)$ dimensional $V \subset H^0(X, E)$, we have a short exact sequence of locally free sheaves

$$0 \to V \otimes \mathcal{O}_X \to E \to E' \to 0,$$

where E' is locally free of rank d.

Suppose X is a smooth surface. Let E be a globally generated rank 2 vector bundle on X. Then for general $s \in H^0(X, E)$ we have by the above discussion that

$$\mathcal{O}_X \xrightarrow{s} E$$

is generically injective, except over a codimension 2 locus where s vanishes. Note that locally over some U_i we have $s|_{U_i} = (s_1, s_2)$ where $s_i \in \mathcal{O}_X(U_i)$. So $Z := X_s$ is a local complete intersection of codimesion 2. In particular, if we begin with the Koszul resolution

$$0 \to \bigwedge^2 E^\vee \to E^\vee \to \mathcal{I}_Z \to 0$$

resolving the ideal sheaf of Z, and let $\mathcal{L}^\vee := \bigwedge^2 E^\vee$, then if we tensor by $\mathcal{L}$ we obtain the short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E^\vee \otimes \mathcal{L} \to \mathcal{L} \otimes \mathcal{I}_Z \to 0.$$

Making the identification $E \cong E^\vee \otimes \mathcal{L} = E^\vee \otimes \det(E)$, we obtain the short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0.$$

A good reference for the subtle details involving these short exact sequences is in [OSS, Chapter 1, Section 5.1]. Thus, from E we have produced the data of $(\mathcal{L}, Z)$. Serre asked for the converse of this construction:

Question 20.4. *Let X be a nonsingular surface. Fix $\mathcal{L} \in \mathrm{Pic}(X)$ and a finite local complete intersection of codimension 2 subscheme Z. When does there exist a short exact sequence*

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

such that E is a rank 2 vector bundle?

It turns out that we have a converse exactly when the data of $\mathcal{L}$ and Z satisfy the Cayley-Barach property. Here is the theorem in generality.

Theorem 20.5. [HL, Chapter 5.1] *Let X be a nonsingular projective surface over an algebraically closed field of characteristic 0. Fix $\mathcal{L} \in \mathrm{Pic}(X)$ and a finite local complete intersection of codimension 2 subscheme Z. Then there exists an extension*

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

where E is locally free if and only if $(K_X \otimes \mathcal{L}, Z)$ satisfies the Cayley-Barach property:

- *for all $Z' \subset Z$ such that $\ell(Z') = \ell(Z) - 1$ and $s \in H^0(X, K_X \otimes \mathcal{L})$ such that $s|_{Z'} = 0$, then $s|_Z = 0$.*

We will now motivate the Cayley-Barach condition of Theorem 20.5. In particular, we will prove Theorem 20.5 at least in the case when Z is reduced, since I think this is where most of the intuition is – the general case including non-reduced subschemes will follow by some commutative algebra technicalities. This exposition follows Lazarsfeld's exposition on the Serre construction [Laz94, Section 3].

For now, let Z still be any arbitrary finite local complete intersection of codimension 2 subscheme of X. Fix $\mathcal{L} \in \mathrm{Pic}(X)$. First, we might look at some

$$e \in \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X).$$

If there is a non-trivial element, then we will obtain an extension

$$0 \to \mathcal{O}_X \to \mathcal{F}_e \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

where $\mathcal{F}_e$ is forced to be a torsion-free coherent sheaf of rank 2. What conditions are sufficient to force $\mathcal{F}_e$ to be locally free?

Proposition 20.6. *Let X be a smooth projective surface. Let $\mathcal{L} \in \mathrm{Pic}(X)$ and Z be a finite local complete intersection of codimension 2. Then $\mathcal{F}_e$ is not locally free if and only if $e \in Ext^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ in the image of the map*

$$Ext^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to Ext^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$$

induced by inclusion, for some proper $Z' \subset Z$ (possibly the empty subset).

Proof. First, suppose $\mathcal{F}_e$ is not locally free. Then note the double dual $\mathcal{F}_e^{\vee\vee}$ is reflexive on X, and thus is locally free. Since $\mathcal{F}_e$ is torsion-free, by definition the canonical map $\mathcal{F}_e \to \mathcal{F}_e^{\vee\vee}$ is injective. Then $\mathcal{O}_X \to \mathcal{F}_e$ induces an injection $\mathcal{O}_X \to \mathcal{F}_e^{\vee\vee}$. Note we have $\mathcal{F}_e|_{X\setminus Z} \cong \mathcal{F}_e^{\vee\vee}|_{X\setminus Z}$, but since $\mathcal{F}_e$ is not locally free we must have $\frac{\mathcal{F}_e^{\vee\vee}}{\mathcal{F}_e}$ is supported on some nonempty codimension 2 subscheme. In particular, the locus where $\mathcal{O}_X \to \mathcal{F}_e^{\vee\vee}$ vanishes on fibers is a local complete intersetion of codimension 2 which is properly contained in Z. Thus, again using the Koszul resolution we have there is a short exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{F}_e^{\vee\vee} \to \mathcal{L} \otimes \mathcal{I}_{Z'} \to 0,$$

and it turns out then we have a diagram

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] \arrow[d, "\mathrm{id}"'] & \mathcal{F}_e \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_Z \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] & \mathcal{F}_e^{\vee\vee} \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_{Z'} \arrow[r] \arrow[d] & 0 \\
 & & \frac{\mathcal{F}_e^{\vee\vee}}{\mathcal{F}_e} \arrow[r, "\cong"] \arrow[d] & \mathcal{L} \otimes \underline{\mathbb{C}} \arrow[d] & \\
 & & 0 & 0 &
\end{tikzcd}$$

Note the morphism $\mathcal{L} \otimes \mathcal{I}_Z \to \mathcal{L} \otimes \mathcal{I}_{Z'}$ is obtained by tensoring the short exact sequence

$$0 \to \mathcal{I}_Z \to \mathcal{I}_{Z'} \to \underline{\mathbb{C}} \to 0$$

by $\mathcal{L}$. By functoriality of Ext^1 in the first entry and how that translates in terms of extensions, this diagram exactly implies that $e \in \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ is in the image of the morphism

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X)$$

induced by $\mathcal{L} \otimes \mathcal{I}_Z \to \mathcal{L} \otimes \mathcal{I}_{Z'}$; in particular, it is in the image of the element of $\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ corresponding to the extension in the diagram exhibiting $\mathcal{F}_e^{\vee\vee}$.

We now prove the converse. Suppose $e \in \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ is in the image of

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$$

for some proper $Z' \subset Z$. Suppose it is in the image of e'. Then by functoriality there exists a diagram

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] \arrow[d, "\mathrm{id}"'] & \mathcal{F}_e \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_Z \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] & \mathcal{F}_{e'} \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_{Z'} \arrow[r] \arrow[d] & 0 \\
 & & \frac{\mathcal{F}_{e'}}{\mathcal{F}_e} \arrow[r, "\cong"] \arrow[d] & \mathcal{L} \otimes \underline{\mathbb{C}} \arrow[d] & \\
 & & 0 & 0 &
\end{tikzcd}$$

Now if $\mathcal{F}_{e'}$ is locally free, then note $\mathcal{F}_e$ cannot be locally free. This is because if $\mathcal{F}_e$ were locally free, then an injective morphism of sheaves $\mathcal{F}_e \to \mathcal{F}_{e'}$ must be generically injective on fibers and thus either injective on fibers everywhere or must fail to be injective along a divisor. If it were injective on fibers everywhere, then $Z = Z'$, contradicting our assumption. Then $\frac{\mathcal{F}_{e'}}{\mathcal{F}_e}$ would have to be supported on a divisor of X. But in fact, it is supported on a codimension 2 subscheme. This shows $\mathcal{F}_e$ could not be locally free if $\mathcal{F}_{e'}$ were locally free. Then it remains to show $\mathcal{F}_e$ still cannot be locally free even if $\mathcal{F}_{e'}$ is not locally free. If $\mathcal{F}_{e'}$ is not locally free, then we can consider its double dual $\mathcal{F}_{e'}^{\vee\vee}$. Since $\mathcal{F}_{e'}$ is torsion free we have $\mathcal{F}_{e'} \to \mathcal{F}_{e'}^{\vee\vee}$ is still injective morphism of sheaves, and we can again build a diagram

$$\begin{tikzcd}
 & & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] \arrow[d, "\mathrm{id}"'] & \mathcal{F}_e \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_Z \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathcal{O}_X \arrow[r] & \mathcal{F}_{e'}^{\vee\vee} \arrow[r] \arrow[d] & \mathcal{L} \otimes \mathcal{I}_{Z"} \arrow[r] \arrow[d] & 0 \\
 & & \frac{\mathcal{F}_{e'}^{\vee\vee}}{\mathcal{F}_e} \arrow[r, "\cong"] \arrow[d] & \mathcal{L} \otimes \underline{\mathbb{C}} \arrow[d] & \\
 & & 0 & 0 &
\end{tikzcd}$$

and we see that by the same argument as before, since $\mathcal{F}_{e'}^{\vee\vee}$ is locally free and $\mathcal{F}_e \to \mathcal{F}_{e'}^{\vee\vee}$ is generically injective on fibers but the quotient is supported on a codimension 2 subset then we cannot have $\mathcal{F}_e$ is locally free. $\square$

Now suppose Z is reduced. We claim that there exists

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

where E is locally free if and only if the images of

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$$

is proper for every proper (possibly empty) $Z' \subset Z$.

If there is such a locally free E, represented by $e \in \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$, then this means that all the induced maps on Ext^1 groups fail to include e in the image, so the image is a proper subspace for every proper (possibly empty) $Z' \subset Z$. On the other hand, suppose for every proper (including empty) $Z' \subset Z$ we have that the image of

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$$

is a proper subspace. Here comes the important point of our assumption: since Z is a finite reduced subscheme, there is only finitely many proper $Z' \subset Z$. Then in the affine space $\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$, there is certainly an element which avoids the finitely many proper subspaces that are images, which must then give rise to a locally free E.

Corollary 20.7. *Let X be a nonsingular projective surface. Fix $\mathcal{L} \in \mathrm{Pic}(X)$ and Z a finite reduced subscheme local complete intersection of codimension 2. Then there exists an extension*

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

where E is locally free if and only if the image of

$$Ext^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to Ext^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$$

is proper for every $Z' \subset Z$ such that $\ell(Z') = \ell(Z) - 1$.

Proof. Any proper $Z"$ must be so that $Z" \subseteq Z' \subset Z$ for some $Z' \subset Z$ where $\ell(Z') = \ell(Z) - 1$. Then we have the factorization

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z"}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X),$$

so if the image of $\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ is a proper subspace for every $Z' \subset Z$ with $\ell(Z') = \ell(Z) - 1$, then so is the image of $\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z"}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, \mathcal{O}_X)$ for every $Z" \subset Z$. ◻

We are now in a position to prove Theorem 20.5 in the case that Z is finite reduced. For all $Z' \subset Z$ such that $\ell(Z') = \ell(Z) - 1$, consider the short exact sequence

$$0 \to \mathcal{I}_Z \to \mathcal{I}_{Z'} \to \mathbb{C} \to 0.$$

Tensor by $\omega_X \otimes \mathcal{L}$ to obtain

$$0 \to \omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z \to \omega_X \otimes \mathcal{L} \otimes \mathcal{I}_{Z'} \to \mathbb{C} \to 0.$$

Taking cohomology LES, we obtain

$$0 \to H^0(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z) \to H^0(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_{Z'}) \to \mathbb{C} \to H^1(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z) \to H^1(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_{Z'}) \to 0.$$

Note that $(\omega_X \otimes \mathcal{L}, Z)$ has the Cayley-Barach property if and only if $H^0(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z) \cong H^0(\omega_X \otimes \mathcal{L} \otimes \mathcal{I}_{Z'})$, if and only if

$$\mathrm{Ext}^1(\mathcal{O}_X, \omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z) \to \mathrm{Ext}^1(\mathcal{O}_X, \omega_X \otimes \mathcal{L} \otimes \mathcal{I}_{Z'})$$

fails to be injective if and only if (taking Serre duality)

$$\mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_{Z'}, \mathcal{O}_X) \to \mathrm{Ext}^1(\mathcal{L} \otimes \mathcal{I}_Z, O_X)$$

fails to be surjective. This completes the proof of Theorem 20.5 in the case that Z is reduced.

Remark 20.8. *Everything we have down shows the forward direction of Theorem 20.5, and also explains the phrasing of the Cayley-Barach property in terms of sections of $\omega_X \otimes \mathcal{L}$ and the testing against subschemes of Z of length one shorter. The crux of showing the reverse direction is that when Z is not reduced, there may be infinitely many $Z' \subset Z$ with $\ell(Z') = \ell(Z) - 1$. The crucial point is that a local Gorenstein ring – in particular, a local ring of Z – has a unique minimal non-zero ideal of dimension one. Hence a finite local complete intersection scheme does not contain continuous families of maximal proper subschemes* [Laz94, Exercise 3.15].

Using the Serre correspondence, we will now show that for large enough c_2, there exists μ-stable rank 2 vector bundles on any smooth projective surface.

Proposition 20.9. *Let (X, H) be a smooth projective surface. Let $L \in \mathrm{Pic}(X)$. Then there exist c_0 such that for all $c \geq c_0$, there exist μ-stable rank 2 vector bundle E such that $\det E \cong \mathcal{L}$ and $c_2(E) = c$.*

Proof. First, we show that it suffices to prove the claim for $\deg \mathcal{L} > 0$. Suppose we have shown the claim in the positive degree case. We show it implies the general claim.

If $\deg \mathcal{L} \leq 0$, then there exists $n \gg 0$ such that $\deg \mathcal{L}' := \deg \mathcal{L} \otimes \mathcal{O}_X(2nH) > 0$. Then there exist c_0 such that for all $c \geq c_0$, there exist μ-stable rank 2 vector bundle E' such that $\det E' \cong \mathcal{L}'$ and $c_2(E') = c$. Now define $E := E' \otimes \mathcal{O}_X(-nH)$. Then $\det E \cong \mathcal{L}$ and

$$c_2(E) = c_2(E') - (nH.c_1(E')) + n^2(H.H) = c - n(H.c_1(\mathcal{L}) - 2nH) + n^2(H.H)$$

$$= c - n \deg \mathcal{L} + (2n + n^2)(H.H).$$

This shows that for all $\lambda \geq c_0 - n \deg \mathcal{L} + (2n + n^2)(H.H)$, there exist μ-stable E such that $\det E \cong \mathcal{L}$ and $c_2(E) = \lambda$. This complets our claim of reduction.

Thus, suppose $\deg \mathcal{L} > 0$. Now consider $h := \dim H^0(X, \omega_X \otimes \mathcal{L})$. For a general 0-dimensional subscheme Z of length h, we have $H^0(X, \omega_X \otimes \mathcal{L} \otimes \mathcal{I}_Z) = 0$. This is because everytime we ask for vanishing of global sections at a general point, we cut down the space of global sections by at least 1. Then the same holds for any general 0-dimensional subscheme Z of length $h+1$. Then let Z be a general 0-dimensional subscheme of length $\ell \geq h+1$. Then $(\omega_X \otimes \mathcal{L}, Z)$ will satisfy the Cayley-Barach property. Thus, by the Serre correspondence (20.5) there exists

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0$$

where E is locally free. Note that $\det E \cong \mathcal{L}$ and $c_2(E) = \ell(Z) \geq h+1$.

We are halfway there. The next step will be to show that we can increase the lower bound on the length of Z so that for general Z of sufficiently high length, E is forced to be μ-stable. Note that

$$\mu(E) = \frac{c_1(E).H}{2} = \frac{\deg \mathcal{L}}{2} > 0.$$

Then if E were not μ-stable, then there exist maximal destablising $F \subset E$. This is a saturated torsion free rank 1 subsheaf. Since X is a surface, this implies F is line bundle. Note that by construction of the maximal destablising subsheaf, since $\mu(F) \geq \mu(E) > \mu(\mathcal{O}_X) = 0$, we have then $F \not\subset \mathcal{O}_X$. This implies that

$$F \to E \to \mathcal{L} \otimes \mathcal{I}_Z$$

is a nontrivial morphism. In particular, It defines a nonzero element of $H^0(X, F^\vee \otimes \mathcal{L})$. Thus, there is an effective divisor D of degree $\deg \mathcal{L} - \deg F^\vee \leq \frac{1}{2} \deg \mathcal{L} = d$ containing Z. Then we will show that we can increase the lower bound on $\ell(Z)$ so that the general Z is not contained in any effective divisor of degree $\leq d$.

Let Hilb_X^t denote the Hilbert scheme of effective divisors of degree t. Then let $Y = \bigsqcup_{t \leq d} \mathrm{Hilb}_X^t$ denote the Hilbert scheme of effective divisors of degree $t \leq d$. There is a relative Hilbert scheme $\widetilde{Y}$ parameterizing tuples of ℓ' points on effective divisors of degree $t \leq d$ of X. This relative Hilbert scheme admits a map $\widetilde{Y} \to Y$, and we see that since its fiber dimension is ℓ', we have $\dim \widetilde{Y} = \dim Y + \ell'$. Now we consider the projection $\widetilde{Y} \to \mathrm{Hilb}(X, \ell')$, the Hilbert scheme of ℓ' points on X.

$$\begin{tikzcd}
\widetilde{Y} \arrow[r] \arrow[d] & \mathrm{Hilb}(X, \ell') \\
Y &
\end{tikzcd}$$

The image of the projection has dimension at most $\ell' + \dim Y$. Note that $\dim \mathrm{Hilb}(X, \ell') = 2\ell'$. Thus we pick ℓ' so that $\ell' > \max\{\dim Y, h+1\}$. This implies that the general 0-dimensional subscheme Z of length $\ell \geq \ell'$ will not be contained by an effective divisor of degree $t \leq d$. Then since $(K_X \otimes \mathcal{L}, Z)$ satisfies the Cayley-Barach property, we have there exists

$$0 \to \mathcal{O}_X \to E \to \mathcal{L} \otimes \mathcal{I}_Z \to 0,$$

so $\det E \cong \mathcal{L}$ and $c_2(E) = \ell \geq \ell'$. Furthermore, E is μ-stable since Z is chosen generally. $\square$

Remark 20.10. *The Cayley-Bacharach property is connected to this paper by Harris, Eisenbud, and Green (https://www.ams.org/journals/bull/1996-33-03/S0273-0979-96-00666-0/S0273-0979-96-00666-0.pdf).*

21. A beautiful example due to Tyurin and Beauville

Under the umbrella desire to see some concrete examples of moduli of sheaves, I'd like to recount a beautiful example due to Tyurin and Beauville. Tyurin in [Tyu08] considers the general situation in which restriction of vector bundles on a three-fold to an embedded surface provides a local isomorphism onto a Lagrangian between moduli of vector bundles. Beauville [Bea] added explicit examples of such phenomena to the literature, and here we recount one of these.

Let $X \subset \mathbb{P}^4$ be a smooth cubic three-fold. Fix very ample $\mathcal{O}_X(1) := \mathcal{O}_{\mathbb{P}^4}(1)|_X$ and note by the adjunction formula that $\omega_X \cong \mathcal{O}_X(-2)$. Note that X is a Fano 3-fold of Picard rank 1. We can use the Serre correspondence in the setting of nonsingular 3-dimensional projective varieties to produce certain rank 2 vector bundles on X:

Theorem 21.1 (Hartshorne-Serre correspondence)**.** *Let X be a nonsingular three dimensional projective algebraic variety. Fix an invertible sheaf $\mathcal{L}$ on X such that $H^1(\mathcal{L}^{-1}) = H^2(\mathcal{L}^{-1}) = 0$. Then there is a bijective correspondence between*

1. *the set of triples $(\mathcal{E}, s, \phi)$ where $\mathcal{E}$ is a rank 2 locally free sheaf on X, $s \in H^0(\mathcal{E})$ is a global section whose zero locus $(s)_0$ has codimension 2 and $\phi : \det \mathcal{E} = \wedge^2\mathcal{E} \to \mathcal{L}$ is an isomorphism of invertible sheaves modulo the equivalence relation $(\mathcal{E}, s, \phi) \sim (\mathcal{E}', s', \phi')$ which means there exists an isomorphism $\psi : \mathcal{E} \to \mathcal{E}'$ and non-zero $a \in k$ such that $s' = a\psi(s)$ and $\psi' = a^2\phi(\wedge^2\psi)^{-1}$, and*
2. *the set of pairs (Y, ξ) where Y is a closed subscheme of X of pure dimension 1, locally complete intersection and $\xi : \mathcal{L} \otimes \omega_X \otimes \mathcal{O}_Y \to \omega_Y$ is an isomorphism of invertible sheaves.*

Proof. A proof can be found in [Val04, Theorem 2]. □

In the setting of our smooth cubic three-fold $X \subset \mathbb{P}^4$, we let $\mathcal{L} := \mathcal{O}_X(1)$. If we were to have a smooth curve $C \subset X$ such that $\mathcal{L} \otimes \omega_X|_C \cong \mathcal{O}_X(-1)|_C \cong \omega_C$, note C would have to have genus 0, and since $\omega_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(-2)$, this implies that $C \cong \mathbb{P}^1$ would have to be embedded onto $X \subset \mathbb{P}^4$ as a rational conic. Note being a local complete intersection is an intrinsic property of a variety (Stacks project 23.8), and all rational conics are local complete intersections. So for any rational conic $C \subset X$, we obtain by the Serre correspondence a short exact sequence

$$0 \to \mathcal{O}_X \to E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C \to 0.$$

In particular, $c_1(E) = c_1(\mathcal{O}_X(1)) := h, c_2(E) = [C] \in \mathrm{CH}^2(X)$, and $c_3(E) = 0$. Note that any such E is actually μ-stable, because if E were not μ-stable, then there would exist a rank 1 maximal destablizing subsheaf $F \subset E$ with $\mu(F) \geq \mu(E) = \frac{(h^3)}{2}$. But in our situation and with F being rank 1, it is forced to be a line bundle. Then $F \cong \mathcal{O}_X(n)$ for $n > 0$ since $\mathrm{Pic}(X) \cong \mathbb{Z}\langle \mathcal{O}_X(1)\rangle$. So if E were not μ-stable, we must have $H^0(X, E \otimes \mathcal{O}_X(-1)) \neq 0$, but this is impossible because the short exact sequence obtained by Serre correspondence forces $H^0(X, E \otimes \mathcal{O}_X(-1)) \cong H^0(X, \mathcal{I}_C) = 0$. Altogether, if we let $\mathrm{Hilb}_X^{\mathrm{conics}}$ denote the Hilbert scheme of rational conics on X, and we let $M_X^s(2, h, [C], 0)$ denote the moduli space of Gieseker stable sheaves with the indicated Chern class information and $M_X \subset M_X^s(2, h, [C], 0)$ be the open loci of vector bundles, then the Serre correspondence induces a map

$$\mathrm{Hilb}_X^{\mathrm{conics}} \to M_X.$$

In fact, we claim this map is surjective. Let $E \in M_X(\mathbb{C})$ be a rank 2 Gieseker stable vector bundle with the specified Chern class information. By Hirzebruch-Riemann-Roch, we have

$$\chi(E) = \int_X ch(E)\cdot td(T_X) = \int_X (2+h+\frac{1}{2}h^2-[C]+\frac{1}{6}h^3-\frac{1}{2}h.[C])(1+h+\frac{1}{3}h^2+\frac{1}{12}c_2(T_X)+\frac{1}{12}h.c_2(T_X))$$

$$= \frac{1}{6}h.c_2(T_X) + 1 + \frac{1}{12}h.c_2(T_X) + \frac{3}{2} - 2 + \frac{1}{2} - 1 = \frac{1}{4}h.c_2(T_X).$$

Note that

$$\chi(\mathcal{O}_X(1)) = 5 = \int_X (1 + h + \frac{1}{2}h^2 + \frac{1}{6}h^3)(1 + h + \frac{1}{3}h^2 + \frac{1}{12}c_2(T_X) + \frac{1}{12}h.c_2(T_X))$$

$$= \frac{1}{12}h.c_2(T_X) + 1 + \frac{1}{12}h.c_2(T_X) + \frac{3}{2} + \frac{1}{2} \implies \frac{1}{6}h.c_2(T_X) = 2.$$

Altogether, this implies that $\chi(E) = 3$. The generic vector bundle in M_X has $h^0(E) = 3$ and higher cohomology vanishing. This implies that for every $E \in M_X(\mathbb{C})$ we have $h^0(E) \geq 3$. Furthermore, note that for any nonzero $s \in H^0(E)$, if we let D denote the zero loci of s, the Koszul resolution

$$0 \to \bigwedge^2 E^\vee \to E^\vee \to \mathcal{I}_D \to 0$$

implies we will have a short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_D \to 0.$$

We claim that D cannot be a divisor. Indeed, if it were, then since it's an ideal sheaf it would have negative slope and since $\mathrm{Pic}(X) \cong \mathbb{Z}\langle \mathcal{O}_X(1)\rangle$, we'd then have that $\det E \ncong \mathcal{O}_X(1)$, contradiction. So D must be a local complete intersection curve C. But then we're back in the situation of the Serre correspondence. And note that since $h^0(E) \geq 3$ we must have $h^0(\mathcal{O}_X(1) \otimes \mathcal{I}_C) \geq 2$, but we must have $h^0(\mathcal{O}_X(1) \otimes \mathcal{I}_C) = 2$ since C cannot be contained in a line, which forces $h^0(E) = 3$. Altogether, what we have shown is that the Serre correspondence map

$$\mathrm{Hilb}_X^{\mathrm{conics}} \xrightarrow{\mathbb{P}^2} M_X$$

is surjective[12] and for every $E \in M_X(\mathbb{C})$ we automatically have $h^0(E) = 3$ with $h^1(E) = h^2(E) = 3$. Furthermore, every $0 \neq s \in H^0(E)$ corresponds to a rational conic unique up to scaling s, so we see that the above map is a surjection with $\mathbb{P}^2$ fibers.

Next, we claim that $\mathrm{Hilb}_X^{\mathrm{conics}}$ is smooth. Indeed, for each rational conic $C \subset X$, notice that it spans a 2-plane, which must intersect X at a degree 3 curve. This implies each C is associated to a residual line. Then there is a map $\mathrm{Hilb}_X^{\mathrm{conics}} \to \mathrm{Hilb}_X^{\mathrm{lines}} =: F(X)$ to the Fano scheme of lines on X, which is nonsingular of dimension 2 [Huy]. Clearly this map is surjective and there is a $\mathbb{P}^2$ worth of 2-planes containing each line on X, which implies that the fiber of this map is $\mathbb{P}^2$. This implies that $\mathrm{Hilb}_X^{\mathrm{conics}}$ is smooth of dimension 4.

Note that for any rational conic $C \subset X$, if we restrict the associated short exact sequence $0 \to \mathcal{O}_X \to E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C$ to C, we obtain $E^\vee \otimes \mathcal{O}_X(1)|_C \cong E|_C \cong \mathcal{N}_C$. This tells us the Chern class information of $\mathcal{N}_C$, and using Riemann-Roch we find $\chi(\mathcal{N}_C) = 4$. Since $\mathrm{Hilb}_X^{\mathrm{conics}}$ is smooth of dimension 4, this implies $H^1(\mathcal{N}_C) = 0$. By [Bea, Proposition 1], this implies that $H^2(\mathscr{E}\mathrm{nd}(E)) = 0$. Note that a priori there is an open loci of M_X of those vector bundles with have $H^2(\mathscr{E}\mathrm{nd}(E)) = 0$. But what we have just shown is that this open locus is all of M_X. In particular, $H^2(\mathscr{E}\mathrm{nd}(E)) = 0$ means that M_X is smooth everywhere.

Next, we claim there is a well-defined map $M_X \to F(X)$ which is actually an isomorphism, so that we have a diagram

$$\begin{tikzcd} \mathrm{Hilb}_X^{\mathrm{conics}} \arrow[r] \arrow[dr] & M_X \arrow[d] \\ & F(X) \end{tikzcd}$$

We define the map $M_X \to F(X)$ in the following way: suppose $0 \neq s \in H^0(E)$. Recall s is associated to a rational conic $C \subset X$ by the short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C \to 0.$$

Furthermore, C is associated to a residual line ℓ. We claim the residual line ℓ does not depend on the choice of s, i.e. for $s, s' \in H^0(E)$ associated to C and C', the residual lines ℓ and ℓ' are actually the same line! To prove this, we'll need the following two lemmas which are deep facts about the geometry of cubic three-folds.

Lemma 21.2. [Huy, Chapter 5.3, Corollary 3.5] *The Abel-Jacobi map for a smooth cubic three-fold to its second intermediate Jacobian*[13]

$$\theta : CH^2(X) \to \mathcal{J}^2(X) = \frac{H^{2,1}(X)^\vee}{H_3(X, \mathbb{Z})}$$

[12] In particular, note this means every Gieseker stable vector bundle with this Chern class information is also μ-stable.

[13] A quick and clean construction of intermediate Jacobians is given in [Zak12].

is an isomorphism, and choosing a line ℓ_0 as a basepoint, the map embeds $F(X)$ into the intermediate Jacobian via $\ell \mapsto \theta(\ell - \ell_0)$.

The above result is due to Beauville. Note that $c_2(E) = [C] = [C'] \in \mathrm{CH}^2(X)$, and $[C] = h^2 - \ell$ and $[C'] = h^2 - \ell'$. Thus $\ell = \ell' \in \mathrm{CH}^2(X)$, which by the above lemma implies that $\ell = \ell'$. This establishes the diagram

$$\begin{tikzcd} \mathrm{Hilb}_X^{\mathrm{conics}} \arrow[r] \arrow[dr] & M_X \arrow[d] \\ & F(X) \end{tikzcd}$$

The map $M_X \to F(X)$ is clearly a surjection on $\mathbb{C}$-points. Furthermore, it is an injection on $\mathbb{C}$-points since if $E, E' \in M_X(\mathbb{C})$ map to the same residual line $\ell \in F(X)(\mathbb{C})$, then we'd have short exact sequences

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C \to 0$$
$$0 \to \mathcal{O}_X \xrightarrow{s'} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_{C'} \to 0,$$

but since C and C' share the same residual line, there must exist $s" \in H^0(E)$ whose vanishing loci is C', so that we have short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s"} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_{C'} \to 0,$$

but by uniqueness of the Serre correspondence this forces $E \cong E'$. Then since M_X and $F(X)$ are both smooth, the bijection on $\mathbb{C}$-points forces $M_X \to F(X)$ to be an isomorphism of schemes.

Let us clarify this geometric phenomenon even more. Note that since any $E \in M_X(\mathbb{C})$ is witnessed by a short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C \to 0,$$

we have $\det E \cong \mathcal{O}_X(1)$. Note that there is a natural map

$$\phi : \bigwedge^2 H^0(E) \to H^0(\bigwedge^2 E) \cong H^0(\mathcal{O}_X(1)) \cong H^0(\mathcal{O}_{\mathbb{P}^4}(1))$$

If we let $H^0(E) \cong \mathbb{C}\langle s_0, s_1, s_2\rangle$ where locally $s_j = (f_j, g_j)$, then $\bigwedge^2 H^0(E) \cong \mathbb{C}\langle s_0 \wedge s_1, s_0 \wedge s_2, s1 \wedge s_2\rangle$. Note $s_i \wedge s_j = f_i g_j - f_j g_i$ locally. The map ϕ is injective, and so to any E we have an associated $\mathbb{C}^3$ space of linear forms, or a $\mathbb{P}^2$ space of hyperplanes. One sees that in the short exact sequence

$$0 \to \mathcal{O}_X \xrightarrow{s} E \to \mathcal{O}_X(1) \otimes \mathcal{I}_C \to 0,$$

we have $H^0(\mathcal{O}_X(1) \otimes \mathcal{I}_C)$ is exactly the space of hyperplanes which can be written in the form $s \wedge -$. Thus our geometric situation is the following: on the smooth cubic 3-fold X, there is a bijection between lines on X and rank 2 vector bundles $E \in M_X(\mathbb{C})$ on X. There is a $\mathbb{P}^2$ worth of hyperplanes in $\mathbb{P}^4$ containing this line, and choosing a pencil of hyperplanes is the same as choosing a section $s_0 \in H^0(E)$. The pencil of hyperplanes $\{\lambda_1 s_0 \wedge s_1 + \lambda_2 s_0 \wedge s_2\}$ intersect to give a 2-plane which intersects X at the residual line and the rational conic associated to the section of E.

Now let $S = X \cap Q$ be a very general smooth K3 surface on X, where Q is a quadric hypersurface. Note S is an anticanonical K3 surface of the Fano 3-fold X. The quadric Q should be chosen very generally so that Q does not contain any of the lines on X and $\mathrm{Pic}(S) = \mathbb{Z}\langle \mathcal{O}_S(1)\rangle$. Then consider the restriction of $E \in M_X(\mathbb{C})$ to S: the rank 2 vector bundle $E|_S$ will have Chern classes $c_1(E|_S) = c_1(\mathcal{O}_S(1))$ and $c_2(E|_S) = [S.C] = [Z] = 4 \in \mathrm{CH}^2(S)$. We also obtain short exact sequences of the form

$$0 \to \mathcal{O}_S \to E|_S \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0$$

where $Z = C \cap S$ is a length 4 finite subscheme of S. By a similar argument as we did before, the above short exact sequence and the fact that $\mathrm{Pic}(S) \cong \mathbb{Z}\langle \mathcal{O}_S(1)\rangle$ forces

- $E|_S$ to be μ-stable
- every $0 \neq s \in H^0(E|_S)$ vanishes as expected: along a length 4 finite scheme $Z \cap C$. Note every section of $E|_S$ comes from the restriction of a section of E.

If we let $M_S^s(2, \mathcal{O}_S(1), 4)$ denote the moduli space of Gieseker stable rank 2 sheaves with the prescribed Chern classes, and let $M_S \subset M_S^s(2, \mathcal{O}_S(1), 4)$ denote the open loci of those which are vector bundles, then we have a restriction map

$$\begin{tikzcd} \mathrm{Hilb}_X^{\mathrm{conics}} \arrow[r] \arrow[dr] & M_X \arrow[r, "\mathrm{res}"] \arrow[d] & M_S \\ & F(X) & \end{tikzcd}$$

The Hirzebruch-Riemann-Roch tells us that $\chi(E) \geq 3$ for any $E \in M_S(\mathbb{C})$. Note $h^2(E) \cong h^0(E^\vee) = 0$ since S is a K3 surface and $E^\vee$ is Gieseker stable with negative slope. Then $h^0(E) \geq 3$. For any $0 \neq s \in H^0(S, E)$ we have the short exact sequence

$$0 \to \mathcal{O}_S \xrightarrow{s} E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0$$

where Z must be a length 4 finite scheme; Z cannot jump to have dimension 1, otherwise it will be a divisor - this is the same argument as we did before for $E \in M_X(\mathbb{C})$. Then, again, we have $h^0(\mathcal{O}_S(1) \otimes \mathcal{I}_Z) = 2$ since $Z \subset S = X \cap Q$ and by Bezout's theorem Z cannot be contained in a line; the line would intersect X and Q at more than 3 and 2 points respectively, and thus would need to be contained in X and Q by Bezout's theorem, but this cannot happen by assumption of how Q was chosen. This forces $h^0(E) = 3$. Thus, for every $E \in M_S(\mathbb{C})$ we have E is μ-stable, and $h^0(E) = 3$ and $h^1(E) = h^2(E) = 0$. Furthermore, every $0 \neq s \in H^0(E)$ vanishes at a finite length 4 scheme Z such that we have a short exact sequence induced by the Koszul complex:

$$0 \to \mathcal{O}_S \to E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0.$$

Next, let $S^{[2]}$ denote the Hilbert scheme of 2 points on S. We claim that the diagram can be extended to

$$\begin{tikzcd} \mathrm{Hilb}_X^{\mathrm{conics}} \arrow[r] \arrow[dr] & M_X \arrow[r] \arrow[d] & M_S \arrow[d] \\ & F(X) & S^{[2]} \end{tikzcd}$$

so that the map $M_S \to S^{[2]}$ is not only well-defined but actually an isomorphism. The map $M_S \to S^{[2]}$ is defined as follows: for any $E \in M_S(\mathbb{C})$ if we pick a section $s \in H^0(E)$ then it vanishes along a length 4 scheme Z. Note $h^0(\mathcal{O}_S(1) \otimes \mathcal{I}_Z) = 2$, so Z spans a plane Λ. Then $\Lambda \cap S$ is a length 6 finite scheme, and let R denote the residual length 2 finite scheme of $Z \subset \Lambda \cap S$. So the map $M_S \to S^{[2]}$ is defined as sending E to this residual length 2 finite scheme R.

We claim that $M_S \to S^{[2]}$ is well-defined. For any $E \in M_S(\mathbb{C})$, the short exact sequences

$$0 \to \mathcal{O}_S \xrightarrow{s} E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0$$

implies that $\det E \cong \mathcal{O}_S(1)$ so we consider the natural map

$$\phi : \bigwedge^2 H^0(S, E) \to H^0(S, \bigwedge^2 E) \cong H^0(S, \mathcal{O}_S(1)) \cong H^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(1)).$$

This map is analogous to the map we considered for $E \in M_X(\mathbb{C})$; it is also injective and if we identify $H^0(S, E) \cong \mathbb{C}\langle s_0, s_1, s_2\rangle$ then $\mathbb{C}\langle \phi(s_0 \wedge s_1), \phi(s_0 \wedge s_2), \phi(s_1 \wedge s_2)\rangle$ is a net of hyperplanes in $\mathbb{P}^4$. Furthermore, the hyperplanes in $H^0(\mathcal{O}_S(1) \otimes \mathcal{I}_Z)$ intersect to give the plane that Z spans. This pencil of hyperplanes is given by linear forms of the form $\phi(s \wedge -)$.

Now consider the line which is the intersection of all hyperplanes in the net:

$$\phi(s_0 \wedge s_1) \cap \phi(s_0 \wedge s_2) \cap \phi(s_1 \wedge s_2) = \ell.$$

Then $\ell \cap S$ will be some finite scheme. Note that over $\ell \cap S$, the evaluation morphism

$$H^0(S, E) \otimes \mathcal{O}_S \xrightarrow{\mathrm{ev}} E$$

fails to be surjective. Note that the evaluation morphism can only drop to rank 1 and not rank 0 on fibers, since otherwise this would imply E would have a base point; but it is 0-regular and thus globally generated so this cannot happen. Then the kernel of ev is a torsion free of rank 1; over $S \setminus (\ell \cap S)$ it is isomorphic to $\mathcal{O}_S(-1)$. By a Hartog extension property for torsion free rank 1 sheaves, the kernel of ev is in fact $\mathcal{O}_S(-1)$ everywhere:

$$0 \to \mathcal{O}_S(-1) \to H^0(S, E) \otimes \mathcal{O}_S \xrightarrow{\mathrm{ev}} E \to \mathcal{O}_{\ell \cap S} \to 0$$

and the cokernel of ev is $\mathcal{O}_{\ell \cap S}$ by the 'drop by only rank 1 argument.' If we dualize ev, note

$$0 \to E^\vee \xrightarrow{\mathrm{coev}} H^0(S, E)^\vee \otimes \mathcal{O}_S$$

is an injection of sheaves because coev is an injective on $S \setminus (\ell \cap S)$, so the kernel would is torsion free sheaf of rank 0, and thus must be the 0. The cokernel of coev on $S \setminus (\ell \cap S)$ is $\mathcal{O}_S(1)$. However, coev should fail to be injective on some finite scheme A. Using Chern classes, one calculates that

$$0 \to E^\vee \xrightarrow{\text{coev}} H^0(S,E)^\vee \otimes \mathcal{O}_S \to \mathcal{O}_S(1) \otimes \mathcal{I}_A \to 0$$

where A is a finite scheme of length 2. Note $A \subset \ell \cap S$. We claim that A is the residual scheme of length 2 R.

Indeed, for general $s \in H^0(E)$ we have $Z(s) = Z$ is disjoint from A. If this were not the case for some $p \in A$, then we could pick sections s_0^p, s_1^p, s_2^p spanning $H^0(E)$ and $\phi(s_0^p \wedge s_1^p), \phi(s_0^p \wedge s_1^p)$ and $\phi(s_1^p \wedge s_2^p)$ would all vanish at p at order least 2, contradiction since their intersection should be a line which is reduced.

So for general $s \in H^0(E)$, we have $Z(s) = Z$, but if we consider the 2-plane $\Lambda = Z(\phi(s \wedge s')) \cap Z(\phi(s \wedge s''))$, note $\ell \cap S \subset \Lambda \cap S$ and in particular $A \subset \Lambda \cap S$. Since A and Z are disjoint, this implies $A + Z = \Lambda \cap S$, so we see that A is the residual finite scheme of length 2. The expression that we have found for A in terms of the coevaulation also implies that A is the residual scheme of length 2 for any $0 \neq s \in H^0(E)$, not just the general section.

Thus, the map $M_S \to S^{[2]}$ is well-defined. Note for any $E \in M_S(\mathbb{C})$ and nonzero section $s \in H^0(E)$, we have the short exact sequence

$$0 \to \mathcal{O}_S \to E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0$$

where Z is finite of length 4. By Theorem 20.5, the Serre correspondence on smooth projective surfaces, Z has the Cayley-Bacharach property. Thus, for any $s \in H^0(S, \mathcal{O}_S(1))$ such that $s|_{Z'} = 0$ for some $Z' \subset Z$ where $\ell(Z') = 3$, we have $s|_Z = 0$ as well. In particular, this implies that any $Z' \subset Z$ of length 3 cannot be contained in a line. Then consider the locally closed subscheme $\mathcal{H} \subset S^{[4]}$ parameterizing length 4 finite Z satisfying the Cayley-Bacharach property and satisfying $H^0(\mathcal{O}_S(1) \otimes \mathcal{I}_Z) = 2$. For any such Z, we have $\text{ext}^1(\mathcal{O}_S(1) \otimes \mathcal{I}_Z, \mathcal{O}_S) = h^1(S, \mathcal{O}_S(1) \otimes \mathcal{I}_Z)$ by Serre duality and

$$0 \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to \mathcal{O}_S(1) \to \mathcal{O}_S(1)|_Z \to 0$$

which implies $h^1(S, \mathcal{O}_S(1) \otimes \mathcal{I}_Z) = 1$, so there exists a unique rank 2 torsion free E with relation

$$0 \to \mathcal{O}_S \to E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0$$

and since Z satisfies the Cayley-Bacharach property, this E is locally free by the Serre correspondence on smooth projective surfaces. Therefore, we have a map $\mathcal{H} \to M_S$ which by the residual length 2 subscheme argument factors through

$$\mathcal{H} \to S^{[2]} \to M_S \to S^{[2]}.$$

Note that the map $\mathcal{H} \to S^{[2]}$ is surjective on $\mathbb{C}$-points.

- To see this, first consider $(p,q) \in S^{[2]}(\mathbb{C})$ for distinct p and q. Consider the line $\ell = \overline{pq}$. Any 2-plane Λ containing p and q must contain ℓ, and $\Lambda \cap S$ is a length 6 subscheme of S. One must show that Λ can be picked so that the residual length 4 subscheme satisfies the Cayley-Bacharach property. Note that if a residual length 4 subscheme failed to satisfy the Cayley-Bacharach property, then there is a hyperplane section vanishing at three of the points but not the fourth – since they are all contained in a 2-plane, this implies that there is a line through three of the points but not the fourth. Then it suffices to show that for any p and q, there exists a 2-plane Λ containing $\ell = \overline{pq}$ such that no three of the points in the residual length 4 subscheme lie on a line not containing the fourth.
- Note Q is chosen very generally so that X and Q do not share any lines. Note there are infinitely many lines on Q. First, suppose ℓ is not a line on Q. Consider any 2-plane Λ containing ℓ. Then Q and X define a conic curve and cubic curve, respectively, in the 2-plane Λ. If three of the residual four points lie on a line ℓ', then ℓ' must be an irreducible component of the conic curve. Then the conic curve must be two lines (either intersecting transversely at one point, or a double line). One can choose Λ so that the conic curve is two lines intersecting at a point. Note that ℓ' cannot go through p or q, otherwise then ℓ' is contained in X and then X and Q share a line. But then the other line component of the conic curve in Λ must intersect p and q. But then the line must equal ℓ, but by assumption ℓ cannot lie on Q. Thus, any such $p + q \in S^{[2]}(\mathbb{C})$ is in the image of $\mathcal{H} \to S^{[2]}$. One sees that $h^0(\mathcal{O}_S(1) \otimes \mathcal{I}_Z) = 2$ in this case as well, since the the residual 4 points are contained in a 2-plane.
- Now suppose $\ell = \overline{pq}$ does lie on Q. Then any 2-plane Λ containing ℓ will intersect Q at a conic on Λ which will be two lines. It is totally possible that if the intersection is two lines intersecting transversely at a single point, then there may be three points lying on the other line ℓ' and the fourth lies on ℓ. But we can choose Λ to give double line 2ℓ, which to satisfy the Cayley-Bacharach property and h^0 requirement.
- Finally, if we choose $2p \in S^{[2]}(\mathbb{C})$, we can choose a 2-plane Λ containing 2p such that $\Lambda \cap Q$ is a double line, which is tangent to Q at p.

Then the composition implies that $M_S \to S^{[2]}$ is surjective on $\mathbb{C}$-points. By uniqueness of Serre correspondence, it is also injective. One can also show that $H^2(S, \mathscr{E}\mathrm{nd}(E)) = 0$ for $E \in M_S(\mathbb{C})$ so that M_S is smooth. Since $S^{[2]}$ is smooth by Fogarty's result (Proposition 15.8), the bijection on $\mathbb{C}$-points implies that $M_S \to S^{[2]}$ is an isomorphism.

The overall geometric situation of $M_S \cong S^{[2]}$ here is the following. There is a bijection between $E \in M_S(\mathbb{C})$ and tuples of 2 points on S. If we let $H^0(E) = \mathbb{C}\langle s_0, s_1, s_2\rangle$, then note any 2 points on S (including those of multiplicity 2) will span a line, and there will be

3-dimensional vector space of linear forms restricted to S which vanish on this line. This 3-dimensional vector space is identified with $\mathbb{C}\langle \phi(s_0 \wedge s_1), \phi(s_0 \wedge s_2), \phi(s_1 \wedge s_2)\rangle$. Any particular section s we choose so that we obtain short exact sequence

$$0 \to \mathcal{O}_S \to E \to \mathcal{O}_S(1) \otimes \mathcal{I}_Z \to 0,$$

the 2-plane spanned by Z is the common vanishing loci of the pencil described by $\{\phi(s \wedge -)\}$.

The picture we have achieved is the following:

$$\begin{tikzcd} \mathrm{Hilb}_X^{\mathrm{conics}} \arrow[r] \arrow[dr] & M_X \arrow[r, "\mathrm{res}"] \arrow[d] & M_S \arrow[d] \\ & F(X) \arrow[r] & S^{[2]} \end{tikzcd}$$

We have concretely identified M_X and M_S with the geometry of X and S, and the restriction map $\mathrm{res} : M_X \to M_S$ is very clear: simply intersect a line $\ell \subset X$ with the quadric Q to obtain $\ell \cap Q \in S^{[2]}$. Note we see again it was important that Q is chosen very generally so as not to contain any line on X.

Finally, res is actually a closed embedding onto a Lagrangian. First, we should note that M_S is a symplectic manifold: it carries a non-degenerate symplectic 2-form. Recall that the tangent space of M_S at $E \in M_S(\mathbb{C})$ is $\mathrm{Ext}^1(E, E) \cong H^1(S, \mathscr{E}\mathrm{nd}(E))$. The symplectic form on M_S at least at the level of tangent spaces $T_{[E]}M_S$ is given by the composition

$$H^1(S, \mathscr{E}\mathrm{nd}(E)) \otimes H^1(S, \mathscr{E}\mathrm{nd}(E)) \xrightarrow{\cup} H^2(S, \mathscr{E}\mathrm{nd}(E)) \xrightarrow{\mathrm{tr}_*} H^2(S, \mathcal{O}_S) \cong \mathbb{C}$$

where $\cup$ is the cup product and tr_* is induced by the surjective trace morphism $\mathrm{tr} : \mathscr{E}\mathrm{nd}(E) \to \mathcal{O}_S$. Viewing sheaf cohomology as twisted Dolbeaut cohomology, this pairing is in essence given by wedging 1-forms and integration. The non-degeneracy of the pairing is exactly given by Serre duality, which identifies $H^1(S, \mathscr{E}\mathrm{nd}(E)) \cong H^1(S, \mathscr{E}\mathrm{nd}(E))^\vee$. Importantly, the pairing works out here because E is locally free so $\mathscr{E}\mathrm{nd}(E)^\vee \cong \mathscr{E}\mathrm{nd}(E)$ and S is a K3 surface so $\omega_S \cong \mathcal{O}_S$. Thus, that M_S is symplectic is fundamentally because S is a K3 surface.[14]

Now recall that for every $E \in M_X(\mathbb{C})$ we had $H^2(X, \mathscr{E}\mathrm{nd}(E)) = 0$. Note that $S \subset X$ is an anticanonical K3, so that tensoring its ideal sheaf short exact sequence by $\mathscr{E}\mathrm{nd}(E)$ yields:

$$0 \to \mathscr{E}\mathrm{nd}(E) \otimes \omega_X \to \mathscr{E}\mathrm{nd}(E) \to \mathscr{E}\mathrm{nd}(E|_S) \to 0.$$

Then $H^2(X, \mathscr{E}\mathrm{nd}(E)) = 0 \implies H^1(X, \mathscr{E}\mathrm{nd}(E) \otimes \omega_X) = 0$ by Serre duality, which implies there is a short exact sequence

$$0 \to H^1(X, \mathscr{E}\mathrm{nd}(E)) \to H^1(S, \mathscr{E}\mathrm{nd}(E|_S)) \to H^2(X, \mathscr{E}\mathrm{nd}(E) \otimes \omega_X) \to 0.$$

[14]To argue that the symplectic pairing can be globally defined is a bit subtle and is treated in [HL]. The original treatment was given by Mukai on moduli spaces of simple sheaves on K3 surfaces.

Note that $T_{[E]}M_X = H^1(X, \mathscr{E}\mathrm{nd}(E))$ and $T_{[E|_S]}M_S = H^1(S, \mathscr{E}\mathrm{nd}(E|_S))$. Furthermore, $T_{[E]}M_X^\vee \cong H^2(X, \mathscr{E}\mathrm{nd}(E) \otimes \omega_X)$. This shows that $\mathrm{res} : M_X \to M_S$ is a local isomorphism. Furthermore, by functoriality, we have the diagram

$$\begin{tikzcd}
H^1(X,\mathscr{E}\mathrm{nd}(E)) \otimes H^1(X,\mathscr{E}\mathrm{nd}(E)) \arrow[r] \arrow[d] & H^1(S,\mathscr{E}\mathrm{nd}(E|_S)) \otimes H^1(S,\mathscr{E}\mathrm{nd}(E|_S)) \arrow[d] \\
0 = H^2(X,\mathscr{E}\mathrm{nd}(E)) \arrow[r] \arrow[d] & H^2(S,\mathscr{E}\mathrm{nd}(E|_S)) \arrow[d] \\
0 = H^2(X,\mathcal{O}_X) \arrow[r] & H^2(S,\mathcal{O}_S)
\end{tikzcd}$$

and since $H^2(X, \mathscr{E}\mathrm{nd}(E)) = 0$ for every $E \in M_X(\mathbb{C})$, the diagram implies that res is a local isomorphism onto a Lagrangian subvariety of M_S. In fact, $\mathrm{res} : M_X \to M_S$ is an embedding onto a Lagrangian of M_S which can be most easily observed from the fact that $\mathrm{res} : F(X) \to S^{[2]}$ must be injective since there is a unique line through any 2 points.

22. Appendix

22.1. $\mathbf{Ext}^1$ correspondence.

For any abelian category $\mathcal{A}$ with enough projectives and injectives, one has the correspondence

$$\mathrm{Ext}^1(A, B) = \{ \text{ extensions of A by B } \}/\sim$$

which is in fact an isomorphism of abelian groups. By the Freyd-Mitchell embedding theorem, any small abelian category admits a fully faithful exact embedding into the category of R-modules for some ring with unit R (not necessarily commutative).

Thus, we prove the correspondence in the category of R-modules, since the proof for the correspondence in any small abelian category with enough projectives and injectives not only 'feels the same,' it almost literally is the same, by the Freyd-Mitchell embedding theorem.

Proposition 22.1. *Let A and B be R-modules. Then there is an isomorphism of abelian groups*

$$\{ \textit{ isomorphism classes of extensions of } A \textit{ by } B\} \longleftrightarrow \mathit{Ext}^1(A, B).$$

Recall that two extensions $0 \to B \to X \to A \to 0$ and $0 \to B \to X' \to A \to 0$ are isomorphic if there exists a diagram

$$\begin{tikzcd}
0 \arrow[r] & B \arrow[r] \arrow[d, "\mathrm{id}"'] & X \arrow[r] \arrow[d, "\cong"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

Let us first discuss what the correspondence is set-wise. We'll worry about the group structures later. Define a map

$$\Theta : \{ \text{ isomorphism classes of extensions of A by B} \} \to \mathrm{Ext}^1(A, B)$$

in the following way: suppose ξ is an isomorphism class represented by some extension $0 \to B \to X \to A \to 0$, from which we obtain a long exact sequence

$$0 \to \mathrm{Hom}_R(A, B) \to \mathrm{Hom}_R(X, B) \to \mathrm{Hom}_R(B, B) \xrightarrow{\delta} \mathrm{Ext}^1(A, B) \to \cdots$$

Then define

$$\Theta(\xi) := \delta(\mathrm{id}_B).$$

Note that this is well-defined: if two extensions are in the same isomorphism class, then there are identity maps between their respective copies of B and A. These identity maps lift to identity maps between projective resolutions on each respective B and A, which induces an identity identification between the two connecting morphisms $\mathrm{Hom}_R(B, B) \xrightarrow{\delta} \mathrm{Ext}^1(A, B)$.

Now we show that Θ is surjective. Suppose $x \in \mathrm{Ext}^1(A, B)$. How can we access this element x? Note we have a short exact sequence

$$0 \to M \to P \to A \to 0$$

where P is a projective R-module. Then part of the long exact sequence is the exact sequence

$$0 \to \mathrm{Hom}_R(A, B) \to \mathrm{Hom}_R(P, B) \to \mathrm{Hom}_R(M, B) \xrightarrow{\delta} \mathrm{Ext}^1(A, B) \to 0$$

Note $\mathrm{Ext}^1(P, B) = 0$ since P is projective, we can simply take $\cdots \to 0 \to P_0 \to P \to 0$ to be a projective resolution of P where $P_0 = P$. Then let $\beta \in \mathrm{Hom}_R(M, B)$ be a lift so that $\delta(\beta) = x$. Then consider the pushout diagram

$$\begin{tikzcd} M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[d, "\sigma"] \\ B \arrow[r, "\iota"'] & X \end{tikzcd}$$

where X may be explicitly written as

$$X = \frac{B \oplus P}{\langle \iota\beta(m) - \sigma j(m) \rangle}.$$

Here $\iota : B \to X$ and $\sigma : P \to X$ descend from the inclusion maps $\iota : B \to B \oplus P$ where $\iota(b) = (b, 0)$ and $\sigma : P \to B \oplus P$ where $\sigma(p) = (0, p)$. Note that $\iota : B \to X$ is injective. Because if $\iota(b) \in X$ was 0, this would mean we could write $\iota(b) = (b, 0) = (\beta(m), -j(m))$ for some $m \in M$. But then $j(m) = 0 \implies m = 0 \implies b = 0$. Furthermore, the quotient of X by $B \times 0$ is A. Because then any element $(b, p) \in X$ could be written as $(0, p)$, and under the equivalence relation on X we have $(0, p) \sim (\beta(m), p - j(m))$, which is then equivalent to $(0, p - j(m))$. Thus, the quotient looks like $0 \times A$. We see then we have a morphism of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[r] \arrow[d, "\sigma"] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r, "\iota"'] & X \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

So to recap, we fixed $x \in \mathrm{Ext}^1(A, B)$, then lifted it to an element $\beta \in \mathrm{Hom}_R(M, B)$ such that $\delta(\beta) = x$. Then from β we obtain the above diagram and in particular the extension

$$\xi : 0 \to B \xrightarrow{\iota} X \to A \to 0.$$

We claim that $\Theta(\xi) = x$. Indeed, by naturality, the element $\mathrm{id}_B \in \mathrm{Hom}_R(B, B)$ is sent to $\beta \in \mathrm{Hom}_R(M, B)$, which is sent to $x \in \mathrm{Ext}^1(A, B)$ (the one corresponding to $0 \to M \to P \to A \to 0$), which is sent to $x \in \mathrm{Ext}^1(A, B)$ (the one corresponding to ξ) since in the morphism of short exact sequences the map between the two copies of A is identity. We are almost done showing surjectivity; to conclude, we should verify that this whole process is well-defined with respect to the choice β lifting x. Suppose $\beta' \in \mathrm{Hom}_R(M, B)$ is some other element such that $\delta(\beta') = x$. Then note $\beta - \beta' \in \ker \delta$, so there exists some $f \in \mathrm{Hom}_R(P, B)$ such that

$$\beta' = \beta + f \circ j.$$

From β' we obtain the diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r, "j"] \arrow[d, "\beta'"'] & P \arrow[r] \arrow[d, dashed, "\sigma"] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r, "\iota"] & X' \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

where $X' = \frac{B \oplus P}{\langle \iota\beta'(m) - \sigma j(m) \rangle}$. We would like to show that there exists a diagram

$$\begin{tikzcd}
0 \arrow[r] & B \arrow[r] \arrow[d, "\mathrm{id}"'] & X \arrow[r] \arrow[d, "\phi"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

where ϕ is an isomorphism. Indeed, we can define $\phi(b, p) = (b - f(p), p)$. Alternatively, this can be written as $\phi : \iota(b) + \sigma(p) \mapsto \iota(b - f(p)) + \sigma(p)$, where $x = \iota(b) + \sigma(p)$. Note that such a ϕ is surjective. It is also well-defined because $\phi : \iota\beta(m) - \sigma j(m) \mapsto \iota\beta(m) + \iota f j(m) - \sigma j(m) = \iota\beta'(m) - \sigma j(m)$, and it is injective because if $\phi(b, p) = 0 \in X'$, then $\phi(b, p) = (\beta'(m), -j(m))$ for some $m \in M$, which means $(b - f(p), p) = (\beta'(m), -j(m))$, so $p = -j(m)$, and $b = \beta'(m) - f j(m) = \beta(m)$, so $(b, p) = (\beta(m), -j(m))$. Thus, different lifts β' produce the same isomorphism class of extensions. Note what we've effectively done is defined a well-defined map from

$$\Gamma : \mathrm{Ext}^1(A, B) \to \{ \text{ isomorphism classes of extensions of A by B } \}$$

so that $\Theta \circ \Gamma$ is identity on $\mathrm{Ext}^1(A, B)$, thereby showing surjectivity of Θ.

To show injectivity of Θ, it suffices to show that $\Gamma \circ \Theta$ is identity on the collection of isomorphism classes of extensions of A by B. Suppose we have some isomorphism class ξ represented by extension $0 \to B \to T \to A \to 0$. Suppose that $\Theta(\xi) = x \in \mathrm{Ext}^1(A, B)$. Then choosing a lift β, we have the diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r, "j"] \arrow[d, "\beta"'] & P \arrow[r] \arrow[d, "\sigma"] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r, "\iota"'] & X \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

Note by the projectivity of P, there is a lift $P \to T$ of the surjection $T \to A$, so that $P \to A$ factors through $P \to T$. Since the composition of $M \to P \to T$ is 0, there is also a map $M \to B$ so we get the diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta_T"'] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & T \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

We claim that T is the pushout of $M \to P$ and $\beta_T : M \to B$. If we take the pushout, we will obtain X_T, and by universal property a unique map $X_T \to T$

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta_T"'] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] \arrow[d, "\mathrm{id}"'] & X_T \arrow[r] \arrow[d, "\exists!"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & T \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

But by the five lemma, the map $X_T \to T$ must be an isomorphism. Then since T is a pushout, and since β_T and β are lifts of the same $x \in \mathrm{Ext}^1(A, B)$, by our previous work when studying surjectivity of Θ, we conclude that $0 \to B \to T \to A \to 0$ is isomorphic to $0 \to B \to X \to A \to 0$. This concludes injectivity.

At this point, we've established the bijective correspondence

$$\{ \text{ isomorphism classes of extensions of A by B} \} \longleftrightarrow \mathrm{Ext}^1(A, B).$$

We'd also like to show that it is an isomorphism of abelian groups. First, let's endow the former set with a group structure. The way we'll add extensions is via the Baer sum.

Suppose $0 \to B \to X \xrightarrow{\phi} A \to 0$ and $0 \to B \to X' \xrightarrow{\phi'} A \to 0$ are two extensions. Consider the pullback

$$\begin{tikzcd}
X' \times_A X \arrow[r] \arrow[d] & X \arrow[d, "\phi"] \\
X' \arrow[r, "\phi'"'] & A
\end{tikzcd}$$

Note $X'\times_A X = \{(s,t) \in X'\times X \mid \phi'(s) = \phi(t)\}$. Note that $X'\times_A X$ has 3 copies of B. Consider the following three sub R-modules of $X' \times_A X$: $B \times 0$, $0 \times B$ and $\Delta_{\text{skew}} = \{(-b, b) \mid b \in B\}$. Note all 3 are isomorphic to B. What's special about these copies of B? Well they give extensions.

- Consider the quotient $X' \times_A X/B \times 0$. One sees this is isomorphic to X, and gives back the extension
$$0 \to B \to X \to A \to 0.$$
- Consider the quotient $X' \times_A X/0 \times B$. This is isomorphic to X' and gives back the extension
$$0 \to B \to X' \to A \to 0.$$
- Consider the quotient $Q = X' \times_A X/\Delta_{\text{skew}}$. Note that under this quotient, one has that $B \times 0$ becomes identified with $0 \times B$. So there is an injection $0 \to B \to Q$ where $b \mapsto (b, 0) \sim (0, b)$. What's the quotient? Well under Q/B we have (s,t) such that $\phi'(s) = \phi(t)$. But note $(b, 0) \sim (0, 0) \sim (0, b)$. So we see that each element (s,t) is just represented by the image $\phi'(s) = \phi(t) \in A$. So we obtain a short exact sequence
$$0 \to B \to Q \to A \to 0.$$
This is the **Baer sum** of the two extensions we started with.

One can tediously check that the Baer sum is well-defined with respect to isomorphism classes of extensions. The identity element is given by the trivial extension
$$0 \to B \to B \oplus A \to A \to 0.$$
Indeed, given any other extension $0 \to B \to X \xrightarrow{\phi} A \to 0$, the resulting Baer sum is the extension
$$0 \to B \to Q \to A \to 0$$
where $Q = \frac{X\times_A(B\oplus A)}{\Delta_{\text{skew}}}$. Note $X \times_A (B \oplus A) = \{[x, (b, a)] \mid \phi(x) = a\}$. We have $\Delta_{\text{skew}} = \{[b, (-b, 0)]\}$. Note every element of Q can then be represented as $[x, (0, a)]$ where $\phi(x) = a$. From this one can deduce that the Baer sum is isomorphic to the extension $0 \to B \to X \xrightarrow{\phi} A \to 0$ we started with. On the other hand, given $0 \to B \xrightarrow{\iota} X \xrightarrow{\phi} A \to 0$, the inverse extension is the extension
$$0 \to B \xrightarrow{-\iota} X \xrightarrow{\phi} A \to 0.$$
Indeed, the Baer sum in this case will be
$$0 \to B \to Q \to A \to 0,$$
where $Q = \frac{X\times_A X}{\Delta_{\text{skew}}}$, where $X \times_A X = \{(s,t) \mid \phi(s) = \phi(t)\}$ and $\Delta_{\text{skew}} = \{(b, b) \mid b \in B\}$. Then there is a splitting $0 \to A \to Q \to A$, since one can define $A \to Q$ where $a \mapsto (s, s)$ where $s \in X$ such that $\phi(s) = x$. Since we're in an abelian category, the splitting implies the extension $0 \to B \to Q \to A \to 0$ is isomorphic to the trivial extension.

Now that we've endowed the set of isomorphism classes of extensions with a group structure, we show that

$$\Theta : \{ \text{ isomorphism classes of extensions of A by B}\} \to \mathrm{Ext}^1(A,B)$$

is a group homomorphism. Suppose ξ and ξ' are isomorphism classes of extensions, represented by extensions $0 \to B \to X \to A \to 0$ and $0 \to B \to X' \to A \to 0$, respectively. Let the Baer sum $\xi + \xi'$ be represented by $0 \to B \to Q \to A \to 0$. With respect to

$$0 \to M \to P \to A \to 0$$

where P is projectivce R-module, we have that ξ and ξ' can be thought of as a part of the diagrams

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta"'] & P \arrow[r] \arrow[d, "\tau"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & X \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

and

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta'"'] & P \arrow[r] \arrow[d, "\tau'"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & X' \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

We claim the Baer sum will admit the diagram

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta+\beta'"'] & P \arrow[r] \arrow[d, "\tau_Q"'] & A \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & B \arrow[r] & Q \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

Indeed, $\tau_Q : P \to Q$ is the map desending from the unique map $P \to X' \times_A X$, which exists by universal property of pullback since $P \xrightarrow{\tau} X \to A$ and $P \xrightarrow{\tau'} X' \to A$ are equal. Since the composition $M \to P \xrightarrow{\tau_Q} Q$ is 0, there is a map $M \to B$, which we claim is $\beta + \beta'$. Note τ_Q sends $p \in P$ to the equivalence class of $(\tau'(p), \tau(p))$ in Q, and note $B \to Q$ is induced by $b \mapsto (b, b) \in X' \times X$. Note $M \to P \xrightarrow{\tau_Q} Q$ is given by $m \mapsto (\tau'(m), \tau(m)) = (\beta'(m), \beta(m)) \sim (\beta'(m) + \beta(m), 0)$, so we see that $\beta + \beta'$ makes the square commute. This implies that the Baer sum maps to the element $\delta(\beta + \beta') = \mathrm{Ext}^1(A,B)$, and since the connecting homomorphism is a group homomorphism, we have $\delta(\beta + \beta') = \delta(\beta) + \delta(\beta')$ in $\mathrm{Ext}^1(A,B)$, showing that Θ is a group homomorphism and thus an isomorphism. This concludes the proof of Proposition 22.1.

Now that we've proven Proposition 22.1 in any category of R-modules, by the Freyd-Mitchell embedding theorem, we know that the correspondence holds for any small abelian category with enough projectives and injectives.

Corollary 22.2. *Let X be a scheme. Note $Coh(X)$ is a small abelian category with enough projectives and injectives. Let $\mathcal{F}$ and $\mathcal{G}$ be coherent sheaves on X. Then there is a group isomorphism between*

$$Ext^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G}) \cong \{0 \to \mathcal{G} \to \mathcal{M} \to \mathcal{F} \to 0\}/\sim$$

We can go slightly further.

Proposition 22.3. *Let X be a $\mathbb{C}$-scheme. Let $\mathcal{F},\mathcal{G} \in Coh(X)$. Then the bijective correspondence*

$$Ext^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G}) \cong \{0 \to \mathcal{G} \to \mathcal{M} \to \mathcal{F} \to 0\}/\sim$$

is an isomorphism of $\mathbb{C}$-vector spaces.

Proof. Note that the Hom sets of $Coh(X)$ are $\mathbb{C}$-vector spaces, since X is a $\mathbb{C}$-scheme. Then $\mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ is a $\mathbb{C}$ vector space. Suppose we have some element $x \in \mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ and, with respect to the short exact sequence

$$0 \to M \to P \to \mathcal{F} \to 0$$

where P is a projective, let $\beta \in \mathrm{Hom}_{\mathcal{O}_X}(M,\mathcal{G})$ be a lift of $x \in \mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$. Recall that in the bijective correspondence, x maps to the extension given by pushout:

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta"'] & P \arrow[r] \arrow[d] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & \mathcal{G} \arrow[r] & X \arrow[r] & \mathcal{F} \arrow[r] & 0
\end{tikzcd}$$

Let $\lambda \in \mathbb{C}^\times$. Then $\lambda \cdot x \in \mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ lifts to $\lambda \cdot \beta \in \mathrm{Hom}_{\mathcal{O}_X}(M,\mathcal{G})$. We see that the corresponding extension is

$$\begin{tikzcd}
0 \arrow[r] & M \arrow[r] \arrow[d, "\beta"'] & P \arrow[r] \arrow[d] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & \mathcal{G} \arrow[r, "j"] \arrow[d, "\cdot\lambda"'] & X \arrow[r] \arrow[d, "\mathrm{id}"] & \mathcal{F} \arrow[r] \arrow[d, "\mathrm{id}"] & 0 \\
0 \arrow[r] & \mathcal{G} \arrow[r, "\frac{j}{\lambda}"'] & X \arrow[r] & \mathcal{F} \arrow[r] & \bullet
\end{tikzcd}$$

And if we have $x, x' \in \mathrm{Ext}^1_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ with lifts $\beta,\beta' \in \mathrm{Hom}_{\mathcal{O}_X}(M,\mathcal{F})$, then $\lambda \cdot (x + x') = \lambda \cdot x + \lambda \cdot x'$ since the former corresponds to the extension given by $\lambda \cdot (\beta + \beta')$ which corresponds to the Baer sum of the extensions corresponding to $\lambda \cdot \beta$ and $\lambda \cdot \beta'$. □

The ideas here extend to Ext^n for $n \geq 2$.

Proposition 22.4. *Suppose A and B are R-modules and let $n \geq 2$. Then $Ext^n(A,B)$ is in bijective correspondence with extensions*

$$\xi : 0 \to B \to X_n \to \cdots \to X_1 \to A \to 0$$

with equivalence relation $\xi \sim \xi'$ if there exists a diagram

$$\begin{tikzcd} 0 \arrow[r] & B \arrow[r] \arrow[d, "id"'] & X_n \arrow[r] \arrow[d] & \cdots \arrow[r] & X_1 \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, "id"] & 0 \\ 0 \arrow[r] & B \arrow[r] & X'_n \arrow[r] & \cdots \arrow[r] & X'_1 \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

In particular, by the Freyd-Mitchell embedding theorem, we have the above correspondence for small abelian categories with enough projectives and enough injectives. Sometimes in the literature, if an abelian category does not have enough projectives and injectives, one directly defines Ext^n to be these equivalence classes of relations.

The correspondence in Proposition 22.4 remains a group isomorphism. See [Wei94, Chapter 3.4, Page 79] for how the Baer sum in these cases of $n \geq 2$ is defined, as well as for references with details on the proof of the above correspondence.

22.2. **Automorphisms of coherent sheaf as group scheme.** Let $(X, \mathcal{O}_X(1))$ be a projective $\mathbb{C}$-variety. Let $F \in Coh(X)$.

Proposition 22.5. *The moduli functor $T \mapsto \mathrm{Hom}_{\mathcal{O}_{T\times X}}(\pi_X^* F, \pi_X^* F)$ is represented by the affine scheme $\mathrm{Hom}_{\mathcal{O}_X}(F, F)$.*

Proof. Let π_T and π_X denote the canonical projections from $T \times X$. Since F is coherent, we have ([Bel])

$$\pi_X^* \mathcal{H}(F, F) \cong \mathcal{H}(\pi_X^* F, \pi_X^* F).$$

Now we have that

$$\mathrm{Hom}_{\mathcal{O}_{T\times X}}(\pi_X^* F, \pi_X^* F) \cong H^0(T \times X, \mathcal{H}(\pi_X^* F, \pi_X^* F)) \cong H^0(T \times X, \pi_X^* \mathcal{H}(F, F))$$

$$\cong H^0(T, \pi_{T,*}\pi_X^* \mathcal{H}(F, F)) \cong H^0(T, \mathcal{O}_T \otimes Hom_{\mathcal{O}_X}(F, F)) \cong H^0(T, \mathcal{O}_T) \otimes \mathrm{Hom}_{\mathcal{O}_X}(F, F) \cong \mathrm{Hom}_{\mathcal{O}_X}(F, F)(T),$$

which shows the moduli functor is representable by $\mathrm{Hom}_{\mathcal{O}_X}(F, F)$. $\square$

Proposition 22.6. *The moduli functor*

$$T \mapsto \{\phi \in \mathrm{Hom}_{\mathcal{O}_{T\times X}}(\pi_X^* F, \pi_X^* F) \mid \forall t : \phi_t : F \to F \text{ iso } \}$$

is representable by $\mathrm{Aut}(F)$ and is a group scheme.

Proof. It suffices to show that $\mathrm{Aut}(F)$ defines an open subscheme of the affine scheme $H := \mathrm{Hom}_{\mathcal{O}_X}(F, F)$. Let $\phi : \pi_X^* F \to \pi_X^* F$ be the universal homomorphism, where $\pi_X^* F \in Coh(H \times X)$. Then if we take the cokernel

$$\pi_X^* F \xrightarrow{\phi} \pi_X^* F \to Q,$$

we have that since Q is coherent, then the support of Q is closed. Then since X is proper, note that under the projection $H \times X \to H$ we have $\mathrm{Supp}\, Q$ maps to a closed subset $V \subset H$. Now we use the following crucial fact: since F is coherent, then an endomorphism is an automorphism if and only if it is a surjection. Then the loci of $t \in H$ where $\phi_t : F \to F$

is a surjection and thus an automorphism is exactly $H \setminus V$, which is open. This shows that the moduli functor in question is an open subfunctor of $\mathrm{Hom}_{\mathcal{O}_X}(F, F)$ and represented by $\mathrm{Aut}(F)$, and it's clear the moduli functor is a group scheme, so $\mathrm{Aut}(F)$ is a group scheme. □

22.3. **Some lemmas for moduli spaces of Gieseker (semi)stable sheaves construction.**

Lemma 22.7. *Let $(X, \mathcal{O}_X(1))$ be a projective scheme. Let $r \in \mathbb{Z}^+$ and ρ be a reduced Hilbert polynomial of degree d. Then for all sufficiently large m, the following are equivalent:*

- *F is (semi)stable with Hilbert polynomial $r \cdot \rho$.*
- *$r \cdot \rho(m) \leq h^0(F(m))$ and $h^0(F'(m))(\leq)\alpha_d(F') \cdot \rho(m)$ for all $F' \subset F$ with $0 < \alpha_d(F') < r$.*
- *$\alpha_d(Q)\rho(m)(\leq)h^0(Q(m))$ for all quotient sheaves $F \to Q$ where $0 < \alpha_d(Q) < r$,*

and we have equality in the latter two if and only if F' is maximal destablising subsheaf / maximal destablising quotient.

Recall that if F is (semi)stable if and only if $\rho(F')(\leq)\rho(F)$ for all $0 < \alpha_d(F') < \alpha_d(F)$. Then for each F', we have for sufficiently large m we will have $\rho(F')(m)(\leq)\rho(F)(m) \implies h^0(F'(m))(\leq)\alpha_d(F')\rho(m)$. So in a specific instance of semistable F and a subsheaf F', for sufficiently large m we can obtain a clear bound on the number of global sections of $F'(m)$ in terms of $\alpha_d(F')$ and the reduced Hilbert polynomial at hand. The point of lemma 22.7 is that we can uniformly pick this sufficiently large m across all $F' \subset F$, and across all F with $\alpha_d(F) = r$ and $\rho(F) = \rho$.

Lemma 22.8. *If $F \in Coh(X)$ has dimension d which can be deformed to a pure sheaf, then there exist homomorphism*

$$\phi : F \to E$$

where $\ker \phi = T_{d-1}(F)$ and E is pure with $P(E) = P(F)$.

Note that a priori $F/T_{d-1}(F)$ is pure, so $P(F/T_{d-1}(F)) \leq P(F), P(E)$. But the point is that one can find E which remains pure and $P(F) = P(E)$.

22.4. **A question of Maestrano and Simpson.**

Question 22.9. *What is the smallest value of d for which, for a very general hypersurface $X \subset \mathbb{P}^3$ of degree d and for some c_2, there are two or more components of the moduli space of vector bundles on X?*

In $d \leq 4$, fall into situation of K3 or del Pezzo surfaces. Maestrano and Simpon analyze the $d = 5$ case. They show that the moduli space of stable rank 2 vector bundles of degree one on a very general quintic hypersurface is irreducible for all $c_2 \geq 4$, and empty otherwise. `https://arxiv.org/abs/1302.3736`

In the $d = 6$ case, they show when $c_2 = 11$, the moduli space of stable rank 2 vector bundles of degree one has at least two irreducible components `https://msp.org/pjm/2018/293-1/pjm-v293-n1-p04-s.pdf`. But in this paper, apparently for $d = 6$ and $c_2 \geq 27$ it is irreducible `https://arxiv.org/abs/2003.07036`.

The question itself is resolved. But probably can ask what happens in $d = 7$ case and try to obtain effective bounds on the c_2.

Another thing to do is study jumping loci. Consider vector bundle E on $\mathbb{P}^2$. Any line L, have $E|_L$ grothendieck decomposition. For general L restriction is well-balanced, but we can ask for what special values of L the decomposition is unbalanced. This gives a subvariety of $(\mathbb{P}^2)^\vee$ of those points $[L] \in (\mathbb{P}^2)^\vee$ where unbalanced. For $r = 2$ and $c_1 = 0$ it is a divisor. The map $E \mapsto D_E$ is called a Barth morphism.

Jumping curve construction closely related to strange duality conjecture for sheaves on $\mathbb{P}^2$, in Le Potier's point of view.

Vector bundles on $\mathbb{P}^1$ classified. Then Atiyah classified vector bundles on elliptic curve `https://math.berkeley.edu/~nadler/atiyah.classification.pdf`. Then Barth classified vector bundles on $\mathbb{P}^2$ (`https://eudml.org/doc/142496`). natural generalization: looking at hypersurfaces $X \subset \mathbb{P}^3$ of degree 2. General hyperplane cuts out a plane conic $\cong \mathbb{P}^1$ on X. Can again study Grothendieck decomposition of $E|_C$. Sukmoon huh `https://arxiv.org/pdf/0910.2053` shows that the set of planes in $\mathbb{P}^3$ which cut out conics on which decomposition jumps is hypersurface of degree $c_2(V) - 1$. Torelli type theorem, jumping hypersurfaces determine the bundle generically. Uses this to describe the moduli spaces $M_X(\mathcal{O}_X(1), c_2)$. Further directions(?):

- Look at hypersurfaces X of degree 3 in $\mathbb{P}^3$. general hyperplane cuts out an elliptic curve on X. Can one study restrictions of vector bundles to these elliptic curves? Use Atiyah's classification of vector bundles on elliptic curves. Something about semistability of indecomposable vector bundles degree 0 on elliptic curves here `https://webdoc.sub.gwdg.de/ebook/dissts/Duisburg/Lekaus2001.pdf`.
- Look at Fano 3 fold, general antincaonical K3 surface. Infintely many smooth rational curves on K3 surface. Study vector bundle restrictions in this case?

23. Interesting references

(1) `https://www.numdam.org/item/AST_1982__96__1_0.pdf`
(2) Strange Duality, resolved for curves. Work ongoing in higher dimensions?
   - Beauville strange duality conjectures `https://math.univ-cotedazur.fr/~beauvill/pubs/berkeley.pdf` `https://www.numdam.org/item/10.5802/aif.2216.pdf`
   - Belkale resolves for generic curve `https://arxiv.org/pdf/math/0602018`
   - The marian-oprea trick `https://arxiv.org/abs/1212.6533`

  – elliptic surfaces https://arxiv.org/abs/2103.16417
  – https://arxiv.org/pdf/1807.09172

(3) Local study of moduli space of vector bundles over curves https://arxiv.org/pdf/alg-geom/9405012
(4) A NEU-MIT seminar https://gauss.math.yale.edu/~il282/MSK3.html
(5) Connections of moduli sheaves with arithmetic questions, Brauer groups, twisted sheaves, etc. https://arxiv.org/pdf/1708.00504
(6) Moduli of VB with respect to Fano 3 folds, anticanonial K3, and curves: Beauville's paper https://arxiv.org/pdf/1906.03594 and Tyurin's paper starting page 176/177 of https://univerlag.uni-goettingen.de/bitstream/handle/3/isbn-978-3-938616-74 tyurin_1.pdf;jsessionid=34CAEADC198CAA8C974065B027D89E6A?sequence=1
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